

MEDICAL SURGICAL NURSING

Disorders of the Cardiovascular System

By Afework E.

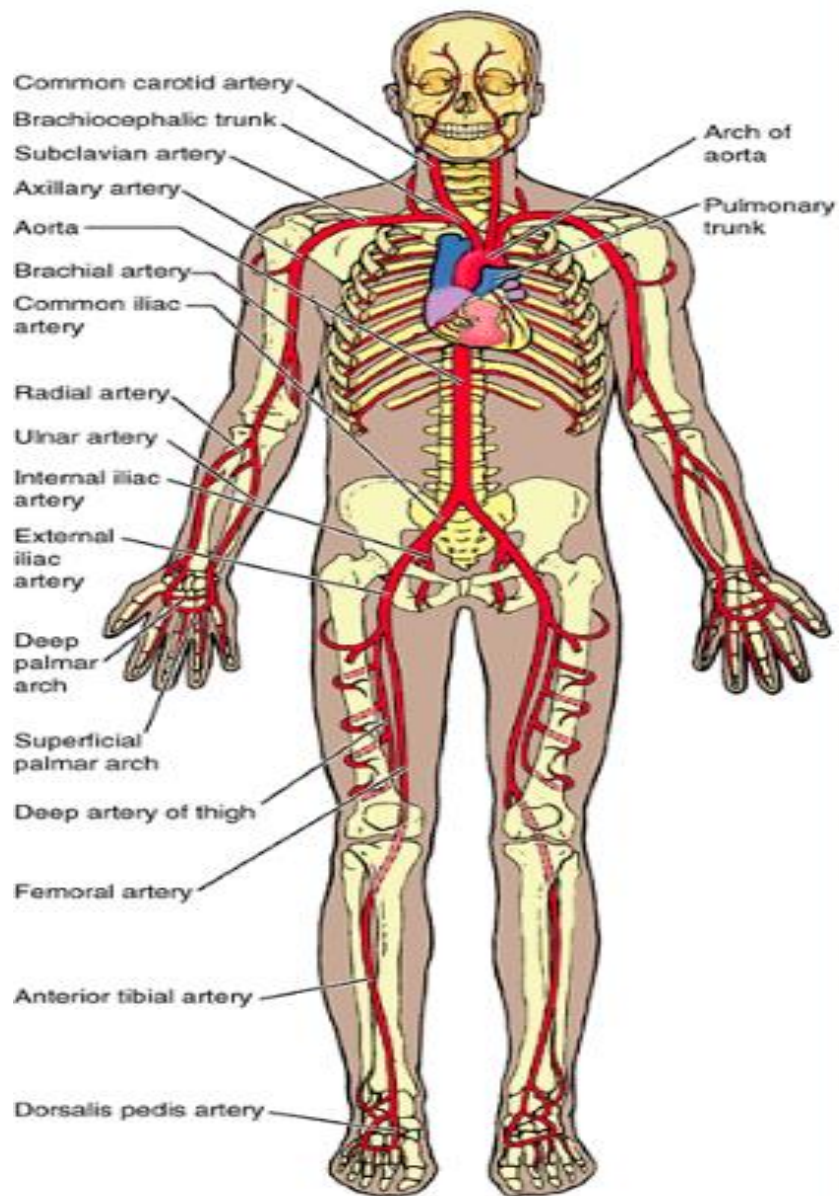
(BSN, HD, MSN, Assistant Professor)

Anatomy & physiology of circulatory system

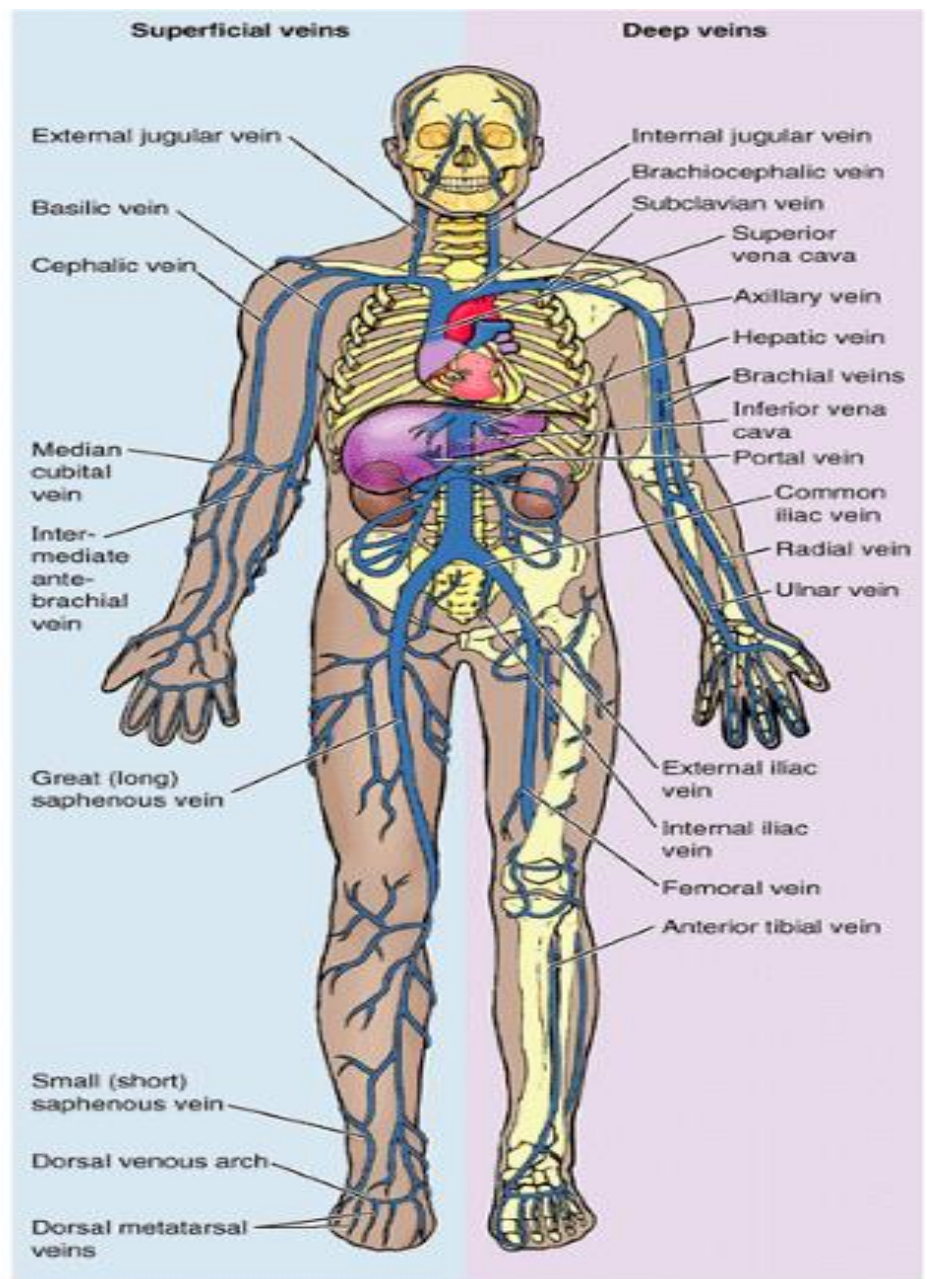
- ✓ Acts as a **transport service** for the cells
- ✓ Contains two fluids: blood & lymph (made up of two systems)

Major Components of the Circulatory System

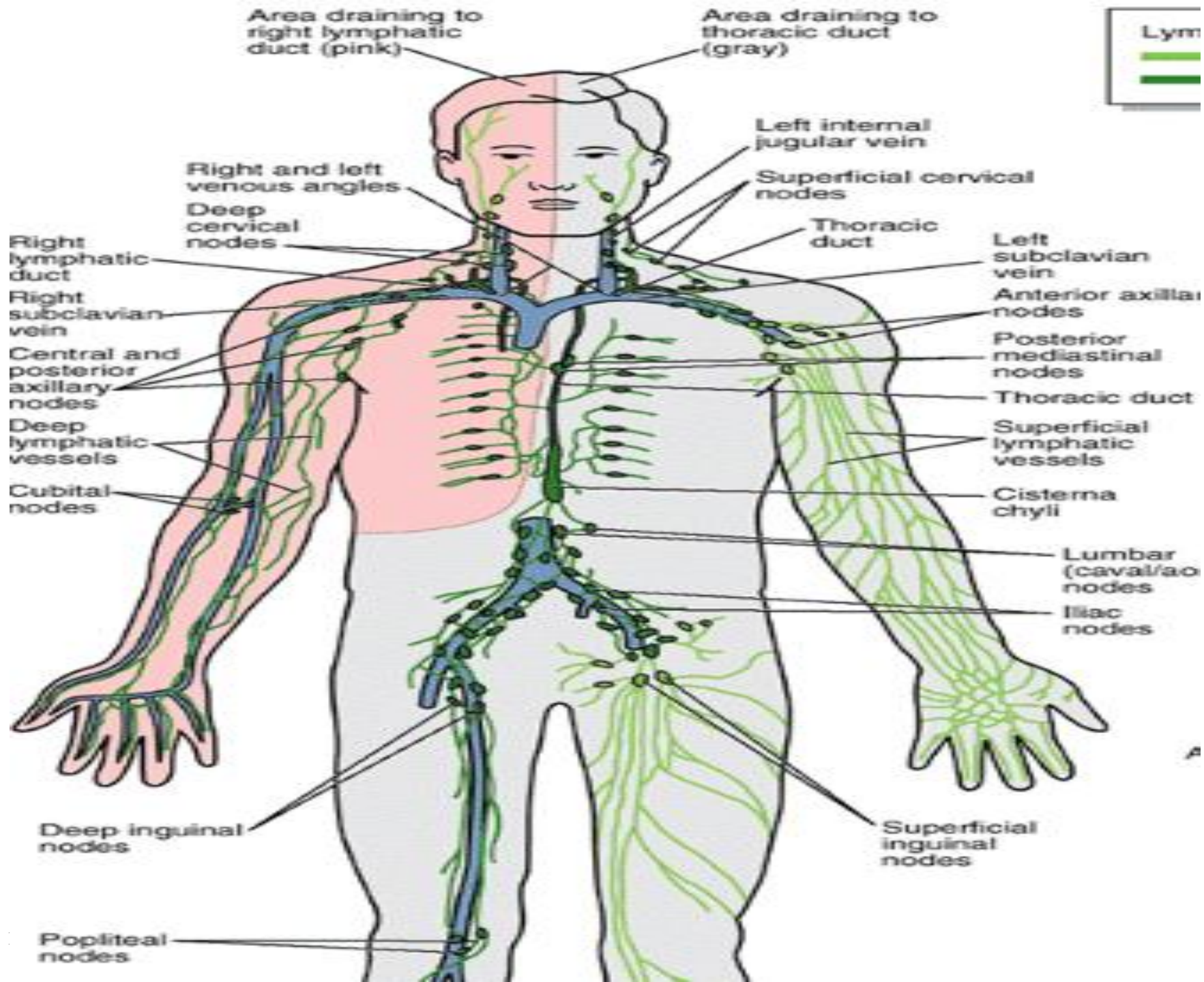
- **Cardiovascular System:** consist of **heart** and **blood vessels** (arteries, veins and capillaries) and **blood**.
- **Lymphatic System:** consists of the **lymph, lymph nodes** and **lymph vessels**.



Principal arteries



Principal veins



Division of the Circulation

➤ In the CVS, blood passes through two (double) circulations:

- **Systemic circulation**
- **Pulmonary circulation**

➤ **Systemic circulation:**

- Starts in the LV → Aorta → Systemic arteries → Systemic capillaries → Veins → SVC & IVC → ends in RA

➤ **Pulmonary circulation:**

- Starts in the RV → Pulmonary trunk → Pulmonary arteries → Pulmonary capillaries → Pulmonary veins → ends in the LA.

LA=left atrium

LV=left ventricle, SVC=superior vena cava

IVC=inferior vena cava

The Heart

- Heart is the **hollow, muscular organ** that plays a central pumping role
- Vertically divided into left & right sides by a structure called **septum**
- Composed of 4-chambers: **2 atria** and **2 ventricles**
- **Has 4 valves**: 2 AV valves (TCV &BCV) and 2 SLV (PV &AV)
- Size: Approximately equivalent to clenched fist
- Weight: 250 to 320 grams in average adults (250 in F,320 in M)
- **Located in the mediastenum**

...

➤ **Heart is made up of 3 distinct layers:**

1 - Endocardium - innermost layer; epithelial tissue that lines the entire circulatory system

2 - Myocardium - thickest layer; consists of cardiac muscle

3 - Epicardium - thin, external membrane around the heart

➤ **Pathway of blood through the heart and lungs**

- Left atrium ➡ bicuspid valve ➡ left ventricle ➡ aortic semilunar valve ➡ aorta ➡ systemic circulation ➡ vena cava ➡ right atrium ➡ tricuspid valve ➡ right ventricle ➡ pulmonary semilunar valve ➡ pulmonary arteries ➡ lungs ➡ pulmonary veins ➡ left atrium

Heart sound

- Heart sounds (lubb-dubb) associated (usually) with valve closure. Measured by stethoscope.

1. First heart sound(S_1)

- ➔ Due to closure of AV valves
- ➔ is always audible, has a LUBB-sound

2. Second heart sound(S_2):

- ➔ Occur at end of ejection (at onset of diastole)
- ➔ Closure of Aortic & Pulmonic valves (Semilunar valves)
- ➔ Is always audible, has DUBB-sound
- ➔ Duration of Lubb is longer than that of Dubb

...

3. Third heart sound (S₃)

- ➔ S3: an abnormal heart sound detected **early in diastole** as resistance is met to blood entering either ventricle; most often due to volume overload associated with heart failure
- ➔ is **audible in children** and in **adults during exercise**

4. Fourth heart sound(S₄) (occasional)

- ➔ S4: an abnormal heart sound detected **late in diastole** as resistance is met to blood entering either ventricle during atrial contraction; most often caused by hypertrophy of the ventricle
- ➔ is audible very rarely

Blood supply to the heart

- The heart receives arterial blood from the **coronary artery** which is the branch of **ascending aorta**.
- Resting coronary blood flow = 250 ml/min, 5% CO

Conduction system of the heart

- The **specialized heart cells** of the cardiac conduction system methodically generate and coordinate the transmission of electrical impulses to the myocardial cells
- **Three physiologic** characteristics of the cardiac conduction cells account for this coordination:
 - **Automaticity**: ability to initiate an electrical impulse
 - **Excitability**: ability to respond to an electrical impulse
 - **Conductivity**: ability to transmit an electrical impulse from one cell to another
 - **Elasticity**

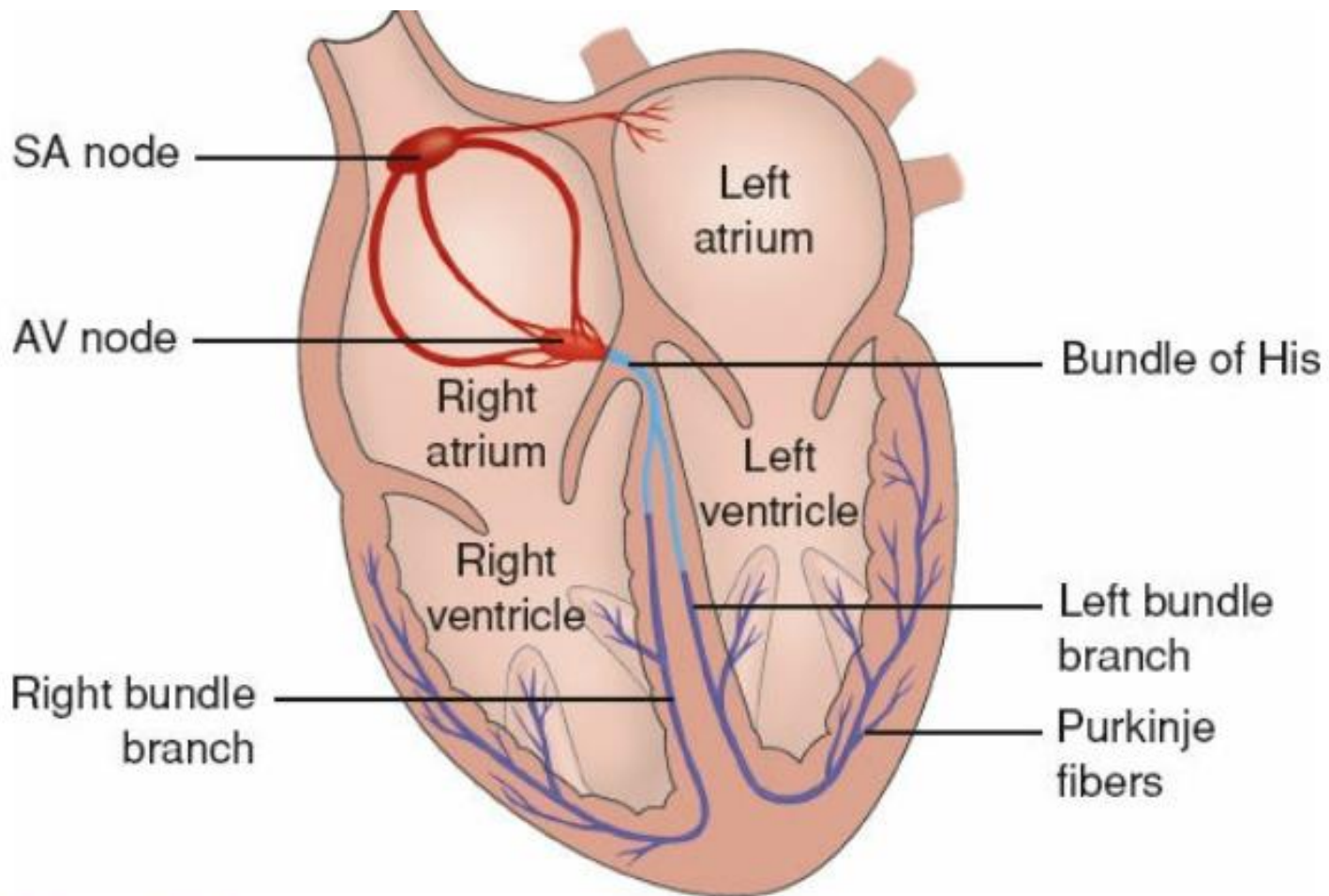


Figure 25-3 • Cardiac conduction system.

Sinoatrial node(SA)

- **Primary pacemaker** of the heart
- Has inherent firing rate of **60 to 100 impulses per minute**
- Conduct impulses along the myocardial cells causing electrical stimulation/ depolarization/ **atrial systole**
- Conduct to atrioventricular (AV) node

AV node

- ✓ Coordinate the incoming electrical impulses from the atria and **after a slight delay (allowing the atria time to contract and complete ventricular filling)** relays the impulse to the ventricles.
- ✓ Has inherent firing rate of **40 to 60 impulses per minute**

Bundle of His

- Divides into the **right and left bundle branch**
- Impulse travel through the bundle branches to reach the terminal point in the conduction system, called **Purkinje fibers**
- Then myocardial cells are stimulated → **ventricular contraction**
- **If the SA node malfunctions**, the **AV node generally takes over** the pacemaker function of the heart at its inherently lower rate.
- Should both the SA and the AV nodes fail in their pacemaker function, **a pacemaker site in the ventricle will fire** at its inherent bradycardic rate of **30 to 40 impulses per minute**

Important terms

- **Ventricular volumes:** The volume of blood in the ventricles
- **End diastolic volume (EDV):** The volume of blood in the ventricle at the end of ventricular diastole (relaxation phase)

EDV = about 120 ml

- **End systolic volume (ESV):** The volume of blood that remains in the ventricle at the end of ventricular systole (contraction phase).

ESV = about 50 ml

- **Stroke volume (SV):** the volume of blood ejected from the ventricle during ventricular systole.

$SV = EDV - ESV,$ about 70 ml

...

- **Ejection fraction:** the blood proportion that the amount ejected from ventricle to enters the ventricles during diastole.

$$EF = SV/EDV = \text{about } 60\%$$

- **Cardiac output:** the volume of blood ejected from the heart per minute.

$$CO = SV \times HR = 5\text{-}6 \text{ L/min}$$

- **Cardiac reserve:** is the difference between maximal exercise and resting CO

- CO = 5-6 liters per minute at rest
- CO = 20-40 liters per minute at maximum exercise

Functions of the Circulatory System

- **Transportation**
 - **Respiratory**: red blood cells carry oxygen. CO₂ is carried by blood to the lungs for elimination
 - **Nutritive**
 - **Excretory**: capillaries in kidney
- **Regulation**: Circulating blood helps maintain homeostasis of all body fluids
 - Blood helps **regulate pH** through the use of buffers.
 - It also helps **adjust body temperature** through the heat absorbing and coolant properties of the water
 - The blood **carries hormones** and **other regulatory molecules** from their site of origin to distant target tissues.
- **Protection**: **Blood can clot**, which protects against its excessive loss from the cardiovascular system after an injury. In addition, **its white blood cells** protect against disease **by carrying on phagocytosis**

Assessment

A. Health history: The following are the most common signs and symptoms of CVD :

- **Chest pain or discomfort** (angina pectoris, ACS, dysrhythmias, valvular heart disease)
- **Pain or discomfort** in other areas of upper body, including one or both arms, back, neck, jaw, or stomach (ACS)
- **Shortness of breath or dyspnea** (ACS, cardiogenic shock, HF, valvular heart disease)
- **Peripheral edema, weight gain, abdominal distention** due to enlarged spleen and liver or ascites (HF)
- **Palpitations** (tachycardia from a variety of causes, including ACS, caffeine or other stimulants, electrolyte imbalances, stress, valvular heart disease, ventricular aneurysms)



- **Unusual fatigue**, sometimes referred to as vital exhaustion (an **early warning symptom** of ACS, HF, or valvular heart disease, characterized by feeling unusually tired or fatigued, irritable, and dejected)
- **Dizziness, syncope, or changes in level of consciousness** (cardiogenic shock, cerebrovascular disorders, dysrhythmias, hypotension, postural hypotension, vasovagal episode)

B. Past Health, Family, and Social History

- Assess all patients with cardiovascular symptoms for coronary artery disease, regardless of age.
- Inquire about a **family history of sudden death** or unexplained death.
- Assess for signs and symptoms of **hyperlipidemias**
- Obtain **previous an electrocardiogram and an echocardiogram** results
- Assess for **muscular weakness**
- Assess for episodes of shortness of breath, dizziness, or palpitations.
- **Review laboratory data** for abnormal values.
- Gather dietary history.
- Assess for secondary risk factors (e.g., **diet, smoking, overweight, high stress, and alcohol use**)

C. Physical Assessment

- **The heart as a pump** (reduced pulse pressure, displaced PMI from fifth intercostal space midclavicular line, gallop sounds, murmurs)
- **Atrial and ventricular filling volumes and pressures** (**elevated jugular venous distension**, **peripheral edema**, ascites, crackles, **postural changes in BP**)
- **Cardiac output** (**reduced pulse pressure**, hypotension, tachycardia, reduced urine output, lethargy, or disorientation)
- **Compensatory mechanisms** (peripheral vasoconstriction, tachycardia).
- **Changes in level of consciousness** and mental status may be attributed to inadequate perfusion of the brain from a compromised cardiac output or thromboembolic event (stroke).
- Patients are observed for **signs of distress**, which include pain or discomfort, shortness of breath, or anxiety.

• • •

- General appearance
- Blood Pressure, Pulse pressure
- **Pulse rate, rhythm, amplitude, deficit**
- Jugular Venous Pulsations and pressure
 - ❖ Right-sided heart function can be estimated by observing the pulsations of the jugular veins of the neck, which reflects central venous pressure (CVP).
 - ❖ CVP is the pressure in the right atria or the right ventricle at the end of diastole.
- Heart **Inspection and Palpation**
- Heart Auscultation (**heart sounds**)
- Assessment of other systems: **lung, abdomen, urine out put, hepatojugular reflux**

Assessment Findings	Associated Causes and Conditions
Clubbing of the fingers or toes (thickening of the skin under the fingers or toes)	Chronic hemoglobin desaturation most often due to congenital heart disease, advanced pulmonary diseases
Cool/cold skin and diaphoresis	Low cardiac output (e.g., cardiogenic shock, acute myocardial infarction) causing sympathetic nervous system stimulation with resultant vasoconstriction
Cold, pain, pallor of the fingertips or toes	Intermittent arteriolar vasoconstriction (Raynaud disease). Skin may change in color from white, blue, and red accompanied by numbness, tingling, and burning pain
Cyanosis, central (a bluish tinge observed in the tongue and buccal mucosa)	Serious cardiac disorders (pulmonary edema, cardiogenic shock, congenital heart disease) result in venous blood passing through the pulmonary circulation without being oxygenated
Cyanosis, peripheral (a bluish tinge, most often of the nails and skin of the nose, lips, earlobes, and extremities)	Peripheral vasoconstriction, allowing more time for the hemoglobin molecules to become desaturated. It can be caused by exposure to cold environment, anxiety, or ↓ cardiac output
Ecchymosis or bruising (a purplish-blue color fading to green, yellow, or brown)	Blood leaking outside of the blood vessels Excessive bruising is a risk for patients on anticoagulants or platelet-inhibiting medications
Edema, lower extremities (collection of fluid in the interstitial spaces of the tissues)	Heart failure and vascular problems (PAD, chronic venous insufficiency, deep vein thrombosis, thrombophlebitis)

D. Electrocardiography (ECG)

- It is a **graphic recording of the electrical activity of the heart.**
- The ECG is obtained by placing disposable electrodes in standard positions on the skin of the chest wall and extremities.
- The heart's electrical impulses are recorded as a tracing on special graph paper.
- used to diagnose **dysrhythmias**, **conduction abnormalities**, and **chamber enlargement**, as well as myocardial ischemia, injury, or infarction.
- It can also suggest cardiac effects of **electrolyte disturbances** (high or low calcium and potassium levels) and the effects of antiarrhythmic medications

E. Echocardiography

- Echocardiography is a **noninvasive ultrasound** test that is used to examine **the size, shape**, and **motion of cardiac structures**.
- It is a particularly useful tool for diagnosing **pericardial effusions**, determining the etiology of heart murmurs, evaluating the function of prosthetic heart valves, determining chamber size, and evaluating ventricular wall motion.
- It involves transmission of high-frequency sound waves into the heart through the chest wall and recording of the return signals.
- The ultrasound is generated by a hand-held transducer applied to the front of the chest.
- The transducer picks up the echoes, converts them to electrical impulses, and transmits them to the echocardiography machine for display on an oscilloscope and recording on a videotape.
- An ECG is recorded simultaneously to assist with interpreting the echocardiogram.

F. Cardiac catheterization

- Is an **invasive diagnostic procedure** in which radiopaque arterial and venous catheters are **introduced into** selected blood vessels of the right and left sides of the heart.
- Catheter advancement is guided by **fluoroscopy**.
- Most commonly, the catheters are inserted **percutaneously** through the blood vessels, or via a cut down procedure if the patient has poor vascular access.
- **Pressures and oxygen saturations** in the four heart chambers are measured.
- Is used to **diagnose CAD**, **assess coronary artery patency**, and determine the extent of atherosclerosis based on the percentage of coronary artery obstruction.

G. Chet X-Ray and fluoroscopy

Chest X-Ray

- **For the size, contour, and position of the heart**
- Revels **cardiac and pericardial calcification**
- Demonstrate physiological alteration in the pulmonary circulation
- Is not help in the diagnosis of MI, but can help in the diagnosis of some complications like, CHF
- It also helps to **confirm correct placement of cardiac catheters.**

Fluoroscopy

- Allows visualization of the heart on X-ray screen.
- It shows cardiac and vascular pulsations and unusual cardiac contours.
- Useful for guiding catheter insertion during cardiac catheterization.

H. Laboratory tests

i. Lipid profile

- TC < 200 mg/dl
- HDL >60 mg/dl(40-60)
- LDL <100 mg/dl

- **Good indicator for risk of cardiovascular disease, as well as to manage medications to prevent hyperlipidemia**

ii. Complete Blood Count (CBC)

- Males
 - o Hg; 13.5–17.5 g/dl
 - o Hct; 38–54%
- Females
 - o Hg 12 -16 g/dl
 - o Hct 37–47%
- WBC 4,500–10,500 cells/mm³
- **Tests for red blood cell (hg, hct, ESR) function and white blood cell function (leukocytes) to determine ability of red blood cells to carry oxygen and white blood cell role in infection**

...

iii. Glucose and Hemoglobin A1C (HgA1C)

- o Glucose (fasting) 70–105 mg/dl
- o HgA1C < 7%

- Used to evaluate blood sugar levels and effectiveness of glucose management medications on glucose function among older adults.

iv. Serum Iron 35–165 ug/L: Plays a role in hemoglobin and red blood cell function. Low iron is diagnostic for iron-deficiency anemia.

v. Kidney Function Tests (BUN) and Creatinine

➤ Males

- o BUN 8–35 mg/dl
- o Serum CR 0.4–1.1 mg/dl

➤ Females

- o BUN 6–30 mg/dl
- o Serum CR 0.4–1.1 mg/dl

...

vi. Cardiac enzyme analysis: for acute MI

- **Creatine kinase (CK)** and its iso-enzyme are the **most specific enzymes analyzed in acute MI at the first level.**
- **Myoglobin**: An early marker of MI due to the damage myocardial tissue (within 1 to 3 hours after the onset of an acute MI). **Not specific for MI.**
- **Troponin (I or T)**: this is the **most commonly ordered and most specific of the cardiac markers.**
- It is elevated (positive) within a few hours of heart damage and **remains elevated for up to two weeks.**
- Rising levels in a series of troponin tests performed over several hours can help diagnose a heart attack.



- **Natriuretic peptides**

- Brain natriuretic peptide, also called B-type natriuretic peptide (BNP), is a protein that heart and blood vessels produce.

- **BNP helps body eliminate fluids, relaxes blood vessels and funnels sodium into urine.**

- **Results**

- ☐ **BNP levels below 100 pg/mL indicate no heart failure.**
- ☐ **BNP levels of 100-300 pg/mL suggest heart failure is present.**
- ☐ **BNP levels above 300 pg/mL indicate mild heart failure.**
- ☐ **BNP levels > 600 pg/mL indicate moderate heart failure.**
- ☐ **BNP levels above 900 pg/mL indicate severe heart failure.**

...

➤ **Coagulation studies**

- Should be done before invasive procedures such as cardiac catheterization, or cardiac surgery
- Monitor partial thromboplastin time (PTT) and prothrombin time(PT) helps to assess the cause of CAD.

I. Cardiac Stress Testing

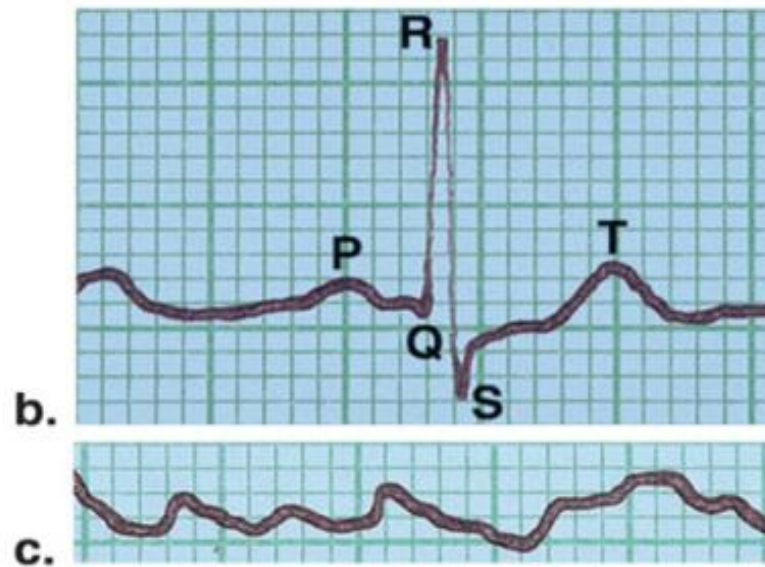
- During an exercise stress test, the patient walks on a treadmill (most common) or pedals a stationary bicycle.
- During the test, the following are monitored:
 - 2 or more ECG leads for heart rate, rhythm, and ischemic changes;
 - BP;
 - skin temperature;
 - physical appearance;
 - perceived exertion; and symptoms, including chest pain, dyspnea, dizziness, leg cramping, and fatigue.
- The test is terminated when the target heart rate is achieved or if the patient experiences signs of myocardial ischemia

Analysis of the Electrocardiogram

Electrocardiography

- Is a **graphic recording of the electrical activity of the heart.**
- is obtained by placing disposable electrodes in standard positions on the skin of the chest wall and extremities, and the heart's electrical impulse is recorded as a tracing on special graph paper.
- It has different leads but **the most common and standard lead** is 12-lead ECG, (10 electrodes, six on the chest and four on the limbs placed on the body).

- Used to diagnose **dysrhythmias, conduction abnormalities, enlarged heart chambers, myocardial ischemia or infarction**



The “PQRST”

- ➔ P wave: Atrial depolarization
- ➔ QRS : Ventricular depolarization
- ➔ T wave: Ventricular repolarization

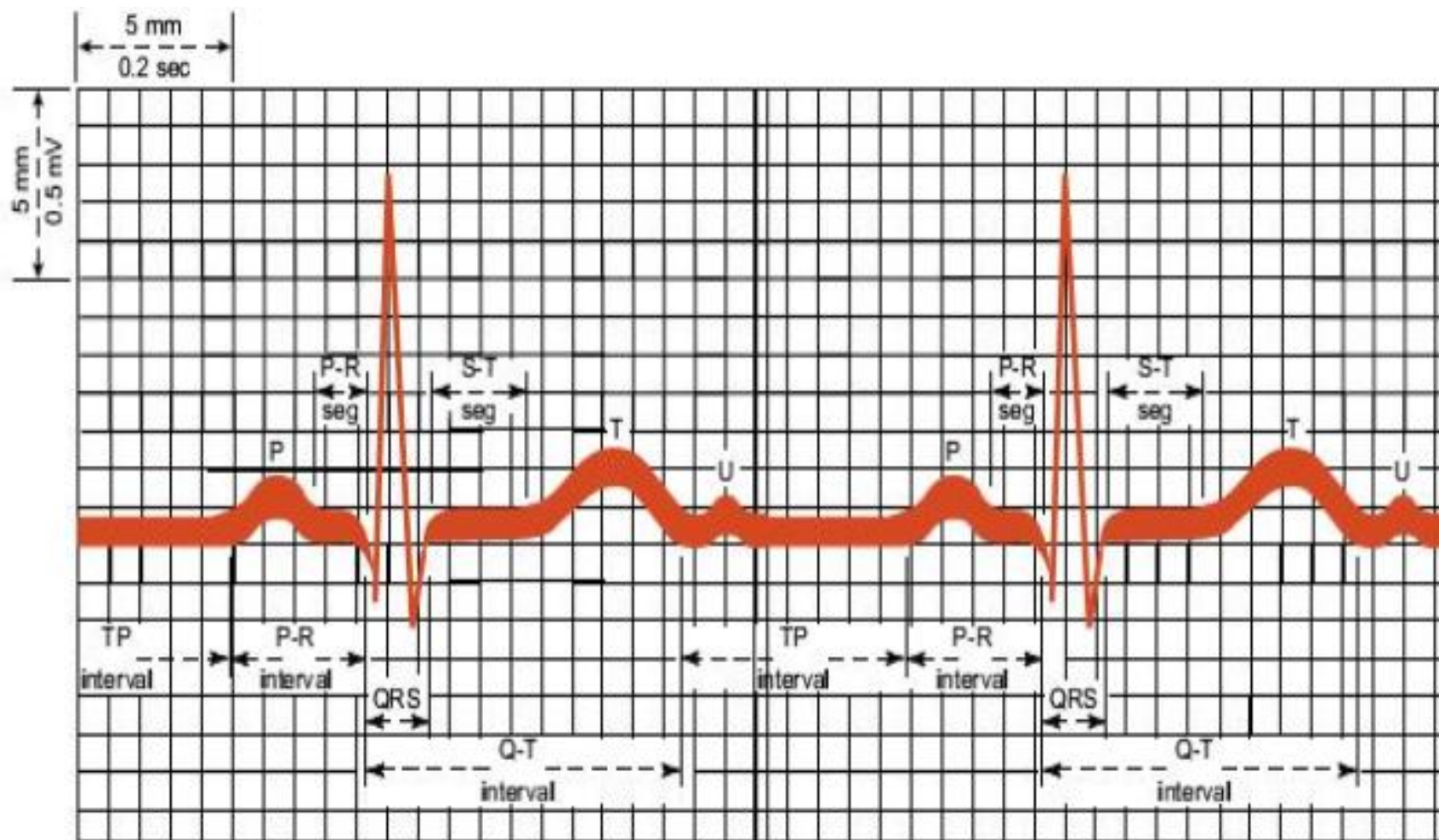
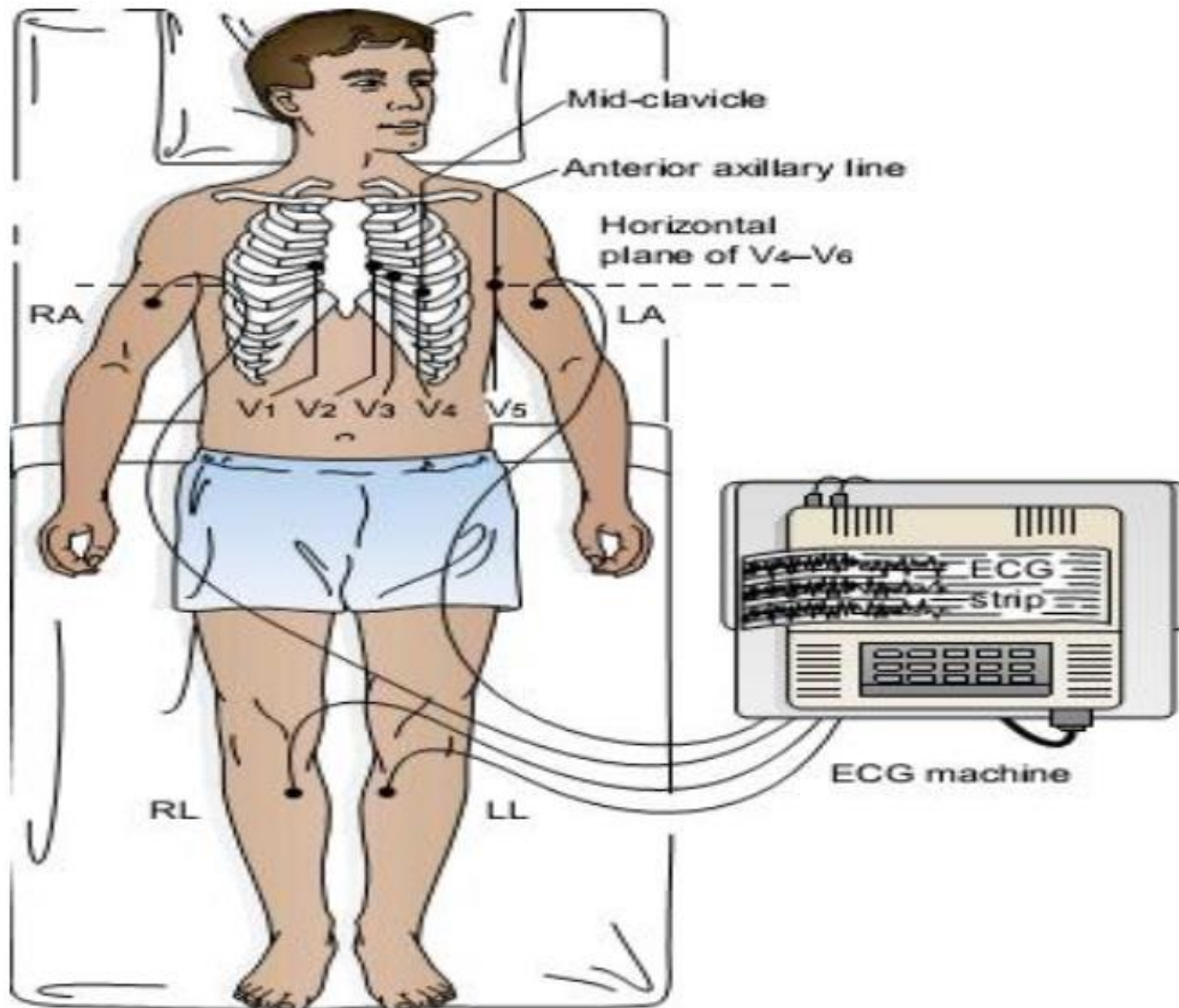


FIGURE 27-3 ECG graph and commonly measured components. Each small box represents 0.04 seconds on the horizontal axis and 1 mm or 0.1 millivolt on the vertical axis. The PR interval is measured from the beginning of the P wave to the beginning of the QRS complex; the QRS complex is measured from the beginning of the Q wave to the end of the S wave; the QT interval is measured from the beginning of the Q wave to the end of the T wave; and the TP interval is measured from the end of the T wave to the beginning of the next P wave.

...

- ECG Electrode placement: the standard left pericardial leads are:
 1. V1- 4th intercostals space, **right sternal border**.
 2. V2- 4th intercostals space **left sternal border**.
 3. V3 -Diagonally between V2 and V4
 4. V4- 5th intercostal space, **left mid clavicular**.
 5. V5- same level as V4, **anterior axillary line**.
 6. V6- same level as V4 and V5 **mid axillary line**.
 7. And four on the limbs.

...



Components of ECG

➤ ECG presents different type of cardiac events in a wave like motion. These includes:

- P-wave
- QRS- complex
- T-wave
- U-wave
- P-R interval
- S-T segment
- Q-T interval,
- T-P interval and
- P-P,
- R-R interval.

• • •

- ***P- wave***: represents the electrical impulse starting in the sinus node and spreading through the atria, therefore it represents **depolarization of the atrial muscle**.
 - Normal: height < 2.5mm and duration: < 0.11second.
- ***QRS complex***: Represents **ventricular muscle depolarization**.
 - Normal: duration <0.12 second.)
- ***T- wave***: Represents **ventricular muscle repolarization**, also called resting state, because the cells regain a negative charge.
- ***U- wave***: represents **repolarization of the purkinje fibers**, and commonly seen in-patient with hypokalemia (low potassium levels, hypertension or heart disease).
- ***P-R interval***: measured from the beginning of the P wave to the beginning of the QRS complex.
 - Normal in adult: duration 0.12 to 0.20 seconds.

• • •

- **S-T segment:** period between ventricular depolarization and repolarization.
- Lasts from the **end of QRS complex to the beginning of the T wave.**
- **Q-T interval:** represents the total time for **ventricular depolarization and repolarization.**
- Starts at the beginning of the QRS complex to the end of the T wave. (**Normal duration: 0.32 to 0.40 seconds**).
- **T-P interval:** starts from the end of T wave to the beginning of the next P wave.
- **P-P and R-R intervals:** P-P starts from the beginning of one P wave to the beginning of the next, and used to determine atrial rhythm and atrial rate.
- **R-R interval:** measured from one QRS complex to the next QRS complex, and used to determine ventricular rate and rhythm. (time between two ventricular depolarization).

Rhythm Analysis

- Step 1: Calculate rate.
- Step 2: Determine regularity.
- Step 3: Assess the P waves.
- Step 4: Determine PR interval.
- Step 5: Determine QRS duration

Example

Step 1: Calculate Rate



- Option 1

- **Count the # of R waves in a 6 second rhythm strip, then multiply by 10.**

Interpretation?

$$9 \times 10 = 90 \text{ bpm}$$

Step 2: Determine regularity

R R



- Look at the R-R distances (using a caliper or markings on a pen or paper).
- Regular (are they equidistant apart)? Occasionally irregular? Regularly irregular? Irregularly irregular?

Interpretation?

Regular

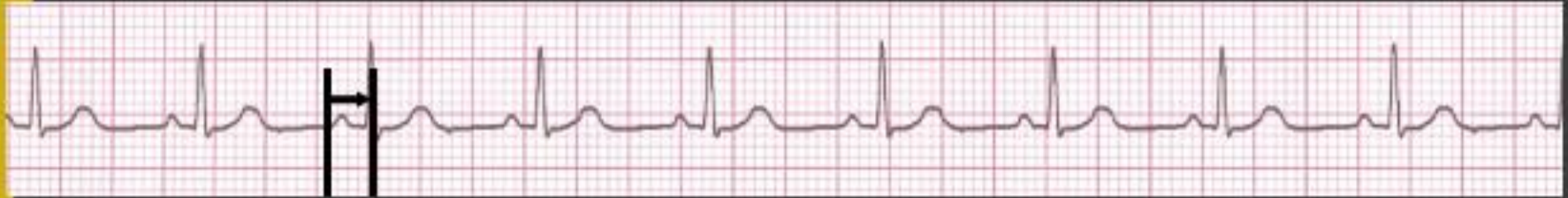
Step 3: Assess the P waves



- Are there P waves?
- Do the P waves all look alike?
- Do the P waves occur at a regular rate?
- Is there one P wave before each QRS?

Interpretation? *Normal P waves with 1 P wave for every QRS*

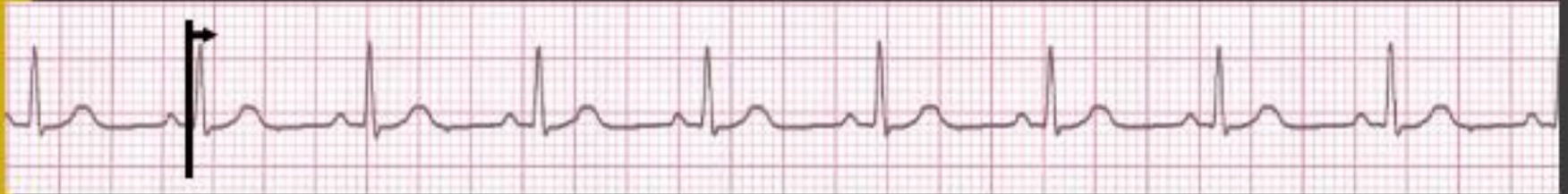
Step 4: Determine PR interval



- Normal: 0.12 - 0.20 seconds.
(3 - 5 boxes)

Interpretation? *0.12 seconds*

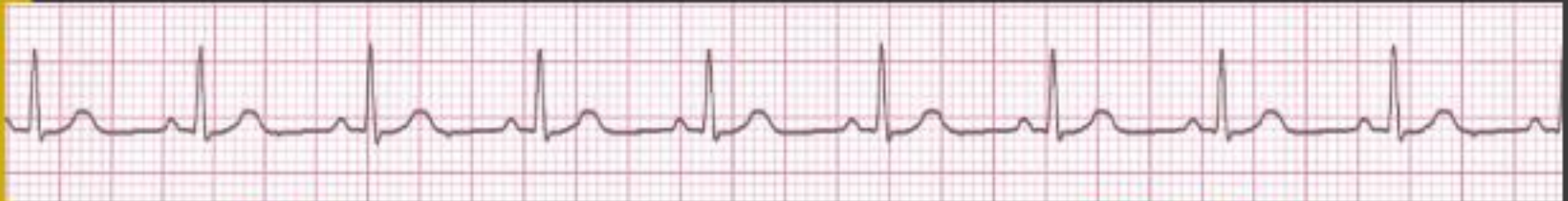
Step 5: QRS duration



- Normal: 0.04 - 0.12 seconds.
(1 - 3 boxes)

Interpretation? *0.08 seconds*

Rhythm Summary



- Rate 90-95 bpm
- Regularity regular
- P waves normal
- PR interval 0.12 s
- QRS duration 0.08 s

Interpretation? *Normal Sinus Rhythm*

Normal Sinus Rhythm

- Normal sinus rhythm occurs when the electrical impulse **starts at regular rate and rhythm in the sinus node** and travels through the normal conduction pathway
- **Characterized by:**
 - ✓ Ventricular and atrial rate : 60 to 100/ minute (adult)
 - ✓ Ventricular and atrial rhythm : regular
 - ✓ P wave is normal and consistent; always in front of QRS
 - ✓ QRS shape and duration normal
 - ✓ PR interval : consistent interval b/n 0.12 and 0.20 seconds
 - ✓ P: QRS ratio is 1:1

Dysrhythmias/ arrhythmia

- Are **disorders of the formation or conduction** (or both) of the electrical impulse within the heart.
- These disorders can cause **disturbances** of **the heart rate**, the heart **rhythm**, or both.
- May initially be evidenced by the hemodynamic effect they cause (eg, a change in conduction may change the pumping action of the heart and cause decreased blood pressure).
- They are **named according to the site of origin** of the impulse and the mechanism of formation or conduction involved
 - For example, an impulse that originates in the sinoatrial (SA) node and that has a slow rate is called sinus bradycardia.

Identifying Dysrhythmias

Sites of Origin

Sinus (SA) node

Atria

Atrioventricular (AV) node or junction

Ventricles

Mechanisms of Formation or Conduction

Normal (idio) rhythm

Bradycardia

Tachycardia

Dysrhythmia

Flutter

Fibrillation

Premature complexes

Blocks

Types of dysrhythmias

- Dysrhythmias include :
 - Sinus node
 - Atrial ,
 - Junctional and
 - Ventricular dysrhythmias

Sinus arrhythmia

- Occurs when the sinus node creates an impulse at an **irregular rhythm**; the rate usually increases with inspiration and decreases with expiration.
- Non respiratory causes include **heart disease and valvular disease**, but these are rarely seen.

Characteristics:

- *Ventricular and atrial **rate**: 60 to 100 in the adult*
- *Ventricular and atrial **rhythm: Irregular***
- *PR interval: Consistent interval between 0.12 and 0.20 seconds*
- *P wave: Normal and consistent shape; always in front of the QRS*
- *QRS shape and duration: Usually normal, but may be regularly abnormal*
- *P:QRS ratio: 1:1*

Sinus arrhythmia

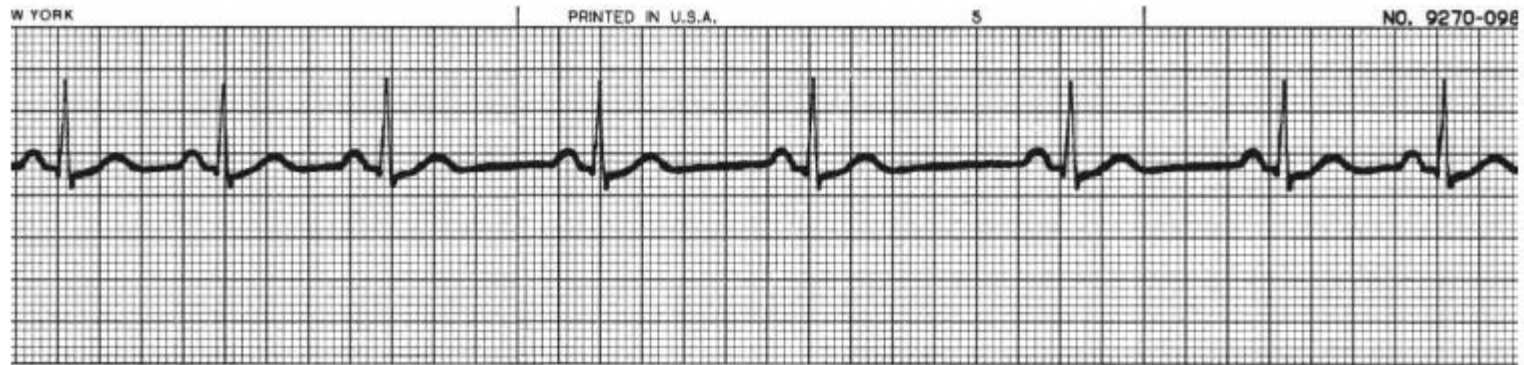


Figure 26-8 • Sinus arrhythmia in lead II.
Note irregular RR and PP intervals.

Sinus node dysrhythmias

/ Sinus bradycardia

- occurs when the sinus node creates an impulse **at a slower-than-normal rate**.
- Causes include:
 - **Lower metabolic needs** (eg, sleep, athletic training, hypothermia, hypothyroidism),
 - **Vagal stimulation** (eg, from vomiting, suctioning, severe pain, extreme emotions),
 - **Medications** (eg, calcium channel blockers, amiodarone, beta-blockers),
 - **Increased intracranial pressure**, and
 - **Myocardial infarction** (MI), especially of the inferior wall.
- The following are characteristics of sinus bradycardia
 - *Ventricular and atrial rate: **Less than 60 in the adult***
 - *Ventricular and atrial rhythm: Regular*
 - *PR interval: Consistent interval between 0.12 and 0.20 seconds*
- **Mgt: atropine, prevent vagal stimulation**

Sinus bradycardia

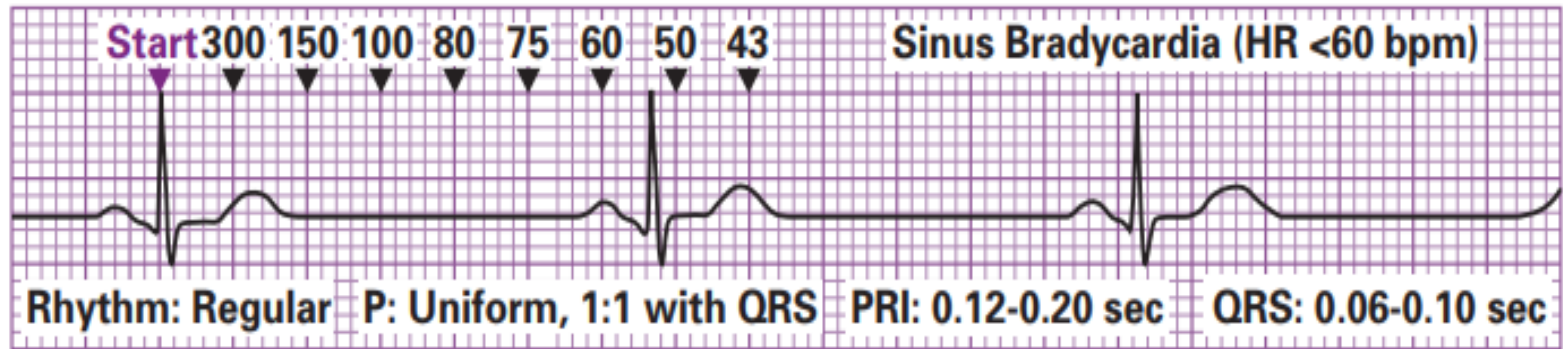


Figure 2 Sinus Bradycardia

2/ Sinus tachycardia

- Occurs when the sinus node creates an impulse **at a faster-than-normal rate**.
- Caused by **acute blood loss, anemia, shock, hypervolemia, hypovolemia, congestive heart failure, pain, hypermetabolic states, fever, exercise, anxiety**, medications that stimulate the sympathetic response (e.g., catecholamines, aminophylline, atropine), stimulants (e.g., caffeine, nicotine), and illicit drugs (e.g., cocaine)
- Characteristics:
 - *Ventricular and atrial rate: **Greater than 100 in the adult***
 - *Ventricular and atrial rhythm: Regular*
 - *QRS shape and duration: Usually normal, but may be regularly abnormal*
 - *P wave: Normal and consistent shape; always in front of the QRS, but may be buried in the preceding T wave*
 - *P:QRS ratio: 1:1*
 - *PR interval: Consistent interval between 0.12 and 0.20 seconds*
- **Mgt : Ca channel blocker, beta blockers, treat underline cause**

Sinus tachycardia

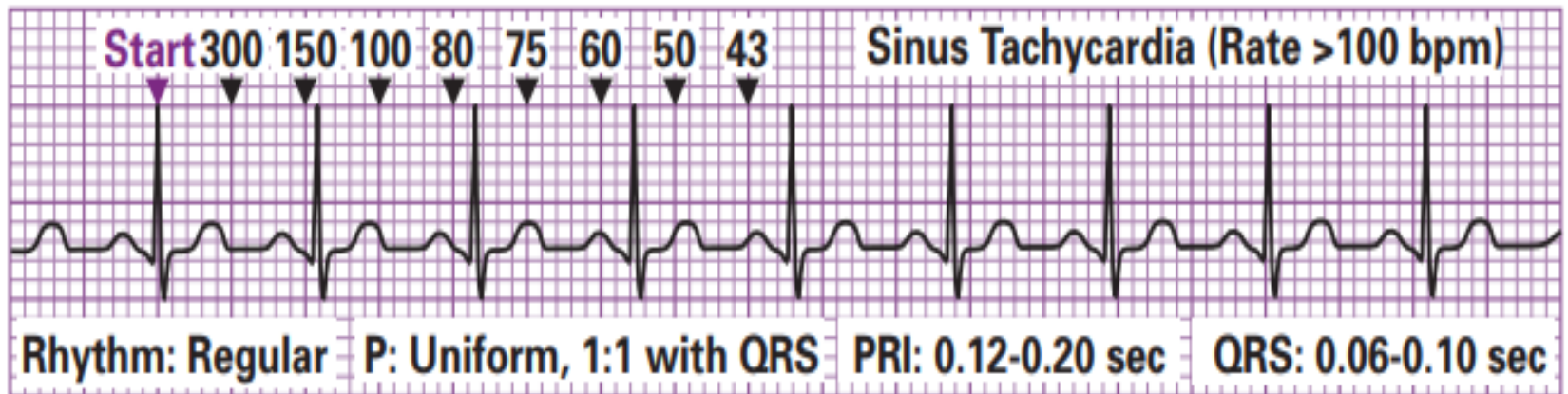


Figure 3 Sinus Tachycardia

Atrial dysrhythmias

A. Premature Atrial Complex(PAC)

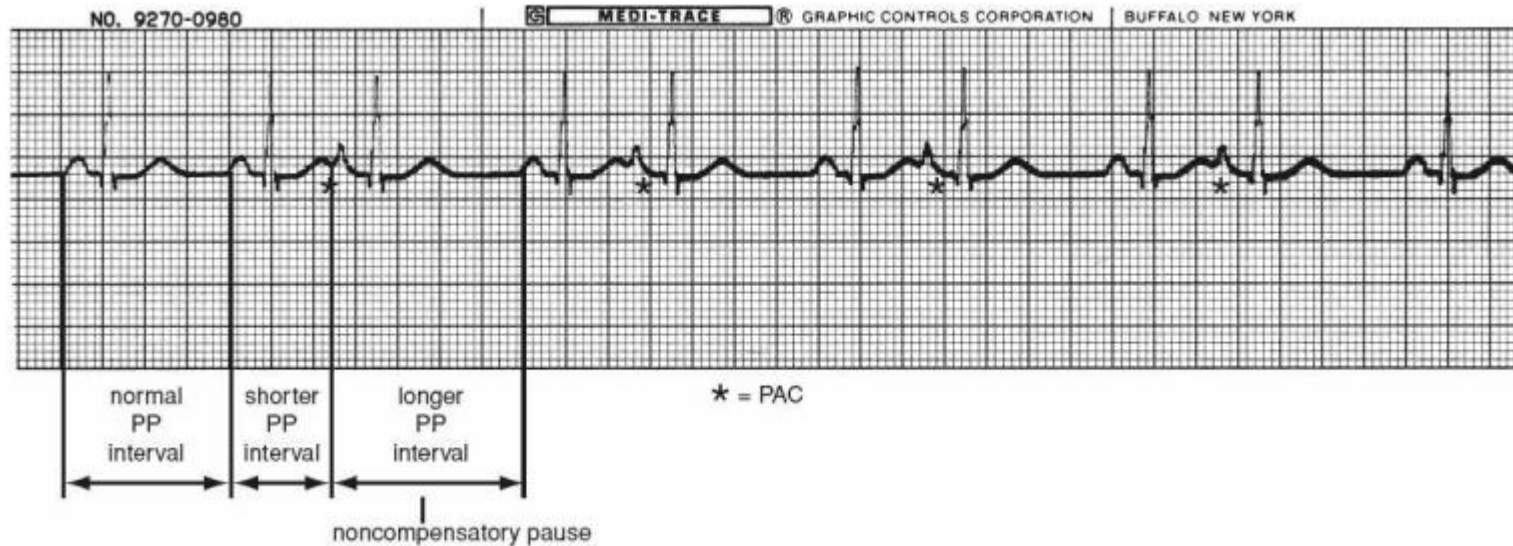
- Is a single ECG complex that occurs when an electrical impulse **starts in the atrium** before the next normal impulse of the sinus node.
- Caused by **caffeine**, alcohol, nicotine, stretched atrial myocardium (as in **hypervolemia**), anxiety, hypokalemia (**low potassium level**), hypermetabolic states, or atrial ischemia, injury, or infarction.
- **PACs are often seen with sinus tachycardia.**

...

PACs have the following characteristics

- Ventricular and atrial rate: **Depends on the underlying rhythm** (eg, sinus tachycardia)
- *P wave*: **An early and different P wave** may be seen or may be hidden in the T wave; other P waves in the strip are consistent.
- **Ventricular and atrial rhythm: Irregular** due to early P waves, creating a PP interval that is shorter than the others.
- **PR interval: The early P wave has a shorter-than-normal PR interval, but still between 0.12 and 0.20 seconds.**
- PACs are common in normal hearts.
- The patient may say, “**My heart skipped a beat.**”
- **A pulse deficit** (a difference between the apical and radial pulse rate) **may exist.**

Premature Atrial Complex(PAC)



B. Atrial fibrillation

- Causes a **rapid, disorganized**, and **uncoordinated** twitching of atrial musculature.
- It is the **most common dysrhythmia** that causes patients to seek medical attention.
- It may **start and stop suddenly**.
- May occur for a **very short time**; it may be chronic.
- Is usually associated with **advanced age, valvular heart disease, coronary artery disease, hypertension, cardiomyopathy, hyperthyroidism, pulmonary disease, acute moderate to heavy ingestion of alcohol (“holiday heart” syndrome)**, or the **aftermath of open heart surgery**.

Characteristics

- Ventricular and atrial rate: **Atrial rate is 300 to 600**. Ventricular rate is usually **120 to 200** in untreated atrial fibrillation
- Ventricular and atrial rhythm: **Highly irregular**
- P wave: **no visible p waves**; irregular undulating waves
- PR interval: cannot be measured
- P: QRS – **Many: one**

Mgt

- ✓ Beta blockers, Ca channel blockers
- ✓ Antithrombotic medications to decrease **risk of embolism**
- ✓ Flecainide, dofetilide, propafenone, amiodarone
 - ➔ Medications that may be given to achieve pharmacologic cardioversion to sinus rhythm within 7 days of onset

Atrial fibrillation

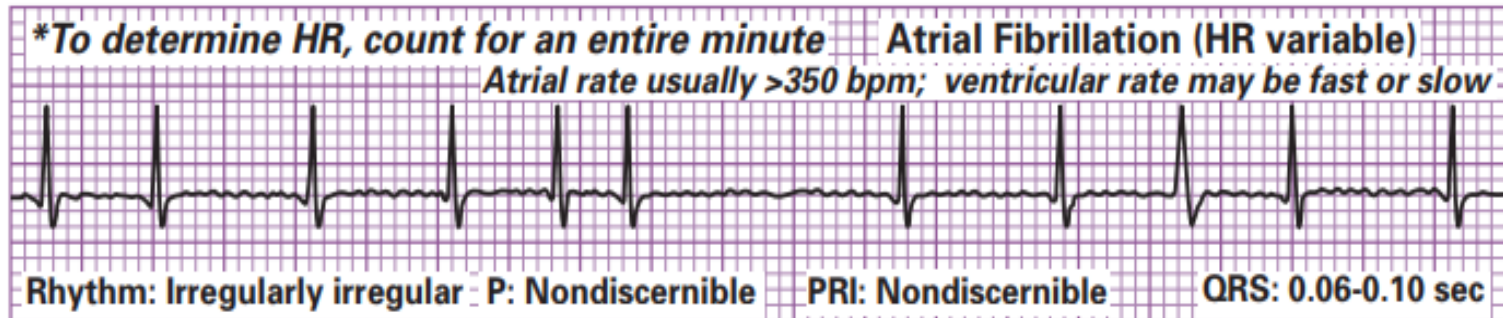


Figure 4 Atrial Fibrillation



Figure 26-10 • Atrial fibrillation in lead II.

C. Atrial flutter

- Occurs in the atrium and creates impulses at an atrial rate between **250 and 400 times per minute**.
- Because the atrial rate is faster than the AV node can conduct, not all atrial impulses are conducted into the ventricle, causing a **therapeutic block at the AV node**.
- If all atrial impulses were conducted to the ventricle, the ventricular rate would also be 250 to 400, which would result in **ventricular fibrillation, a life-threatening dysrhythmia**.
- Causes are similar to that of atrial fibrillation.

...

- *Ventricular and atrial rate: Atrial rate ranges between **250 and 400**; ventricular rate usually ranges between **75 and 150**.*
- *Ventricular and atrial rhythm: **The atrial rhythm is regular**; the ventricular rhythm is usually regular but may be irregular because of a change in the AV conduction.*
- *PR interval: it difficult to determine*
- *P wave: **Saw toothed shape(F waves)***
- *P: QRS: **2:1, 3:1 or 4:1***
- *Atrial flutter can cause **serious signs and symptoms**, such as
– **chest pain, shortness of breath, and low blood pressure.***
- *Mgt: similar with atrial fibrillation*

Atrial flutter

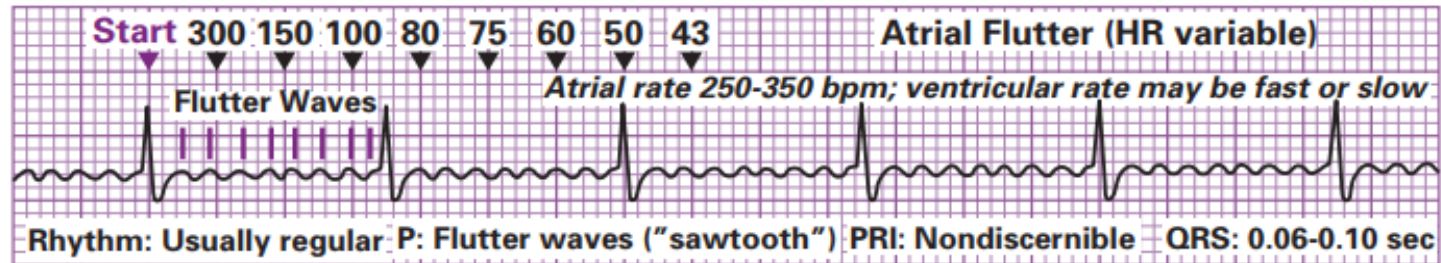


Figure 5 Atrial Flutter



Figure 26-12 • Atrial flutter in lead II.



Junctional dysrhythmia

- Occurs **when the AV node**, instead of the sinus node, **becomes the pacemaker of the heart**
- When the **sinus node slows** or when the **impulse can not be conducted through the AV node** (e.g **because of complete heart block**), the AV node discharges the impulse
- Characteristics:
 - ✓ Rate: ventricular rate - **40 to 60, atrial rate – 40 to 60**
 - ✓ Rhythm : regular
 - ✓ QRS shape and duration: usually normal
 - ✓ **P wave : may be absent or after QRS complex**
 - ✓ *PR interval*: If the P wave is in front of the QRS, the PR interval is less than 0.12 seconds.
 - ✓ **P : QRS – 1:1 or 0:1**
- Mgt : atropine

Junctional dysrhythmia



Figure 26-13 • Junctional rhythm in lead II; note short PR intervals and inverted P wave.

Ventricular dysrhythmia

A. Ventricular Tachycardia(VT)

- Due to **myocardial irritability**
- Associated with coronary artery disease
- Need medical emergency
- Characteristics:
 - ✓ Rate : ventricular rate **100 to 200 /minute**
 - ✓ Rhythm : usually regular
 - ✓ QRS shape and duration: **bizarre** shape; 0.12 second or more
 - ✓ P wave: very difficult to detect
 - ✓ QRS: P: difficult to determine; **more QRS than P wave**
- Mgt : cardioversion

Ventricular Tachycardia(VT)

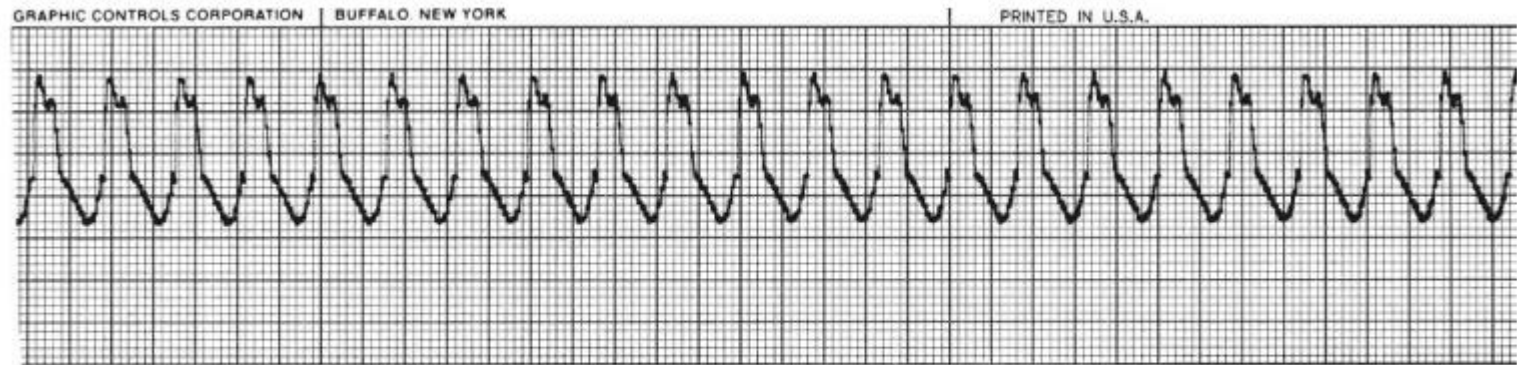


Figure 26-16 • Ventricular tachycardia in lead V_1 .

B. Ventricular fibrillation

- A **rapid but disorganized ventricular** rhythm that causes ineffective quivering of the ventricles
- Most common cause of ventricular fibrillation is **coronary artery disease** and resulting acute MI
- **Atrial activity is not seen on ECG.**
- No audible heart beat
- **No palpable pulse**

Characteristics :

- ✓ rate: ventricular rate is **>300/minute**
- ✓ Rhythm: **extremely irregular**
- ✓ QRS: **Unrecognized QRS**
- Mgt: defibrillation

Ventricular fibrillation

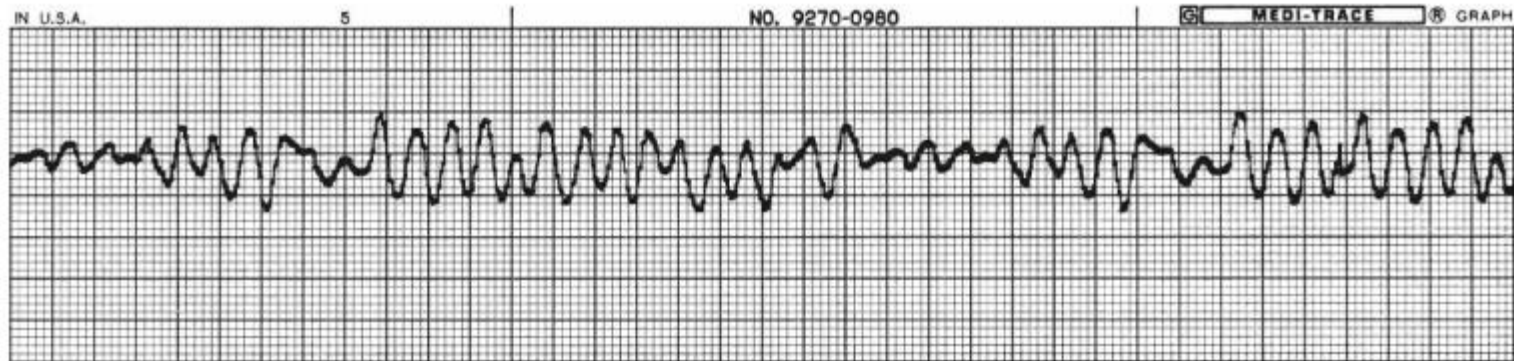


Figure 26-17 • Ventricular fibrillation in lead II.

Nursing diagnoses

- Based on the assessment data, major nursing diagnoses may include:
 - ✓ Decreased cardiac output related to inadequate ventricular filling or altered heart rate
 - ✓ Anxiety related to fear of the unknown outcome of altered health state
 - ✓ Deficient knowledge about the dysrhythmia and its treatment
- The major goals for the patient may include:
 - eliminating or decreasing the occurrence of the dysrhythmia (by decreasing contributory factors) to maintain cardiac output;
 - verbalizing reduction in anxiety;
 - verbalizing an understanding about the dysrhythmia, tests used to diagnose the problem, and its treatment; and developing or maintaining self-management skills.

Nursing interventions

1. Monitoring and managing the dysrhythmia

- The nurse regularly evaluates blood pressure, pulse rate and rhythm, rate and depth of respirations, and breath sounds to **determine the dysrhythmia's hemodynamic effect.**
- The nurse also asks patients about episodes of lightheadedness, dizziness, or fainting as part of the ongoing assessment.
- If a patient with a dysrhythmia is hospitalized, the nurse may obtain a 12-lead ECG, continuously monitor the patient, and **analyze rhythm strips to track the dysrhythmia.**
- In addition to medication, the nurse **assesses for factors** that contribute to the dysrhythmia (eg, caffeine, stress, non adherence to the medication regimen) and assists the patient in developing a plan to make lifestyle changes that eliminate or reduce these factors.

2. Minimizing anxiety

3. Promoting home and community-based care

Coronary Artery Disease(CAD)

- **Coronary artery disease** is the **most prevalent** type of cardiovascular disease.
- **Atherosclerosis** is an abnormal accumulation of lipid, or fatty substances and fibrous tissue in the vessel wall.



Create blockages or narrow the vessel in a way that reduces blood flow to the myocardium.

- It is the most common heart disease in the United States
- It is a progressive disease

Risk factors

- Major risk factors include
 - Use of tobacco
 - Hypertension
 - Elevated blood lipid levels,
 - Obesity,
 - Lack of physical activity,
 - Excessive alcohol use, and
 - Mental stress
 - Family history of premature cardiovascular disease and
 - Age (>45 years for men; >55 years for women).

Pathogenesis

- Coronary atherosclerosis produces symptoms and complications according to
 - **Location and degree of narrowing of the arterial lumen,**
 - **Thrombus formation,** and
 - **Obstruction of blood flow to the myocardium**
- This impediment to blood flow is usually progressive, causing an inadequate blood supply that deprives the muscle cells of oxygen needed for their survival. **The condition is known as ischemia.**
- **Angina pectoris:** refers to chest pain that is brought about by myocardial ischemia; usually is caused by significant coronary atherosclerosis.
- **Myocardial infarction**(irreversible damage and death of myocardial cells): occurs if the decrease in blood supply is great enough, of long enough duration.

...

A/ Angina pectoris

- Is a clinical symptom characterized by **pain or a feeling of pressure in the anterior chest**

**Coronary
artery
spasm,**

**Fixed stenosing
plaques,
disrupted plaques**

**=> Thrombosis,
=> Platelet aggregation,
=> Embolization**

**inadequate oxygen supply to the myocardium →
Ischemia → pain**

Types of angina

- **Stable angina(TYPICAL):**
 - Predictable and consistent pain(10-15 minutes)
 - **Caused by reduction of coronary perfusion** by chronic stenosing coronary atherosclerosis;
 - Related to activities that increase myocardial demand
 - **Relieved by rest**
- **Unstable angina(*preinfarction angina* or *crescendo angina*)**
 - Symptoms occur **more frequently and last longer** than stable angina (over 20 minutes).
 - **Caused by disruption of an atherosclerotic plaque** or vasospasm or both
 - The threshold for pain is lower, and **pain may occur at rest.**
 - **May not be relieved by NTG or rest**

...

- **Intractable or refractory angina:**
 - severe incapacitating chest pain
- **Variant angina (*Prinzmetal's angina*):**
 - Occurs at rest, often during sleep in early morning hours
 - Caused by coronary artery vasospasm,
 - may subside with exercise ;
 - associated with reversible ST segment elevation,
- **Silent ischemia:**
 - Objective evidence of ischemia but patient reports no symptoms
 - Transient ST depression persisting for at least 1 min
 - Usually unrecognized by patients

Table 28-3 • Canadian Cardiovascular Society Classification of Angina

CLASS	ACTIVITY EVOKING ANGINA	LIMITS TO ACTIVITY
I	Prolonged exertion	None
II	Walking >2 blocks	Slight
III	Walking <2 blocks	Marked
IV	Minimal or rest	Severe

Adapted from Campeau L. (1976). Grading of angina pectoris. *Circulation* 54, 596-599.

Factors that produce angina pain

- **Physical exertion:** precipitate on attack by increasing myocardial demand
- **Exposure to cold:** vasoconstriction
- **Eating a heavy meal:** decreases available blood flow to the heart as the mesenteric blood flow increases
- **Any emotion provoking situation:** causing the release of adrenalin & increase blood pressure, may accelerate the heart rate, thus decreasing the available blood supply

Clinical features of angina

- Ischemia of the coronary arteries cause **pain**
- Location: usually **felt deep in the chest** behind the upper and middle third of sternum
- **Severe apprehension and a feeling of impending death**
- A feeling of **weakness in the arms, wrists & hands**
- Sensation of **pressure, heaviness, or squeezing** in the anterior chest
- **Pain may radiate** to the neck, jaw, shoulder, back, or arm.
- Pain may be accompanied by dyspnea, nausea, vomiting, or **diaphoresis**.
- Symptoms are often **provoked by exertion** (e.g., walking, climbing stairs, and doing yard or house work) or emotional stress and **relieved within minutes by rest or nitroglycerin**

Diagnosis

- Clinical manifestation of pain and patients history
- Abnormal heart sounds, such as paradoxical splitting of the second heart sound, a third heart sound, or a loud fourth heart sound
- CK-MB, troponin I and troponin T are elevated in MI (ST segment elevation), **but in unstable angina, only CK-MB**

Management

- The objective of treatment is to **decrease the oxygen demand** and to **increase the oxygen supply of the myocardium**
- Medically: through pharmacologic therapy and control of risk factors

Control risk factors

1. Non modifiable risk factors

- Positive family history
- Gender (High in men-3x & premenopausal women)
- Race (higher in African American)
- Age (M > 45, F > 55)

2. Modifiable risk factors

- High blood cholesterol
- Elevated blood pressure,
- Elevated blood glucose
- Cigarette smoking: produces tachycardia and raises the B/P
- Obesity,
- Physical inactivity

Pharmacologic therapy

A/ Nitroglycerin

- To reduce myocardial oxygen consumption, which decreases ischemia and **relieves pain within 3 minutes**, the route is **sublingually**.
- It helps to increase coronary blood flow **by preventing vasospasm** and increasing perfusion through the collateral vessels
- **Nitroglycerin dilates primarily the veins and, in higher doses, also dilates the arteries** ➔ Dilation of the veins causes venous pooling of blood throughout the body ➔ As a result, less blood returns to the heart, and filling pressure (**preload**) **is reduced**.
- If the patient is hypovolemic (does not have adequate circulating blood volume), the decrease in filling pressure can cause a significant decrease in cardiac output and blood pressure.



- **Nitrates in higher doses also relax the systemic arteriolar bed and lower blood pressure (decreased after load).**
- **Nitrates may increase blood flow to diseased coronary arteries** and through collateral coronary arteries, arteries that have been under used until the body recognizes poorly perfused areas. These effects decrease myocardial oxygen requirements and increase oxygen supply, bringing about a more favorable balance between supply and demand.
- It usually is **not given if the systolic blood pressure is 90 mm Hg or less**

...

B/ Beta-adrenergic blocking agents

- To **reduce myocardial oxygen consumption** by blocking the **beta-adrenergic** sympathetic stimulation to the heart. The result is
 - **A reduction in heart rate,**
 - **Slowed conduction of an impulse** through the heart,
 - **Decreased blood pressure,** and
 - Reduced myocardial **contractility (force of contraction)** **that establishes** a more favorable balance between myocardial oxygen needs (demands) and the amount of oxygen available (supply).
- This helps to control chest pain and delays the onset of ischemia during work or exercise.
- Beta-blockers reduce the incidence of recurrent angina, infarction, and cardiac mortality.
- The dose can be titrated to achieve a resting heart rate of 50 to 60 beats per minute



C/ Calcium channel blockers

- **Decrease sinoatrial node automaticity** and
- **Decrease atrioventricular node conduction**
 → slower heart rate
- **decrease in the strength of the heart muscle contraction**
(negative inotropic effect).
 → These effects decrease the workload of the heart.
- **relax the blood vessels**, causing a decrease in blood pressure and an increase in coronary artery perfusion.
- Calcium channel blockers increase myocardial oxygen supply by dilating the smooth muscle wall of the coronary arterioles;
- They decrease myocardial oxygen demand by reducing systemic arterial pressure and the workload of the left ventricle.

...

D/ Antiplatelet agents and anticoagulant medications

- **Aspirin:** prevents platelet activation and reduces the incidence of MI and death in patients with CAD.
- **Heparin:** prevents the formation of new blood clots.

E/ Oxygen administration

F. Anti Pain

Intervention	Rationale
Oxygen	Increases myocardial oxygen supply.
Beta-blocker	Antidysrhythmic, antihypertensive, reduces afterload to decrease myocardial oxygen consumption, decreases infarct size, and improves survival rates.
ACE inhibitor	Decreases afterload and therefore myocardial oxygen consumption.
Stool softener	Reduces straining which could precipitate MI, vasovagal effect.
Nitroglycerin	Reduces preload and afterload and dilates coronary arteries, increases myocardial oxygen supply and reduces oxygen consumption.
Morphine	Reduces preload by venous vasodilation decreasing myocardial oxygen consumption, relieves pain, decreases sympathetic nervous system stimulation.
Thrombolytic agent	Removes clot that is causing ischemia/infarction.

Nursing Diagnosis

- **DIAGNOSIS:** **Acute or chronic pain** related to narrowing of the coronary artery(ies) and associated with atherosclerosis, spasm, or thrombosis as evidenced by self-reported pain, chest pressure, shortness of breath, and/or tachycardia; anxiety, fear
- **OUTCOMES:** Cardiac pump effectiveness; Circulation status; Comfort level; Pain control behavior; Pain level; Tissue perfusion:
- **INTERVENTIONS:**
 - Cardiac care;
 - Cardiac precautions;
 - **Oxygen therapy;**
 - **Pain management;**
 - Medication administration;
 - **Circulatory care; Positioning**

...

- **DIAGNOSIS.** **Decreased cardiac tissue perfusion** related to narrowing of the coronary artery(ies) as evidenced by chest pain, shortness of breath, pallor, diaphoresis, and/or palpitations
- **OUTCOMES.** Cardiac pump effectiveness; circulation status; comfort level; pain control behavior; pain level; tissue perfusion:
- **INTERVENTIONS.**
 - ✓ Cardiac care;
 - ✓ Cardiac precautions;
 - ✓ **Oxygen therapy;**
 - ✓ Pain management;
 - ✓ Medication administration;
 - ✓ Circulatory care;
 - ✓ **Positioning**

B/ Myocardial Infarction(MI)

- Is the death of a portion of heart muscle in an area where there is sudden loss of blood supply due to occlusion of a major coronary artery or one of its branches.
- Is sometimes called a **heart attack** or a **coronary thrombosis**.
- Refers to the process by which areas of myocardial cells in the **heart are permanently destroyed**.

Pathophysiology:

- **Plaque rupture and subsequent thrombus formation** → complete occlusion of the artery → ischemia and necrosis of the myocardium supplied by that artery.
- **Vasospasm** (sudden constriction or narrowing)
- **Increased demand** for oxygen from acute blood loss, low blood pressure, anemia, or (e.g., from a rapid heart rate, thyrotoxicosis, or ingestion of cocaine) → ischemia and necrosis

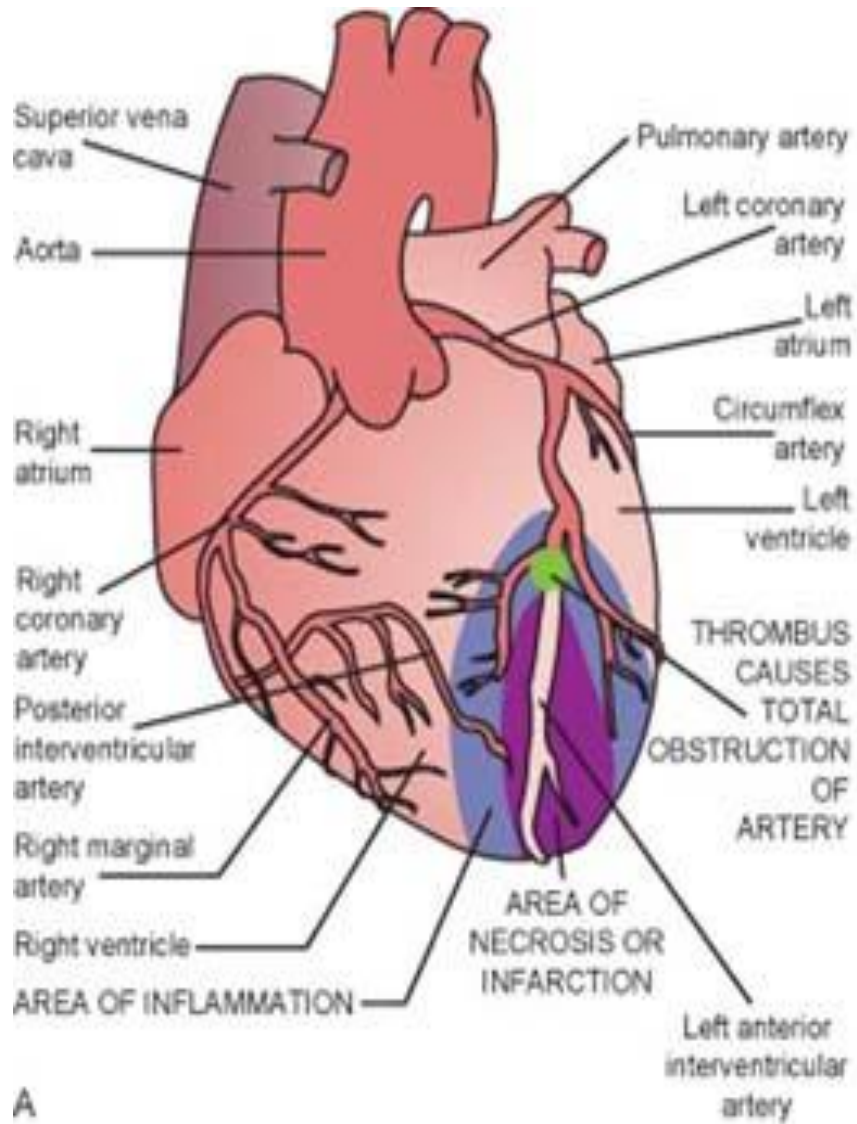
...

- Coronary artery completely obstructed
 - Prolonged ischemia and cell death of myocardium
- **Most common cause is atherosclerosis with thrombus**
- **3 ways it may develop:**
 - ❖ Thrombus obstructs artery
 - ❖ Vasospasm due to partial occlusion
 - ❖ Embolus blocks small branch of coronary artery
- **Majority involve Lt ventricle**
 - Size and location of infarction determine severity of damage
- Function of myocardium contraction and conduction quickly lost
 - Oxygen supplies depleted

• • •

- Time Line
 - 1st 20 min critical
 - 48 hrs inflammation begins to subside
 - 7th day necrosis area replaced by fibrous tissue
 - 6-8 weeks scar forms

Myocardial Infarction



W.B. Saunders Company items and derived items copyright © 2002 by W.B. Saunders Company

Risk factors for acute MI

Modifiable risk factors

- Hyperlipidemia
- Smoking & alcoholism
- Diabetes mellitus
- Hypertension
- Obesity
- Physical inactivity
- **Oral contraceptives**

Non modifiable risk factors

- Age
- Gender
- Genetic/family history
- Pre-existing coronary heart diseases

Etiology -MI

- The **major** cause of MI is coronary artery occlusion by **thrombosis** or **atheroma**
- **Uncommon causes**
 - Inflammation of the coronary arteries (rare);
 - A stab wound to the heart;
 - A blood clot forming elsewhere in the body (for example, in a heart chamber) and traveling to a coronary artery where it gets stuck;
 - Cocaine abuse which can cause a coronary artery to go into **spasm**; complications from heart surgery; and some other rare heart problems.

Signs and Symptoms

➤ Pain

- Sudden/sub-sternal area/, **more severe & lasts longer than** angina pectoris
- **Radiates to Lt arm and neck**

➤ Pallor (conjunctiva), dyspnea, sweating, nausea, dizziness, palpitations, loss of consciousness

➤ Anxiety and fear

➤ Hypotension, rapid and weak pulse (low CO)

➤ **Low grade fever after 12 hrs of infarction**

➤ **Sudden death**

➤ **Chest pain:**

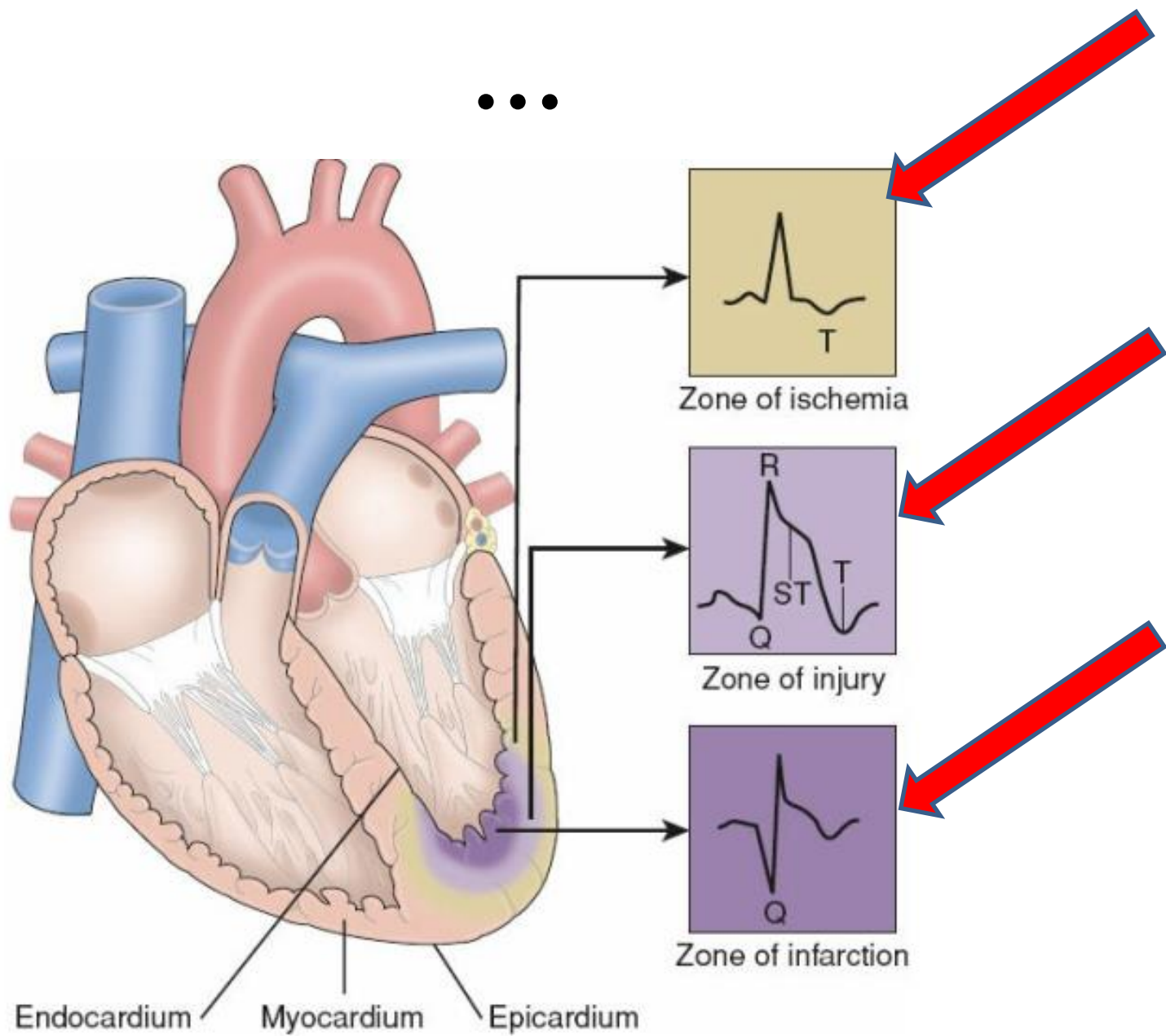
- It is a heavy which may radiate to the shoulder and down the arms, usually the left arm.
- In some cases the pain may radiate to the **jaw & neck**.

➤ **Pain is often accompanied by pallor, diaphoresis, dizziness, nausea and vomiting.**

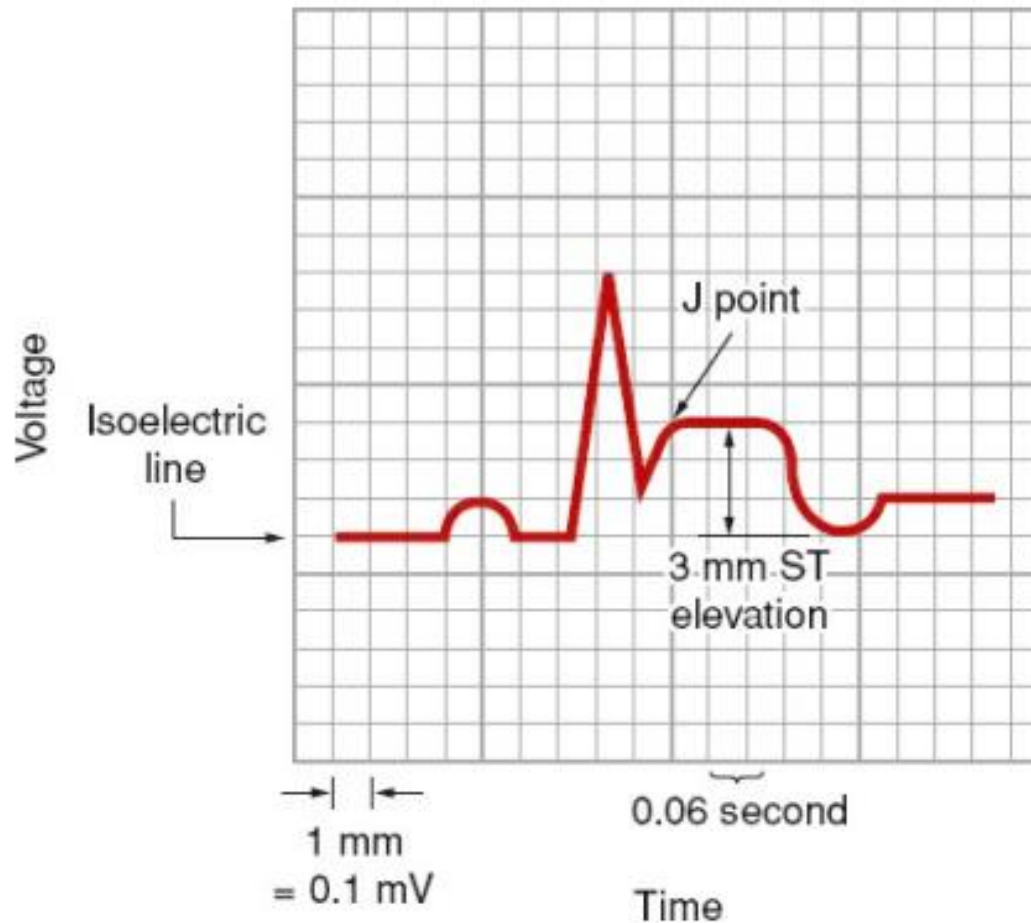
...

Diagnosis

- Patient Hx, ECG, serum enzymes and isoenzymes
- Cardiac troponin levels have the highest sensitivity and specificity (better than CK-MB levels) in detecting MI; used to differentiate between angina and MI
- Creatine kinase isoenzyme (CK-MB) elevated
- development of an **abnormal Q wave**;
- decrease in the height of the R wave.
- **ST-segment elevation (for acute MI)**, and
- T-wave inversion,
- **WBC 12000-15000, high ESR**



Acute myocardial infarction (MI)



Management

- ❖ **Vasodilators:** Nitroglycerine 0.5mg sublingual Q.5min
- ❖ **Anti coagulants:** heparin **reducing** the probability of **thrombus formation** and the subsequent diminished blood flow (Heparin: IV bolus 60-70 u/kg, then 12-15u/kg/hr, Aspirin, clopidogrel)
- ❖ **Thrombolytic** - to dissolve any thrombus in a coronary artery (**streptokinase** is the known agent), not given for unstable angina.
- ❖ **High flow O₂** – at the onset of chest pain
- ❖ **Analgesics** - Morphine sulphate IV 2-4 mg . The need for analgesia is limited to those patients in whom nitrates and anticoagulants are ineffective in relieving pain
- ❖ **Cardiac rehabilitation**

Management of acute MI

- Assess circulation: Pulse, BP, Capillary refill
 - *Use B-Blockers: Atenolol 50 mg orally and 12 hours later then 100 mg per day or metoprolol*
 - Morphine 2-4mg I/V, repeat every 5-10 minutes
 - Other analgesics- pethidine, tramadol, diclofenac
 - Aspirin immediately
 - Nitrates – sublingual or skin patches

Nursing care/ interventions/

- **Preventing pain:** avoid activities known to cause Angina pectoris
- **Reducing anxiety:** Physical presence of another alleviate fear of death
- **Patient education:** home care considerations to improve the quality of life and promote health
- **Relieving chest pain:** vasodilator, anticoagulant, physical rest
- **Importing adequate tissue perfusion:** keeping the pt on bed or chair that he may rest and administer O_2
- Monitoring and managing potential complications

Infectious/inflammatory disorders

Rheumatic Fever

- Rheumatic fever is **an autoimmune disorder** that follows an upper respiratory infection with group A beta-hemolytic streptococci (**GABHS**) and causes an inflammatory response.
- Rheumatic fever affects the **heart, central nervous system, skin, and musculoskeletal system**, and many people experience recurrences after the initial illness.
- Acute rheumatic fever is most destructive to the heart.
- **Rheumatic heart disease (RHD)** occurs in up to 60% of patients with acute rheumatic fever and may affect any of the layers of the heart during the acute phase.



- Rheumatic fever causes endocarditis, myocarditis, Pericarditis , pericardial effusion, valvular heart disease and heart failure.
- Caused by a prior streptococcal infection and is often associated with nasopharyngitis or upper respiratory infections.
- **Rheumatic fever is an autoimmune response triggered by antibodies that are produced in response to the streptococcal infection.**
- **The antibodies react with the body's cells and produce characteristic lesions in the target organs.**
- Risk factors include family history, skin infections, and environmental factors such as overcrowding and poor sanitation.
- Children who have untreated streptococcal throat infections are at high risk.

Clinical Manifestations

- Usually, the patient has a sore throat and a fever of at least 100.4F (38C) **for a few days to several weeks before the onset.**
- The latent period between the acute infection of streptococcal pharyngitis and acute rheumatic fever **averages 18 days**
- The patient **may not have been treated** with antibiotics or may not have completed a full course of treatment.
- Determine if the patient has experienced migratory joint tenderness (polyarthrititis), chest pain, fever, and fatigue.
- shortness of breath, cough, and right upper quadrant abdominal pain
- Fatigue or activity intolerance, along with periorbital, abdominal, or pedal edema.

• • •

- distinctive red rash (*erythema marginatum*: nonpruritic rash appears primarily on the trunk of the body, the buttocks, and the extremities)
- **Chorea**: produces hyperirritability, problems concentrating, and illegible handwriting.
- Severe chorea causes purposeless, uncontrollable, jerky movements and muscle spasms, speech disturbances, muscle fatigue, and incoordination.

Diagnosis

- History
- Prolonged PR-interval on ECG
- Throat culture
- **ASO**

Management

- Bed rest
- Emotional support, especially crucial when Sydenham's chorea is present
- Antibiotics:
 - **Benzathine penicillin G**, 1.2M IU for weight $> 27\text{kg}$ and 600,000IU IM for weight $\leq 27\text{kg}$)
 - **Erythromycin** 40mg/kg/24 orally divided into 2-4 doses for 4 days *OR*
Azithromycin 500mg orally on the first day, then 250mg daily for 4 days.

Infective Endocarditis

- is an inflammatory process of the endocardial lining of the heart.
- Endocarditis occurs when an invading organism enters the bloodstream and attaches to a sterile fibrin clot already present on the leaflets of the valves or the endocardium.
- Most commonly, the **mitral or aortic valve is involved**
- Causes of IE have been **nosocomial infections** from IV catheters, IV drug abuse, and prosthetic valve, dialysis shunts
Skin, bone, and pulmonary infections
- **Staphylococcus aureus** is the primary pathogen of endocarditis followed by **Streptococcal infections**.

Clinical Manifestations

- Anorexia and weight loss, chills, malaise, headache, muscle and joint aches, shortness of breath, and cough.
- **Continuous fever (103F–104F [39.3C–40C]) in acute IE**
- History (3–6 months) of **dental procedures**, tonsillectomy, bronchoscopy, endoscopy, colonoscopy.
- Splenomegaly
- Heart murmur (most commonly mitral and aortic regurgitation murmurs)

Diagnosis

- CBC
- Blood culture

Management

- Provide adequate rest by assisting the patient with daily hygiene
- Penicillin G - 2 million units IV q 4 hr for 4 wk
- Oxacillin; ceftriaxone, gentamicin, vancomycin, or tobramycin
- Acetaminophen

Myocarditis

- an **inflammatory condition of the cardiac muscle**, with infiltration of the myocardium with lymphocytes, eosinophils, neutrophils, or giant cells that leads to necrosis or degeneration of myocytes (heart muscle cells)
- Myocarditis generally occurs as a result of an infectious agent.
- Common cause of myocarditis is **viral infection** (e.g., coxsackie [group B], adenovirus, hepatitis C, cytomegalovirus, Epstein-Barr virus)
- It also can be **caused by an autoimmune response** in conditions such as celiac disease, Crohn disease, rheumatoid arthritis, and systemic lupus erythematosus.
- **Toxic reactions to radiation or other physical agents**, such as lead, may lead to the condition, as may a variety of drugs, such as phenothiazines, lithium, and chronic use of cocaine

Clinical Manifestations

- Most common symptoms are **malaise, fatigue, chest pain, dyspnea, palpitations, myalgias, and fever**
- Determine if the patient has experienced excessive tachycardia, both at rest and with effort.
- Elicit a history of medication use, particularly the use of phenothiazines, lithium, or cocaine.

Diagnosis

- Cardiac troponin I and cardiac troponin T , Creatine kinase isoenzyme (CK-MB) **elevated**
- Presence of **abnormal numbers of lymphocytes**; degeneration of muscle fibers

Management

- Oxygenation and rest are prescribed in order to prevent dysrhythmias and further damage to the myocardium.
- Antibiotics or antivirals: penicillin for hemolytic streptococci

Pericarditis

- Pericarditis refers to an inflammation of the pericardium, which is the membranous sac enveloping the heart.
- It may be a primary illness or secondary
- Pericarditis also may occur 10 days to 2 months after acute myocardial infarction
- Pericarditis may be acute, chronic, or recurring.

Causes:

- ✓ Infection: **usually viral** (e.g., human immune deficiency virus, coxsackievirus, influenza); rarely bacterial (e.g., staphylococci, streptococci, meningococci, gonococci, gram-negative rods)
- ✓ Hypersensitivity states
- ✓ Trauma: chest injury, cardiac surgery, cardiac catheterization,

Pathophysiology

- The inflammatory process of pericarditis may lead to an accumulation of fluid in the pericardial sac (**pericardial effusion**) and **increased pressure on the heart, leading to cardiac tamponade**
- Frequent or prolonged episodes of pericarditis also may lead to **thickening and decreased elasticity of the pericardium**, or scarring may fuse the visceral and parietal pericardium → restrict the heart's ability to fill with blood (**constrictive pericarditis**).
- The pericardium may become **calcified**, further restricting ventricular expansion during ventricular filling(diastole).
- Restricted diastolic filling may result in **increased systemic venous pressure, causing peripheral edema and hepatic failure.**

Clinical Manifestations

- Pericarditis may be asymptomatic.
- The most characteristic symptom of pericarditis is **chest pain**, although pain also may be located beneath the clavicle, in the neck, or in the left trapezius (scapula) region.
- Patients may have a nonproductive cough
- Dyspnea
- The heart rate may increase to maintain cardiac output

Diagnosis

- history, signs, and symptoms.
- **concave ST elevations**
- **depressed PR segments**

Management

- bed rest until fever, chest pain, and friction rub have subsided.
- **NSAIDs such as aspirin or ibuprofen; Indomethacin is contraindicated because it may decrease coronary blood flow**
- Colchicine or corticosteroids (e.g., prednisone) may be prescribed if the pericarditis is severe
- Pericardiocentesis

Valvular Heart Disease

Mitral stenosis

- a **pathological narrowing of the orifice of the mitral valve**, occurs when the mitral valve is **unable to open fully**.
- The mitral valve leaflets fuse together and become stiff and thickened by fibrosis and calcification.
- When mitral stenosis occurs, blood can flow from the left atrium to the left ventricle only if it is moved forward by an abnormally elevated left atrial pressure.
- The **elevated left atrial pressure** leads to **increased pulmonary venous and capillary pressures**, decreased pulmonary compliance, and **exertional dyspnea**.

• • •

- Left atrial dilatation, an increase in pulmonary artery pressure, and right ventricular hypertrophy follow as the heart compensates for the stenotic valve.
- **The predominant cause of mitral stenosis is rheumatic fever**

C/M

- Excessive fatigue, malaise, decreased tolerance to exercise, dyspnea on exertion, orthopnea, paroxysmal nocturnal dyspnea, and dry cough.
- **Neck vein distention and pitting peripheral edema.**
- **Pulmonary edema** may also occur and lead to orthopnea, tachypnea, diaphoresis, pallor, cyanosis, and pink frothy sputum.

Mgt

- Definitive therapy for mitral stenosis is surgical replacement of the stenotic valve, particularly when the valve has marked stenosis with an orifice less than 1 cm².

Mitral Insufficiency (Regurgitation)

- is the **inadequate closure of the mitral valve**, which interferes with expulsion of cardiac output from the left ventricle.
- **Mitral insufficiency causes an increase in blood volume in both the left atrium and ventricle.**
- The increased volume of blood in chronic mitral insufficiency accumulates slowly, allowing the left atrium and left ventricle to increase in size.
- **Common causes** of chronic mitral insufficiency are **rheumatic heart disease**, endocarditis, congenital anomaly, and idiopathic calcification of the mitral
- Acute mitral insufficiency can occur with myocardial infarctions that have been caused by ischemia or necrosis of the papillary muscle or by chordae tendineae that support the mitral leaflets.

C/M

- Patients may remain asymptomatic for years
- **Fatigue** and **shortness of breath**.
- **soft first heart sound** and a **systolic murmur**, which is loudest at the apex.
- Fine crackles (rales) if pulmonary congestion is present
- Enlarged liver if the patient has severe right-sided heart failure.
- The patient may also have jugular vein distention

Mgt

- Most patients with mitral insufficiency can compensate or be stabilized with medical treatment for their entire lives.
- Surgical repair or valve replacement is considered in patients with progressive severe disease.
- **Mitral valve repair (valvuloplasty)** is preferred over replacement whenever possible.

Pharmacologic Highlights

Medication or Drug Class	Dosage	Description	Rationale
Diuretics	Varies with drug	Thiazides; loop diuretics	Manage fluid overload and congestive symptoms
Coronary vasodilators	Varies with drug including angiotensin-converting enzyme inhibitors	Nitroglycerine, nitroprusside, captopril, enalapril, lisinopril, hydralazine	Decrease preload and afterload; decrease regurgitant blood flow; reduce ventricular size
Warfarin	Initially 10–15 mg, then 2–10 mg/day maintenance	Anticoagulant	Prevents thrombi from forming on the synthetic valve
Heparin or low-molecular-weight heparin	Initially 80 units/kg IV bolus, followed by an infusion of 18 units/kg; serial monitoring of activated partial thromboplastin time to guide future doses	Anticoagulant	Prevents thrombi initially until warfarin therapy is well regulated

Aortic Valve Stenosis

- Aortic stenosis (AS) is a **narrowing of the aortic valve** orifice, which obstructs outflow from the left ventricle during systole.
- The left ventricle must overcome the increased resistance to ejection by generating a higher-than-normal pressure during systole, which is achieved by stretching and generating a more forceful contraction.
- The blood is propelled through the narrowed aortic valve at an increased velocity.

• • •

- Left ventricular end-diastolic pressure rises, **myocardial oxygen demand increases**, and left ventricular mass and wall stress are increased.
- The increase in left ventricular pressure is reflected **backward into the left atrium**.
- Because the left atrium is unable to empty adequately, the **pulmonary circulation becomes congested**.
- The predominant causative factor in AS is congenital malformation of the aortic valve.
- Congenital causes can lead to unicuspid and bicuspid or malformed valves.

• • •

- Acquired valvular stenosis results from rheumatic heart disease and degenerative calcification.
- Risk factors for degenerative stenosis include advanced age, hypertension, elevated cholesterol, diabetes mellitus, and smoking.
- Atherosclerosis and degenerative calcification of the aortic valve are the predominant causes for stenosis in people older than 70 years.

C/M

- People often have no signs and symptoms for 10 to 20 years.
- **Chest pain, fainting on exertion (syncope), and labored breathing on exertion** due to heart failure.
- Patients may be easily fatigued and report difficulty with effort and exercise. Inspection of the thoracic wall may reveal a thrusting apical pulsation.
- A systolic thrill (vibrations felt from turbulent blood flow) may be palpated over the second intercostal space to the right of the sternum.
- A lift or heave may be palpated over the apex of the heart.
- Fine crackles (rales) if pulmonary congestion is present in left-sided heart failure.

Dx

- Cardiac catheterization

Mgt

- Activity restrictions to decrease cardiac workload.
- Supplemental oxygenation will enhance oxygen levels in the blood to decrease labored breathing and chest pain.
- Beta-adrenergic receptor blockers (esmolol, metoprolol)
- Open surgical
- Transcatheter aortic valve replacement (TAVR)

Aortic Valve Insufficiency

- Aortic insufficiency (AI), also called aortic regurgitation, **is the incomplete closure of the aortic valve leaflets**, which **allows blood to regurgitate backward** from the aorta into the left ventricle.
- The backflow of blood into the ventricle decreases forward flow in the aorta and increases left ventricular volume and pressure.
- In compensation, **the left ventricle dilates and hypertrophies** to accommodate the increased blood volume.
- Eventually, the increase in left ventricular pressure is reflected backward into the left atrium and pulmonary circulation.
- AI may result either from an abnormality of the aortic valve or from dilation and distortion of the aortic root. AI can be congenital or acquired.

• • •

- Dilation or distortion of the aortic root may be due to systemic hypertension, aortic dissection, syphilis, ankylosing spondylitis.
- **Rheumatic heart disease and endocarditis cause the valve cusps to become thickened and retracted**, whereas an aortic aneurysm causes dilation of the annulus (the valve ring that attaches to the leaflets).

C/M

- **labored breathing on exertion**, which may be present for many years before progressive symptoms develop. **Angina with exertion, orthopnea, and paroxysmal nocturnal dyspnea** are also principal complaints
- a thrusting apical pulsation. Palpation of the precordium reveals the apical pulse to be bounding and displaced to the left.
- Auscultation of heart sounds reveals the classic decrescendo diastolic murmur.

Dx

- Cardiac catheterization and aortic angiography

Mgt

- Treating the underlying cause, such as endocarditis or syphilis.
- Patients are encouraged to limit strenuous physical activity.
- Fluid restrictions and diuretics may be ordered to reduce pulmonary congestion.
- Supplemental oxygenation will enhance oxygen levels in the blood to decrease labored breathing and chest pain
- Most patients can be stabilized with medical treatment, but early elective valve surgery should be considered
- Surgical repair or replacement is the most common

Pharmacologic Highlights

Medication or Drug Class	Dosage	Description	Rationale
Digoxin	0.25 mg PO qd	Cardiotonic	Increases force of contraction in people with left ventricular dysfunction
Vasodilators such as nifedipine, hydralazine, prazosin, nitroprusside	Nifedipine, 10–30 mg tid PO or SL	Calcium channel blocker; systemic vasodilators	Decrease afterload (pressure that the left ventricle has to pump against) and decreases regurgitant blood flow
Diuretics	Varies with drug	Thiazides; loop diuretics	Enhance pumping ability of the heart
Angiotensin-converting enzyme (ACE) inhibitors such as captopril, enalapril, lisinopril	Varies with drug	Competitive inhibitors of ACE	Reduces angiotensin II levels and aldosterone secretion to control blood pressure and proteinuria; increases systemic blood flow and reduces potassium loss

Cardiomyopathy

- is **disease of the heart muscle** that is associated with cardiac dysfunction.
- It is classified according to the structural and functional abnormalities of the heart muscle:
 - dilated cardiomyopathy (DCM),
 - hypertrophic cardiomyopathy (HCM),
 - restrictive or constrictive cardiomyopathy (RCM)
- ✓ **Primary cardiomyopathies** (genetic, nongenetic, and acquired),
- ✓ **Secondary cardiomyopathies**, which show myocardial involvement secondary to the influence of disease processes

Pathophysiology

- The pathophysiology of all cardiomyopathies is a series of events that culminate in impaired cardiac output.
- Decreased stroke volume stimulates the **sympathetic nervous system** and **the renin–angiotensin–aldosterone response** → **increased systemic vascular resistance** and **increased sodium and fluid retention** → increased workload on the heart.
- These alterations can **lead to heart failure**

Dilated Cardiomyopathy

- is the most common form of cardiomyopathy.
- DCM is **distinguished by significant dilation** of the ventricles without simultaneous hypertrophy (i.e., increased muscle wall thickness) and systolic dysfunction
- The ventricles have elevated systolic and diastolic volumes but a **decreased ejection fraction**(stasis of blood in the left ventricle)
- Caused by different conditions and disease such as **pregnancy, thyrotoxicosis, myxedema, persistent tachycardia.**
- When the causative factor cannot be identified, the diagnosis is idiopathic DCM

• • •

- Increasing the amount of blood **remaining in the ventricle after contraction.**
- Less blood is then able to enter the ventricle during diastole, increasing end-diastolic pressure and eventually **increasing pulmonary and systemic venous pressures.**
- Poor blood flow through the ventricle may also cause **ventricular or atrial thrombi**

Restrictive Cardiomyopathy

- **Both ventricles become rigid**, which **distorts the filling** phase of the heart. The contraction phase remains normal.
- is characterized by **diastolic dysfunction** caused by rigid ventricular walls that impair diastolic filling and ventricular stretch
- The result is that ventricular walls become fibrotic, cardiac filling diminishes, and **cardiac output decreases**
- RCM may be **associated with amyloidosis**
- **Dyspnea, nonproductive cough, and chest pain**

Hypertrophic Cardiomyopathy

- known as hypertrophic obstructive cardiomyopathy or idiopathic hypertrophic subaortic stenosis
- consists of **ventricular hypertrophy**, **rapid contraction of the left ventricle**, and **impaired relaxation**.
- It is commonly the result of **hypertension or valvular heart disease**.
- In HCM, the heart muscle asymmetrically increases in size and mass, especially along the septum
- The increased thickness of the heart muscle reduces the size of the ventricular cavities and causes the ventricles to take a longer time to relax after systole.

...

- The **hypertrophied cardiac muscle cells** are **disorganized, oblique, and perpendicular** to each other → decreasing the effectiveness of contractions → possibly increasing the risk of dysrhythmias such as **ventricular tachycardia and ventricular fibrillation.**

Causes

- The etiology of many cases remains unclear.
 - coronary artery ischemia, hypertension, or valvular disease.
 - Infections such as viruses, bacteria,
 - genetic cause.
 - Inflammatory conditions such as rheumatoid arthritis or systemic lupus erythematosus may lead to the condition,
 - metabolic and endocrine disorders.
 - Toxins such as alcohol, cocaine, antineoplastic agents, poisons, or anesthesia agents
 - Amyloidosis,
 - Fibrosis secondary to radiation,
 - Vitamin deficiencies, or pregnancy.
- ➔ Cardiomyopathy

TYPE	SIGNS AND SYMPTOMS
Dilated cardiomyopathy	Fatigue and generalized weakness Narrow pulse pressure, peripheral edema, and neck vein congestion Orthopnea, dyspnea on exertion, paroxysmal nocturnal dyspnea—signs of left-sided CHF Palpitations and chest pain Pulmonary congestion and pleural effusions Syncope Tachycardia
Hypertrophic cardiomyopathy	Angina and palpitations Dysrhythmia, possibly leading to sudden death Loud systolic murmur caused by obstruction of outflow Orthopnea and dyspnea on exertion Pulmonary congestion Sudden loss of pulses Syncope
Restrictive cardiomyopathy	Bradycardia Dyspnea Fatigue and generalized weakness Peripheral edema, liver congestion, abdominal discomfort, and neck vein distention—signs of right-sided CHF

Mgt

Pharmacologic Highlights

Medication or Drug Class	Dosage	Description	Rationale
Angiotensin-converting enzyme (ACE) inhibitors	Varies by drug	Inactivates renin-angiotensin system, thereby decreasing vascular resistance and ventricular afterload	ACE inhibitors decrease mortality rates in patients with left ventricular dysfunction; reduces readmissions caused by heart failure
Digoxin	0.25 mg qd	Direct action on cardiac muscle to increase contractility	Idiopathic dilated cardiomyopathy: Improves contractility and slows the renin-angiotensin response
Vasodilators	Varies by drug	Drugs such as nitrates and hydralazine dilate arteries	Reduce both preload and afterload by causing venous and arterial vasodilation

Other Drugs: Beta-adrenergic antagonists have also been known to be beneficial in the treatment of dilated cardiomyopathy because of the resultant decrease in myocardial oxygen demand, improved ventricular filling, and inhibition of sympathetic vasoconstriction. Medical interventions for HCM are aimed at decreasing the force of ventricular contraction and decreasing the outflow obstruction. Agents commonly used to achieve this goal include beta-adrenergic antagonists or calcium channel blockers. Di-

Heart Failure(HF)

- HF is the inability of the heart to maintain adequate circulation to meet tissue needs for oxygen and nutrients
- HF occurs **when the heart muscle is unable to pump** effectively, resulting in:
 - ❖ inadequate cardiac output,
 - ❖ myocardial hypertrophy, and
 - ❖ pulmonary/systemic congestion
- HF is the result of an **acute** or **chronic** cardiopulmonary problem, such as systemic hypertension, myocardial infarction, pulmonary hypertension, dysrhythmias, valvular heart disease, pericarditis, and cardiomyopathy

...

- Severity of HF is graded on the **New York Heart Association's** functional classification scale indicating how little, or how much, activity it takes to make the client symptomatic (chest pain, shortness-of-breath)
 - ❖ **Class I:** Client exhibits **no symptoms with activity**
 - ❖ **Class II:** Client has symptoms **with ordinary exertion**
 - ❖ **Class III:** Client displays symptoms with minimal exertion
 - ❖ **Class IV:** Client has **symptoms at rest**

Stages of Heart Failure (ACC/AHA)

Classification	Criteria	Patient Characteristics	Treatment Recommendations for Appropriate Patients
Stage A	Patients at high risk for developing left ventricular dysfunction but without structural heart disease or symptoms of HF	Hypertension Atherosclerotic disease Diabetes Metabolic syndrome	Heart healthy lifestyle Risk factor control of hypertension, lipids, diabetes, obesity
Stage B	Patients with left ventricular dysfunction or structural heart disease who have not developed symptoms of HF	History of myocardial infarction Left ventricular hypertrophy Low ejection fraction	Implement stage A recommendations, plus: • ACE inhibitor or ARB for low EF or history of MI • Beta blocker • Statin
Stage C	Patients with left ventricular dysfunction or structural heart disease with current or prior symptoms of heart disease	Shortness of breath Fatigue Decreased exercise tolerance	Implement stage A and B recommendations, plus: • Diuretics • Aldosterone antagonist • Sodium restriction • Implantable defibrillator • Cardiac resynchronization therapy
Stage D	Patients with refractory end-stage HF requiring specialized interventions	Symptoms despite maximal medical therapy Recurrent hospitalizations	Implement stage A, B, and C recommendations, plus: • Fluid restriction • End-of-life care • Extraordinary measures:

Different forms of heart failure

A. High out put failure:

- The cardiac out put is **normal or above normal** but is unable to meet the body's need
- An **uncommon** form of heart failure
- Causes: Anemia, pregnancy, Hyperthyroidism, atrioventricular fistula, beriberi

B. Low out put failure:

- Cardiac out put are **below normal**
- Causes: Hypertension, MI, arteriosclerosis, dilated cardiomyopathy, valvular & pericardial disease

...

- **Low output HF** can initially occur on either the left or right side of the heart

A. Left-sided heart (ventricular) failure results in **inadequate left ventricle (cardiac) output** and consequently in inadequate tissue perfusion. Forms include:

- **Systolic heart (ventricular) failure (ejection fraction below 40%,** pulmonary and systemic congestion)
- **Diastolic heart (ventricular) failure** (inadequate relaxation or “stiffening” **prevents ventricular filling**), ejection fraction is normal

B. Right-sided heart (ventricular) failure results in **inadequate right ventricle output** and systemic venous congestion (for example, peripheral edema)

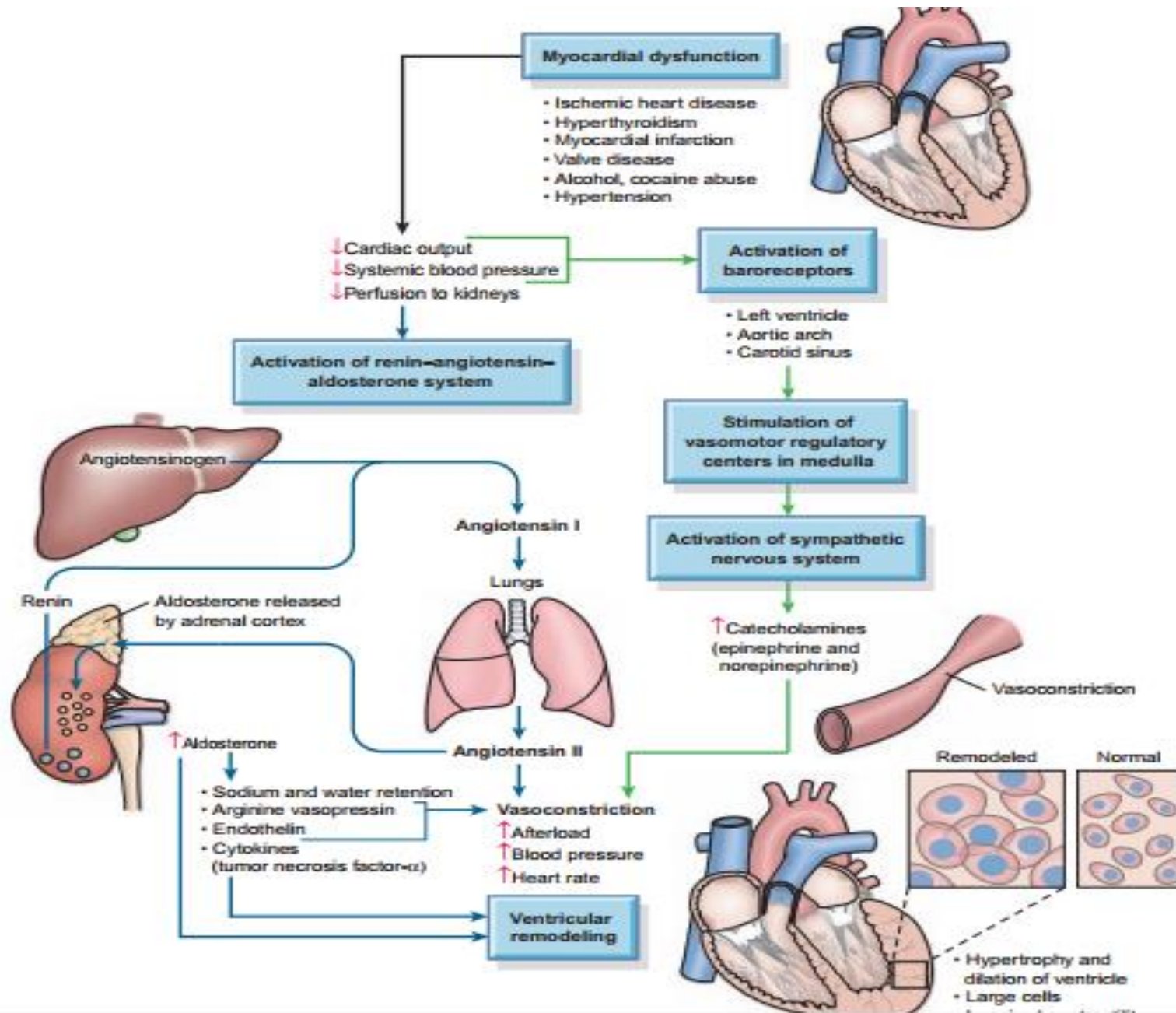
➤ **Acute Vs chronic HF**

...

Risk Factors/Causes/

- **Left-Sided Heart (Ventricular) Failure**
 - Hypertension, CHD
 - Valvular disease (mitral and aortic)
- **Right-Sided Heart (Ventricular) Failure**
 - Left-sided heart (ventricular) failure
 - Right ventricular myocardial infarction
 - Pulmonary problems (COPD, ARDS)
- **High-Output Heart Failure**
 - Increased metabolic needs, Septicemia (fever)
 - Anemia, Hyperthyroidism
- **Cardiomyopathy**
 - Coronary artery disease
 - Infection or inflammation of the heart muscle
 - Various cancer treatments, Prolonged alcohol abuse, Heredity

Pathophysiology



...

Signs and symptoms

Left-sided failure

- Dyspnea on exertion, orthopnea, nocturnal dyspnea,
- Fatigue, pallor, cyanosis
- Displaced apical pulse,
- S3 heart sound (gallop), tachycardia
- **Pulmonary congestion (dyspnea, cough, bibasilar crackles)**
- Frothy sputum (may be blood-tinged)
- Altered mental status (confusion, disorientation)
- Symptoms of organ failure, such as **oliguria**

Hemodynamic findings:

- **CVP/right atrial pressure** (normal = 1 to 8 mm Hg): Normal or elevated
- PAP (normal = 15 to 26 mm Hg/5 to 15 mm Hg): Elevated
- **CO** (normal = 4 to 7 L/min): Decreased

Signs and symptoms

Right-sided failure

- ✓ **Jugular vein distention**
- ✓ **Ascending dependent edema** (legs, ankles, sacrum, buttocks)
- ✓ Abdominal distention(bloating), **ascites**
- ✓ Fatigue, weakness
- ✓ Nausea and anorexia
- ✓ Nocturnal diuresis
- ✓ **Liver enlargement (hepatomegaly)** and tenderness
- ✓ **Weight gain**

Hemodynamic findings

- CVP/right atrial pressure (normal = 1 to 8 mm Hg): Elevated

Cardiomyopathy

- ✓ Fatigue, weakness
- ✓ Heart failure (left with dilated type, right with restrictive type)
- ✓ Dysrhythmias (for example, heart block)
- ✓ Cardiomegaly

Pitting edema

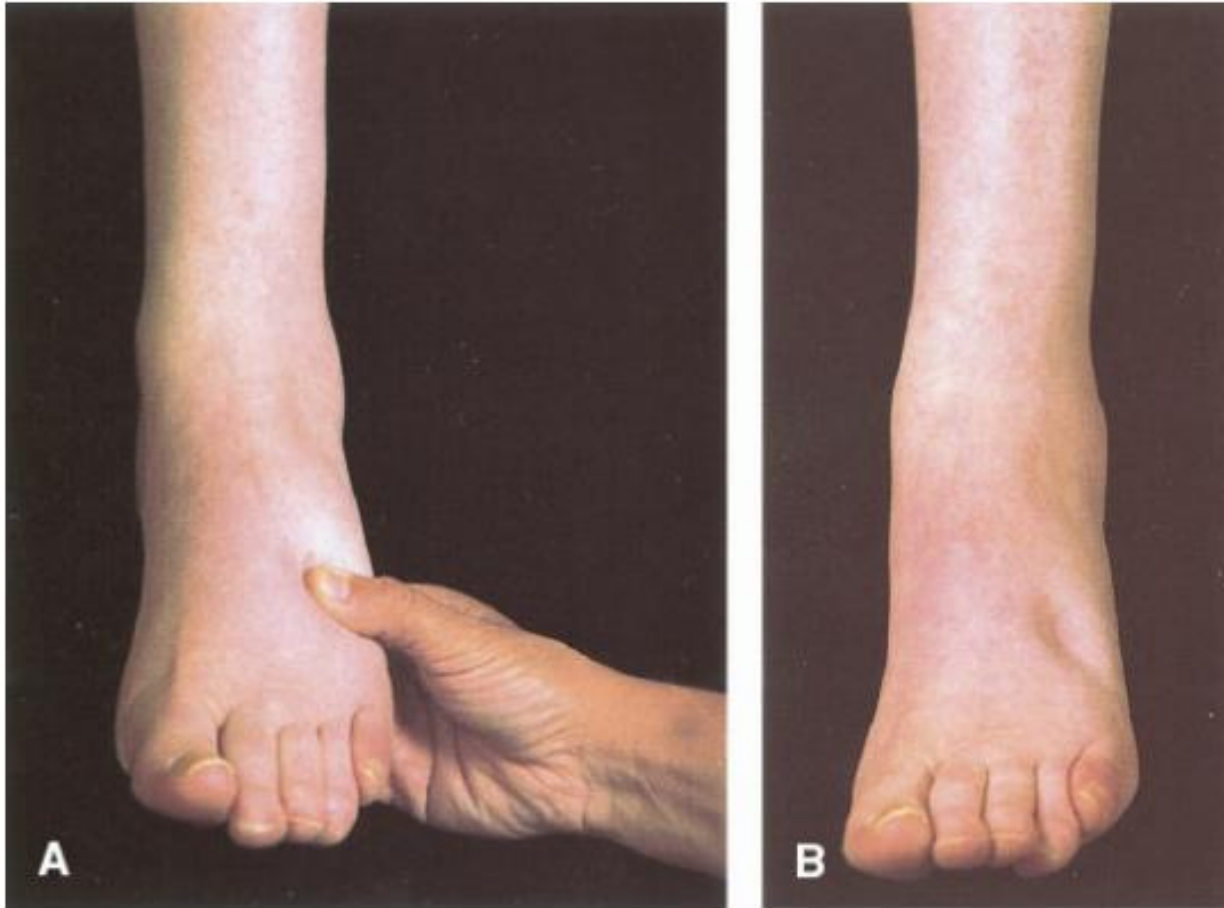


Figure 29-2 • Example of pitting edema. A.

Framingham's criteria for diagnosis of CHF

At least **One major & two minor** criteria

Major

- Neck vein distension
- Cardiomyopathy
- Acute pulmonary congestion
- Increased CVP
- Paroxysmal nocturnal dyspnea
- Weight loss of 4.5kg in 5 days in response to Rx
- Pulmonary edema
- **Hepatogular reflux**
- Pulmonary edema

Minor

- Peripheral edema
- Night cough
- Dyspnea on exertion
- Hepatomegaly
- Pleural effusion
- Tachycardia (> 120)

Diagnostic procedures

- BNP(B-type Natriuretic Peptides) < 100 pg/mL = no HF
- BNP levels of 100 to 300 pg/mL suggest heart failure is present;
- BNP > 300 pg/mL = mild HF
- BNP > 600 pg/mL = moderate HF
- BNP > 900 pg/mL = severe HF
- **Chest X-ray** : Cardiomegaly and pleural effusions
- **Electrocardiogram** (ECG),
- Cardiac enzymes, electrolytes, and arterial blood gases
- Assess factors contributing to heart failure and/or the impact of heart failure.

...

Diagnostic procedures

- **Ultrasound** to measure both systolic and diastolic function of the heart.
- LVEF : The volume of blood pumped from the left ventricle into the arteries upon each beat. Normal is 55 to 70%
- RVEF: The volume of blood pumped from the right ventricle to the lungs upon each beat. Normal is 45 to 60%.
- CBC,
- Electrolytes,
- RFT,
- LFT,
- Thyroid function tests,

...

Assess/monitor

- ✓ Oxygen saturation
- ✓ Vital signs
- ✓ Heart rhythm
- ✓ Lung sounds for crackles, wheezes
- ✓ Level of dyspnea upon exertion
- ✓ Serum electrolytes (especially potassium if receiving diuretics)
- ✓ Daily weight
- ✓ Changes in level of consciousness
- ✓ Intake and output
- ✓ For signs of drug toxicity
- ✓ Coping ability of client and family

Nursing interventions

- If a client is experiencing respiratory distress, place the client in **high-Fowler's position and administer oxygen** as prescribed
- Encourage **bed rest until the client is stable**
- Encourage energy conservation by assisting with care and activities of daily living
- Maintain dietary restrictions as prescribed (**restricted fluid intake, restricted sodium intake**)
- Monitor the patient for common signs and symptoms of heart failure, such as chest discomfort, shortness of breath, and paroxysmal nocturnal dyspnea
- Watch for signs and symptoms of left-sided heart failure, such as **anxiety, orthopnea, and abnormal breath sounds**

...

- Monitor for signs and symptoms of right-sided heart failure, such as jugular venous distension, hepatomegaly, splenomegaly, peripheral edema, and bounding peripheral pulses
- Encourage bed rest in **semi-Fowler's position** for ease of breathing
- Provide rest intervals between periods of activity
- Restrict fluids & salt as prescribed
- Administer medications as prescribed, and monitor for their therapeutic and adverse effects

Administer medications as prescribed

❖ Diuretics: **To decrease preload**

- Loop diuretics, such as furosemide , bumetanide
- Thiazide diuretics, such as hydrochlorothiazide
- Potassium-sparing diuretics, spironolactone
 - Teach the client to take foods and drinks that are high in potassium
 - Potassium supplementation may be required
 - (Lasix) no faster than 20 mg/min

❖ Inotropic agents, such as **digoxin, dopamine, dobutamine** , milrinone (Primacor): To increase contractility and thereby improve cardiac output

❖ Vasodilators, such as nitrates: **To decrease preload and afterload**

...

❖ **Afterload-reducing agents**

- **Angiotensin converting enzyme (ACE) inhibitors**, such as enalapril, captopril ; monitor for initial dose hypotension
- **Beta-blockers**, such as carvedilol, metoprolol
- **Angiotensin receptor II blockers**, such as losartan

❖ **Anticoagulants:** warfarin, heparin, clopidogrel: **To prevent thrombus formation** (risk associated with congestion/stasis and associated atrial fibrillation)

...

➤ **Teach clients who are self-administering digoxin to:**

- Count pulse for one full minute before taking the medication.
- If **the pulse rate is irregular** or the pulse rate is outside of the limitations set by the provider (usually less than 60 or greater than 100), instruct the client **to hold the dose** and to contact the primary care provider.
- Take digoxin dose **at same time each day**
- **Do not take digoxin** at the similar time with **antacids. separate by 2 hr**
- Report **signs of toxicity**, including **fatigue, muscle weakness, confusion, loss of appetite, bradycardia, nausea, and blurred vision.**
- Regularly have digoxin and potassium levels checked.
- Provide emotional support to the client and family

...

✓ Dosage of digoxin

- Start with 1 mg over 24 – 48 hrs, when rapid effect is required & then 0.25mg po daily.
- Monitoring of serum digoxin is important

✓ Digoxin is **contraindicated** in:

- Acute MI
- AV conduction disturbance
- Hypomagnesemia
- Chronic lung disease
- Hypokalemia

Complications

- Cardiogenic shock
- Pulmonary edema
- Thromboembolism
- Pulmonary Embolism
- Pericardial Effusion
- Cardiac Tamponade
- Cardiac arrest

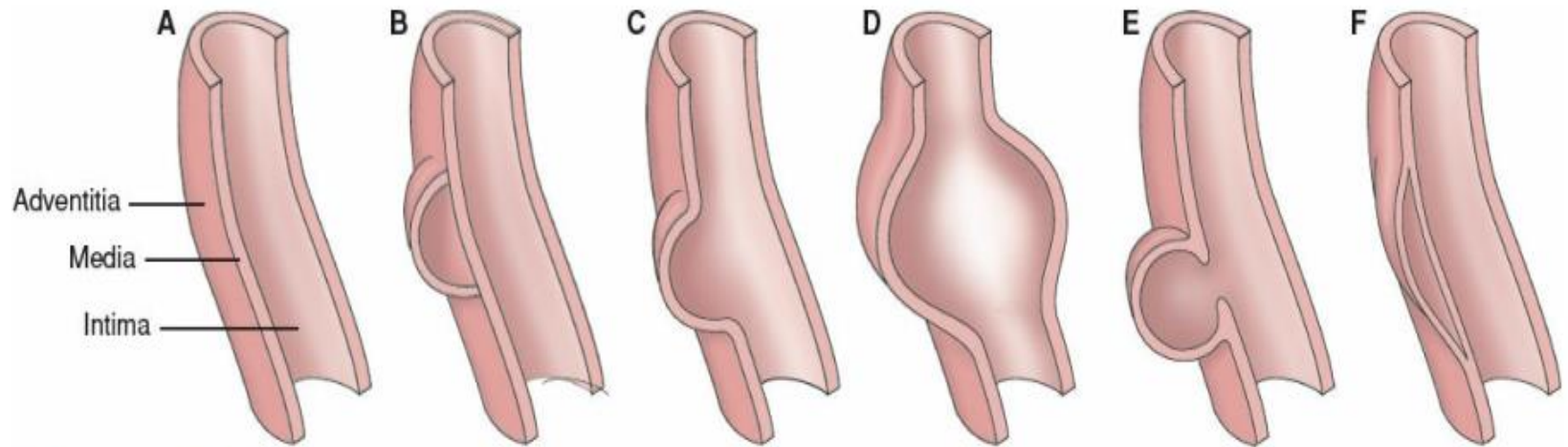
Vascular disorders

- A. Disorders of the arteries
- B. Disorders of the veins
- C. Disorders of the lymphatic system

A. Disorders of the arteries

Aneurysm

- An aneurysm is **distension of an artery** caused by structural weakening of the arterial wall.
- Under hemodynamic pressure the weakened area enlarges, causing serious complications by **compressing surrounding structures**.
- It results from **degeneration of the medial wall**, which occurs as a normal part of the **aging process** as well as with **hypertension, atherosclerosis, trauma or infection, immunologic conditions**.



A. Normal artery.

B. False aneurysm

C. True aneurysm.

D. Fusiform aneurysm—symmetric

E. Saccular aneurysm: bulbous protrusion of one side of the arterial wall

F. Dissecting aneurysm : a hematoma that splits the layers of the arterial wall.



➤ **Thoracoabdominal aneurysm**

- May originate in the **ascending aorta & aortic arch** (frequent site of dissection) or in the lower descending aorta & upper abdominal aorta.

➤ ***Abdominal Aorta Aneurysm***

- Originate in the abdominal aorta, typically ***b/n the renal arteries & iliac branches.***
- ***It is Sacular type***
- AAAs are the **most common type of arterial aneurysms**
- *Many of these pts (mostly male) are asymptomatic.*

• • •

- Aneurysm may also occur in **peripheral arteries** (femoral, popliteal, renal, subclavian) or any major artery.
- The distension may occur **from one side or entire** circumference, but the directing type is tear of the intima & separation of the medial layers **causing hemorrhage or intramural hematoma.**

➤ **Intracranial aneurysm**

- *Is a **sacular dilation of a cerebral artery** whose walls are congenitally weakened, resulting in hemorrhage (stroke), increased ICP, vasospasm, ischemia.*

...

Predisposing factors:

- Local infections: pyogenic or fungal
- Congenital weakness of vessels
- Arteriosclerosis, trauma, **syphilis**

Complications:

- **Fatal hemorrhage**
- **Paraplegia** due to interruption of anterior spinal artery
- **Ischemia of abdomen, myocardium, lower extremity, stroke, renal failure**

...

Clinical manifestations

- **Thoracoabdominal aneurysm:**
 - Constant, boring pain or pressure in chest
 - Intermittent neurologic pain due to nerve compression
 - Dyspnea, cough, & **hoarseness b/s** of pressure against trachea & laryngeal nerve
 - Dysphagia
 - Dilated superficial veins of chest & cyanosis due to compression of chest vessels
 - **Pulse or B/P variations b/n arms**
- **Abdominal aneurysm**
 - Persistent or Intermittent abdominal pain, often localized to middle or lower Lt side of abdomen
 - **Pulsating mass with bruits**
 - B/P elevated in arm more than in thigh

Aneurysm

Diagnostic evaluation:

- Abdominal or chest X-ray
- CT scan or Ultrasonography
- Arteriography

Management

- **Surgery** to remove the aneurysm & restore vascular continuity
- Endovascular grafting
- Nursing interventions for complications & postoperative cases

Hypertension

- Is a systolic blood pressure greater than 140 mm Hg and a diastolic pressure greater than 90 mm Hg **over a sustained period**, based on the average of **two or more** blood pressure **measurements taken in two or more contacts** with the health care provider **after an initial screening**.
- Is the product of cardiac output multiplied by peripheral resistance

$$BP = CO * PR$$

- Cardiac output is the product of the heart rate multiplied by the stroke volume. $CO = HR * SV$
- In normal circulation, pressure is exerted by the flow of blood through the heart and blood vessels.

• • •

- **High blood pressure**, known as hypertension, can result from
 - a change in cardiac output
 - a change in peripheral resistance,
 - The amount, velocity, or viscosity of blood, or
- The medications used for treating hypertension decrease peripheral resistance, blood volume, or the strength and rate of myocardial contraction

• • •

Pathophysiology

- Although **the precise cause for most cases of hypertension cannot be identified**, it is understood that hypertension is a multifactorial condition
- For hypertension to occur there must be a change in one or more factors affecting peripheral resistance or cardiac output
- In addition, there must also be a problem with the control systems that monitor or regulate pressure
- ✚ **Hypertension may be caused by one or more of the following:**
 - **Increased sympathetic nervous system** activity related to dysfunction of the autonomic nervous system, leading to increased stress responses.



- **Increased renal reabsorption** of sodium, chloride, and water related to a genetic variation in the pathways by which the kidneys handle sodium
- **Increased activity of the renin-angiotensin-aldosterone system**, resulting in expansion of extracellular fluid volume and increased systemic vascular resistance, leads to **vasoconstriction** and **retention of sodium and water**. The increase in blood volume leads to hypertension
- **Decreased vasodilatation** of the arterioles related to **dysfunction of the vascular endothelium**
- **Resistance to insulin action**, which may be a common factor linking hypertension, type 2 diabetes mellitus, hypertriglyceridemia, obesity, and glucose intolerance

Hypertension Stages

Stages	Systolic blood pressure (mmHg)	Diastolic blood pressure(mmHg)
Normal	< 120	< 80
Pre hypertension	120 - 139	80 -89
Stage I	140 - 159	90 - 99
Stage II	≥ 160	≥ 100
10/10/2022	By Afework E.	179



Types of hypertension

There are two types of hypertension

➤ **Primary hypertension or essential hypertension :**

- The reason for the elevation in blood pressure **cannot be identified.**
- Is **more common in adolescents**
- Has multiple risk factors, including obesity and a family history of hypertension

➤ **Secondary hypertension**

- Is the term used to signify high blood pressure from an identified cause.
- Is more common in preadolescent children, with most cases caused by **renal disease.**



Risk factors

1. Non modifiable risk factors: Unable to be changed

- Age - The amount of **collagen in arteries increases** with age, causing the blood vessels to **get stiff**.
- Family history of hypertension
- Sex
- Race and ethnicity



2. Modifiable risk factors: Changeable

- **Tobacco smoking:** Smoking **increases blood cholesterol**, creating **small plaques in arteries**
- **Stress** - Stress reduction and relaxation can significantly improve your general health state as well as lower your blood pressure
- **Diet with high levels of saturated fat in it** - Fats will stimulate the process called **atherosclerosis** which makes a man more prone to all kinds of cardiovascular diseases
- **Alcohol abuse**
- **Obesity** - One of the most important risk factors. According to some researches done in the past, **every extra kilogram generally will increase your blood pressure by two mmHg**

NB. High blood pressure increases morbidity and mortality from:
Cardio-vascular diseases, Stroke, Congestive heart failure and many renal diseases

• • •

Clinical manifestations

- Hypertension is sometimes **called the silent killer** because people who have it are often symptom free
- Rare: **Headache, bloody nose, blurred vision, dizziness-** late signs, **calf pain** aggravated by walking and relieved by rest
- Physical examination may reveal no abnormalities other than high blood pressure
- Occasionally, **retinal changes such as hemorrhages**, exudates (fluid accumulation), arteriolar narrowing.
- In severe hypertension: **Papilledema** (swelling of the optic disc) may be seen,
- Target organ disease: Damage to blood vessels of **heart, kidney, brain** (cerebral hemorrhage & hypertensive encephalopathy)



- **Assessment and diagnostic evaluation**

- A thorough health history and physical examination are necessary.
- The retinas are examined, and laboratory studies are performed to assess possible target organ damage
- Routine laboratory tests include:
 - Urinalysis
 - Blood chemistry (ie, analysis of sodium, potassium, creatinine, fasting glucose, and total and high-density lipoprotein [HDL] cholesterol levels),
- Electrocardiogram - **Left ventricular hypertrophy** can be assessed
- Renal damage may be suggested by elevations in BUN and Creatinine levels

Management of hypertension

- Goal of Treatment: To Keep BP **below 140/90** and **below <150/90** for elderly (>75years)

I. Non-pharmacologic treatment

➤ **Dietary Changes:**

- Restrict salt (sodium chloride) and reduce salt when cooking, limit processed and fast foods.
- Limit fatty meat, dairy fat and cooking oil (less than two table- spoons per day).
- Replace palm or coconut oil with soya/corn/safflower oil, if available.
- Replacing other meat with chicken (without skin) and fish, if affordable, can be useful.
- Also limit intake of free sugars (e.g. sugar sweetened beverages, calorie rich, but nutritionally poor, snacks such as sweets, cakes, and crisps).



➤ **Exercise**

- All individuals should be encouraged to take at least 30 minutes of moderate physical activity (e.g. brisk walking, cycling) a day for at least five days a week.

➤ **Weight loss**

- All individuals who are overweight or obese should be encouraged to lose weight through a combination of reduced-energy diet (dietary advice) and physical activity.

➤ **Limited alcohol intake**

- Women who consume two or more alcoholic beverages per day and men who have three or more drinks per day should be advised to reduce alcohol consumption.

➤ **Smoking cessation**

- All nonsmokers should be encouraged not to start smoking and all smokers should be strongly encouraged to quit smoking by a health professional and supported in their efforts to do so.

B. Specific drugs

- **Diuretics**
 - ❖ Thiazide (hydrochlorothiazide)
 - ❖ Loop diuretics (lasix)
 - ❖ Potassium-sparing diuretics (Aldestron antagonists) (Sprinolactone)
- **Sympatholytics**
 - ❖ Adrenergic neuron blockers e.g. Reserpine, Guannethedine)
 - ❖ Alpha adrenergic blockers e.g. Prazosin
 - ❖ Beta adrenergic blockers : e.g. Propanolol
- **Vasodilators:** e.g. Hydralizine, Sodium nitroprusside
- **Calcium channel blockers :** e.g. Verapamil, Nifedipine, Diltiazem
- **Angotensin converting enzyme inhibitors:** e.g. Captopril, Enalapril



Table 4.1: Compelling indications for the use of specific antihypertensive drugs

Indication	Antihypertensive drugs
Proteinuric CKD	ACE inhibitor, ARB
Systolic heart failure	ACE inhibitor or ARB, Beta-blocker, diuretic, aldosterone antagonist
Atrial fibrillation/flutter rate control	Beta-blocker, non-dihydropyridine calcium channel blocker
Post-myocardial infarction	ACE inhibitor, beta-blocker, aldosterone antagonist
Angina pectoris	Beta-blocker, calcium channel blocker

Table 4.2 Drugs: dosage and contraindications for use

Class of drug	Drug	Daily dosage	Contraindications
ACE inhibitors	Enalapril	Initial dose 2.5-5 mg BID, increasing to 10-20 mg BID	
	Lisinopril	Initial dose of 5-10 mg daily increasing to 20-40 mg daily	
ARBs (Angiotensin receptor blockers)	Losartan	Initial dose of 50 mg daily, increased to 100 mg daily	Pregnancy
	Valsartan	Initial dose of 80 mg daily , increased to 320 mg daily	Hyperkalemia Bilateral renal artery stenosis
CCBs (calcium channel blockers)	Nifedipine (sustained release formulation)	Initial dose of 30 mg daily (XL nifedipine), increasing to 60-120 mg daily For other preparations starting dose of 10-20 mg BID increased to 20-40 mg BID	Tachyarrhythmias Heart failure
	Amlodipine	Starting at 2.5 mg daily and increasing to 10 mg daily	
Thiazide diuretics	Hydrochlorothiazide	Starting at 12.5 mg increasing to 25 mg once daily	Gout
Beta-blockers	Propranolol	80 mg BID	Asthma COPD
	Atenolol	50mg -100mg/d	AV block (grade 2 or 3)

...

Complications of hypertension

- ❖ Stroke
- ❖ Atherosclerosis
- ❖ Coronary artery disease
- ❖ Myocardial infarction or heart failure
- ❖ Left ventricular hypertrophy
- ❖ Kidney/eye damage

Hypertensive crises

- Hypertensive **emergency** and hypertensive **urgency**
 - May occur in patients whose hypertension has been poorly controlled or in those who have **abruptly discontinued** their medications.
- **Hypertensive emergency:**
 - Hypertensive emergency means severely raised blood pressure (**>200/120**) **in the presence of end organ damage.**
 - Is a situation in which blood pressure must be lowered **immediately** (not necessarily to less than 140/90 mm Hg) to halt or **prevent damage to the target organs**
 - Conditions associated with hypertensive **emergency** include: **Acute myocardial infarction, intracranial hemorrhage, PE, HF, stroke**
 - Are acute, life threatening blood pressure elevations that require prompt treatment in an intensive care setting because of the serious target organ damage that may occur.

...

- The **medications** of choice in **hypertensive emergencies** are those that have an **immediate effect**. Intravenous vasodilators, including:
 - **Sodium nitroprusside / hydralazine**
 - Nicardipine hydrochloride
 - Fenoldopam mesylate , enalaprilat (I.V.), and
 - Nitroglycerin have an immediate action that is short lived (minutes to 4 hours), and they are therefore used as the initial treatment
 - Hydralazine, 5-10 mg IV/IM initially and refer to higher health facility after giving one oral dose of long-acting antihypertensive like Atenolol (50mg)

...

- **Hypertensive urgency:**

- Hypertensive urgency occurs when blood pressure **readings are 180/110** or higher -- **but there is no damage to the body's organs.**
- Is a situation in which blood pressure must be **lowered within a few hours**
- Hypertensive urgencies are managed with oral doses of fast-acting agents such as loop diuretics like furosemide [Lasix]), beta-blockers propranolol (Inderal).

...

Prevention

- **The most important fact about this condition is that we are able to prevent it.**
- The degree to which hypertension can be prevented depends on a number of features including:
 - Current blood pressure level
 - Changes in end/target organs (retina, kidney, heart - among others)
 - Risk factors for cardiovascular diseases and
 - The age at presentation.
- Unless the presenting patient has very severe hypertension, there should be a relatively prolonged assessment period within which should be repeated measurements of blood pressure.

• • •

- Following this, **lifestyle advice and non-pharmacological options should be offered to the patient, before any initiation of drug therapy.**
- These measures include;
 - ❖ Weight reduction and regular aerobic exercise
 - ❖ Reducing sodium (salt) in the diet
 - ❖ Additional dietary changes: beneficial to reducing blood pressure includes the DASH diet (dietary approaches to **stop hypertension**), which is **rich in fruits and vegetables and low-fat or fat-free dairy foods.**
 - ❖ Discontinuing tobacco use and alcohol consumption.
 - ❖ **Reducing stress**

Shock

- **Shock 1:** **inadequate organ perfusion** and substrates to meet the tissue's oxygenation demand
- **Shock 2:** a condition in which system B/P is too low to deliver oxygen & nutrients to support vital organs & cellular function
- **Shock 3:** a life threatening medical condition that occurs due to **inadequate substrate for aerobic cellular respiration**
- **Shock 4:** **acute circulatory failure** leading to inadequate tissue perfusion & end organ injury

Shock

- **SUPPLY < DEMAND**
- **Failure to remove metabolic end-products**
- Result of inadequate blood flow and/or oxygen delivery

Shock is not a synonym to hypotension!

Hypotension: is an indication of:

1/ An abnormality of HR, SV, or PR

2/ Failure of the others to compensate

○ In adults:

- Systolic **BP < 90 mmHg & Diastole < 60 mmHg**), specially in the arteries of the systemic circulation + S & Sxs
- Mean arterial pressure (**MAP**) **≤ 60 mm Hg**
- Reduction **of systolic BP > 40** mm Hg from the patient's baseline pressure

$$\text{MAP} = \frac{\text{SBP} + 2\text{DBP}}{3} = \text{DBP} + 1/3 \text{ pulse pressure}$$

What is shock?

Generalized State of Hypoperfusion

Inadequate oxygen delivery



Catecholamines and other responses



Anaerobic metabolism



Cellular dysfunction



Cell death

Pathophysiology of shock syndrome

Cells switch from aerobic to anaerobic metabolism

lactic acid production



Cell function ceases & swells



membrane becomes more permeable



electrolytes & fluids seep in & out of cell



Na⁺/K⁺ pump impaired



mitochondria damage



cell death

Shock Classifications

1/ Hypovolumic(the fluid) shock

- Loss of blood, plasma, electrolytes
- Burn pts, dehydration(vomiting, diarrhea, diaphoresis, DKA, ascites)

2/ Cardiogenic(the pump) shock

- Assoc w/ cardiac impairment(MI, stenosis, regurgitation, arrhythmia, cardiomyopathy, HF)

3/ Distributive (the tubing)shock

- Occur with profound **peripheral vasodilatation**
- **CO & total body fluid volume** are **normal**, but organ perfusion pressure are markedly compromised
 - Anaphylactic shock, septic shock /**Toxic Shock Syndrome(TSS)**
 - Neurogenic shock
 - Endocrine Ex. Hypothyroidism, Thyrotoxicosis, Addison's, etc

4/ Obstructive shock

- **Obstruction of the blood flow outside of the heart** but CO is normal

Ex. Cardiac Tamponade, Pneumothorax, **pericarditis**, pulmonary HPN

Effects of shock

- ❖ Decreased oxygen delivery
- ❖ Fall in body temperature
- ❖ Secretion of adrenalin
- ❖ Metabolic disorder
- ❖ Impairment of liver function
- ❖ Shock kidney
- ❖ Shock lung
- ❖ Reduced coronary perfusion
- ❖ Hypoxic brain damage

Stages of shock

- Pre-shock
- Shock
- End-organ dysfunction

Stages: Pre-shock

- Warm or **compensated** shock
- **Regulatory mechanisms are able to compensate for diminished perfusion**
- Low-preload:
 - Tachycardia
 - Peripheral vasoconstriction
 - Decrease in blood pressure
- Low-after load:

Stages: Shock

- Compensatory mechanisms become **overwhelmed**, resulting in:
 - Tachycardia
 - Tachypnea
 - Metabolic acidosis
 - Oliguria
 - Cool, clammy skin
- Usually occur with loss of **20-25% of effective blood volume**

Irreversible stage

- The irreversible (or refractory) stage of shock represents the point along the shock continuum at which **organ damage is so severe that the patient does not respond to treatment and cannot survive**



- **Compensation mechanism**
 - SNS, adrenal medulla stimulated
 - Renin secreted
 - Increased secretion of ADH
 - Secretion of glucocorticoids
 - Acidosis stimulates respiration
- **Complications of decompensation of shock**
 - Acute renal failure
 - Adult respiratory distress syndrome (ARDS)
 - Hepatic failures
 - Hemorrhagic ulcers
 - Infection of septicemia
 - Decreased cardiac function

Shock signs and symptoms

- **1st signs**
 - **Hypotension**, thirst, agitation, restlessness
 - Often missed
- **2nd signs**
 - Cold, moist, pale skin; diaphoretic skin, tachycardia; oliguria
 - Tachypnea, shallow respirations
 - Compensation
 - Vasoconstriction
 - Acidosis
- **Prolonged**
 - Alteration in level of consciousness, anxiety
 - Compensated metabolic acidosis progresses to decompensated
 - Acute renal failure
 - Monitoring(decreased urinary output)

Empiric criteria for shock

4 out of 6 criteria have to be met

- i. Altered mental status
- ii. Heart rate >100 bpm
- iii. Respiratory rate > 22 b/m (or $\text{PaCO}_2 < 32$ mmHg)
- iv. Urine output < 0.5 ml/kg/hr
- v. Arterial hypotension > 20 minutes duration
- vi. Lactate > 4 mmol/l (Pyruvate Dehydrogenase $\rightarrow$ lactate)

OR

- $\text{CVP} < 8$ (N = 8 -12 mmHg)
- $\text{MAP} < 65$ mmHg (N = 65 – 95mmHg)

Shock treatment

- Start treatment as soon as you suspect Pre-shock or Shock
- **DO NOT WAIT for hypotension**
- Primary problem must be treated
- **Position the pt**
- Prevent metabolic disturbances
- **Prevent/treat hypothermia**
- Treat **pain** if it is there
- **Maximize oxygen supply**
- ✓ Hypovolemic shock
 - Whole blood, plasma, electrolytes, bicarbonate required
- ✓ Anaphylactic shock
 - Antihistamines, corticosteroids
- ✓ Septic
 - Antimicrobials, glucocorticoids

...

- Intubation
- Warm the Pt
- Manage pain and anxiety
- Fluids - 20 ml/kg bolus x 3 (1-2 lit over 10 min for adults)
 - Normal saline, Ringer's lactate
 - (Until MAP 60 mmHg or adequate peripheral pulses)**
 - Use packed red blood cells if no improvement to fluids, If due to haemorrhage, apply transfusion of packed Red Blood Cells (RBC) or whole blood 20ml/kg over 4 hrs, repeat as needed until hgb level reaches 10gm/dl and the vital signs are corrected.
 - Monitor the pts progress**

Hypovolemic shock

- *Causes*

- Hemorrhage
- Vomiting
- Diarrhea
- Dehydration
- Third-space loss
- Burns

Hypovolemic shock - S & Sxs

- Tachycardia and tachypnea
- Weak, tready pulses
- Hypotension
- Skin cool & clammy
- Mental status changes
- Decreased urine output: dark & concentrated

Classes of Hemorrhagic Shock

	<u>Class I</u>	<u>Class II</u>	<u>Class III</u>	<u>Class IV</u>
Blood Loss	< 750	750-1500	1500-2000	> 2000
% Blood Vol.	< 15%	15 – 30%	30 – 40%	> 40%
Pulse	< 100	> 100	> 120	> 140
Blood Pressure	Normal	Normal	Decreased	Decreased
Pulse Pressure	Normal	Decreased	Decreased	Decreased
Resp. Rate	14 – 20	20 – 30	30 – 40	> 40
UOP	> 30	20 – 30	5 – 15	negligible
Mental Status	sl. Anxious	mildly anx	confused	lethargic
Fluid	crystalloid	crystalloid	blood	blood

Treatment – Hypovolemic shock

- Reverse hypovolemia vs. hemorrhage control
- Transport times < 15 minutes showed pre-hospital fluids were ineffective, however, if transport time > 100 minutes fluid was beneficial.

Fluid administration

- **Crystalloids** are cheaper, 2-3 litres in 20-30 minutes
- **Blood** must supplement
- One big vein, or smaller veins
- Use larger Vigo/canula
- Warming fluid may be needed

Effects of acute blood loss

- Reduced **oxygen transfer from the lungs to the red cells**- due to decreased venous return, and CO
- Reduced **oxygen storage by the red cells**- reduced total amount of Hg
- Reduced oxygen transport and delivery to the tissues
mismatching of the pulmonary blood flow and ventilation

Compensatory response to acute blood loss

- Restoration of plasma volume
- Restoration of CO- activation of sympathetic NS,
- Increase the rate and force of contraction
- Circulatory compensation- vasoconstriction
- Stimulation of ventilation
- Hormonal changes

Blood volume

- ❖ Neonate- 85-90ml/kg
- ❖ Children and adolescents-70-80ml/kg
- ❖ Adult- 60-70ml/kg
- Compensatory vasoconstriction may hide the signs of acute blood loss until at least 10% of blood volume is lost
- Healthy pts may lose up to 20% of their blood before signs of hypovolemia occur
- Blood loss $> 30\%$ may exceed the ability of crystalloids to replace blood volume without jeopardizing the O_2 carrying capacity of the blood

Blood loss estimation

- Clinical findings- history, V/S, capillary refill, mental state, urine out put, response to IV fluids, level of Hct ...
- Amount and speed of blood loss- **soaked towels and swabs, contents of suction machine, inspection of the surgical area**

Interventions

1. Stabilization
2. Monitoring
3. Etiologic diagnoses
4. Specific management

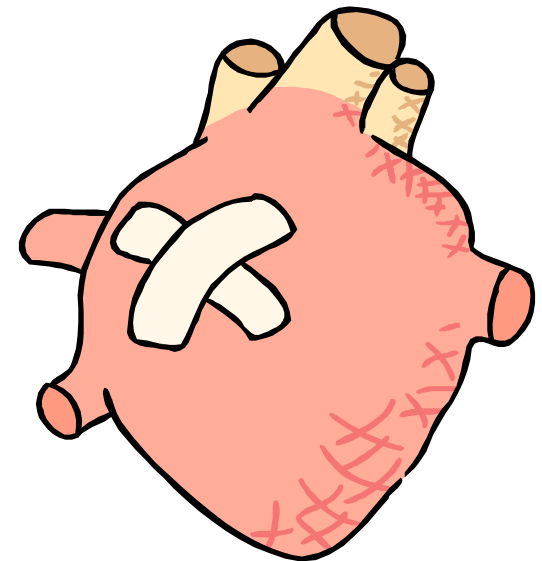


Cardiogenic shock

- *Cause*
 - Results from pump failure and decreased cardiac output
- Main categories:
 - Myopathies
 - Arrhythmia
 - Extracardiac/obstructive
- *Signs*
 - ↓ Cardiac output, **hypotension in the absence of hypovolemia** as well as **oliguria, cyanosis, cool extremities, and reduced mental status.**

Cardiogenic shock

- **Preload augmentation** - Consider Fluids
 - Morphine as needed (Decreases preload, anxiety)
- **Contractility**
 - Dopamine
 - Dobutamin
- **Afterload reduction**
 - Nitroglycerin



Distributive shock

- *Types*
 - Sepsis
 - Anaphylactic
 - Acute adrenal insufficiency
 - Neurogenic
- *Signs*
 - $\pm$ Cardiac output ,
 - **SIRS - Distributive Shock**
- Prompt volume replacement - fill the tank
- Early antibiotic administration
- Inotropes - first try Dopamine
- If MAP < 60
 - Dopamine = 2 - 3 $\mu\text{g/kg/min}$
 - Nor epinephrine- 0.5- 1microgram/kg/min

Sepsic Shock

Fluids- Initial resuscitation until

- CVP: 8- 12 mm Hg
- MAP $\geq$ 65 mm Hg
- UOP $\geq$ 0.5 ml/kg/hr

Consider:

- Transfusion to Hgb $\geq$ 10
- Dobutamine up to 20 μ g/kg/min

Correct the cause

- Antibiotics, Debridement

Vasopressors

- Phenylephrine , Levophed

Vasopressors

- Assure adequate fluid volume
- Do not use dopamine for renal protection
- Requires arterial line placement
- **Vasopressin:**
 - Refractory shock
 - Infusion rate 0.01 – 0.04 Units/min

Steroid use in sepsis

- Refractory shock 200-300 mg/day of hydrocortisone in divided doses for 7 days
- ACTH test
- Once septic shock resolves, taper dose
- Add fludrocortisone 50 µg po q day

Adrenal Crisis- distributive shock

- ***Causes***
 - Autoimmune adrenalitis
 - Adrenal apoplexy = hemorrhage or infarct
 - Heparin may predispose bleeding.
- **Steroids** may be lifesaving in the patient who is unresponsive to fluids, **inotropic, and vasopressor support.**
- **Mgt**
 1. Normal saline Large volumes (1 to 3 liters) or 5 % dextrose in 0.9% percent saline should be infused intravenously /infusion like hypovolemic shock.
 2. Hydrocortisone, 200mg, IV, stat then 100mg, IV, QID

Obstructive shock

- *Causes*
 - Cardiac Tamponade
 - Tension Pneumothorax
 - Massive Pulmonary Embolus
- *Signs*
 - ↓ cardiac output
 - ↑ Pulmonary Artery Obstructive Pressure
 - ↑ Systemic Vascular Resistance

Vasopressor agents

- **Augments contractility, after preload established**, thus improving cardiac output.
- Risk tachycardia and increased myocardial oxygen consumption if used too soon

Vasopressors & Inotropic agents

A. Dopamine

- Low dose (0.5 - 2 $\mu\text{g/kg/min}$) = dopaminergic
- Moderate dose (3-10 $\mu\text{g/kg/min}$) = **β -effects**
- High dose ($> 10 \mu\text{g/kg/min}$) = **α -effects**

Side effects

- Tachycardia

B. Dobutamine

- **β -agonist** 5 - 20 $\mu\text{g/kg/min}$
- Potent inotrope
- Caution in hypotension (inadequate volume) may precipitate tachycardia or worsen hypotension

C. Norepinephrine

- Potent α -adrenergic vasopressor
- Some β -adrenergic, inotropic,
- Dose 1 - 100 $\mu\text{g}/\text{min}$
- Unproven effect with low-dose dopamine to protect renal and mesenteric flow.

D. Epinephrine

- α - and β -adrenergic effects
- Potent inotrope
- Dose 1 - 10 $\mu\text{g}/\text{min}$
- Increases myocardial oxygen consumption particularly in coronary heart disease

Disorders of the veins

Phlebitis

- Phlebitis is defined as **inflammation of a vein** related to a chemical or mechanical irritation, or both
- It is characterized by
 - a **reddened, warm area** around the insertion site or along the path of the vein,
 - **Pain or tenderness** at the site or along the vein, and **swelling**
- The incidence of phlebitis increases with
 - The **length of time** the intravenous line is in place,
 - The **composition of the fluid or medication** infused
 - The **size and site of the cannula** inserted
 - **Improper anchoring** of the line, and
 - The **introduction of micro-organisms** at the time of insertion.

...

- **Treatment consists of**
 - **Discontinuing the IV and restarting it in another site, and**
 - **Applying warm moist compress to the affected site.**
- **Phlebitis can be prevented by using**
 - Aseptic technique during insertion
 - The appropriate size catheter and needle size for the vein,
 - Considering the composition of fluids and medications when selecting a site,
 - Observing the site for any complications every hour, and
 - Anchoring the catheter or needle well.

Thrombophlebitis

- Refers to the presence of **a clot plus inflammation** in the vein
- Is marked by inflammation of the venous wall and thrombus formation of the deep or superficial veins
- It is evidenced by
 - Localized pain, redness, warmth, and swelling around the insertion site or along the path of the vein, immobility of the extremity because of discomfort and swelling, sluggish flow rate, fever, malaise, and leukocytosis.
- **Deep vein thrombophlebitis** may lead to occlusion of the vessels or systemic embolization such as **pulmonary embolism**

...

- Several conditions may lead to thrombophlebitis, including
 - **hypercoagulability** (such as from cancer, blood dyscrasias, or oral contraceptives);
 - **injury to the venous wall** (such as **from I.V. injections, fractures, antibiotics, or infection**); and
 - **venous stasis** (such as from **varicose veins, pregnancy, heart failure, or prolonged bed rest**)

Diagnosis and treatment

- Diagnostic tests may include **Doppler ultrasonography**, and **venography**; laboratory tests include a **CBC**

...

Signs and symptoms

- *Deep vein thrombophlebitis* will sometimes cause no clinical symptoms or physical findings; when they do occur, they may include
 - **Cramping pain, edema, tenderness to touch, fever, chills, and malaise**
- *Superficial thrombophlebitis* produces visible and palpable signs, such as
 - **heat, pain, swelling, tenderness, and induration along the affected vein's length**

...

- **Treatment includes**
 - ❖ discontinuing the IV infusion,
 - ❖ **applying a cold compress first** to decrease the flow of blood and increase platelet aggregation followed by a **warm compress**,
 - ❖ Elevating the extremity, and
 - ❖ Restarting the line in the opposite extremity.
- If the patient has signs and symptoms of thrombophlebitis, the IV line **should not be flushed** (although flushing may be indicated in the absence of phlebitis to ensure cannula patency and to prevent mixing incompatible medications and solutions).
- Thrombophlebitis can be **prevented by avoiding trauma to the vein at the time the IV is inserted**, observing the site every hour, and checking medication additives for compatibility

...

- ❖ **An anticoagulant** (initially I.V. heparin or low-molecular-weight heparin followed by oral warfarin is administered to prolong clotting time
- ❖ **Thrombolytic therapy** (such as streptokinase [Streptase]) is indicated for acute, extensive deep vein thrombophlebitis
- ❖ **Embolectomy, venous ligation**, or insertion of a vena caval umbrella or filter may also be indicated



Nursing interventions

- If the patient is receiving a thrombolytic, heparin, or warfarin, **monitor him for signs and symptoms of bleeding**
- Assess the patient for signs and symptoms of **pulmonary embolism**, such as crackles, dyspnea, tachypnea, hemoptysis, tachycardia, and chest pain
- Make sure the patient **maintains bed rest** and elevates the affected extremity
- **Apply moist, warm compresses** to improve circulation to the affected area and relieve pain
- Tell the patient to **avoid prolonged sitting and standing** to help prevent recurrence
- Teach the patient how to properly **apply and use antiembolism stockings**

Deep Vein Thrombosis (DVT)

- **Deep veins**
 - **Obstruction of the deep veins of the legs produces edema** & swelling of the extremity because the out flow of venous blood is inhibited
 - The amount of swelling can be determined by measuring the circumference of the leg at various levels with a tape measure.
 - One leg is compared with the other at the some level to determine size may be difficult to detect
 - The **affected leg, may feel warmer than the unaffected leg**, & the superficial veins may appear to be more prominent.
 - **Tenderness**, which usually occurs later, is produced by inflammation of the vein wall and can be detected by gently palpating the leg

...

- **"Homans' sign** (pain in the calf after the foot is sharply **dorsiflexed**) is not specific for deep venous thrombosis because it can be elicited in any painful condition of the calf.
- In same cases, **signs of a pulmonary embolus** are the first indication of a DVT.

Interventions

- **Encourage rest**
- Facilitate bed rest and **elevation of the extremity** above the level of the heart
- Administer intermittent or continuous **warm moist compresses** as ordered (to **prevent thrombus from dislodging and becoming an embolus, do NOT massage the affected limb**)

...

- Administer medications as prescribed. Ex. **Anticoagulants**
- **Thrombolytic therapy** is effective in dissolving thrombi quickly and completely. It must be initiated **within 5 days** after the onset of symptoms to be most effective. The advantage is the prevention of valvular damage and consequential venous insufficiency, or “postphlebitis syndrome.”
- **Analgesics**

Varicose veins (varicosities)

- Are **abnormally dilated, tortuous superficial veins caused by incompetent venous valves.**
- Most commonly, this condition **occurs in the lower extremities, the saphenous veins**, or the lower trunk; however, it can occur else where in the, such as esophageal varices
- The condition is most common in **women** and in people whose occupations require **prolonged standing**, such as **salespeople, hair stylists, teachers, nurses, and construction workers,**
Pregnancy
- A **hereditary weakness** of the vein wall may contribute to the development of varicosities, and it is not uncommon to see this condition occur in several members of the same family

...

Pathophysiology

- Varicose veins may be considered
 - **Primary** (without involvement of deep veins) or
 - **Secondary** (resulting from obstruction of deep veins)
- A reflux of venous blood in the veins results in venous stasis.
- If only the **superficial** veins are affected, the person may have **no symptoms** but may be troubled by the appearance of the dilated veins

...

Clinical manifestations

- Symptoms, if present, may take the form of **dull aches, muscle cramps, and increased muscle fatigue in the lower legs**
- **Ankle edema** and a **feeling of heaviness** of the legs may occur.
- Nocturnal cramps are common.
- When deep venous obstruction results in varicose veins, patients may develop the signs and symptoms of chronic venous insufficiency: **edema, pain, pigmentation, and ulcerations**
- **Susceptibility to injury and infection** is increased

Prevention

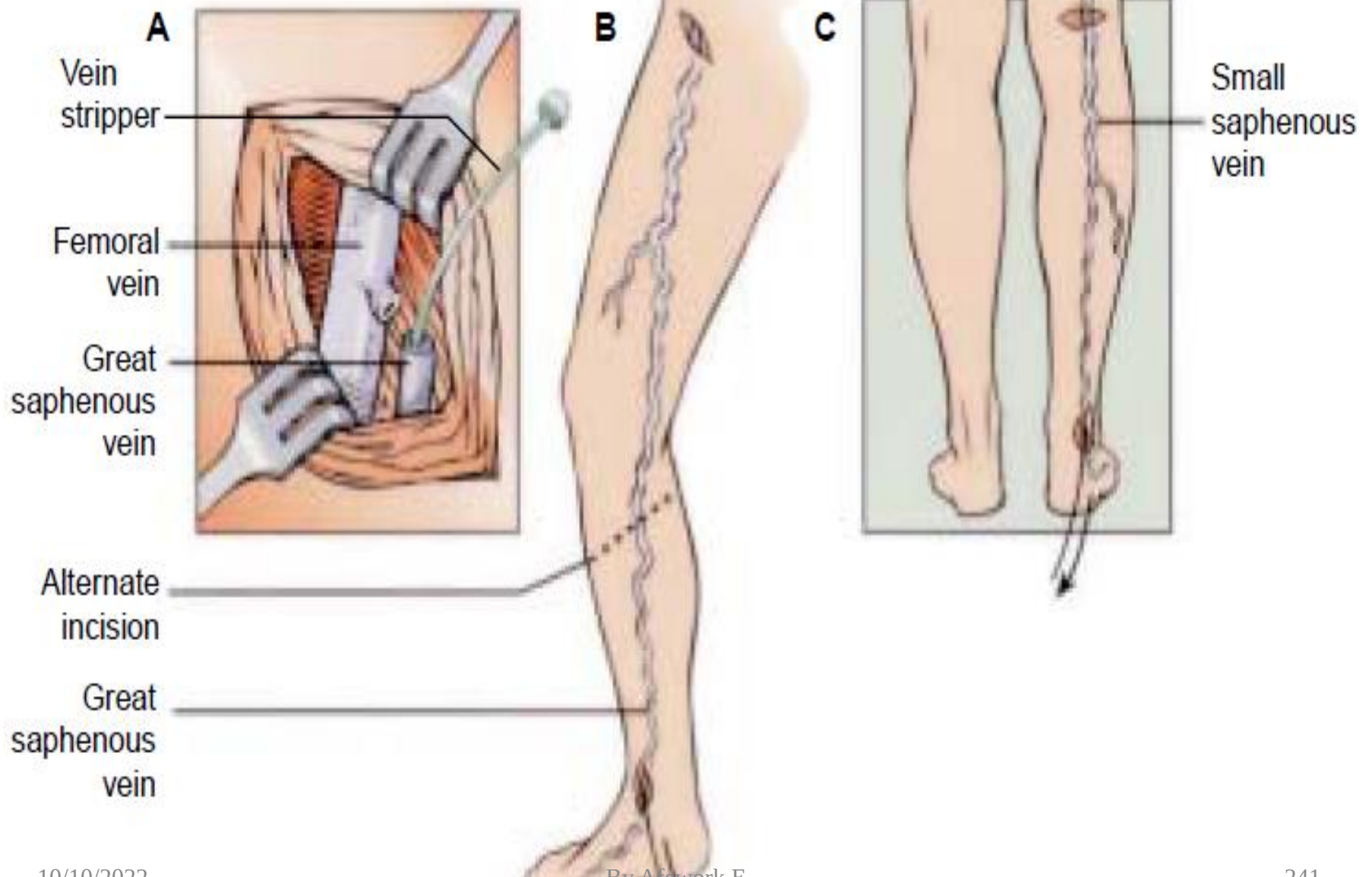
- The patient should **avoid activities** that cause venous stasis, such as
 - Wearing tight socks or a constricting panty girdle,
 - Avoid crossing the legs at the thighs, and
 - Avoid sitting or standing for long periods.
- Changing position frequently
- Elevating the legs when they are tired, and getting up to walk for several minutes of every hour promote circulation.
- The patient should be encouraged to walk 1 or 2 miles each day if there are no contraindications.
- Walking up the stairs rather than using the elevator or escalator is helpful in promoting circulation.
- **Swimming** is also good exercise for the legs.
- The overweight patient should be encouraged to begin a weight-reduction plan.

Surgical managemet

- Surgery for varicose veins requires that the deep veins be patent and functional.
- The patient is placed under general anesthesia, and the saphenous vein is ligated and divided.
- The vein is ligated high in the groin where the saphenous vein meets the femoral vein. Additionally, the vein may be removed (stripped). After the vein is ligated, an incision is made in the ankle, and a metal or plastic wire is passed the full length of the vein to the point of ligation.
- The wire is then withdrawn, pulling (removing, “stripping”) the vein as it is removed.

Sclerotherapy

- In sclerotherapy, a chemical is injected into the vein, **irritating the venous endothelium and producing localized phlebitis and fibrosis**, thereby obliterating the lumen of the vein.



Disorders of Lymphatic system

Lymphedema

- Characterized by **tissue swelling occurs in the extremities** because of an increased quantity of lymph that results **from obstruction of lymphatic vessels**.
- It is especially marked when the extremity is in a dependent position.
- **Initially, the edema is soft and pitting**. As the condition progresses, the **edema becomes firm, nonpitting**, and unresponsive to treatment.
- The most common type is congenital lymphedema (lymphedema praecox), which is caused by **hypoplasia (underdevelopment) of the lymphatic system** of the lower extremity.

Elephantiasis

- Lymphatic obstruction caused by **a parasite (filaria)** is most frequently seen in the tropics.
- Usually **unilateral**, affects the groin
- When chronic swelling is present, there may be frequent bouts of acute infection characterized by **high fever** and **chills** and **increased residual edema after the inflammation has resolved.**
- These lead to **chronic fibrosis, thickening of the subcutaneous tissues, and hypertrophy of the skin**
 ➔ elephantiasis
- This condition, in which chronic swelling of the extremity recedes only slightly with elevation.

Management

- **Active and passive exercises** assist in moving lymphatic fluid into the bloodstream.
- **External compression devices** milk the fluid proximally from the foot to the hip or from the hand to the axilla.
- When the leg is affected, **continuous bed rest** with the **leg elevated** may aid in mobilizing the fluids.
- **Medical mgt**
 - Diuretic furosemide along with elevation of the leg
 - Antibiotic if **lymphangitis** or **cellulitis** is present.
- **Surgical Management**: excision of the affected subcutaneous tissue and fascia, with skin grafting to cover the defect.

Lymphatic Filariasis

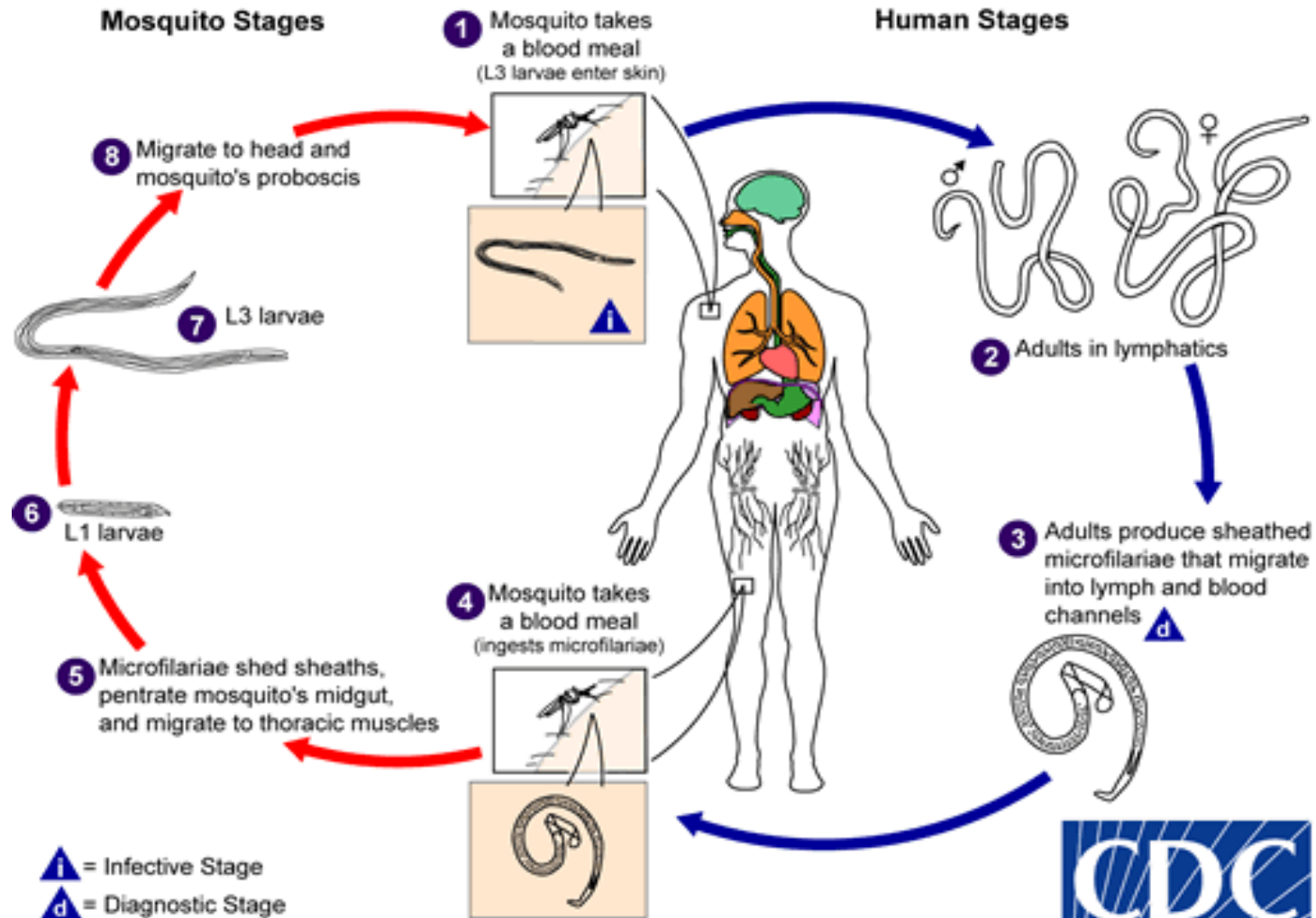
- A disease caused by **the reaction of the body to worms in the lymphatic system**
- **Agents**
 - **Wuchereria bancrofti**
 - Brugia malayi
 - Brugia timori

• • •

- **Reservoir:** Humans are definitive hosts
- **Mode of transmission:** **by bite of mosquito** harboring infective larvae
- **Incubation period-** **one month**
- **Period of communicability:** Not directly transmitted from person to person
 - Humans may infect mosquitoes when microfilaria are present in the peripheral blood

Life cycle

Wuchereria bancrofti



Clinical Manifestation

Three phases may be distinguished

A. Acute phase

- Starts within a **few months** after infection
- **Toxic products of living or dead worm** → Inflammatory manifestations: **due to hypersensitivity reaction**
- Lymphadenopathy (neck and axilla)
- Fever, chills
- **Eosinophilia**

B. Sub--acute phase

- This occurs **after about one year**
- **Microfilaria: peripheral blood**
- Reactions to the adult worms cause attacks of fever with **lymphangitis or Epididymitis**
- Recurrent attacks lead to **hydrocele**
- Lesions associated with hypereosinophilia

C. Chronic phase

- Lymph vessels and glands: **obstructed →** as a result **lymph edema develops**
- Lymph edema mostly in **legs or scrotum (elephantiasis)**
- Hard, thick, rough and fissured skin
- Worms **died**, no microfilariae in the blood



Diagnosis

- Obstructive signs with history and residence in endemic areas
- microfilariae in the peripheral blood (blood film)

Treatment

- a) Diethyl carbamazoin Citrate (DEC) 8 to 10 mg/kg for two to three weeks for filarial infections
- b) Ivermectin (200 mcg/kg), albendazole
- c) Doxycycline
- d) Refer the patient for surgical treatment of hydrocele

Prevention

- Reducing the vector population
- Mass and selective treatment
- Personal protection against mosquito bite

Podoconiosis

- Podoconiosis is a geochemical disease occurring in individuals exposed **with bare feet to red clay soil or irritant soil.**
- Is **non infectious** or **non filarial elephantiasis** **bilateral** with asymmetrical

Pathogenesis

- Is the result of genetically determined **abnormal inflammatory reaction to minerals particles** in irritant clay soils derived from volcanic deposits.
- This occurs when irritant soil particles penetrate the feet and collect in lymphatic vessels.
- This inflammatory condition overtime causes the lymphatic vessels to be undergo **fibrosis and occlusion.**

Podoconiosis cause, prevention and disease management

What is podoconiosis?

Non-infectious elephantiasis caused by long-term exposure of bare feet to irritant soil.

Clinical Features of Podoconiosis



Below knee swelling, on foot or leg; knobs or bumps present above or below the ankle. Disease is bilateral but asymmetrical.

Podoconiosis prevention is simple!

Podoconiosis is preventable by avoiding contact with irritant soil which is done primarily by

- Keeping good foot hygiene (washing with soap and water);
- Wearing clean socks and shoes daily; and
- Covering floor in the home to avoid contact with irritant soil.

Basic Disease Management Package

- Foot hygiene and skin care
- Bandaging
- Footwear
- Elevation and movement
- Minor surgery to remove nodules
- Acute attack management

Foot Hygiene and Skin Care

- Wash feet in clean, cool water with soap.
- Lather soap onto hands, clean leg from top to bottom.
- Rinse soap with clean water.
- Air dry leg and feet carefully.
- Keep skin soft and flexible by rubbing a small amount of Whitfield ointment, Vaseline, or local oil onto skin of lower leg after washing and air drying.



Bandaging

- Bandage from toe to knee using "V" shape over foot and use spiral up lower leg; overlap 50% when bandaging.
- Patients should learn how to use bandages themselves.
- Two bandages should be used per affected leg.



Elevation and Movement

- It helps to improve lymph circulation and reduce swelling.
- Ankle circles are done when sitting by raising the leg and rotating the foot in a circular motion while pointing the toes.
- The foot and leg have to be elevated while sitting and sleeping.



Footwear	Acute Attack Management
- Shoes are vital for treatment as well as prevention! - Patients can be trained as shoemakers. - Clean socks (2 pairs) and large size/custom made shoes are needed.	- Early detection. - Hygiene. - Elevation and complete rest to the legs. - Use cold application. - Treat headache, pain and fever. - Drink plenty of fluids. - Avoid hot application, exercise, and bandaging. - Referral if patient has no relief.
Surgery Surgery is not commonly used for podoconiosis treatment.	Minor surgery is only done to remove nodules that prevent patients from wearing shoes.



Simple Disease Management is Effective				Other possible causes of foot swelling
				- Filarial lymphoedema - Onchocerciasis - Leprosy - Malnutrition (children) - Rheumatic heart disease - Post surgery

Podoconiosis Clinical Disease Staging

Stage 1	Swelling is reversible overnight	
Stage 2	Persistent below-knee swelling; if present, knobs or bumps are below the ankle ONLY	 A photograph of a person's foot showing persistent swelling below the knee. A red starburst with the number '2' is in the top right corner.
Stage 3	Persistent below-knee swelling; knobs or bumps present above the ankle	 A photograph of a person's foot showing persistent swelling below the knee. A red starburst with the number '3' is in the top left corner.
Stage 4	Persistent above-knee swelling; knobs or bumps present at any location	 A photograph of a person's legs showing persistent swelling above the knee. A red starburst with the number '4' is in the top left corner.
Stage 5	Joint fixation; swelling at any place in the foot or leg	 A photograph of a person's feet showing joint fixation and swelling at any place in the foot or leg. A red starburst with the number '5' is in the top left corner.



Thank you!!!

Disorders of the Hematological System

Anatomy and Physiology Review

- The **hematologic system** consists of the **blood** and the sites where blood is produced, including the bone marrow and the **reticuloendothelial system** (RES).
- Blood is a **specialized organ** that differs from other organs in that it exists in a fluid state.
- Blood is composed of plasma and various types of cells.
- **Plasma** is the fluid portion of blood; it contains various proteins, such as albumin, globulin, **fibrinogen**, and other factors necessary for clotting, as well as electrolytes, waste products, and nutrients.
- About 55% of blood volume is plasma

Blood

- The cellular component of blood consists of three primary cell types: **erythrocytes** (red blood cells [RBCs], red cells), **leukocytes** (white blood cells [WBCs]), and **thrombocytes** (platelets).
- These cellular components of blood normally make up 40% to **45% of the blood volume.**
- Because most blood cells have a short lifespan, the need for the body to replenish its supply of cells is **continuous**; this process is termed **hematopoiesis.**
- **The primary site for hematopoiesis is the bone marrow.**

Types of cells and functions

Cell Type	Major Function
WBC (Leukocyte)	Fights infection
Neutrophil	Essential in preventing or limiting bacterial infection via phagocytosis
Monocyte	Enters tissue as macrophage; highly phagocytic, especially against fungus; immune surveillance
Eosinophil	Involved in allergic reactions (neutralizes histamine); digests foreign proteins
Basophil	Contains histamine; integral part of hypersensitivity reactions
Lymphocyte	Integral component of immune system



T lymphocyte	Responsible for cell-mediated immunity; recognizes material as “foreign” (surveillance system)
B lymphocyte	Responsible for humoral immunity; many mature into plasma cells to form antibodies
Plasma cell	Secretes immunoglobulin (antibody); most mature form of B lymphocyte
RBC (Erythrocyte)	Carries hemoglobin to provide oxygen to tissues; average lifespan is 120 days
Platelet (Thrombocyte)	Fragment of megakaryocyte; provides basis for coagulation to occur; maintains hemostasis; average lifespan is 10 days

• • •

- The danger that trauma can lead to excess blood loss always exists.
- To prevent this, an intricate clotting mechanism is activated when necessary to seal any leak in the blood vessels.
- Excessive clotting is equally dangerous, because it can obstruct blood flow to vital tissues.
- To prevent this, the body has a fibrinolytic mechanism that eventually dissolves clots (thrombi) formed within blood vessels.
- The balance between these two systems: **clot (thrombus)** formation and clot dissolution or **fibrinolysis** is called **hemostasis**.

Physical characteristics and volume of blood

- Blood is a sticky, opaque fluid with a metallic taste
- It is a viscous fluid connective tissue, which is heavier and thicker than water.
- Color varies from scarlet (oxygen-rich) to dark red (oxygen-poor)
- The pH of blood is **7.35–7.45** and its **salt content is 0.9%**
- Temperature is **38°C, slightly higher than “normal”** body temperature
- Blood accounts for approximately **8% of body weight**
- Average volume of blood is 5–6 L for males, and 4–5 L for females

Functions of Blood

- Blood performs a number of functions dealing with:
 - Substance distribution
 - Regulation of blood levels of particular substances
 - Protection

- **Blood transports:**
 - Oxygen from the lungs and nutrients from the digestive tract
 - Metabolic wastes from cells to the lungs and to kidneys for elimination
 - Hormones from endocrine glands to target organs

...

➤ **Blood maintains:**

- Body temperature by absorbing and distributing heat
- Normal pH in body tissues using buffer systems (e.g., Hb, HCO_3^-)
- Adequate fluid volume in the circulatory system

➤ **Blood prevents blood loss** by:

- Activating plasma proteins and platelets
- Initiating clot formation when a vessel is broken

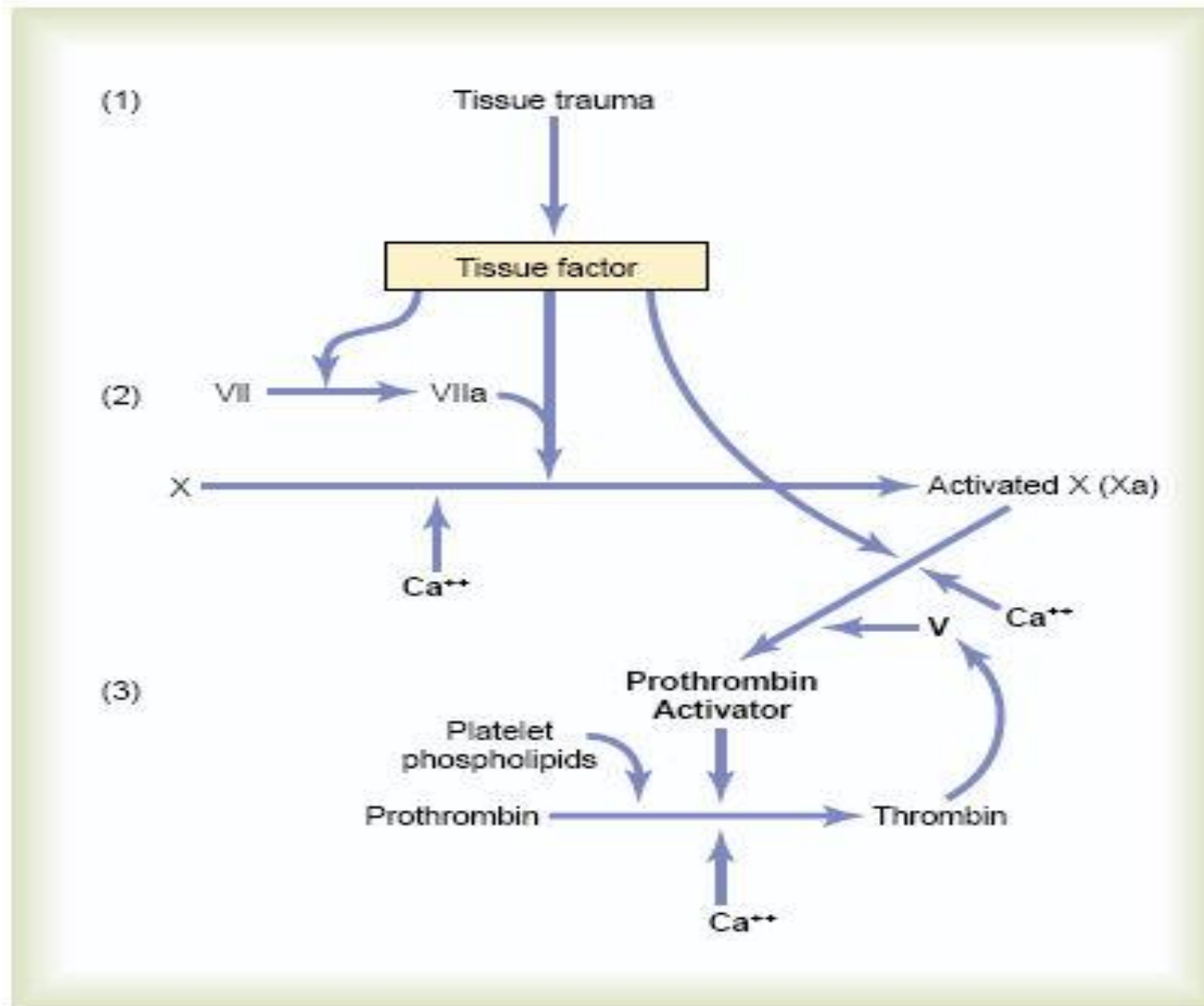
➤ **Blood prevents infection** by:

- Synthesizing and utilizing antibodies
- Activating WBCs to defend the body against foreign invaders

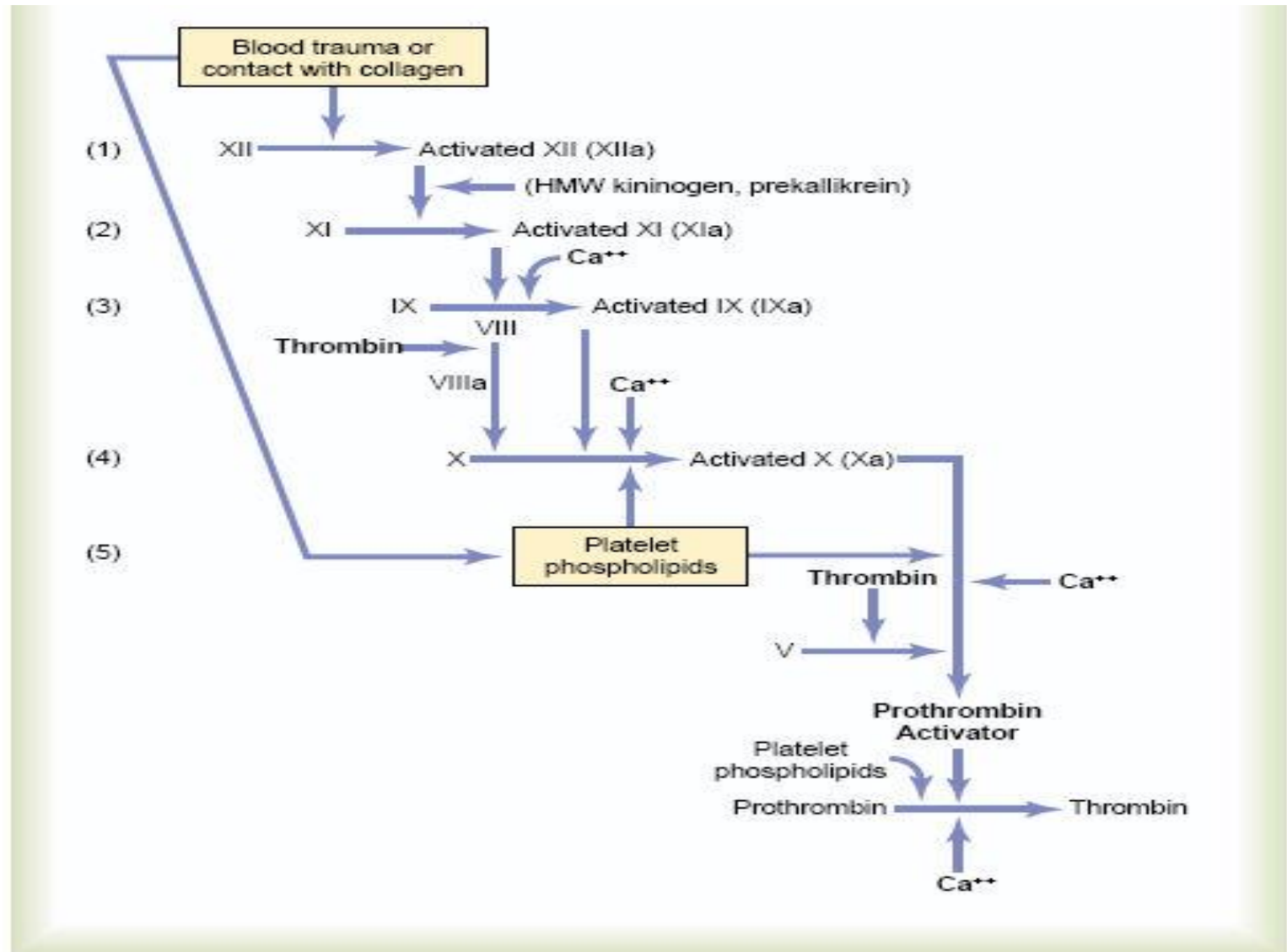
Bone Marrow

- The bone marrow is the site of hematopoiesis, or blood cell formation.
- In adults, blood cell formation is usually limited to the **pelvis, ribs, vertebrae, and sternum**.
- The marrow is highly vascular. Within it are primitive cells called **stem cells**.
- The stem cells have the ability to self-replicate, thereby ensuring a continuous supply of stem cells throughout the life cycle.
- When stimulated to do so, stem cells can begin a process of **differentiation** into either **myeloid** or **lymphoid** stem cells
- **Lymphoid stem** cells produce either T or B **lymphocytes**.
- **Myeloid stem** cells differentiate into three broad cell types: erythrocytes, leukocytes, and platelets.

Blood coagulation: extrinsic pathway



Blood coagulation: intrinsic pathway



Assessment

1. Health History and physical examination

- Because many hematologic disorders are more prevalent in certain ethnic groups, assessments of **ethnicity and family history are useful**
- Similarly, obtaining a **nutritional history** and assessing the use of prescription and over-the-counter medications, as well as herbal supplements,
- **Careful attention to the onset of a symptom** or finding (e.g., rapid vs. gradual; persistent vs. intermittent), its severity, and any contributing factors can further differentiate potential causes.

• • •

- Assess for specific symptoms of hematologic diseases:
 - **Extreme fatigue (the most common symptom of hematologic disorders)**
 - Delayed clotting of blood
 - Easy or deep bruising
 - Abnormal bleeding (e.g., frequent nosebleeds)
 - Abdominal pain (hemochromatosis) or joint pain (sickle cell disease)
- Careful attention to the skin, oral cavity, lymph nodes, and spleen

2. Diagnostic Evaluation and Hematologic Studies

- **Complete blood count (CBC)** and the **peripheral blood smear**.
- Prothrombin time (PT)
- International Normalized Ratio(INR)

3. Bone Marrow Aspiration and Biopsy

- also used to document infection or tumor within the marrow.
- The **posterior superior iliac crest** is the preferred site for bone marrow aspiration and biopsy because no vital organs or vessels are nearby.
- The patient is placed either in the lateral position with one leg flexed or in the prone position. The anterior iliac crest or sternum may also be used.

Hematologic disorders

A/ RBC disorders

B/ WBC disorders

C/ Clotting disorders

A/ RBC disorders

Anemia

- is defined by **reduction in Hgb concentration**, Hct concentration or **RBC count**
- As a result the amount of oxygen delivered to body tissues is also diminished
- It is **not a specific disease** state by itself but a sign of an underlying disorder
- It is by far **the most common** hematologic condition.
- **Anemia is Hematocrit**
 - ✓ < 40% (**hemoglobin < 13.0 g/dL**) in men or
 - ✓ < 37% (**hemoglobin < 12.0 g/dL**) in women. (**WHO definition**)
- Revised WHO criteria for patient's with malignancy Hgb < 14 in M and Hgb < 12 in F

Classifications of anemia

1. Hypoproliferative anemia (resulting from defective RBC Production):

- Iron deficiency anemia
- **Megaloblastic anemia**: Vitamin B12 deficiency, & folate deficiency
- Decreased erythropoietin production (e.g. from Renal dysfunction)
- Cancer/inflammation

2. Bleeding (resulting from RBC loss):

- Bleeding from GI tract, menorrhagia, epistaxis, trauma

3. Hemolytic anemia:

- Altered erythropoiesis: sickle cell anemia, thalassemia
- Hypersplenism (hemolysis)
- Drug-induced anemia
- Autoimmune anemia
- Mechanical heart valve related anemia

20-2 Anemia Classification by Etiology

Type of Anemia	Pathophysiology	Lab Findings	Treatment
Hypoproliferative (Decreased Production)			
Pernicious, megaloblastic	Vitamin B ₁₂ deficiency; lack of intrinsic factor required for B ₁₂ absorption	Decreased vitamin B ₁₂ level; increased MCV Elevated methylmalonic acid and homocysteine levels Low reticulocyte count	B ₁₂ replacement Cyanocobalamin 1,000 µg IM monthly
Folate deficiency	Decreased folic acid intake; impaired absorption; alcohol impairs folate metabolism in the liver, causing a significant depletion in folate reserves Decreases in folate impair DNA synthesis and increase erythroblast cell deaths	Decreased folate levels Increased MCV	Oral folic acid 1 to 5 mg/day PO
Iron deficiency	Chronic blood loss, failure to recapture iron from recycled RBCs Demands for iron exceed iron intake. Decreased iron intake results in use and depletion of iron stores, resulting in decreased Hgb production	Decreased reticulocytes, iron, ferritin, iron saturation, MCV Increased TIBC	Iron replacement therapy: ferrous sulfate 325 mg PO t.i.d. Parenteral iron therapy reserved for those who cannot take oral replacement due to intolerance, poor enteral absorption, or continued blood loss IV iron therapy requires careful monitoring for signs of hypersensitivity

• • •

Decreased erythropoietin production	Renal disease can impair erythropoietin production (due to renal disease, malignancy), which in turn decreases RBC production	Decreased erythropoietin level; normal MCV and MCH; increased creatinine level	Erythropoietin replacement
Anemia of chronic disease (seen in AIDS, malignancy, SLE, chronic renal disease, chronic liver disease, rheumatoid arthritis)	Marked by three defects: Decreased erythrocyte lifespan, ineffective bone marrow response to erythropoietin, altered iron metabolism	Normal MCV and MCHC Low serum iron Low or normal TIBC Normal or high ferritin	Does not respond to iron replacement therapy due to altered iron transport in the bone marrow Primary treatment is to treat the underlying cause
Aplastic anemia	An immune response mediated by cytotoxic T cells; targets the cells in the bone marrow, causing cell death and bone marrow failure Hereditary or acquired (exposure to benzene, pesticides)	Decreased reticulocyte count Decreased white blood cells, Hgb, Hct, and platelets	Immunosuppressive therapy (e.g., cyclosporine, corticosteroids, antithymocyte globulin) Stem cell transplant

Blood Loss

Bleeding	Bleeding from gastrointestinal tract, menorrhagia (excessive menstrual flow), epistaxis (nosebleed), trauma results in decrease in circulating blood volume and decreased hemoglobin and hematocrit	Increased reticulocyte level Normal Hgb and Hct if measured soon after bleeding starts, but levels decrease thereafter Normal MCV initially but later decreases Decreased ferritin and iron levels (later)	Packed RBC transfusions Identify and control source of blood loss
----------	---	---	--

Hemolysis (Destruction of RBCs)

Autoimmune hemolytic anemia Acquired; caused by extrinsic insults such as infection, medications, venom, trauma	Both inherited and acquired forms result in hemolysis within blood vessels and/or lymphoid tissue	Presence of schistocytes Increased spherocyte level Increased reticulocytes	Acquired: remove insult, treat underlying disorder Immunosuppressive therapy (i.e., corticosteroids)
Thalassemias A group of inherited forms of autoimmune hemolytic anemia	Caused by cellular abnormalities of the Hgb structure, marked by an imbalance between the beta chain and alpha chain of hemoglobin, resulting in RBC membrane damage, ineffective RBC production, and hemolysis	Decreased MCV; fragmented RBCs Increased reticulocyte level	Transfusion therapy with PRBCs

Kinetic approach

- **Decreased RBC production**
 - Lack of nutrients (B12, folate, iron)
 - Bone marrow disorder
 - Bone marrow suppression
- **Increased RBC destruction**
 - Inherited and acquired hemolytic anemias
- **Blood Loss**

Morphological approach

- Microcytic ($MCV < 80$ fl, femtoliters (10^{-15})
 - Reduced iron stores cause small RBCs
 - Reduced heme synthesis
 - Reduced globin production
- Normocytic ($80 < MCV < 100$)
- Macrocytic ($MCV > 100$)
 - Liver disease, B12, folate

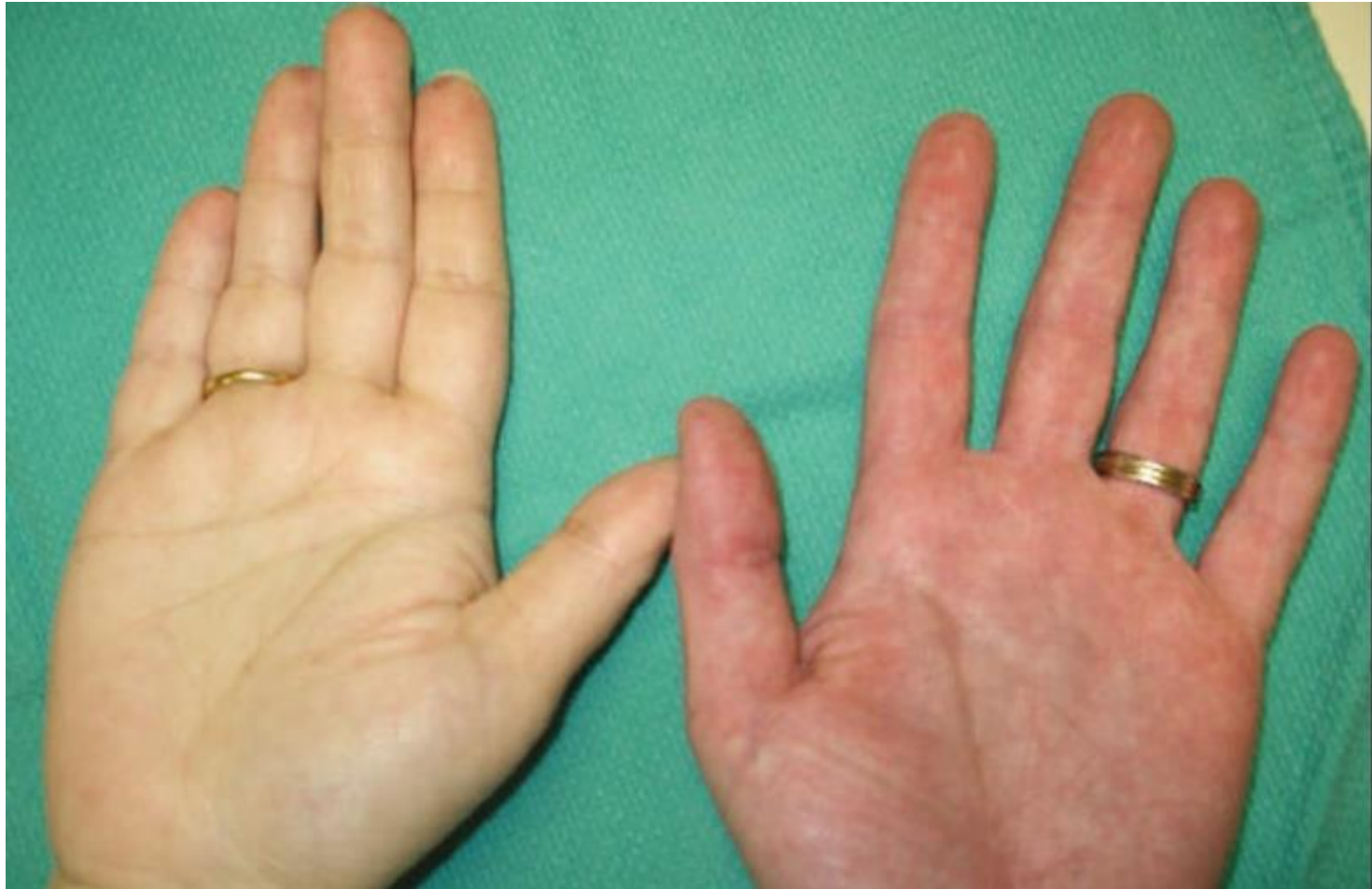
C/M

- **Pallor** of the skin and mucous membranes (conjunctivae, oral mucosa)
- **Fatigue,**
- Dyspnea,
- Jaundice;
- **Angular cheilosis** (ulcerated corners of the mouth); and
- Brittle, ridged, concave nails
- **Restless leg syndrome** with iron deficiency anemia
- Confusion

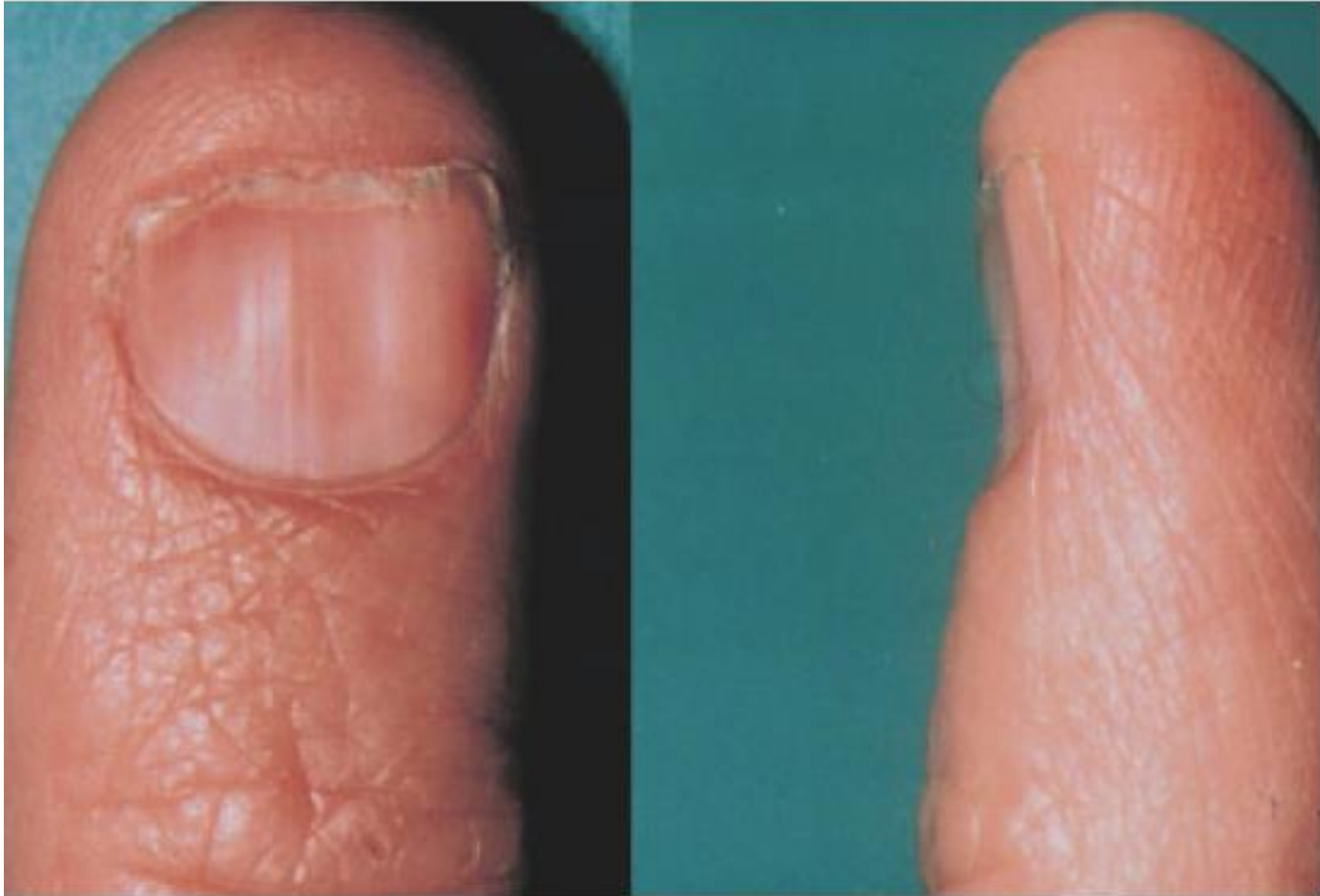
• • •



• • •



Physical manifestation : “Spoon nails” in Iron deficiency



Labs

- ✓ History taking and a physical exam (pallor, jaundice, etc)
- ✓ CBC with Diff
- **Leukopenia with anemia may suggest aplastic anemia**
- **Increased Neutrophils may suggest infection**
- **Increased Monocytes may suggest Myelodysplasia**
- **Thrombocytopenia may suggest hypersplenism**, marrow involvement with malignancy, autoimmune destruction, folate deficiency
- ✓ Reticulocyte count
- ✓ Peripheral smear (biconcave disk, size, color)

Labs

- ✓ **Mean corpuscular hemoglobin (MCH):**
 - To determine the amount of hemoglobin per RBC
 - Normochromic – normal amount of Hgb per cell
 - Hypochromic – decreased Hgb per cell
- **Hemoglobin electrophoresis:**
 - Separates normal hemoglobin from abnormal.
 - It is used to detect sickle cell disease.
- **Sickle cell test:**
 - Evaluates the sickling of red blood cells in the presence of decreased oxygen tension
- **Schilling test:**
 - Measures vitamin B12 absorption with and without intrinsic factor.
 - It is used to differentiate between **malabsorption and pernicious anemia**
- **Bone marrow examination**
 - Diagnoses aplastic anemia (failure of bone marrow to produce RBCs as well as platelets and WBCs)

Iron deficiency anemia

- It is a condition in which total body iron content is inadequate for optimal development of erythrocytes.
- It is **the most common** type of anemia in all age groups & resulted from:
 - Chronic blood loss
 - Iron malabsorption, as in small bowel disease
 - Increased iron requirement, as in pregnancy or periods of rapid growth
 - Insufficient intake caused by inadequate diet or Wt-loss

Complications:

- Growth retardation in children; Heart failure
- Ischemic organ disease such as MI or stroke

...

- Low serum iron level
 - **Low ferritin** (serum ferritin is an indicator of total body iron stores)
 - **Low Hgb**
 - Low reticulocyte count

Gold standard

- **Bone marrow aspiration**
 - Prussian blue staining shows lack of iron in erythroid precursors and macrophages
 - However, it is invasive and costly

What causes Iron deficiency?

- Blood Loss (occult or overt): PUD, Diverticulosis, Colon Cancer
- Decreased Iron absorption: atrophic gastritis, celiac disease
- Foods and medications: Phytate, calcium, soy protein, polyphenols **decrease** iron absorption
- Decreased intake (rare)
 - ❖ Iron deficiency anemia is the most common anemia

...

Pharmacologic & nursing interventions:

- Iron therapy – Oral FeSO_4 , Ferrous gluconate, for 6-12 months
- Iron dextran IV or IM, but prior to parenteral administration of a full dose, a small test dose should be administered to avoid the risk of anaphylaxis.
- Iron rich diets: **organ meat** (beef or calf's, chicken, **liver**), other meats, **beans, leafy green vegetables, fruits**
- Iron rich diets with a source of Vit-C
- **Take iron on empty stomach 1hr before or 2hrs after meal.**
- Take high fiber diets to **minimize constipation**
- Inform the pt that stools will **become dark in color**
- Do not take within **2 hr of milk or antacids**

Aplastic anemia

- Bone marrow hypoplasia resulting in **pancytopenia** (**insufficient numbers of RBCs, WBCs, Platelets**)
- May be idiopathic or caused by exposure to chemical toxins, radiation, viral infections, **certain drugs (CAF)**, or may be congenital → causes damage to marrow stem cells/ replacement of marrow cells by fat.

Treatment:

- Removal of causative agent or toxin
- Allogeneic bone marrow transplantation in severe cases
- Immunosuppressants, bone marrow stimulating factors
- Platelet & RBC transfusions,
- Antibiotics

Megaloplastic anemia

- Results abnormally large red blood cells → these abnormal cells destroyed by in marrow → **fewer mature cells remain** → MCV is high
- Has two causes:
 - ✓ **Pernicious anemia**
 - ✓ *Folate deficiency*

- **Pernicious anemia**
 - Vitamin B12 deficiency from
 - Small bowel disease
 - Gastric resection
 - Absence of intrinsic factor or genetic cause
 - **Strict vegetarians who consume no meat or dairy products**
 - **Features:** GI symptoms & neuropathy
 - **Rx:** Monthly parenteral replacement with Cyanocobalamin
- **Folate deficiency**
 - Folic acid, a vitamin that is necessary for normal RBC production, is stored as compounds referred to as folates.
 - Dietary deficiency, alcoholism, malabsorption disease of the small bowel, pregnancy, & some medications
 - **Rx:** Oral folic acid replacement (**Fefol**) & diet, Rx of underlying cause.
 - **Folate is found in green vegetables and liver**

Hemolytic anemia

- **Inherited**
 - sickle cell anemia
 - Thalassemia
 - G6PD deficiency anemia
- **Acquired**
 - ❖ Auto immune
 - ❖ Hemoglobinopathis
 - ❖ Heart valve hemolysis
 - ❖ Drugs induced

Sickle cell anemia

- Severe hemolytic anemia that results from **inheritance of the sickle hemoglobin gene**.
- This gene causes the hemoglobin molecule to be defective.
- The sickle hemoglobin (HbS) acquires a crystal-like formation when exposed to low oxygen tension.
- The oxygen level in venous blood can be low enough to cause this change; consequently, **the RBC containing (HbS) loses its round, very pliable, biconcave disk shape and becomes deformed, rigid, and sickle-shaped**
- **These red blood cells have short life span.**
- **Mgt :**
 - Bone marrow transplantation
 - Long term RBC transplantation
 - Hydroxyurea → increase hgb F → decrease sickle cell

• • •



Normal Red Blood Cell



Sickle Cell

Thalassemia

- The Thalassemia are a group of hereditary disorders associated with **defective hemoglobin-chain synthesis**
- In Thalassemia, the production of one or more globulin chains within the hemoglobin molecule is reduced. This increases the rigidity of the RBCs and thus the premature destruction of these cells
- **Occurs when the 6th aa valine is changed by glutamine**
- **Mgt :**
 - BMT
 - PRBC

G6DPD

- Occurs due to **deficiency G6PD gene**
- Gene produce an enzyme which is important for membrane stability
- G6DPD → **rigid, deformed shape RBC**
- Mgt
 - ❖ Treat underline cause (discontinue medication)
 - ❖ Blood transfusion

Polycythemia

- Refers to an increased volume of RBCs.
- It is a term used when the hematocrit is elevated (to more than **55% in males, more than 50% in females**).
- Is classified as either **primary** or **secondary**

- **Polycythemia vera**

- **Primary polycythemia**,
- Is a proliferative disorder in which the myeloid stem cells seem to have escaped normal control mechanisms.
- The bone marrow is hypercellular, and the RBC, WBC, and platelet counts in the peripheral blood are **elevated**.
- However, **the RBC elevation is predominant**; the hematocrit can exceed 60%
- **Over time**, the bone marrow may become fibrotic, with a resultant **inability to produce as many cells**

Clinical manifestations

- Patients typically have **splenomegaly (enlarged spleen)**.
- The symptoms result from the increased blood volume:
 - headache, dizziness, tinnitus, fatigue, paresthesias, and blurred vision
- The symptoms result from **increased blood viscosity**:
 - angina, claudication, dyspnea, and thrombophlebitis, particularly if the patient has atherosclerotic blood vessels.
- Another common and bothersome problem is **generalized pruritus**, which may be caused by **histamine release due to the increased number of basophils**.
- **Erythromelalgia**, a burning sensation in the fingers and toes, may be reported and is **only partially relieved by cooling**.

Assessment and Diagnostic Findings

- Diagnosis is made by finding an elevated RBC mass (a nuclear medicine procedure), a normal oxygen saturation level, and an enlarged spleen.
- Other factors useful in establishing the diagnosis include elevated WBC and platelet counts.
- The erythropoietin level is not as low as would be expected with an elevated hematocrit; it is normal or only slightly low.

...

Complications

- Patients with polycythemia vera are at increased risk for **thromboses** resulting in a CVA (brain attack, stroke) or heart attack (MI); thrombotic complications are the most frequent cause of death.
- **Bleeding** is also a complication, possibly due to the fact that the platelets (**often very large**) **are somewhat dysfunctional**.
- The bleeding can be significant and can occur in the form of nosebleeds, ulcers, and frank gastrointestinal bleeding.

Medical management

- The objective of management is
 - To reduce the high blood cell mass.
- **Phlebotomy** is an important part of therapy and can be performed repeatedly to keep the hematocrit within normal range.
 - This is achieved by removing enough blood (**initially 500 ml once or twice weekly**) to deplete the patient's iron stores, thereby rendering the patient iron deficient and consequently unable to continue to manufacture RBCs excessively.
- Patients need to be instructed to **avoid iron supplements**, including those within **multivitamin supplements**.
- If the patient has an elevated uric acid concentration, allopurinol (Zyloprim) is used to prevent gouty attacks.

...

- If the patient develops ischemic symptoms, dipyridamole (eg, Persantine) is sometimes used.
- Radioactive phosphorus (^{32}P) or chemotherapeutic agents (eg, hydroxyurea [Hydrea]) can be **used to suppress marrow function**, but they may increase the risk for leukemia.
- Patients receiving hydroxyurea appear to have a lower incidence of thrombotic complications; this may result from a more controlled platelet count.
- Anagrelide (Agrylin) inhibits platelet aggregation and can also be useful in controlling the thrombocytosis associated with polycythemia vera.

Nursing management

- The nurse's role is primarily that of educator.
- Risk factors for thrombotic complications should be assessed, and patients should be instructed regarding the signs and symptoms of thrombosis.
- Patients with a history of bleeding are usually advised **to avoid aspirin and aspirin-containing medications, because these medications alter platelet function.**
- Minimizing alcohol intake should also be emphasized to further diminish any risk for bleeding.
- For Pruritus, the nurse may recommend bathing in **tepid or cool** water, along with applications of cocoa butter–based lotions and bath products

Secondary Polycythemia

- is caused by **excessive production of erythropoietin**.
- This may occur in **response to a reduced amount of oxygen**, which acts as a hypoxic stimulus,
 - Heavy cigarette smoking,
 - Obstructive sleep apnea (OSA),
 - Chronic obstructive pulmonary disease (COPD), or
 - Cyanotic heart disease,
 - **Living at a high altitude or exposure to low levels of oxygen**
- It can also occur from neoplasms (e.g., renal cell carcinoma) that stimulate erythropoietin production, excess erythropoietin-stimulating agent use, or androgen use.

...

Medical Management

- When secondary polycythemia is mild, treatment may not be necessary; when treatment is necessary, it involves treating the primary condition.
- If the cause cannot be corrected (e.g., by treating the OSA or improving pulmonary function with smoking cessation), therapeutic phlebotomy may be necessary in symptomatic patients to reduce blood viscosity and volume as well as when the hematocrit is significantly elevated.

Blood transfusion

- American Association of Blood Banks Guidelines
 - Hgb < 6 – Transfusion recommended
 - Hgb 6-7 – Transfusion likely recommended
 - Hgb 7-8 – **Restrictive Transfusion Strategy** for stable patients (Strong recommendation).
- Consider transfusion only if post-operative or symptomatic (chest pain, orthostatic hypotension or tachycardia **unresponsive** to fluid resuscitation, or congestive heart failure)
 - Hgb 8 – 10 – transfusion generally not indicated
 - Can consider special circumstances (ie active ischemia, symptomatic anemia, active bleeding, critical ill septic shock).
- Hgb >10 – transfusion not indicated

Compatibility testing

- Compatibility testing involves three separate procedures involving both donor and recipient blood.
 1. ABO & Rh blood type identification both the donor and recipient
 2. Antibody screening of donor plasma
 3. Donor/recipient cross-match
- Most of the fatal hemolytic transfusion reactions result from the transfusion of ABO incompatible blood
- Blood types are defined by the antigens present on the surface of the RBCs
 - Type A has A antigens on the surface of their red cells
 - Type B has B antigens
 - Type AB has both A and B antigens
 - Type O has neither antigen



- The serum contains antibodies to the AB antigens that are lacking on the RBC.
 - Type A has antibodies against the B antigen
 - Type B has antibodies against the A antigen
 - Type AB has no antibodies
 - Type O has both anti-A and anti-B antibodies
- To determine which types are compatible you need to **focus on which antibodies will be present in the recipient plasma.**
- It is the reaction of the antibodies with donor RBC antigens that can activate the complement system and lead to intravascular hemolysis of the red cell.
 - Type O- is the universal donor
 - Type AB+ is the universal recipient

Rhesus (D) antigen

- Patients with the Rhesus (D) antigen are said to be Rh+ and those without are Rh-
- Anti-D antibodies are not constitutively present in the serum of an Rh-negative patient.
- 60-70% of Rh- patients exposed to Rh+ RBCs will develop anti-D antibodies
- There is a latency period before the antibodies are synthesized.

Plasma compatibility table

Recipient	Donor			
	O	A	B	A B
AB				 —
A		 —		 —
B			 —	 —
O	 —	 —	 —	 —

Red blood cell compatibility table

Recipient

Donor

O- O+ A- A+ B- B+ AB- AB+

O-								
O+								
A-								
A+								
B-								
B+								
AB-								
AB+								

Blood Products

- Whole Blood
- Red Blood Cells
- Platelets
- Fresh Frozen Plasma
- Cryoprecipitate

Complications of blood transfusion

- **Infection**
 - Hepatitis B and C, HIV, CMV, Human T-cell lymphotropic viruses
- **Immunological**
 - **Early**: anaphylaxis, acute lung injury, alloimmunization, urticaria, acute haemolysis
 - **Delayed**: delayed haemolysis, immunosuppression
- **Metabolic**
 - Hyperkalaemia, hypocalcaemia, acid–base disturbance, coagulopathy,
- **Physical**
 - Hypothermia, microemboli, air embolus, circulatory overload

Nursing management

- The following steps are taken to determine the type and severity of the reaction:
 - ❖ **Stop the transfusion.**
 - ❖ Maintain the IV line with normal saline solution through new IV tubing, given at a slow rate.
 - ❖ Assess the patient carefully. Compare the vital signs with baseline, including oxygen saturation.
 - ❖ Assess the patient's respiratory status carefully. Note the presence of adventitious breath sounds; the use of accessory muscles; extent of dyspnea; and changes in mental status, including anxiety and confusion.
 - ❖ Note any chills, diaphoresis, jugular vein distention, and reports of back pain or urticaria.
 - ❖ Notify the primary provider of the assessment findings, and implement any treatments prescribed.

...

- ❖ Continue to monitor the patient's vital signs and respiratory, cardiovascular, and renal status. Notify the blood bank that a suspected transfusion reaction has occurred.
- ❖ Send the blood container and tubing to the blood bank for repeat typing and culture.
- ❖ If a hemolytic transfusion reaction or bacterial infection is suspected, the nurse does the following:
 - ❖ Obtains appropriate blood specimens from the patient
 - ❖ Collects a urine sample as soon as possible to detect hemoglobin in the urine
 - ❖ Documents the reaction according to the institution's policy

B. White Blood Cell Disorders

- Important disorders of the white blood cells (WBCs) and lymphoid tissue include the
 - ✓ leukaemias,
 - ✓ multiple myeloma and
 - ✓ malignant lymphomas (Hodgkin's disease and non-Hodgkin's lymphoma).

Neutropenia

- **Neutropenia** (a **neutrophil count of less than 2000/mm³**) results from decreased production of neutrophils or increased destruction of these cells
- A patient with neutropenia is at **increased risk for infection from both exogenous and endogenous sources.**

Causes

- Aplastic anemia, due to medications or toxins
- Chemotherapy
- Metastatic cancer, lymphoma, leukemia
- Radiation therapy
- Increased Destruction of Neutrophils: Bacterial infections, Hypersplenism, Immunologic disorders (e.g., systemic lupus erythematosus), Medication induced, Viral disease

...

Medical Management

- Treatment of the neutropenia varies depending on its cause.
- If the neutropenia is medication induced, the **offending agent is stopped immediately**, if possible.
- Treatment of an underlying neoplasm can temporarily make the neutropenia worse, but with bone marrow recovery, treatment may actually improve it.
- **Corticosteroids may be used if the cause is an immunologic disorder.**

Lymphopenia

- **Lymphopenia** (a **lymphocyte count less than 1500/mm³**) can result from ionizing radiation, long-term use of corticosteroids, uremia, infections (particularly viral infections), some neoplasms (e.g., breast and lung cancers, advanced Hodgkin disease), and some protein-losing enteropathies (in which the lymphocytes within the intestines are lost)
- When lymphopenia is mild, it is often without sequelae; when severe, it can result in bacterial infections (due to low B lymphocytes) or in opportunistic infections (due to low T lymphocytes).

• • •

- T cell lymphocyte depletion is most common, typically due to viral infection, such as the human immune deficiency virus (HIV).
- Excessive or prolonged abuse of alcohol can impair lymphocyte production; lymphocyte counts can improve when alcohol intake ceases

Leukemia

- Leukemias account for **approximately 8% of all human cancers**, and approximately half of these cases are classified as acute leukemia.
- Acute leukemia, a malignant disease of the blood forming organs, **results when white blood cell (WBC) precursors proliferate in the bone marrow and lymphatic tissues.**
- The cells eventually spread to the peripheral blood and all body tissues.
- Leukemia is considered acute when it has a rapid onset and progression and when, if left untreated, it **leads to 100% mortality within days or months.**

• • •

- There are two major forms of acute leukemia:
 - ✓ Lymphocytic leukemia and
 - ✓ Nonlymphocytic leukemia.
- **Lymphocytic leukemia** involves the **lymphocytes** (cells derived from the stem cells and that circulate among the blood, lymph nodes, and lymphatic organs) and lymphoid organs;
- **Nonlymphocytic leukemia** involves hematopoietic stem cells that differentiate into **monocytes, granulocytes, red blood cells (RBCs), and platelets.**

Causes

- **The exact cause of acute leukemia** is unknown, and most patients never have an identifiable cause.
- **Overexposure to radiation** even years before the development of the disease, particularly if the exposure is prolonged, is a major risk factor.
- **Other risk factors include exposure** to
 - Certain chemicals (benzene),
 - Medications (alkylating agents used to treat other cancers in particular) (with in 9 years)
 - Viruses.
 - Down syndrome, albinism, and
 - Congenital immunodeficiency syndrome.

C/m

- The patient appears **acutely ill, short of breath, and pale**, symptoms that occur because of anemia.
- Children are often **febrile**.
- When you inspect the lips and mouth, you may note **bleeding gums and ulcerated areas of the mouth and throat**.
- Lymph node swelling and
- **Enlargement of the liver and spleen**.
- When you auscultate the patient's lungs, you may hear decreased breath sounds, shallow and rapid respirations, a rapid heart rate, and a systolic ejection murmur.
- Patients may have altered mental status from brain infiltration with leukemic cells.

Dx

- Complete blood count and differential
- Bone marrow aspiration/bone marrow biopsy

Mgt

- Given in four phases: induction, consolidation, continuation, and treatment of (CNS) leukemia.
- Adults AML remission induction: Anthracycline drug (idarubicin or daunorubicin) and cytarabine (araC); cladribine may be added
- For ALL, fludarabine and cyclophosphamide and rituximab (FCR) is commonly used
- *Child* AML induction: Daunomycin or idarubicin, cytosine arabinoside, and etoposide along with oral 6-thioguanine and dexamethasone.
- ALL induction: Prednisone, asparaginase, vincristine, and anthracycline

C. Platelet Disorders

Bleeding Disorders

- Bleeding is commonly provoked by trauma; however, in certain circumstances, it can occur spontaneously.
- When the cause is platelet or coagulation factor abnormalities, the site of bleeding can be anywhere in the body.
- When the source is vascular abnormalities, the site of bleeding may be more localized.

Clinical Manifestations

- Abnormalities of the vascular system give rise to local bleeding, usually into the skin.
- Because platelets are primarily responsible for stopping bleeding from small vessels, patients with platelet defects develop **petechiae**, often in clusters; these are seen on the skin and mucous membranes and can occur throughout the body.
- Bleeding from platelet disorders can be severe.

Petechiae



Figure 33-4 • Petechiae. From Wintrobe's

• • •

- Coagulation factor defects do not tend to cause superficial bleeding, because the primary hemostatic mechanisms are still intact.
- Instead, bleeding occurs deeper within the body (e.g., subcutaneous or intramuscular hematomas, hemorrhage into joint spaces).
- External bleeding diminishes very slowly when local pressure is applied; it often recurs several hours after pressure is removed.
- For example, severe bleeding may start several hours after a tooth extraction.

...

Management

- Management varies based on the underlying cause of the bleeding disorder.
- If bleeding is significant, transfusions of blood products are indicated. The specific blood product used is determined by the underlying defect and the extent of the blood loss.
- State changes in lifestyle (e.g., diet, activity) or home environment necessary to decrease risk of bleeding.
- Avoid the use of suppositories, enemas, and tampons
- Avoid constipation

...

- Avoid vigorous sexual intercourse and anal sex
- Avoid contact sports
- Avoid or limit aggressive manual labor
- Use an electric razor for shaving and a soft-bristled toothbrush for teeth brushing
- Notify health care professional before having dental work or other invasive procedures
- Identify medications and other substances to avoid (e.g., aspirin containing medications, alcohol).
- Identify signs and symptoms of bleeding.

Thrombocytopenia

- Thrombocytopenia (**low platelet level**) can result from various factors:
 - ✓ **decreased production of platelets** within the bone marrow,
 - ✓ **increased destruction of platelets**, or
 - ✓ **increased consumption of platelets** (e.g., the use of platelets in clot formation).

Clinical Manifestations

- Bleeding and petechiae usually do not occur with platelet counts greater than 50,000/mm³, although excessive bleeding can follow surgery or other trauma.
- When the **platelet count drops to less than 20,000/mm³, petechiae can appear**, along with **nasal and gingival bleeding, excessive menstrual bleeding**, and excessive bleeding after surgery or dental extractions.
- When the platelet count is **less than 5000/mm³**, spontaneous, potentially **fatal central nervous system** or GI hemorrhage can occur.

Causes and treatments

Cause	Management
Decreased Platelet Production	
Hematologic malignancy, especially acute leukemias	Treat leukemia; platelet transfusion
MDS	Treat MDS; platelet transfusion
Metastatic involvement of bone marrow from solid tumors	Treat solid tumor
Aplastic anemia	Treat underlying condition
Megaloblastic anemia	Treat underlying anemia
Toxins	Remove toxin
Medications (e.g., sulfa drugs, methotrexate)	Stop medication
Infection (especially septicemia, viral infection, tuberculosis, chronic hepatitis C)	Treat infection
Chronic alcohol abuse	Refrain from alcohol consumption
Chemotherapy	Delay or decrease dose; platelet transfusion
Chronic liver disease	Treat underlying disease
Radiation (e.g., pelvic irradiation)	Platelet transfusion
Delayed engraftment after stem cell transplantation	Platelet transfusion

Hemophilia

- Hemophilia refers to a group of congenital coagulation disorders characterized by a deficiency or malfunction of specific clotting factors.
- **Hemophilia A**, or **classic hemophilia**, is caused by a defect in factor VIII (antihemophilic factor).
- **Hemophilia B**, or **Christmas disease**, is caused by a defect in factor IX (plasma thromboplastin component, or Christmas factor).
- Hemophilia A is more common

...

- Persons with hemophilia are able to form a platelet plug but are **unable to form a stable clot.**
- Clinical manifestations and complications of hemophilia are usually **secondary to recurrent bleeding.**
- Complications from subcutaneous and intramuscular hematomas are caused by compression of nerves or other structures, resulting in peripheral neuropathies, pain, compromised airway, muscle atrophy, ischemia, and gangrene.
- **Hemarthrosis, or bleeding into the joint or synovial cavity**, is a common complication that often results in joint deformities.
- Life threatening hemorrhage may result from minor injuries.

C/m

- Note hematomas from **subcutaneous and intramuscular bleeding**.
- Tissue over the bleeding site is hard, raised, and dark purple.
- The hemorrhage extends from this center concentrically, with each successive outer circle becoming lighter in color.
- Intramuscular bleeding usually spreads within a single fascial space.
- Fever or pain may occur, with or without the skin discoloration.
- Signs of excessive blood loss are tachycardia, hypotension, shortness of breath, dizziness, and orthostatic changes in blood pressure.
- Monitor the patient for frank or occult hematuria, melena, or hematemesis.

• • •

- When the patient's extremities are moved, note joint pain and swelling caused by hemarthrosis or spontaneous or trauma-induced bleeding into the joint or synovial cavity.
- Acute hemarthrosis is often preceded by a warm tingling sensation in the affected joint.
- If absorption of the blood from around the periarticular structures is incomplete, the remaining blood can cause chronic inflammation of the synovial membranes.
- Other long-term clinical sequelae of hemarthrosis include impaired joint mobility, bone deformity and demineralization, and stunted growth.
- Approximately 40% of hemophilia patients have splenomegaly; there have been some reported cases of spontaneous splenic rupture.

Hemophilic arthropathy

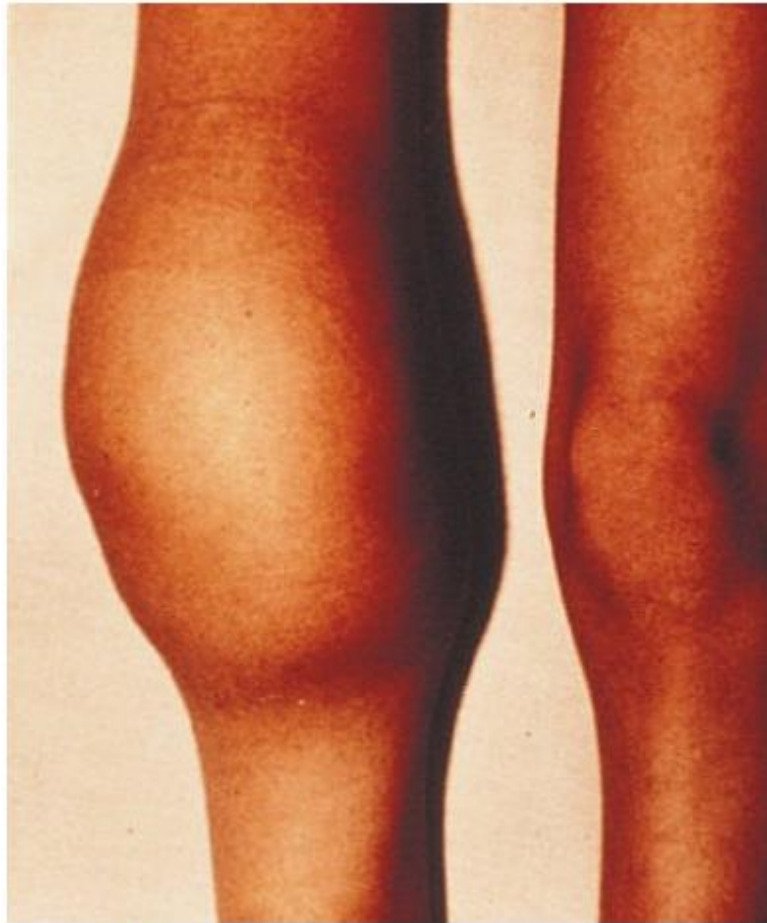


Figure 33-5 • Hemophilic arthropathy:

MGT

- Factor VIII, also found in fresh-frozen plasma (FFP), is usually reserved for patients with hemophilia A who are actively bleeding.
- Bleeding episodes in hemophilia B can be treated with FFP or purified factor IX.
- Cryoprecipitate contains high levels of factor VIII and fibrinogen and may be administered in emergencies.
- Desmopressin: Stimulates rapid release of von Willebrand factor into the blood; treatment of choice in patients with hemophilia A who do not have life threatening bleeding problems
- Analgesia for hemarthrosis; avoid products containing aspirin and NSAIDs; monoclonal antibodies (rituximab); antifibrinolytic agents: aminocaproic acid (Amicar) can be used in oral surgery or bleeding.

Disseminated Intravascular Coagulation

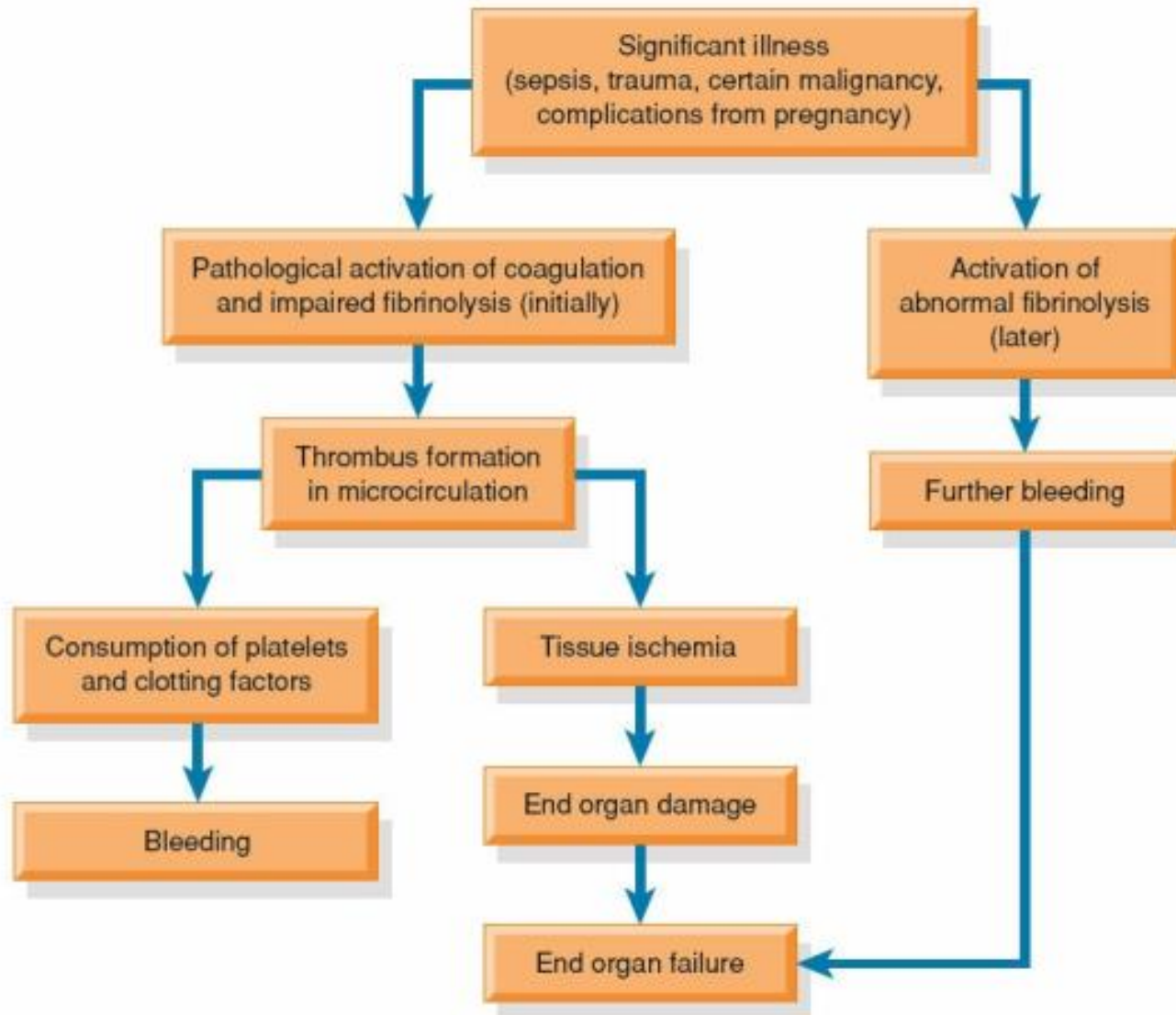
- Disseminated intravascular coagulation (DIC) is not an actual disease but **a sign of an underlying condition.**
- DIC may be triggered by **sepsis, trauma, cancer, shock, abruptio placentae, toxins, allergic reactions**, and other conditions
- DIC is an acquired syndrome characterized by the **intravascular activation of coagulation** with loss of localization arising from different causes.
- The vast majority (two thirds) of cases of DIC are **initiated by an infection or a malignancy.**
- The severity of DIC is variable, but it is **potentially life-threatening**

Pathophysiology

- involves an **overactivation of the clotting mechanisms** with both enhanced fibrin production leading to small clots and fibrinolysis leading to enhanced bleeding.
- As its name implies, tiny clots accumulate in the microcirculation (capillaries) throughout the body, depleting the blood supply of its clotting factors.
- These microemboli interfere with blood flow and lead to **ischemia and organ damage**.
- As the clots begin to lyse, fibrin degradation products (FDPs), which have an anticoagulant property of their own, are released.
- The FDPs, along with decreased levels of clotting factors in the bloodstream, lead to **massive bleeding internally** from the brain, kidneys, adrenals, heart, and other organs or from any wounds and old puncture sites.

• • •

Physiology/Pathophysiology



C/m

- The most common symptoms are
- **blood loss from wounds, gums, and gastrointestinal track, bleeding into the skin, hematuria, and/or epistaxis.**
- Assess the patient's skin for any petechiae, ecchymoses, hematoma formation, epistaxis, bleeding from wounds, vaginal bleeding in the labor or postpartum patient, hematuria, conjunctival hemorrhage, and hemoptysis.

Dx

- Platelet count 150,000–450,000/mm³
- 100,000/mm³ Platelets are consumed during clot formation
- Prothrombin time (PT)
- INR (international normalized ratio)

Mgt

- Medical treatment focuses on correcting the underlying disorder.
- **Packed red blood cells** are used to improve oxygen delivery by increasing the hemoglobin content of the blood.
- **Freshfrozen plasma** replaces many of the clotting factors, whereas cryoprecipitate is the best source of fibrinogen and factors VIII and XIII.
- **Platelet transfusion** is used when the platelet count falls below 100,000/mm³.
- **Heparin**: Varies with patient; patient-based dosing is common, with a starting bolus of 80 units/kg and an infusion of 18 units/kg per hr. Inactivates thrombin and factors X and IX by anti thrombin III

Thank you!!!

HAVE A NICE TIME!!!

Medical Surgical Nursing

DISORDER OF GASTROINTESTINAL SYSTEM

BY AFEWOK E.

Review of anatomy & physiology of GIS

- Gastrointestinal system can be classified into two:
 - ✓ **Gastrointestinal tract (GIT)**
 - ✓ **Accessory Organs:** Liver, Gallbladder & Pancreas
- The gastrointestinal (GI) tract is a pathway that extends from the mouth through the esophagus, stomach, & intestines to the anus.
- **Esophagus** is a collapsible tube, which is about 25cm in length & located in the thoracic cavity posterior to the trachea & larynx at the level of **C6 -T₁₁**.

Stomach:

- is situated in the upper portion of the abdomen to the left of the midline.
- has a **capacity of approximately 1500ml**
- the inlet to the stomach is called ***esophago-gastric junction.***
- It has four anatomical regions: **the cardiac/ entrance, fundus, body & pylorus / outlet/.**

- It is surrounded by a ring of smooth muscle, called **lower esophageal sphincter or cardiac sphincter** which on contraction it closes off the stomach from the esophagus.
- Circular smooth muscle in the wall of the pylorus forms the **pyloric sphincter** & controls the opening between stomach & small intestine.
- The epithelial lining of the stomach contains many glands:
 - **Cardiac glands secrete** - mucus
 - **Peptic /chief cell/ secrete** - mucus & pepsinogen
 - **Parietal cells secrete** – HCl, IF & water
 - **Neck cells secrete** - mucus

Small intestine

- is the longest segment of the GI tract, accounting for **about two thirds of the total length** of the tract.
- It folds back & forth on itself allowing for approximately 7000cm of surface area for secretion & absorption.
- The small intestine is divided into three anatomic parts:
 - **duodenum (upper part) about 25cm,**
 - **jejunum about 2.5m** (middle part), and
 - **ileum about 3.6m** (lower part)
- The common bile duct, which allows for the passage of both bile & pancreatic secretions, empties into duodenum at the **ampulla of Vater.**

- The junction between the small & large intestine is located in the right lower portion of the abdomen. It is called cecum.
- At that junction is the **ileo-cecal valve**, which function to control the passage of intestinal contents in to the large intestine & to prevent reflux of bacteria into the small intestine.
- It is in this area that the **vermiform appendix** is located.

Large Intestine

- It has three parts: **Colon, Sigmoid, & Rectum**
- The colon consists of an ascending segment on the right side of abdomen, a transverse segment that extends from right to left in the upper abdomen, & a descending segment on the left side of abdomen.
- The terminal portion of the large intestine consists of two parts: the sigmoid colon & the rectum.
- The rectum is continuous with the anus.
- The anal outlet is regulated by a network of muscle that form **both internal & external anal sphincter.**

Blood Supply of GIT

- From arteries which originate from thoracic & abdominal aorta. These arteries are:
 - ✓ **Gastric artery**: oxygen & nutrient supply to stomach.
 - ✓ **Superior & inferior mesenteric arteries** to intestine.
 - ✓ **Portal vein** drains blood & nutrient from intestines to liver.

Nerve supply of GIT

- **Enteric Nervous System:** lies entirely in the wall of the gut, in the esophagus to the anus
- **Can act independently to cause motility & secretion of the gut**
- It is also innervated by both sympathetic & parasympathetic portion of the autonomic nervous system.
 - **Sympathetic nerves** exert an inhibitory effect on the GI tract such as decrease gastric secretion & motility.
 - **Parasympathetic nerve** stimulation causes peristalsis to occur & increases secretory activities.
- The only portions of the tract under voluntary control are **the upper esophagus & the external anal sphincter.**

Function of the GIT

The four major activity of the GIT are:

1. **Secretion** of electrolytes, hormones, & enzymes to be used in breakdown of the ingested materials
2. **Movement** of ingested products
3. **Digestion** of food & fluids
4. **Absorption** of end products into blood stream

Accessory Organs

1. Liver

- It is the **largest organ**
- Located in the upper abdominal cavity below diaphragm.
- It is **highly vascularised**. It receives blood from two sources:
 - a) Portal Vein
 - b) Hepatic artery
- Both sources leave the liver through hepatic vein.

Functions of the liver

1. On Carbohydrate Metabolism:

- It is the site of **glycogenesis**, **gluconeogenesis** and **glycogenolysis**
 - Turns proteins into glucose
 - Turns triglycerides into glucose
- } gluconeogenesis
- Turns excess glucose into glycogen & stores (glycogenesis)
 - Turns glycogen back into glucose as needed (glycogenolysis)

2. On Lipid Metabolism

- Formation of **phospholipids**, **lipoproteins**, **synthesis of cholesterol**, and **conversion of CHO into fat**

...

3. Protein Metabolism

- Deamination of amino acids = removes NH_2 (amine group) from amino acids so can use what is left as energy source
- **Converts resulting toxic ammonia (NH_3) into urea** for excretion by the kidney
- **Synthesizes plasma proteins** utilized in the clotting mechanism and immune system

Other Functions of the liver

1. **Inactivation of drugs & hormones** (Sulfonamide, penicillin, thyroid, steroids)
2. **Removes the waste product**; bilirubin
3. **Releases bile**
4. **Stores fat soluble vitamins**: A, B₁₂, D, E, K
5. **Stores iron and copper**
6. **Filtration of blood**: phagocytises worn out blood cells(& bacteria, removes blood clots and toxins from portal circulation
7. **Activates vitamin D**
8. **Storage of blood**: a major blood reservoir
9. **Synthesis of blood clotting factors** (F-I, II, VII, IX, X), Heparin

2. Gall Bladder & Bile duct

- **Function:** Store & concentrate bile salt.

3. Pancreas

- **Function:** Exocrine & endocrine

ACCESSORY ORGANS:

Salivary glands

- Saliva moistens food
- Bicarbonate maintains pH
- Amylase digests starch
- Lysozyme inhibits bacteria

Liver

- Produces bile
- Performs various functions associated with processing and storing nutrients

Gall bladder

- Stores and concentrates bile

Pancreas

- Secretes digestive enzymes into small intestine
- Secretes bicarbonate into small intestine to neutralize stomach acid

ORGANS:

Mouth

- Teeth chew food
- Tongue positions and tastes food

Pharynx

- Passageway for food and air
- Participates in swallowing

Esophagus

- Moves food from pharynx to stomach

Stomach

- Stores and mixes food
- Begins chemical digestion of protein by enzymes and acid
- Regulates delivery to the small intestine

Large intestine

- Absorbs nutrients
- Stores waste material

Small intestine

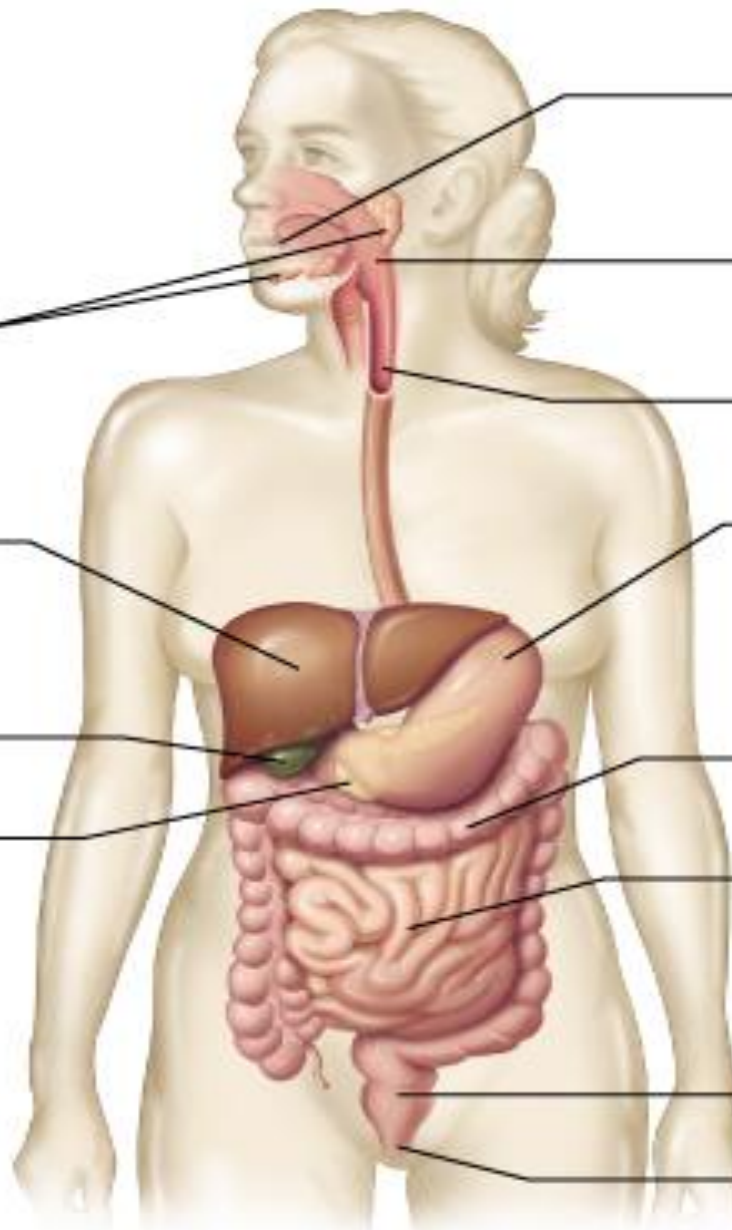
- Digests proteins, fats, and carbohydrates
- Absorbs most of the nutrients
- Secretes digestive hormones and enzymes

Rectum

- Passageway for feces

Anus

- Expells undigested material



Assessment

1. Health History

- Begin by taking a complete history,
- I) **C/C:** Focusing on symptoms common to gastrointestinal dysfunction.
- Symptoms on which the assessment focuses include pain, indigestion, intestinal gas, nausea & vomiting, hematoemesis, & changes in bowel habits & stool characteristics

When conducting a symptom analysis of, ask the client the following questions about the c/c & present illness

- Onset, Duration, Quality & characteristics, Severity, Location, Precipitating factor, Relieving factor,

- ❖ **Pain:** duration, pattern, frequency, location, distribution, aggravating & relieving factors and severity.
- ❖ **Dyspepsia:** upper abdominal discomfort associated with eating (commonly called **indigestion**)
- ❖ **Intestinal gas:**
 - **Bleching:** expulsion of gas from stomach through the mouth OR
 - **Flatulence** (the expulsion of gas from the rectum.
- ❖ **Change in bowel habits:** Diarrhea, Constipation
- ❖ **Characteristics of stool:** it can be brown, contain bright **red blood, black & tarry or pale yellow.**

...

❖ **Vomiting** : The emesis, or vomitus, may vary in color and content and may contain undigested food particles, blood (hematemesis), or bilious material mixed with gastric juices.

Caused by

- Visceral afferent stimulation (i.e., dysmotility, peritoneal irritation, infections, hepatobiliary or pancreatic disorders, mechanical obstruction);
- CNS disorders (i.e., vestibular disorders, increased intracranial pressure, infections, psychogenic disorder);
- irritation of the chemoreceptor trigger zone

II) Past medical & surgical History

III) Family history

IV) Personal & social history

➤ asks about the patient's normal tooth brushing and flossing routine; frequency of dental visits; awareness of any lesions or irritated areas in the mouth, tongue, or throat; recent history of sore throat or bloody sputum; discomfort caused by certain foods; daily food intake; the use of alcohol and tobacco, including smokeless chewing tobacco

2. Physical examination

- Stand on the right side of the patient.
- Place the client in a supine position, elevate the head, arms should be at his sides or folded across his chest.
- The room should be well illuminated.
- Apply draping to expose the area from the sternum to the pubis.

The sequence of examination:

A. Inspection

- Size & shape of abdomen
- Scar, striae
- Abdominal movement
- Umbilical hernia
- Hernial sites
- Visible peristalsis
- Visible distended vein

B. Auscultation: assess the sounds of peristalsis & vascular abnormalities.

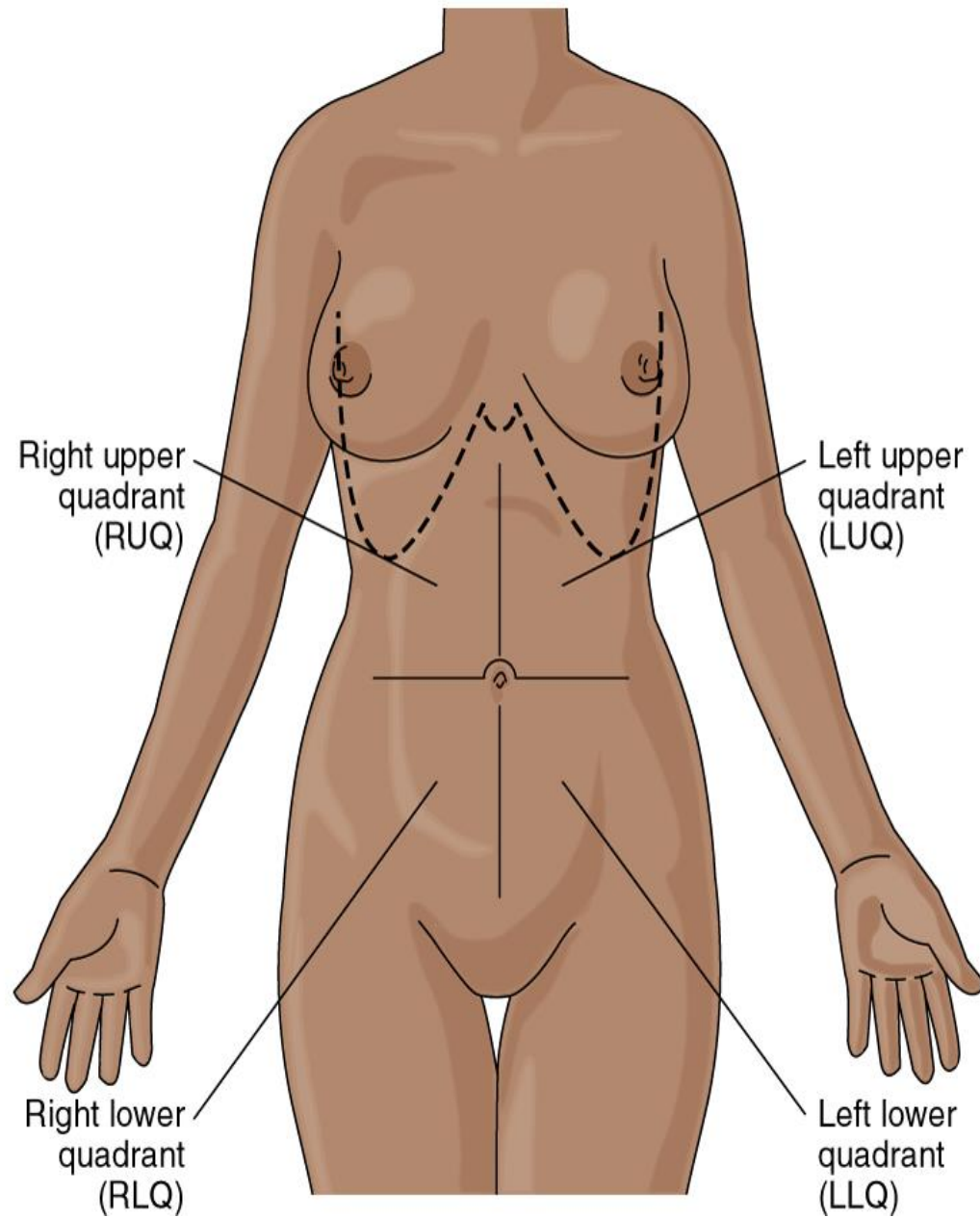
- **Bowel sound**: - frequency & intensity
 - listen in **all quadrants** with stethoscope.

C. Percussion: Tympany is the normal percussion note present throughout the abdomen except for the area over liver & spleen.

- Estimate liver size: normally in adult, it is 6 – 12cm.

D. Palpation

- **Superficial palpation for tenderness** & mass over the four quadrants.
- **Always start from pain free site**
- Perform deep palpation for liver, kidney, & spleen and to identify deep mass.



Diagnosis tests, procedures and nursing responsibilities

- The entire GIT can be delineated by **x-ray studies**, after the introduction of a contrast agent.
- A radi-opaque liquid (such as **barium sulfate**) is one of the most commonly used media.
- Barium is **insoluble substance** & ingested for upper GIT study but instilled rectally (barium enema) for lower GIT studies

GIT Series



A) Upper GIT studies / upper GI series/

- **X-ray of upper GIT is taken after barium is ingested.**
 - This enables the examiner to detect any anatomic & functional derangement of the upper GI organs Or sphincters.
 - It also aids in the diagnosis of ulcers, varices, tumors, regional enteritis, & malabsorption syndromes.
 - The procedure may be extended to examine the duodenum and **small bowel** (small bowel follow-through).

Patient preparation

- Keep the patient NPO after midnight before the test.
- A laxative may be prescribed to clean out the intestinal tract
- Withheld all medications
- Patient should not smoke the morning before the procedure because smoking can stimulate gastric motility.

Procedure

- The patient is required to swallow barium under direct fluoroscopic examination.
- As the barium descends in to stomach, the position, patency, & caliber of the esophagus are visualized
- Fluoroscopic examination next extends to stomach to visualize the motility & thickness of the gastric wall & mucosal pattern, & patency of pylori.

- **Post procedure**

- Follow up care is needed after the procedure to ensure that the **ingested barium has been completely eliminated.**
- Stool must be monitored until they return to their normal color /the barium will look like clay/
- A laxative or enema may be needed.

B) Lower GI studies

- Visualization of the lower GI tract is obtained after rectal installation of barium.
- Barium enema is given before X-ray is taken
 - The purpose of a barium enema is to detect the presence of polyps, tumor, & other lesions of the large intestine & to demonstrate any abnormal anatomy or malfunction of the bowel.
- Nursing care for patient
 - Instruct the patient to eat low - residue diet 1 to 2 days before the test. Give laxatives/colon washout so that colon is absolutely empty
 - Preparation of the patient includes emptying and cleansing the lower bowel.
 - Postprocedural patient education includes information about increasing fluid intake, evaluating bowel movements for evacuation of barium, and noting increased number of bowel movements, because barium, due to its high osmolarity,

C. Endoscopic Procedures

a. Upper Gastrointestinal Fibroscopy / Esophagogastro duodenoscopy(EGD)

Purpose:

- **For direct visualization** of disease of the esophagus, gastric, & duodenal mucosa / upper GI hemorrhage, gastritis, PUD, motility can be evaluated.
- Esophageal & gastric motility can be evaluated.
- For collection of secretions & tissue specimens for further analysis.
- Foreign body removal
- Caution of bleeding point

Gastroscopy



- **Side viewing flexible scopes** are used to visualize the common bile, & pancreatic & hepatic duct through the ampulla of vater in the duodenum.
- This procedure is helpful in evaluation jaundice, pancreatitis, tumor of the pancreas, common duct stones, & biliary tract disease
- This procedure is called **endoscopic retrograde cholangio pancreatography (ERCP)**

Patient Preparation

- Instruct the patient to be NPO 6-12 hour before the examination.
- Spray or gargle with a local anesthesia, along with IV diazepam before the scope is introduced
- Atropine may be administered to reduce secretions.
- Position the pt on left lateral position.

Post procedure care

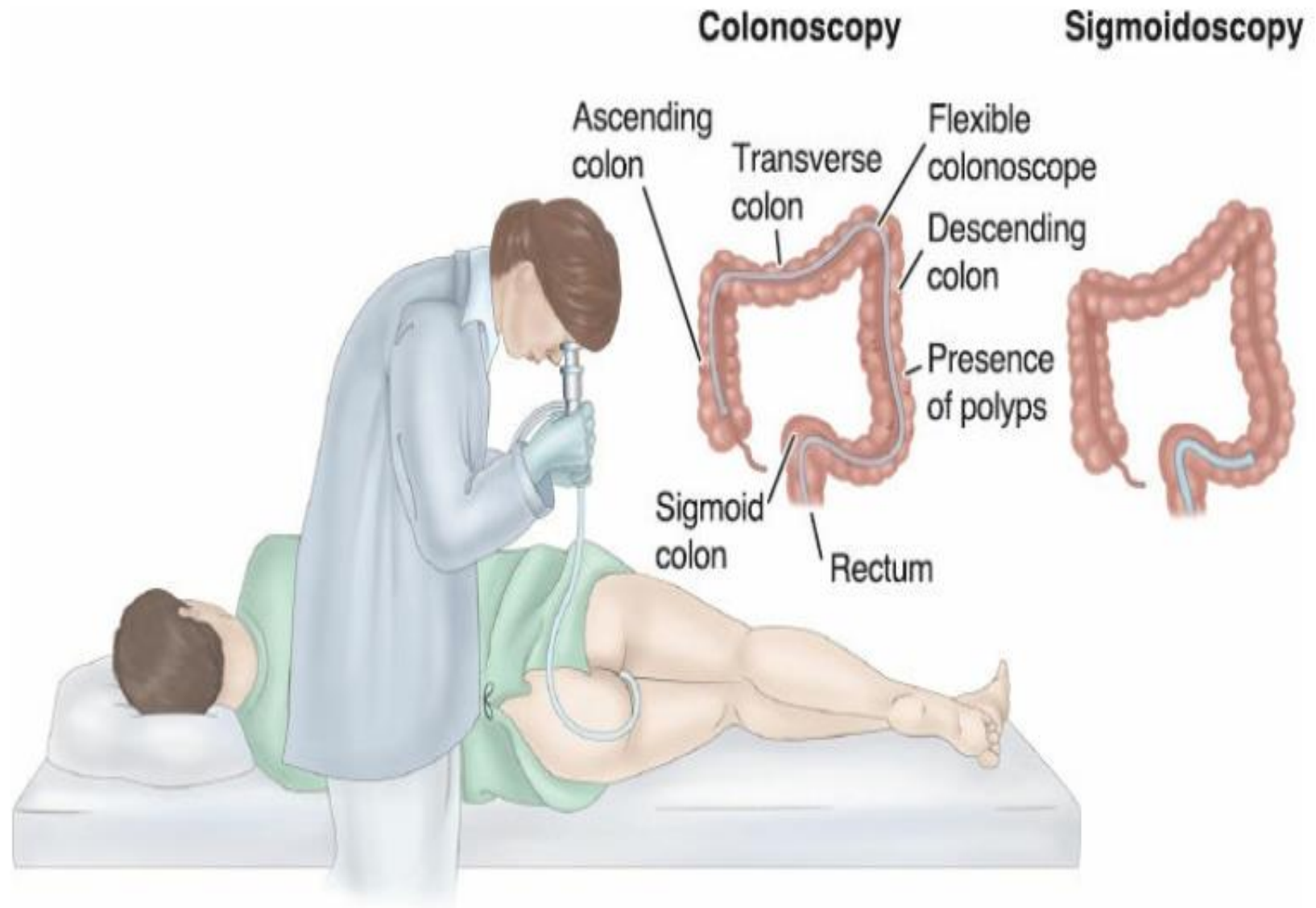
- After a gastroscopy, the patient is **instructed not to eat or drink** until the gag reflex returns (in 1 to 2 hour) to prevent aspiration of foods or fluids in to the lungs.
- Assess the patient for the signs of perforation, such as pain, bleeding, unusual difficulty swallowing, & an elevated temperature.

b) Lower GI fiberscopy

Colonoscopy

- It is **direct visual inspection of the colon to the cecum** by means of a flexible fiber optic colonoscopy
1. Diagnostic colonoscopy
 - lower GI bleeding
 - colonic & ileo-cecal tuberculosis
 - Inflammatory bowel disease & malignancy
 2. Therapeutic colonoscopy
 - Polyp-ectomy, foreign body removal, & control of bleeding by electro coagulation

Colonoscopy



Patient Preparation

- Patient should take only fluids for 1 to 3 days prior to the examination/ or
 - 200cc of mannitol followed by plenty of water after one hour initiates loose motions & clears the bowel or
 - Laxatives may be ordered for 2 nights prior to the examination.
 - Before the examination, a narcotic analgesic may be administered.
 - Position the patient on the left side with the legs drawn up toward the chest.
 - Monitor cardiac & respiratory function.

Post procedure care

- Observe for signs & symptoms of bowel perforation (e.g, rectal bleeding, abdominal pain or distention, fever, or focal peritoneal signs)

Anoscopy, Proctoscopy, & sigmoidoscopy

- **Anoscope** : is a rigid scope used to examine the anal canal.
- **Proctoscopes** & **sigmoidoscopes**: are rigid or flexible scopes used to inspect the rectum & sigmoid colon, respectively, for evidence of ulceration, tumors, polyps, or other pathologic processes.

Patient Preparation

- A warm tap water enema is given until returns are clear.
- Dietary restrictions are not usually necessary
- Sedation is not usually required.

Post procedure care

- Monitor the patient for rectal bleeding & signs of intestinal perforation /e.g. fever, rectal drainage, abdominal distention, & pain/

D. Laparoscopy

- It is **introduction of rigid laparoscope** through a nick in the **anterior abdominal wall** & insufflations of peritoneum with room air or CO₂ make it possible to look at the abdominal viscera & obtain biopsy.
- **Indication:**
 1. GI disease
 - Evaluation of hepatic disease
 - Differential diagnosis of jaundice
 - Cancer
 - Diagnosis of ascites

2. Miscellaneous

- Inspection of pelvic viscera
- Abdominal masses
- Abscure abdominal pain
- Evaluation of abdominal trauma.

E. Abdominal Ultrasonography

- Particularly useful in the detection of an enlarged gallbladder or pancreas, the presence of gallstones, an enlarged ovary, an ectopic pregnancy, or appendicitis.

3. Laboratory test

a) Blood may be tested to determine:

- Haemoglobin level
- Haematocrit
- Leucocyte count
- Serum electrolytes
- Bilirubin levels
- Blood glucose
- Pancreatic enzyme levels

b) Stool examination

- The stool is examined for its amount, consistency, & colour, a screening test for occult blood is also done.
 - Melena / tarry black/ ⇒ **upper GI bleeding**
 - Bright red blood ⇒ **lower GI bleeding**
 - Blood streaking on surface of stool ⇒ **lower rectal or anal bleeding**

Assisting the patient in Gastric analysis

- Gastric fluid analysis involves examination of gastric secretions, **(hydrochloric acid and pepsin in the stomach)**, & it may be performed by examining **a specimen vomitus** or by testing a sample of the gastric contents which have been **aspirated** via a nasogastric tube.
- Two tests are performed in gastric analysis:
 - 1) the basal cell secretion test
 - 2) The gastric acid stimulation test (done for 1hr after Sc injection of drug that stimulate gastric acid secretion).

Purpose

- To diagnose
 - Duodenal ulcer
 - Zollinger Ellison syndrome
 - Gastric carcinoma
 - Pernicious anaemia

Procedure:

- The client should be NPO for 12 hours before the test.
- The client's naso-gastric tube is attached to suction, and stomach contents are collected every 15 minutes for 1 hour
- The nurse must properly label the specimens with the time & volume.

Gastrointestinal Intubation

- It is the insertion of a short or a long flexible rubber or plastic tube into the stomach or intestine by way of the mouth or nose:

Purpose

- ✓ To decompress the stomach & remove gas and fluid.
- ✓ To diagnose gastrointestinal motility
- ✓ To administer medications & feeding
- ✓ To treat an obstruction or bleeding site.
- ✓ To obtain gastric contents for analysis.

Disease of the mouth & related structures

Periodontal Disease

- Periodontum is the tissue that surrounds & supports the teeth.
- Periodontal disease includes:

Gingivitis:

- It is an inflammation of gum
- It is the earliest form of periodontal disease.
- This is the **most common infection of oral tissue.**

Causes:

- Poor oral hygiene: food debris, bacterial plaque, & calculus / tartar/ accumulate.

C/M:

- Painful, inflamed, swollen gums, usually the gums bleed in response to light contact.
- Inflammation causes the gingival to separate from the teeth surface & pockets formed in the gingival that can collect bacteria, foods particles & pus.

Nursing MX

- Brush teeth using a soft toothbrush at least 2 times daily
- Avoid alcohol & tobacco products
- Maintain adequate nutrition & avoid sweets

Medical MX

- Periodic professional dental cleaning

Periodontitis

- It is an inflammation of the **tissue around teeth.**
- May result from untreated gingivitis, Poor or inadequate dental hygiene

C/M:

- May have bleeding
- Infection
- Loosening of teeth
- Loss of teeth /late/

RX:

- Instruction of patient in proper oral hygiene.
- Antibiotic
- analgesics

Peri apical Abscess / Dento-alveolar Abscess/

- Pus collection around the apex of the teeth.
- This is more commonly referred to as an abscessed tooth, involves the collection of pus in the apical dental periosteum & tissue surrounding the apex of the tooth.
- It can be acute or chronic

Acute peri apical abscess

- It is usually secondary to a suppurative pulpitis (a pus producing inflammation of the dental pulp) that arises from an infection extending from dental caries.

C/M:

- dull, gnawing / keep biting/, continuous pain,
- Cellulites, edema of adjacent structures
- Swollen cheek
- Fever, malaise

- **Mx:**

- Incision & drainage

- Antibiotic

Chronic Periapical abscess

- It is slowly progressive infections process
- This can progress to a fully formed “blind dental abscess” without the patient’s knowing it.

DX:

- x- ray

Mx:

- Teeth extraction or root canal therapy, with apicectomy / excision of the apex of the tooth root.

Stomatitis

- It is the inflammation of the mucosa of the oral cavity.

Cause:

- It is associated with **emotional or mental stress, fatigue, hormonal factors, minor trauma, and allergies.**
- Associated with HIV infection / virus/, bacteria / such as, syphilis & streptococcal/ and fungus moniliasis.

C/M:

- shallow ulcers with **white gray center & red border.**
- Seen in the inner side of lip & cheek or on the tongue
- It begins with a burning sensations & slight swelling, painful
- Usually heals without a scar.

Necrotizing Ulcerative gingivitis or ‘Trench mouth, ‘ or Vincent’s Angina

- It is pseudo membranous ulceration affecting **gum, mouth mucosa, tonsil & pharynx.**
- It is an acute bacterial infection of the gingiva.

Etiology:

- combination of a spirochete & a fusiform bacillus.

Risk factors:

- Poor oral hygiene, low tissue resistance, & infection from complex micro- organisms, local tissue damage & diabetes mellitus.

C/M

- **Foul breath**, painful bleeding gums, swallowing & talking are painful
- Mild fever, swelling of neck lymph nodes

Treatment

- Antibiotic
- Teach the patient proper oral hygiene
- Irrigate with 2% to 3% hydrogen peroxide or normal saline.

Dental Caries

- A gradual pathologic disintegration & dissolution of the tooth & dentin, with eventual involvement of the pulp.

Etiology:

- It is the interaction of three factors. These are:
 1. A susceptible tooth surface
 2. proper microflora / streptococcus mutans, Actinomyces Viscosus & lactobacillus/
 3. a suitable substrate for the microflora / carbohydrate/

Predisposing factors

- root exposure, lack of fluoride, inadequate personal hygiene

Mx:

- removal of decay
- restoration
- tooth extraction

Parotitis

- Parotitis (inflammation of the parotid gland) is the most common inflammatory condition of the salivary glands.

Cause:

- Virus in children
- Staphylococcus aureus (except in mumps)

Risk factors:

- elderly, acutely ill & debilitated people with decreased salivary flow due to general dehydration or medication

C/M

- Fever
- Swelling, tenderness & tense gland
- Pain is felt in the ear & swelling gland interfere with swallowing
- Overlying skin soon becomes red & shiny

Medical_MX

- Antibiotic (cloxacillin 500Mg Po Qid for 7 days)
- Massage
- Analgesics
- Incision & drainage (I&D) of the glands if antibiotic therapy is not effective.

Nursing Intervention

- Administer ordered medication
- Reassure the patient
- Tepid sponge bathes.

Cancer of the mouth

- It may occur in any part of the mouth or throat
- It is curable if discovered early.
- This type of cancer is associated with the use of alcohol & tobacco.

Predisposing factors

- Smoking & alcohol
- Old age
- Prolonged exposure to sun & wind

C/M

- It has no symptom during early stages
- Painless sore or mass that will not heal
- A typical lesion in oral cancer is painless indurated (hardened) ulcer with raised edges.
- Any ulcer of the oral cavity that doesn't heal in 2 weeks should be examined through biopsy.
- As the cancer progresses, the patient will complain tenderness; difficulty in chewing, swallowing or speaking, coughing of blood – tinged sputum; or enlarged cervical lymph nodes.

DX

- Clinical (History & P/E)
- Biopsy

Medical Mx

1. Resectional surgery or
2. Radiation therapy or
3. Chemotherapy or
4. combination of these therapies

Nursing Process

Patient with conditions of the Oral cavity

1.Assessment

- **History:** Normal brushing
 - Awareness of any lesion in the mouth, lip, tongue, or throat
 - Recent history of sore throat
 - Discomfort caused by certain foods
 - Daily food intake
 - Use of alcohol & tobacco

P/E

- Inspect and palpate internal & external structure of the mouth & throat.
- **Lip:** Inspect for moisture, colour, symmetry, fissures, & ulceration.
- **Gums:** Inspect for inflammation, bleeding, discolouration & note odor of breath
- **Tongue:** Inspect for colour & lesion

2. Nursing diagnosis

- Altered oral mucous membrane related to infection or disease
- Altered nutrition, less than body requirement
- Pain related to oral lesion
- Body image disturbance related to a physical change in appearance resulting from disease
- Anxiety
- Impaired verbal communication related to Rx
- Risk for infection
- Knowledge deficit

3. Plan

- Improvement in the condition of oral mucous membrane
- Improvement in nutritional intake
- Attainment of comfort
- To attain self image
- Absence of infection
- Understanding of disease process
- Alternative communication

4. Nursing Intervention

- Promoting mouth care
- Advice to avoid smoking & alcohol
- Nutritional intake advice
- Give analgesics
- Provide paper & pen for communication by writing
- Education the patient

5. Expected outcomes

- Shows evidence of intact oral mucous membrane
- Attains & maintains desirable body weight
- Free of anxiety
- Free of infection

Trauma of the Mouth & Jaw

Fracture of the Jaw

- Fracture of the jaw area usually include injury to the soft tissues

Causes

1. Trauma:

- car accident
- Fall
- hit by a fist or a flying object

2. Pathological:

- osteomyelitis
- Large dental cyst or neoplasm

C/M

- Malocclusion, asymmetry, abnormal mobility & crepitus (grating sound with movement) and pain.

Immediate assessment; and Management

A. Determine obstruction to air way.

- Remove any obstruction from pharynx, such as broken teeth, dentures, blood clots or broken bones
- Prepare for emergency tracheostomy

B. Control hemorrhage by direct pressure on vessels supplying the area

- C. Administer analgesics to relieve pain & anxiety but not to depress respiration
- D. Reassure the patient
- E. Position the patient in prone position
- F. Prepare the patient for reduction & fixation

Oesophageal Disease

Achalasia

- It is **absent or ineffective peristalsis distal esophagus** accompanied by **failure of the esophageal sphincter to relax** in response to swallowing.
- Narrowing of the esophagus just above the stomach results in a gradually increasing dilation of esophagus in the upper chest.
- It may progress slowly.
- Affect mostly persons over 40 years of age.

C/M:

- Difficulty in swallowing both liquids & solids
- Patient has a sensation of food “sticking” in lower portion of the esophagus.
- Regurgitation
- Chest pain & heart burn

Diagnostic Evaluation

- X-ray: shows esophageal dilation above the narrowing at the gastroesophageal junction.
- Barium swallow
- Endoscopy

Medical Mx

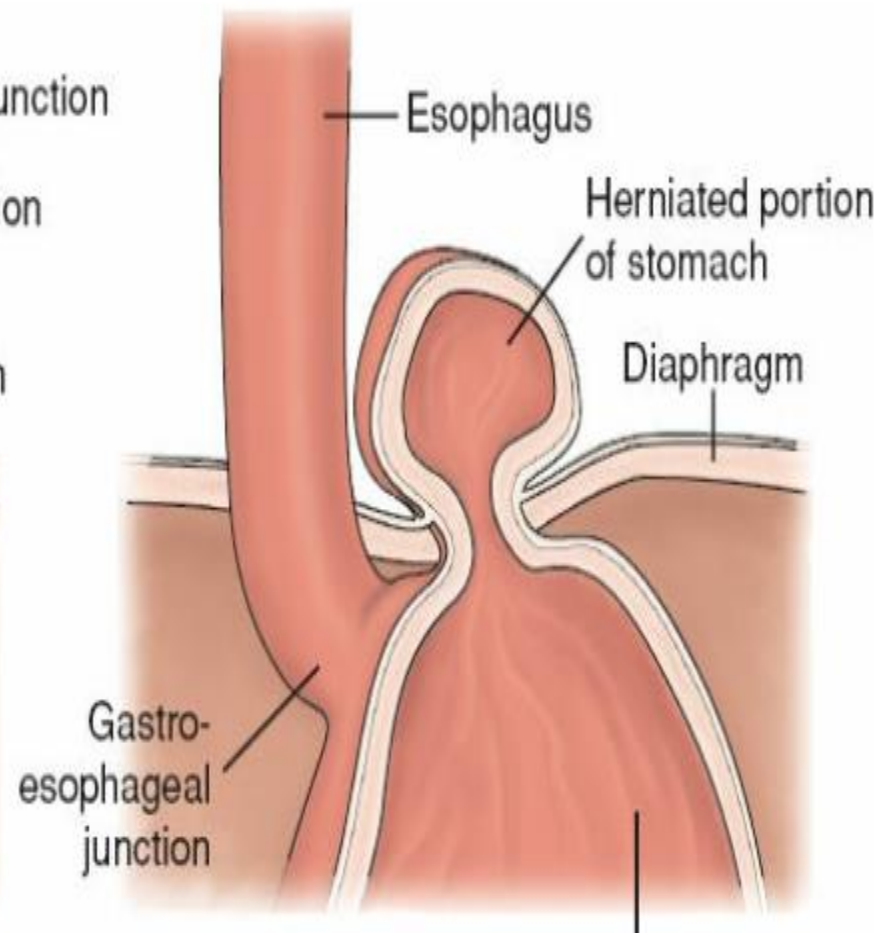
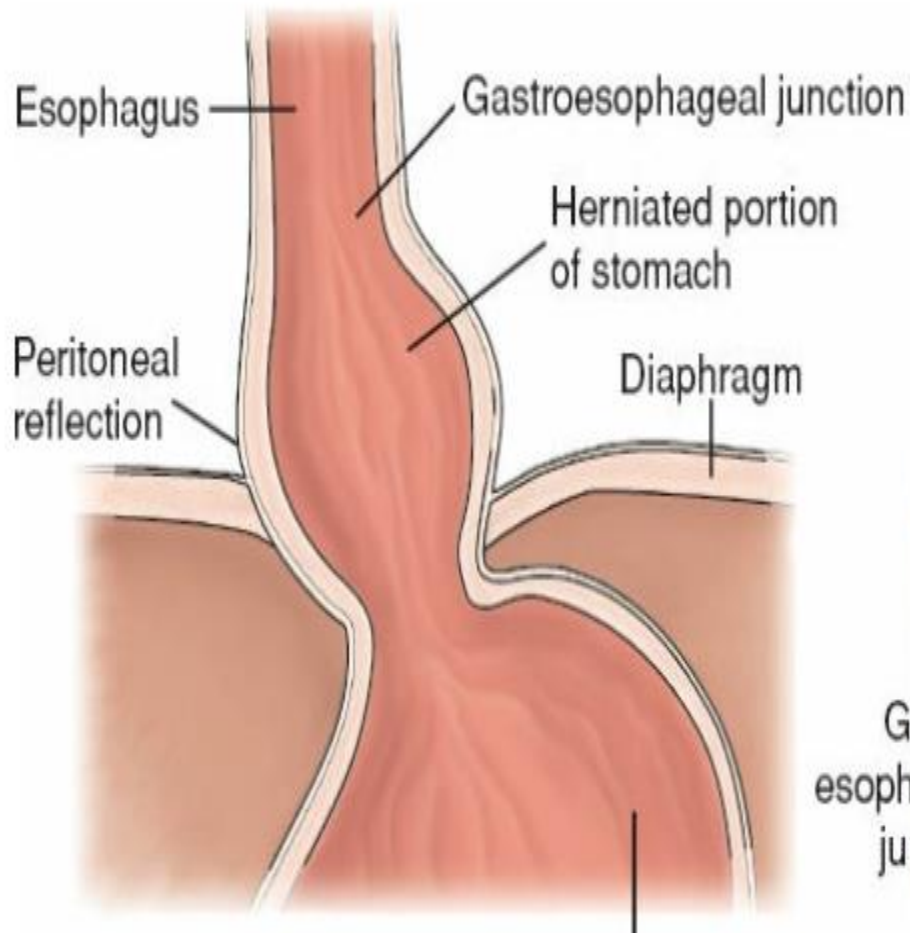
- **Instruct the pt to eat slowly** & drink fluids with meals.
- **Injection of botulinum toxin** (Botox) into quadrants of the esophagus via endoscopy has been helpful because it inhibits the contraction of smooth muscle
- Surgically by esophagomyotomy

Hiatus Hernia

- The oesophagus enters the abdomen through an opening in the diaphragm, & empties at its lower end in to the upper part of the stomach.
- In hiatus hernia, the opening in the diaphragm through which the oesophagus passes becomes enlarged & parts of the upper stomach tends to move up in to lower portion of the thorax
- There are two main types of hiatal hernias: **sliding** and **paraesophageal**.
- It occurs more often in women than men.

Sliding

Paraesophageal



C/M

- 50 % of patients are a symptomatic
- Heartburn
- Regurgitation
- Fullness after eating.
- Dysphagia
- intolerance to food, nausea, and vomiting.
- Sliding hiatal hernias are commonly associated with GERD.
- Hemorrhage, obstruction, and strangulation can occur with any type of hernia

DX:

- Esophagogastroduodenoscopy (EGD)

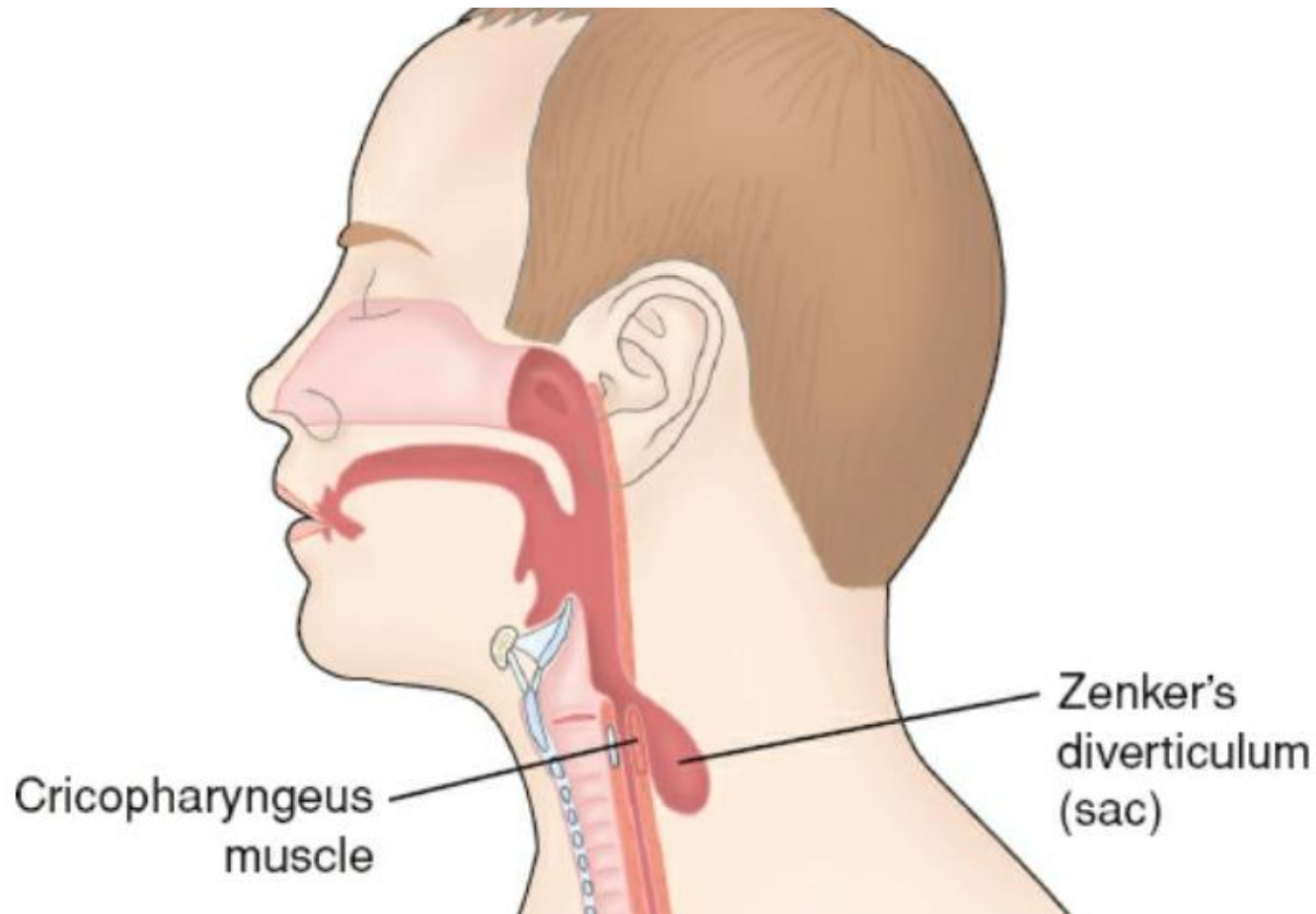
Medical Management

- Frequent, small feedings that can pass easily through the oesophagus.
- Advised not to recline for 1 hour after eating, to prevent reflux or movement of the hernia
- Elevate head of the bed 10- to 20-cm
- Surgery

Oesophageal Diverticula

- It is an **out pouching of mucosa & submucosa** that protrudes through a weak portion of the musculature.
- Diverticula may occur in one of three areas of the oesophagus:
 1. Pharyngoesophageal or upper part of oesophagus
 2. Mid-oesophageal
 3. Lower part of the oesophagus
- Leads to **increased pressure that forces** the mucosa and submucosa to herniate through the esophageal musculature
- Related to the **improper functioning of the lower esophageal sphincter** or to motor disorders of the esophagus.

Oesophageal Diverticula



C/M

- Difficulty in swallowing
- Fullness in the neck & then complain of belching
- Regurgitation of undigested food
- Pouch, becomes filled with food & fluid
- **Sour taste in the mouth**
- Chest pain

Diagnostic Evaluation:

- Barium swallow
- Esophagoscopy is contraindicated because of danger of perforation
- Surgical removal of diverticula

Management:

- Surgical removal (esophagectomy)
- Radiation
- Chemotherapy

Nursing intervention

- Post operatively feed the pt through NG tube
- Food and fluids are withheld until x-ray studies show no leakage at the surgical site.
- The diet begins with liquids and is progressed as tolerated.

Gastroesophageal Reflux Disease

- Common disorder marked by **backflow of gastric or duodenal contents into the esophagus** that causes troublesome symptoms and/or mucosal injury to the esophagus.
- Excessive reflux may occur because of an **incompetent lower esophageal sphincter, pyloric stenosis, hiatal hernia**, or a motility disorder.

Clinical Manifestations

- Symptoms may include pyrosis (heartburn, specifically more commonly described as a burning sensation in the esophagus),
- Dyspepsia (indigestion),
- regurgitation,
- dysphagia or odynophagia,
- hypersalivation, and esophagitis.
- GERD can result in dental erosion, ulcerations in the pharynx and esophagus, laryngeal damage, esophageal strictures, adenocarcinoma, and pulmonary complications

Mgt

- The patient is instructed to **eat a low-fat diet**;
- **avoid caffeine, tobacco, beer, milk**,
- Avoid foods containing peppermint or spearmint, and carbonated beverages;
- avoid eating or drinking 2 hours before bedtime;
- maintain normal body weight;
- **avoid tight-fitting clothes**; and
- elevate the head of the bed by at least 30 degrees



TABLE 45-2

Pharmacologic Management of GERD

Key Examples	Actions/Class	Key Nursing Considerations
Antacids/Acid neutralizing agents • Calcium carbonate (Tums) • Aluminum hydroxide, magnesium, hydroxide, and simethicone (Maalox)	Neutralize acid <i>Therapeutic and Pharmacologic class—Antacid</i>	• Potential risk of gastric acid suppression is the loss of protective flora and an increased risk of infection, especially <i>Clostridium difficile</i>
Histamine-2 (H₂) receptor antagonists • Famotidine (Pepcid) • Ranitidine (Zantac) • Cimetidine (Tagamet)	Decrease gastric acid production <i>Therapeutic class—Antiulcer drugs</i> <i>Pharmacologic class—H₂-receptor antagonists</i>	• Potential risk of gastric acid suppression is the loss of protective flora and an increased risk of infection, especially <i>Clostridium difficile</i> • For direct injection (IVP), dilute 2 mL (20 mg) with compatible solution to a total volume of either 5 or 10 mL; administer over at least 2 minutes • Monitor for QT-interval prolongation in patients with kidney injury
Prokinetic agents Metoclopramide (Reglan)	Accelerate gastric emptying <i>Therapeutic class—GI stimulants</i> <i>Pharmacologic class—Dopamine antagonist</i>	• May cause tardive dyskinesia • Typically used short term

Proton pump inhibitors (PPIs)

- Pantoprazole (Protonix)
- Omeprazole (Prilosec)
- Esomeprazole (Nexium)
- Lansoprazole (Prevacid)
- Rabeprazole (AcipHex)
- Dexlansoprazole (Dexilant)

Decrease gastric acid production

Therapeutic class—Antiulcer drugs

Pharmacologic class—Proton pump inhibitors

- Potential risk of gastric acid suppression is the loss of protective flora and an increased risk of infection, especially *Clostridium difficile*
- For a 2-minute infusion (IVP), give the reconstituted vials (4 mg/mL) over at least 2 minutes
- May increase the risk of hip fractures and interfere with some vitamin and mineral absorption (B12, iron, magnesium)
- Interact with commonly prescribed medications such as diuretics and clopidogrel

Reflux inhibitors

Bethanechol chloride (Urecholine)

Stimulates parasympathetic

Therapeutic and Pharmacologic class—
Cholinergic

- Primary use is for urinary retention
- Do not use with possible GI obstruction or peptic ulcer

Surface Agents/Alginate-based barriers

- Sucralfate

Preserve mucosal barrier

Therapeutic class—Antiulcer drugs

Pharmacologic class—GI protectants

- Give on an empty stomach hour before or 2 hours after meals
- Separate from doses of antacid by 30 minutes

Nursing Process

Patients with conditions of the oesophagus

1. Assessment:

- Complete history, P/E

2. Nursing diagnosis may include:

- Altered nutrition, less than body requirements, related to difficulty swallowing
- Pain, related to a tumor,
- Knowledge deficit about the disease process & Rx

3. Nursing plan

- To attain adequate nutritional intake
- To relief pain
- To increase knowledge level

4. Nursing Intervention

- Encourage the pt to eat slowly & to chew all food thoroughly
- Advice to eat small, frequent food
- Avoid very hot & cold beverages & spicy foods.
- Elevate the head of the bed
- Analgesics
- Educate the patient

5. Expected outcomes

- Achieve an adequate nutritional diet
- Free of pain
- Increased knowledge

Gastric disorders

Gastritis

- is any inflammatory process of **the mucosal lining of the stomach.**
- Gastric structure and function are altered in either the epithelial or the glandular components of the gastric mucosa.
- Gastritis is classified into acute and chronic forms.

Acute gastritis

- It is an inflammation of the stomach mucosa.
- **Lasting several hours to a few days**
- It is most often due to dietary indiscretion.
- Classified as **erosive** or **nonerosive**

Cause

- Alcohol abuse and ingestion of aspirin or other NSAIDs are the most common causes.
 - Other causes are steroid or digitalis medications; ingestion of corrosive agents such as lye or drain cleaners;
 - Ingestion of excessive amounts of tea, coffee, mustard, cloves, paprika, or pepper;
 - Chemotherapy or radiation to the upper abdomen; severe stress related to critical illness;
 - Staphylococcal food poisoning; and infections (candida, cytomegalovirus, herpesvirus) in immunosuppressed patients.
- ➔ **These conditions and substances disrupt the gastric mucosal barrier and make it vulnerable to normal gastric secretions.**

C/M

- a rapid onset of symptoms which can last from a few hours to a few days.
- Epigastric pain or discomfort,
- **dyspepsia** (indigestion),
- anorexia,
- hiccups, or
- nausea and vomiting,
- Erosive gastritis may cause bleeding, which may manifest as blood in vomit or as **melen**a (black, tarry stools) or **hematochezia** (bright red, bloody stools)

Chronic Gastritis

- Prolonged inflammation of the stomach may be caused by either benign or malignant ulcers of the stomach, or by *Helicobacter pylori* (*H. pylori*)

Cause:

- Autoimmune (parietal cell changes, leading to atrophy & cellular infiltration)
- *H. pylori*

Risk factors:

- Hot drinks & spices
- Use of drugs & alcohol, smoking

C/M

- Fatigue,
- **Pyrosis** (a burning sensation in the stomach and esophagus that moves up to the mouth; heartburn) after eating,
- Belching,
- A sour taste in the mouth,
- Early satiety,
- Anorexia, or nausea and vomiting.
- Intolerance to spicy or fatty foods
- **Pernicious anemia or hemorrhagic anemia**

Diagnostic Evaluation

- Endoscopy
- Upper GI x – ray series
- Histologic examination
- Serologic testing for antibodies for the H. pylori antigens.

Management

1. For Acute Gastritis:

- Instruct the pt to refrain from alcohol & food until symptoms subside.
- Recommend non irritant diet
- If symptom persist administer fluids parenterally
- Aluminium hydroxide 2 tsp tid
- Nasogastric intubation
- sedatives

2. For chronic Gastritis

- Modify the pt's diet
- Promote rest
- Reduce stress
- Treat H.pylori with triple or quadruple therapy
 - Clarythromycine 500mg PO BID for 14 days
 - Amoxicilline 1gm PO BID for 14 days
 - Omeprazole 20mg PO BID for 14 days

Nursing Process

The patient with Gastritis

A. Assessment

- Complete history (s/s of gastritis and Aggravating factors)
- P/E (abdominal tenderness, dehydration, any systemic disorder)

B. Nursing diagnosis

- Anxiety
- Altered nutrition, less than body requirement
- Risk for fluid volume deficit
- Pain
- Knowledge deficit

C. Plan

- To reduce anxiety
- To maintain adequate nutritional intake
- To maintain fluid balance
- To increase awareness of dietary Mx.
- To relieve pain

D. Nursing Intervention

- Reassure the pt.
- Advice nutritional intake
- IV fluid administration
- Instruction to avoid coffee, smoking, alcohol, spicy diets
- Educate the patient.

E. Expected out comes

- Anxiety & pain alleviated
- Maintained fluid balance
- Maintained nutritional status

Peptic Ulcer Disease(PUD)

- It is an excavation (hollowed – out area) formed in the mucosal wall of the stomach, the pylorus, the duodenum, or the esophagus.
- It refers to ulcerative disorders in the **lower esophagus, upper duodenum, and lower portion of the stomach.**
- The ulcer may extend through the tissue layers of the **muscle and serosa into the abdominal cavity.**

Comparison of Duodenal & Gastric Ulcer

Duodenal Ulcer

- ✓ Age: Common b/n 30 - 60year
- ✓ Risk factors alcohol, smoking, B/d group O, stress
- ✓ Stomach acid: Hyper secretion
- ✓ Vomiting: Uncommon
- ✓ Weight: gain
- ✓ Pain: occurs 2- 3 hour after meal & relived by food ingestion
- ✓ Perforation: more common
- ✓ Malignancy Possibility rare

Gastric Ulcer

- ✓ Usually 50year & over
- ✓ Gastritis, alcohol, smoking, NSAID, blood group A
- ✓ Normal to: Hypo secretion
- ✓ Vomiting common
- ✓ Loss may occur
- ✓ Pain: ½ to 1hour after meal& relived by vomiting
- ✓ Perforation: Less common
- ✓ Malignancy Occasionally

Etiology

- Etiology of PUD is poorly understood, but it is associated with gram negative **bacteria H.Pylori**

Predisposition

- Stress or anger
- Familial tendency
- Use of NSAIDS,
- Alcohol ingestion &
- Excessive smoking.
- Bacterial infection (H. Pylori)
- **Zollinger–Ellison syndrome (ZES).**

Pathos-physiology

- Peptic ulceration depends on the **defensive resistance** of the mucosa relative to the **aggressive force** of secretory activity.
- The defensive resistance of the mucosa depends on the following factors:
 1. Mucosal integrity & regeneration
 2. Presence of a protective mucus barrier
 3. Adequate blood flow to the mucosa
 4. Ability of the duodenal inhibitory mechanism to regulate secretion
- The aggressive factors relate to the presence of *Helicobacter pylori* & **Volume of HCl** & **biliary acid**.
- Ulceration occurs when aggressive factors exceeds the defensive one.

C/M

- Pain (burning Sensation cramp like, gnawing pain) in the mid-epigastrium or in the back
- Pain left of the midline with gastric ulcers. the right of the midline epigastrium with duodenal ulcers
- Vomiting & Nausea
- Constipation & bleeding
- Abdominal distention,
- Abdominal tenderness
- Bleeding
- Melena (black, tarry stools)
- Hypotension and tachycardia

Diagnostic Evaluation

- History, P/E
- Endoscopy
- X-ray
- **Stool exam for occult blood**
- Gastric Secretary studies (achlorhydria)
- Biopsy & culture for H.pylori
- **Urea breath test**
- CBCs

Management: (Non drug Mx)

- Stress reduction & rest
- Smoking Cessation
- Dietary Modification (advice the Pt. what ever agrees with them)
- Avoid spicy foods
- Avoid highly acidic foods and fruits
- Avoid fatty foods such as butter

Medical Mx of PUD

- Medications are prescribed for clients with PUD for 4 major reasons
 - ✓ To eliminate H. Pylori bacteria from the GIT
 - ✓ To reduce secretion (hypo-secretory drugs),
 - ✓ To neutralize acid (antacids)
 - ✓ To protect the mucosal barrier.

Medication or Drug Class	Dosage	Description	Rationale
H ₂ antagonist	Varies with drug	Cimetidine, rantidine, famotidine, nizatidine	Reduces acid secretion to optimize ulcer healing
Proton pump inhibitors	Varies with drug	Omeprazole, esomeprazole, lansoprazole, rabeprazole, pantoprazole	Optimizes ulcer healing by binding to the proton pump of parietal cell and inhibiting secretion of hydrogen ions into gastric lumen
Triple therapies: Antibiotics (for patients with <i>H. pylori</i> bacteria) and proton pump inhibitor	Omeprazole (Prilosec), 20 mg PO bid for 14 days with clarithromycin (Biaxin) 500 mg PO bid for 14 days and amoxicillin (Amoxil) 1 g PO bid for 14 days and metronidazole or omeprazole, tetracycline, and metronidazole	Other protein pump inhibitors may be used: Lansoprazole (Prevacid); rabeprazole (Aciphex); esomeprazole (Nexium)	Antibiotics eradicate <i>H. pylori</i> and protein pump inhibitor reduces acid secretion; triple therapy may also consist of antibiotics and bismuth-based therapy such as Pepto-Bismol

Surgical Treatment for PUD

Indication of Surgery

- Intractable ulcer (those who fail to heal after 12 to 16wks of medical Rx.)
- Life Threatening Hemorrhage
- Perforation
- Obstruction of gastric out let.
- ZES

Nursing intervention

- **Preoperative nursing care for the patient undergoing surgery for peptic ulcer disease includes:**
 1. Preparing the pt for diagnostic tests (Laboratory analysis, x-ray, & general physical examination).
 2. Attending to the pt's fluid & nutritional level & to maintain an optimal fluid & electrolyte balance.
 3. Clearing & emptying GIT: Naso - gastric Suction often is required to empty the stomach in Pt. with Pyloric obstruction
 4. Limiting oral Intake

Operation

Description

Adverse Effects

Vagotomy



Severing of the vagus nerve. Decreases gastric acid by diminishing cholinergic stimulation to the parietal cells, making them less responsive to gastrin. May be performed via an open surgical approach or laparoscopy. May be performed to reduce gastric acid secretion. A drainage type of procedure (see pyloroplasty) is usually performed to assist with gastric emptying (because there is total denervation of the stomach).

Some patients experience problems with feeling of fullness, dumping syndrome, diarrhea, and gastritis.

Truncal vagotomy

Severs the right and left vagus nerves as they enter the stomach at the distal part of the esophagus; most commonly used to decrease acid secretions.

Some patients experience problems with feeling of fullness, dumping syndrome, diarrhea, or constipation.

Selective vagotomy

Severs vagal innervation to the stomach but maintains innervation to the rest of the abdominal organs.

Fewer associated adverse effects than with truncal vagotomy.

Proximal (parietal cell) gastric vagotomy without drainage

Denervates acid-secreting parietal cells but preserves vagal innervation to the gastric antrum and pylorus.

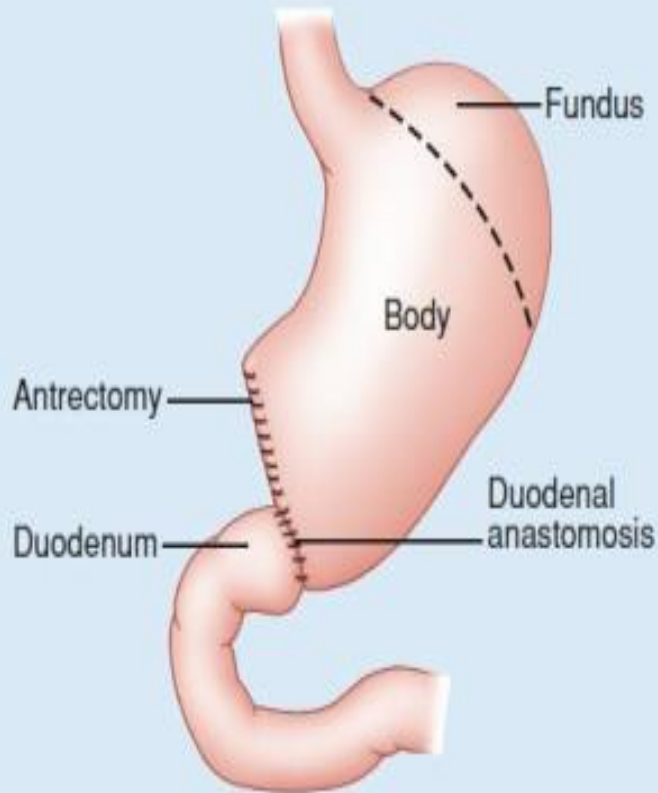
No associated dumping syndrome

Pyloroplasty

Longitudinal incision is made into the

See adverse effects associated with truncal

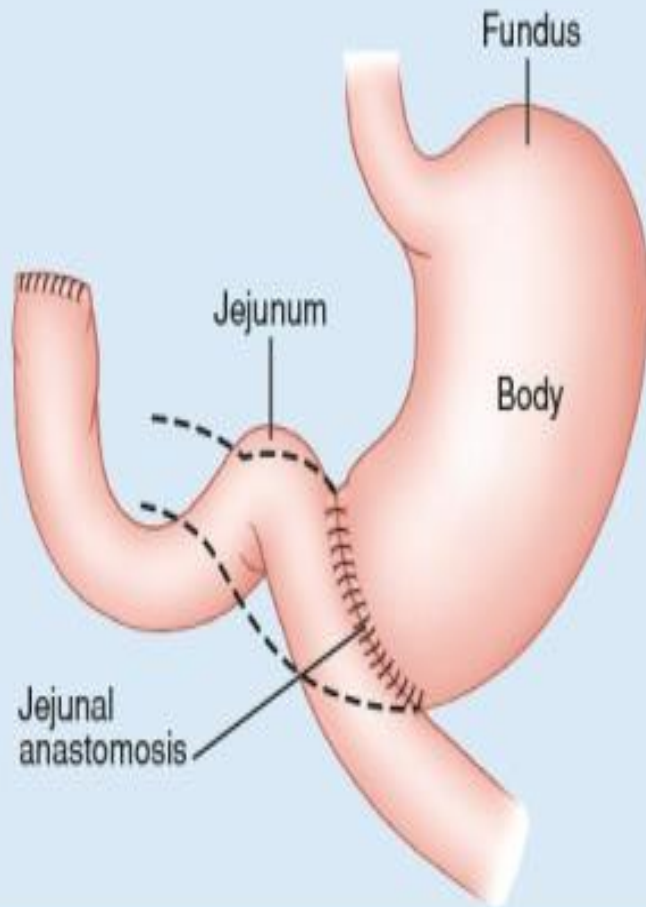
Antrectomy
Billroth I (gastroduodenostomy)



Removal of the lower portion of the antrum of the stomach (which contains the cells that secrete gastrin) as well as a small portion of the duodenum and pylorus. The remaining segment is anastomosed to the duodenum. May be performed in conjunction with a truncal vagotomy.

Patients may have problems with feeling of fullness, dumping syndrome, and diarrhea.

Billroth II (gastrojejunostomy)



Removal of lower portion (antrum) of stomach with anastomosis to jejunum. Dotted lines show portion removed (antrectomy). A duodenal stump remains and is oversewn.

Patients frequently have associated dumping syndrome, anemia, weight loss, and malabsorption.

Nursing Process

- The Patient with PUD

1. Assessment
 - History
 - P/E

2. Nursing Diagnosis

- Pain related to the effect of gastric acid secretion.
- Anxiety related to an able to coping with an acute disease
- Knowledge deficit about prevention & treatment.
- Altered nutrition, less than body requirement, related to pain associated with eating.

3. Plan

- To relief pain.
- " " anxiety
- " Increase pts awareness
- " maintain adequate nutrition

4. Nursing Intervention

- Administer Medication
- Advise the pt to avoid foods that are irritant: Alcohol, caffeine & Cigarettes
- Reassure the pt.
- Teach necessary Information

Complication

1. Perforation (DU > GU)

C/F: abrupt severe pain, tenderness, Rebound tenderness, guarding

2. Gastric outlet obstruction (Pyloric Stenosis)

C/M

- Delayed gastric emptying
- Fullness
- Vomiting
- Weight loss
- Dehydration

3. Hemorrhage

- PUD is the **most common** cause of UGI bleeding
- Occurs in 20% of PUD
- NSAID drugs intake increase risk of bleeding

Cancer of the stomach

- It is most common in people over the age of 70years
- It refers to the malignant neoplasm found in the stomach.

Risk Factors

- Chronic inflammation of the stomach
- Pernicious anemia
- Gastric ulcer
- Heredity
- H.pylori bacteria
- Smoking

Etiology

- unknown

C/M : are vague & indefinite

- Pain
- Indigestion
- Anorexia, nausea & vomiting.
- Dyspepsia
- Weight loss
- Constipation
- The commonest site is pyloric & antral region.

Diagnostic Evaluation

- Endoscopy for biopsy & cytology
- X-ray

Mx

- Surgical removal of the cancer, chemotherapy, Radiation.

Preoperative Nursing Diagnosis

- Knowledge deficit regarding the surgical procedure & postoperative Course

Plan

- To Provide information about the procedure & Postoperative course

Nursing Intervention

- Teach the type of surgery that is planned
- Advise the patient that requests will be made to breathe deeply & Cough, post Operatively.

- Advise the patient that a Nasogastric tube will be placed postoperatively & fluid will be withheld until peristalsis returns.
- Inform the patient that parenteral fluids will be given.
- Inform the patient that wound dressings may have drainage.
- Informed Consent.

Post operative

A. Nursing Diagnosis

- Pain related to the surgical Incision
- Fluid Volume deficit, Related to shock or hemorrhage
- Risk for infection related to surgical incision.

B. Plan

- To relief pain
- To maintain normal fluid volume
- To maintain adequate nutrition
- To prevent infection

C. Nursing Interventions

- Administer analgesics as prescribed
- Withhold oral fluid until prescribed
- Monitor for signs of hemorrhage
 - ❖ Observe Suture line for bleeding
 - ❖ Evaluate B/P, pulse, & RR
 - ❖ Observe gastric aspirate for evidence of bleeding.
- Assess patient for signs of shock
 - ❖ Take B/P, PR, RR frequently
 - ❖ Administer Fluids as prescribed.
- Administer oral & IV fluids as ordered.
- Increase fluids according to pt's tolerance.
- Assess wound for sign & symptom of infection
- Assess abdomen for signs of peritonitis
- Administer Prophylactic antibiotic.

D. Nursing Evaluation

- Pain is relieved
- Remain NPO until physician allow intake of fluids.
- No hemorrhage
- Vital signs within normal ranges
- Bowel sounds audible
- Attains adequate caloric intake
- Absence of signs & symptoms of infection

Acute appendicitis

- It is an inflammation of the vermiform appendix - a narrow, blind tube that extends from the inferior part of the cecum.
- Etiology - not known
- Risk factors: Sex (M>F)
- Age (most common b/n 10 & 30 Years)
- Economic status (high & middle social class)

Pathophysiology:

- Appendix becomes inflamed & edematous as a result of becoming either kicked or occluded, possibly by a fecalith, tumor, or foreign body. This cause ischemia & necrosis.

C/M

- **Right** lower quadrant pain
- a discomfort that creates an **urge to defecate** to obtain relief
- Low grade fever
- Loss of appetite
- Local tenderness at **McBurney's point** when pressure is applied
- Rebound tenderness (production of pain when pressure is released)
- Constipation or diarrhea
- Nausea



- Rebound tenderness, Pain on percussion, and Abdominal guarding/rigidity are specific findings in appendicitis.
- facial grimacing,
- clenched fists,
- diaphoresis,
- tachycardia, and
- shallow but rapid respirations.
- Guarding the abdominal area by lying with the right leg flexed at the knee

- **Check for the Rovsing sign** and for referred rebound tenderness. Press deeply and evenly in the left lower quadrant. Then quickly withdraw your fingers.
 - Pain in the right lower quadrant during left-sided pressure suggests appendicitis (a positive Rovsing sign). So does right lower quadrant pain on quick withdrawal (referred rebound tenderness).
- **Look for a psoas sign.** Place your hand just above the patient's right knee and ask the patient to raise that thigh against your hand. Alternatively, ask the patient to turn onto the left side. Then extend the patient's right leg at the hip. Flexion of the leg at the hip makes the psoas muscle contract; extension stretches it
 - Increased abdominal pain on either maneuver constitutes a positive psoas sign, suggesting irritation of the psoas muscle by an inflamed appendix.

• • •

- **Look for an *obturator sign*.** Flex the patient's right thigh at the hip, with the knee bent, and rotate the leg internally at the hip. This maneuver stretches the internal obturator muscle.
 - Right hypogastric pain constitutes a *positive obturator sign*, suggesting irritation of the obturator muscle by an inflamed appendix.

Diagnostic Evaluation

- Complete Hx & P/E
- X-ray may reveal a right lower quadrant density or localized air - flow levels.
- CBC > Elevated WBC > $10,500/\text{mm}^3$ & neutrophil count > 75%
- Ultrasound

DDx

- Ectopic pregnancy in females of childbearing age,
- Endometriosis,
- Pelvic inflammatory disease,
- Renal calculi,
- Urinary track infection,
- Abdominal abscess,
- Cholecystitis,
- Crohn disease,
- Gastroenteritis, and
- Inflammatory bowel disease.

Management

- Secure IV fluid & administer antibiotics
- Analgesics can be given after diagnosis is made
- Appendectomy (surgical removal of appendix)

Complications

- Ruptured appendix,
- Peritonitis,
- Abscess formation,
- Sepsis,
- Septic shock, and
- Paralytic ileus.

Nursing Intervention

- Reliving pain by administering analgesics
- Preventing fluid volume deficit by giving IV fluids
- Reassure the pt & prevent infection
- Preoperative & postoperative nursing care is the same as that of pt under going major surgery.

Peritonitis

- Is **an inflammation of the peritoneum** (the thin membrane that lines the abdominal wall & covers most of the organs of the body.)
- There are two major types of peritonitis.
 1. **Primary peritonitis**: caused by the spread of infection /bacteria **from the blood & lymph nodes** to the peritoneum. This type peritonitis is rare less than 1% of all cases.
 2. **Secondary peritonitis** the more common type of peritonitis caused **during perforation or rupture** of abdominal organs allowing access to entry of bacteria & irritant digestive juices to the peritoneum.

Etiology:

- The most common cause is infection with *Escherichia coli*, but streptococci, staphylococci, and pneumococci may also cause the inflammation.
- Chemical contamination of peritoneal cavity.
- Risk factors for peritonitis include immune suppression, appendicitis, stomach ulcers, pancreatitis, alcohol abuse, liver cirrhosis, renal failure, inflammatory bowel disease, trauma, peritoneal dialysis, and surgical procedures.
- The major source of inflammations are
 - ❖ From the diseases of GIT
 - ❖ From the external environment (gunshot wound, stab wound)
 - ❖ Through the blood stream septicemia

C/M

- Sharp abdominal pain w/c worsen with movement
- abdominal distention
- diarrhea
- Fever & increased PR
- Abdomen becomes rigid(muscle guarding)
- rebound tenderness with guarding.
- Nausea & vomiting
- Decreased or absent bowel sound
- Peritoneal pain is almost always increased by pressure or tension of the peritoneum, such as coughing, sneezing and palpation.
- The patient is generally in a knee-flexed position with shallow respirations in an attempt to minimize pain.

- **Tenderness**
 - Localized peritonitis: peritonitis confined to a limited space
e.g pelvis
 - Generalized peritonitis: the whole peritoneal cavity involved
- **Respiration may be shallow & rapid**
- Low B/P & sign of dehydration
- dry and swollen tongue, dry mucous membranes, and thirst.

Diagnosis

- Hx
- P/E
- WBC count: elevated
- Laboratory study - elevated WBC & to identify the causative agent
- Abdominal X-ray : show dilation & edema of the intestine or any perforated organ; a free air or fluid in the abdominal cavity
- Serum electrolyte studies - loss of electrolyte

Medical mgt

- Fluid and electrolyte replacement
- IV antibiotics therapy(triple antibiotics) gentamycin, clindamycin/metronidazole & ampicillin
- Analgesics for pain
- NGT: to reduce abdominal distention
- Oxygen therapy by nasal cannula or mask

Surgical mgt

- Drainage of the abscess /excess fluid
- Repair perforated/damaged tissue/organ

Nursing mgt

- Regularly monitor vital signs, fluid intake & output
- Place the pt in fowlers position to promote drainage
- Encourage and assist ambulation
- Watch closely for possible surgical complications
- Close follow up

Complication

- Generalized sepsis
- Shock- may result from septicemia or hypovolemia
- Inflammatory process may cause intestinal obstruction

Disorders of the Lower GIT

Abnormalities of Fecal Elimination

Constipation

- **Fewer than three bowel movements weekly** or
- bowel movements that are **hard, dry, small, or difficult to pass.**
- a symptom and not a disease

Causes:

- Weakness, immobility, debility, fatigue, and an inability to increase intra-abdominal pressure to facilitate the passage of stools
- Many people develop constipation because they do not take the time to defecate or ignore the urge to defecate.
- **low consumption of fiber and inadequate fluid intake), lack of regular exercise, and a stress-filled life**
- Drugs: anticholinergic agents, antidepressants, anticonvulsants, antispasmodics(muscle relaxants), calcium channel antagonists, diuretic agents, opioids, aluminum- and calcium-based antacids, and iron preparations.

...

- The pathophysiology of constipation is poorly understood, but it is thought to include interference with one of three major functions of the colon:
 - **mucosal transport** (i.e., mucosal secretions facilitate the movement of colon contents),
 - **myoelectric activity** (i.e., mixing of the rectal mass and propulsive actions), or
 - **the processes of defecation** (e.g., pelvic floor dysfunction).

Clinical Manifestations

- fewer than three bowel movements per week;
- abdominal distention;
- pain and bloating;
- a sensation of incomplete evacuation;
- 10/10/2021 straining at stool; and

• • •

- Treatment targets the underlying cause of constipation and prevention of recurrence.
- It includes **education, exercise, bowel habit training, increased fiber and fluid intake**
- **bulk-forming agents** (fiber laxatives), saline and osmotic agents, lubricants, stimulants, or fecal softeners.
- Enemas and rectal suppositories are **generally not recommended** for treating constipation unless other medications have failed.

Diarrhea

- an **increased frequency of bowel movements** (more than 3 per day) with **altered consistency** (i.e., increased liquidity) of stool.
- associated with urgency, perianal discomfort, incontinence, nausea, or a combination of these factors
 - ✓ **Increased intestinal secretions,**
 - ✓ **decreased mucosal absorption,** or
 - ✓ **altered motility**

➔ Mechanisms

- **Acute diarrhea** is self-limiting, lasting 1 or 2 days;
- **Persistent diarrhea** typically lasts between 2 and 4 weeks; and
- **Chronic diarrhea** persists for more than 4 weeks

Causes

- chemotherapy,
- antiarrhythmic agents,
- antihypertensive agents,
- metabolic and endocrine disorders (e.g., diabetes, Addison disease, thyrotoxicosis), malabsorptive disorders (e.g., lactose intolerance, celiac disease), anal
- sphincter defect,
- Zollinger–Ellison syndrome,
- acquired immune deficiency syndrome (AIDS), and by
- Parasitic or *Clostridium difficile* infections

CM

- Increased frequency and fluid content of stools,
- The patient usually has abdominal cramps, distention,
- Borborygmus (i.e., a rumbling noise caused by the movement of gas through the intestines),
- Anorexia, and thirst.
- Painful spasmodic contractions of the anus and **tenesmus** (i.e., ineffective, sometimes painful straining with a strong urge)
- **Oil droplets** on the toilet water may be suggestive of pancreatic insufficiency.
- **Nocturnal diarrhea** may be a manifestation of diabetic neuropathy

Mgt

- antibiotics,
- antiinflammatory agents and
- antidiarrheal agents (e.g., loperamide
- diphenoxylate with atropine may be prescribed to reduce the severity of the diarrhea and
- treat the underlying disease
- Intravenous (IV) fluid therapy

Complications

- The most common complication of diarrhea is **dehydration**.
- **Dehydration with electrolyte loss** (especially loss of potassium) may cause cardiac dysrhythmias.
- Loss of bicarbonate with diarrhea can also lead to **metabolic acidosis**.

Fecal Incontinence

- The **recurrent involuntary passage of stool** from the rectum **for at least 3 months**.
- Factors that influence this disorder include the ability of the rectum to sense and accommodate stool, the amount and consistency of stool, the **integrity of the anal sphincters** and **musculature**, and **rectal motility**.
- Fecal incontinence can have a substantially negative impact on quality of life

Irritable Bowel Syndrome (IBS)

- **Irritable bowel syndrome (IBS)** is a chronic functional disorder characterized by **recurrent abdominal pain** associated with disordered bowel movements, which may include **diarrhea, constipation, or both**
- Predisposed by chronic stress, sleep deprivation, surgery, infections, diverticulitis, and some foods (e.g., milk, yeast products, eggs, wheat products, red meat)

Ulcerative Colitis

- It is a recurrent ulcerative & inflammatory disease of the **mucosal and submucosal layer of the colon & rectum.**
- It affects superficial mucosa of the colon & is characterized by **multiple ulcerations, diffuse inflammations desquamation or shedding of the colonic epithelium.**
- Characterized by **unpredictable** periods of remission and exacerbation with bouts of abdominal cramps and bloody or **purulent diarrhea.**
- The inflammatory changes typically begin in the rectum and progress proximally through the colon

Etiology: unknown

Predisposing factors

- Anxiety, tobacco, radiation

C/M:

- **Diarrhea** (10 to 20 liquid stools daily)
- **Left lower quadrant abdominal pain**
- Intermittent tenesmus
- Rectal bleeding
- Anorexia
- Weight loss
- Fever, Vomiting, dehydration
- **Rebound tenderness may occur in the right lower quadrant.**
- Hypoalbuminemia, electrolyte imbalances, and anemia

Diagnostic evaluation

- Colonoscopy: friable, inflamed mucosa with exudate and ulcerations.
- Careful stool exam: to r/o amoeba
- Low hgb & Hct level
- Elevated WBC
- Sigmoidoscopy & barium enema

Mx

- **Oral fluids** & a **low residual, high protein**, high-calorie diet with supplemental vitamin therapy & iron supplement is given.
- Avoid any food which exacerbate diarrhea
- Treat dehydration
- Avoid cold foods & smoking
- Antibiotics can be given for secondary bacterial infection.
- Partial or complete, with ileostomy or anastomosis.

Complication

- **Toxic megacolon**: the inflammatory process extends into the muscularis, inhibiting its ability to contract and resulting in colonic distention.
- **Perforation**,
- **Bleeding** as a result of ulceration,
- Vascular engorgement, and highly vascular granulation tissue.

Crohn's Disease (Regional Enteritis)

- Subacute and chronic inflammation of the GI tract
- **Affects all layers (i.e., transmural lesion)**
- Commonly occurs in the distal ileum and the ascending colon
- Fistulas, fissures, and abscesses form as the inflammation extends into the peritoneum.
- As the disease advances, the bowel **wall thickens** and **becomes fibrotic**, and the intestinal lumen narrows.

CM

- Prominent **right lower quadrant abdominal pain** and diarrhea unrelieved by defecation.
- Scar tissue and the formation of granulomas interfere
- Being thin and emaciated
- diarrhea,
- **steatorrhea** (i.e., excessive fat in the feces),
- anorexia, weight loss, and nutritional deficiencies.
- Abscesses, fistulas, and fissures occur in approximately one third of patients

Dx

- Low Hct, elevated WBC
- Elevated ESR
- Low albumin
- CT scan shows thickened wall
- Abscess, fissure

Intestinal obstruction

- Intestinal Obstruction exists when blockage prevents the normal flow of intestinal contents through the intestinal tract.
- It can be classified as the following
 - A) Mechanical obstruction Vs Functional Obstruction
 - B) Small bowel Obstruction Vs Large bowel obstruction
 - C) Partial Obstruction Vs Complete Obstruction

Causes of Intestinal Obstructions

1. Causes of Small bowel obstruction

- ❖ Adhesion of intestinal wall due to:
 - Surgery
 - Intestinal Tuberculosis
 - Inflammatory Condition of intestine.
- ❖ Paralytic ileus
- ❖ Hernia
- ❖ Gallstones ileus
- ❖ Tumor
- ❖ Ascaris bolus

Cause	Course of Events	Result
Adhesions	Loops of intestine become adherent to areas that heal slowly or scar after abdominal surgery; occurs most commonly in small intestine	After surgery, adhesions produce a kinking of an intestinal loop.
Intussusception	One part of the intestine slips into another part located below it (like a telescope shortening); occurs more commonly in infants than adults	The intestinal lumen becomes narrowed, and blood supply becomes strangulated.
Volvulus	Bowel twists and turns on itself and occludes the blood supply	Intestinal lumen becomes obstructed. Gas and fluid accumulate in the trapped bowel.
Hernia	Protrusion of intestine through a weakened area in the abdominal muscle wall	Intestinal flow may be completely obstructed. Blood flow to the area may be obstructed as well.
Tumor	A tumor that exists within the wall of the intestine extends into the intestinal lumen, or a tumor outside the intestine causes pressure on the wall of the intestine. Most common type is colorectal adenocarcinoma	Intestinal lumen becomes partially obstructed; if the tumor is not removed, complete obstruction results.

C/M:

- Sudden Colicky pain intermittent with 10 -20 minute Interval.
Wavelike and colicky due to persistent peristalsis both above and below the blockage.
- Early Vomiting
- Bile stained vomiting
- Unable to pass feces and flatus rather blood and mucus.
- Restless, dehydration & cry
- Early dehydration
- Distention is late
- Central distension
- Hypovolumic shock
- Metabolic acidosis

2. Cause of large bowel Obstruction

- ❖ Colorectal Cancer
- ❖ Adhesion
- ❖ Paralytic Ileus
- ❖ Inflammatory bowel disease
- ❖ **Volvulus**

Cardinal S/S of large bowel Obstruction

- Colicky lower abdominal pain
- Absolute Constipation (Flatus & Feces)
- Gross abdominal distention
- Nausea and Vomiting
- Abdominal x-ray reveals grossly distended

Comparison

Bases	SBO	LBO
Abdominal Cramp	Yes	Yes
Vomiting	Early	Late
Nature of vomiting	Ingested or Bile stained	Fecal matter
Distension	Central, moderate	Marked and peripheral
Dehydration	Early	Late
Constipation	Late	Early
10/10/2022	By Afework E.	511

Diagnostic evaluation of Intestinal Obstruction

1. Hx
2. P/E: pt is acutely sick looking
 - V/S:
 - ✓ B/P - decrease due to fluid loss & sepsis
 - ✓ PR – Tachycardia
 - ✓ T° - Increases if there is complication
 - HEENT: dry buccal mucosa
 - Abdomen:
 - ✓ Distended
 - ✓ Mild tenderness on palpation
 - ✓ visible loop but not always
 - ✓ Tympanic on percussion
 - ✓ Bowel sound may be absent or increase
 - ✓ **PR** - empty rectum or hard stool
3. Laboratory tests: CBC, BG, electrolytes, abdominal ultra sound,

Medical Management

A. General Management

- Keep the patient NPO
- NG tube should be inserted for small bowel obstruction to aspirate intestinal content.
- Secure IV line (Normal Saline or ringer Lactate)
- Prophylactic antibiotics
- Catheterization
- Deflation for LBO if there is no strangulation
- Perioperative care

B. Surgical management: the definitive management

Hernias

- It is a protrusion of bowel through a weak point in the musculature of the anterior abdominal wall or an existing opening.

Etiology: - Powerful muscular effort or strain
 - Weakness or defect to the wall of abdominal cavity

Predisposing factors

- Constipation, ascites
- Previous abdominal surgery
- Lifting heavy load
- Chronic Cough

Herinia Based on Sites of Hernias

1. Inguinal Hernia

- The protrusion of bowel through the weak point is the inguinal canal which contains the spermatic cord in the male & the round ligament in the female.
- It occurs more commonly in males than females.

2. Femoral Hernia

- More Common in women than men.
- Bowel enters the femoral canal, presenting as a mass in the upper middle thigh or above the inguinal ligament where it points down the leg, unlike an inguinal hernia which points to the groin.
- It is frequently strangulate & irreducible

3. Para-umbilical Hernias

- These occur just above or below the umbilicus.

4. Incisional Hernias

- These follow breakdown of muscle closure after previous Surgery. If obese, repair is not easy.

5. Umbilical Hernia

- Results from failure of umbilical orifice to close.
- Occur most often in obese women & children & in patients with cirrhosis and ascites

Classification of hernia according to Severity

1. Reducible Hernia

- The protruding mass can be replaced in abdomen

2. Irreducible Hernia

- The protruding mass cannot be moved back into abdomen

3. Incarcerated

- An irreducible hernia in which the intestinal flow is completely obstructed.

4. Strangulated

- an irreducible hernia in which the blood & intestinal flow is completely obstructed.

C/F of Strangulation:

- Pain, vomiting
- Swelling of hernial sac,
- fever
- Lower abdominal sign of peritoneal irritation

Management

-repairing the weak abdomen part of abdominal wall.

Disorders of the rectum

Haemorrhoids

- Are dilated portions of veins in the anal canal.
- It is an enlarged & congested patch of mucosa & sub-mucosa at ano-rectal junction or
- Hemorrhoid based on its site:
 - 1. Internal haemorrhoid** (if it is above internal sphincter.)
 - 2. External haemorrhoid** (if it is outside external sphincter)

C/M

- **Bright red blood** occurring at the end of defecation (Late)
- Mass Per rectum
- Peri-anal Discomfort
- Pruritus
- Mucosal Discharge
- Pain when complicated
- ✓ **External hemorrhoids** are associated with **severe pain** due to inflammation & edema caused by thrombosis. Clotting of blood (thrombosis) lead to necrosis & ischemia.
- ✓ **Internal Haemorrhoids** are painless until they bleed.

Classification of hemorrhoids based on its stage(severity)

- 1st degree: Bleed but no prolapsed
- 2nd degree : Prolapsed but reduce spontaneously
- 3rd degree : " but need manual replacement
- 4th degree : " not returned.

Etiology: idiopathic

Predisposing factor

- Chronic Constipation
- Pelvic masses (Pregnancy)
- Portal HTN

Rx

- Regulating bowel by laxatives
- Avoid Constipation
- Advice high - residue diet that contain fruit.
- Sitz bath
- ❖ Good personal hygiene & **by avoiding excessive straining** during defecation hemorrhoid symptoms & discomfort can be relieved.
- ❖ Rx- Bismuth subgallate insert one suppository in the rectum bid for five days.

Conservative Surgical Rx of internal Haemorrhoid

A. Rubber - band ligation procedure:

- The haemorrhoid is visualized through the anoscope, & its proximal portion above the mucocutaneous lines is **grasped with an instrument.**
- A small rubber band is then slipped over the hemorrhoid. Tissue distal to the rubber band becomes necrotic after several days & sloughs off.
- It may cause *infection, pain & hemorrhage*

B. Cryosurgical Hemorrhoidectomy

- Involves freezing the tissue of the hemorrhoid for a sufficient time to cause necrosis.
- Not used widely because the discharge is **very foul-smelling & wound healing is prolonged.**

C. Hemorrhoidectomy, or surgical excision, can be performed to remove all of the redundant tissue involved in the process.

Ano-rectal Abscess

- Is a pus filled cavity caused by bacteria invading a mucus secreting gland in the anus & rectum.

Risk Factors

- Regional enteritis
- Immunodeficient States (HIV/AIDS)
- Many of these abscesses will result in fistulas.

C/M

- Abscess may occur in a variety of spaces in & around the rectum.
- Pain
- Foul - Smelling pus
- In Superficial abscess, (Swelling, redness & tenderness).
- Deeper abscess (Fever, abdominal Pain)
- Fistula

Mx

- Palliative Rx
 - ❖ Sitz Bath
 - ❖ Analgesics
- Surgical Rx
 - ❖ Incision & drainage
- Medical RX
 - ❖ Antibiotics

Anal Fistula

- It is a **tiny, tubular, fibrous tract** that extends into the anal canal from an opening located beside the anus.

Cause

- Fistula usually results from an infection.
- Trauma
- Fissures
- Regional Enteritis

C/M

- Pus or stool may leak constantly from the cutaneous opening
- Passage of flatus or feces from the vaginal or bladder depending on the fistulas tract.
- Fever

Mx:

- Surgery is always recommended
- 10/10/2012 By Anwar H. Fistulectomy (excision of the fistulous tract)

Anal fissure

- It is a **longitudinal tear or ulceration** in the lining of the anal canal

Cause

- Trauma of passing a large firm stool
- Persistent tightening of the anal canal secondary to stress or anxiety (leading to Constipation)
- Child birth
- Trauma

C/M

- Extremely Painful Defecation
- Burning
- Bleeding

Mx

- Increase water intake
- Sitz bath
- Emollient Suppositories
- Corticosteroid Suppositories Relieve Discomfort
- Surgery

*Most of the fissures will heal by conservative measures.

Cancer of the Large Intestine

Colon & Rectum

- Tumors of the small intestine are rare; **conversely tumors of the colon & rectum are relatively common.**

Cause: - Unknown

Risk factors:-

- Age: incidence increases with age (most patients are over age 55).
- It is **the most common cancer** in old age except for prostates cancer in men.
- Family history of colon cancer
- Chronic inflammatory bowel disease, Polyp
- **A diet high in fat, protein, & beef & low in fiber**

C/M

- It is determined by the location, stage of cancer & Function of the intestinal segment.
- Unexplained anemia
- Anorexia
- Weight loss
- Fatigue
- Abdominal Pain
- Melena
- Crampy

- Constipation
- Distention
- Tenesmus (ineffective, painful straining of stool)
- Rectal Pain
- Feeling of incomplete evacuation after a bowel movement
- Alternating Constipation & Diarrhea
- Bloody Stool

Diagnostic Evaluation

- Fecal occult blood testing
- Barium enema
- Procto-sigmoidoscopy
- Colonoscopy
- Biopsy or cytology smears

Medical Mx

- The patient with symptoms of intestinal obstruction is treated with IV fluids & nasogastric Suction.
- Treatment depends on the stage of the disease & related complications. The most widely used staging method is duke's classification:-
- **Class A-** tumor limited to mucosa & Sub-mucosa
- **Class B-** Penetration through bowel wall
- **Class C-** Invasion into regional draining lymph system.
- **Class D-** Advanced & widespread regional metastasis

- * Surgery is indicated for most class A- lesions & all class- B and C.
- Radiation Therapy
- Surgical Removal: is primary treatment
- Segmental Resection with anastomosis
- Temporary Colostomy followed by segmental resection & anastomosis
- Permanent Colostomy or ileostomy

Complications of Colorectal Cancer

- Partial or Complete bowel obstruction
- Hemorrhage
- Perforation

Nursing Process

- **The patient with cancer of the colon or rectum**

1. Assessment

- Hx - Feeling of fatigue
- Presence & Character of abdominal or rectal pain (location, frequency, duration, association with eating or defecation)
- Past & Present elimination pattern
- Description of color, odor, & consistency of stool including presence of blood or mucus
- Hx. of chronic inflammatory bowel disease
- Family hx of colorectal cancer.
- P/E - Auscultation
- Palpation for areas of tenderness, distention & Solid masses
- Inspect the stool for presence of blood.

Nursing Diagnosis

- May include

- Constipation related to obstructing lesion
- Pain related to tissue compression 2^o to obstruction
- Risk for fluid volume deficit related to vomiting & dehydration
- Impaired skin integrity related to the surgical incision
- Body image disturbance related to colostomy
- Altered nutrition, less than body requirements, related to nausea & anorexia
- Anxiety

plan

- To alleviate pain
- To attain optimal level of nutrition
- To maintain normal fluid & electrolyte balance
- To maintain optimal tissue healing, and adequate protection of perio-stomal skin.
- To attain adequate elimination of body waste products.
- To relieve anxiety

Preoperative Nursing Intervention

- Administer laxative & enemas as prescribed.
- Analgesics administration
- Full liquid diet may be prescribed 24 hours before surgery to decrease bulk
- Instruct NPO.
- Parenteral nutrition

Nursing Care for Patient with Colostomy

- **Colostomy** - Is the surgical creation of an opening (stoma) into the colon.
- It can be temporary or permanent diversion.
- It allows for the drainage or evacuation of colon contents to the outside of the body.

Colostomy Irrigation

- a. It is done to permit escape of feces when there is an obstruction of the large bowel or a known lesion, such as cancer, that will eventually cause an obstruction.

- b. It also may be done to permit healing of the bowel distal to it after an infection, perforation or traumatic injury since it diverts the fecal stream from the affected area.
- c. It may be done as a palliative measure in the treatment of an obstruction caused by an inoperable growth of the colon or if the rectum must be removed to treat cancer.
- d. It may be done to provide a permanent means of bowel evacuation.

Purpose of colostomy irrigation

1. To encourage a bowel motion in a recently established colostomy and to ensure that the opening is patent.
2. To relieve constipation in patients who has difficulty managing their colostomy.
3. To teach the patient how to establish regularity of evacuation through the colostomy.
4. To reduce distention before closure of colostomy

Disorders of the Accessory

Diseases of the liver

Patient with hepatitis

- It is an inflammation of the liver
- **Pathophysiology:** Hepatocytes are **damaged & become inflamed & necrosed** by the body's immune response to the virus. This alters cellular function.
- The degree of functional impairment depends on the amount of hepatocellular damage.

- The hepatocytes generally heal in 3 to 4 months

Cause - Virus

- Bacteria
- Toxic substance

Types of Hepatitis

- viral hepatitis
- Toxic hepatitis
- Chronic hepatitis
- Alcoholic hepatitis

Viral hepatitis

- Hepatitis type A
- Hepatitis type B
- Hepatitis type C
- Hepatitis type D
- Hepatitis type E

Hepatitis type A /Infectious hepatitis /

- It is endemic in some areas of the world, especially with poor sanitation
- **Causative agent** - Infectious hepatitis virus
- **Mode of transmission** - The major route of transmission is **through faecal – oral route** (Contaminated food, milk, polluted water)
- Spread of the disease is enhanced by crowding & poor sanitation.
- **Incubation:** 3-7 weeks; average 4 weeks.
- **Occurrence**
 - Worldwide – sporadic or epidemic
 - Autumn & winter months
 - Usually in children & young adults

C/M

1. Pre-icteric (Prior to period of Jaundice) Phase:-

- Headache, Pain over the liver
- Abdominal tenderness, Fever
- Muscle crampy, Anorexia
- vomiting , Backache
- Nausea

2. Icteric phase

- Urine-dark; stool often light for several days
- Liver-enlarged, often tender
- Nausea, vague epigastric distress, heart burn, flatulence, anorexia

Hepatitis B

Causative agent

- Hepatitis B Virus

Mode of transmission

A. Parenteral route

- Blood transfusion from an infected person
- Contaminated needles, syringes

B. Skin puncture – medical equipments

C. Mucosal transmission; dental instruments

- **Incubation period:** 6 weeks to 6 months/ average 2.5 – 3 month
- **Occurrence:** - world wide
- **Diagnosis:** HBSag

C/M:

- S/S similar to infections hepatitis, but usually **more insidious in onset**
- Respiratory manifestations minimal or absent

Rx & Nursing Mx

- Isolate patient to minimize contacts
- Wear gloves; wash hands thoroughly
- Assist with laboratory diagnostic studies
- Handle bed pan carefully & instruct pt. to ensure meticulous personal hygiene habit.
- Use disposable syringe & needles
- Avoid alcohol consumption
- Recognize that recovery is slow & prolonged

Liver cirrhosis

- It is a chronic disease in which there has been diffuse destruction of parenchyma cells followed by liver cell regeneration & an increase in connective tissue.
- These processes result in disorganization of the lobular architecture and obstruction of the hepatic venous & sinusoidal channels, causing portal hypertension.

Classifications

1. Alcoholic cirrhosis of the liver (micro nodular)

- Fibrosis: mainly around central veins & portal area
- Most commonly due to chronic alcoholism.

2. Post necrotic (Macro nodular)

- Due to previous acute viral hepatitis or drug induced massive hepatic necrosis.

3. Biliary cirrhosis

- Scarring around bile ducts & lobes of liver.
- Result from chronic biliary obstruction (With or without infection)
- Much more rare than alcoholic & post necrotic cirrhosis

4. Cardiac cirrhosis is secondary to congestive heart failure with prolonged venous hepatic congestion

Causes

- Alcohol & Hepatitis virus (B & C): **the most common**
- Drugs
 - * Methyldopa
 - * Methotrexate
- Autoimmune chronic active hepatitis
- Chronic hepatic congestion

C/F

- Weight loss, muscle wastage
- Flatulence, palmar erythema
- Jaundice, loss of body hair, gynecomastia
- Oedema , Bleeding
- Anaemia 2⁰ to bleeding
- Abdominal distention (Ascites)
- Hepatomegaly + splenomegally
- Oesophageal varices.
- Sign of hepatic encephalopathy
- Portal hypertension

Diagnosis

- Liver biopsy & Liver function test
- Oesophagoscopy
- Liver scan
- Paracentesis to examine ascetic fluid

Management

A/ Nutrition

- Maintain caloric & vitamin diet intake give protein as tolerated
- Avoid table salt, salty foods & canned foods
- Use ‘Salt’ substitutes such as lemon juice
- Offer small frequent meals
- Eliminate alcohol

B/ Medication

❖ Multivitamins preparation

- Vit. K: if a tendency of bleeding is manifested such as epistaxis, melena, hematoemesis
- Vit B₁₂ to correct anaemia
- Electrolyte fluid balance Eg. Spirinolacton
- Electrolyte fluid balance
- Avoid toxic drugs
 - barbiturates, diazepam
 - Oral contraceptive
 - Alcohol

C/ Rest & activity

- ❖ If there is no ascites & sign of hepatic coma
 - Limit amount of activity
- ❖ In advanced liver impairment
 - Bed rest
 - Frequent change of position
 - Special skin care & passive exercise.

Complication of liver cirrhosis

- Hepatic comma
- Bleeding & oesophageal varices
- Portal hypertension
- Spontaneous peritonitis
- Asites
- Hepatic encephalopathy
- Anaemia

Hepatic Encephalopathy & Hepatic coma

- **Hepatic encephalopathy:** Results from the accumulation of ammonia & other toxic metabolites in the blood.
- **Hepatic coma:** represents the **most advanced stage** of hepatic encephalopathy.
 - Ammonia accumulates because **damaged liver cells fail to detoxify & convert the ammonia to urea** that is constantly entering the blood stream as a result of its absorption from the GIT & its liberation from kidney muscle cells.
 - The increased ammonia concentration in the blood causes brain dysfunction & damage, resulting in hepatic encephalopathy.

C/M

Early Stage

- Euphoria
- Depression
- Apathy
- Irritability
- Memory loss
- Confusion
- Impending coma: Asterixa (flapping tremor of hands)
- Feter hepaticas (Breath odour like freshly mowed grass, acetone, or old wine, may be noticed.
- Disorientation to PPT

Late Stage

- * Drowsiness
- * Insomnia
- * Agitation
- * Slow & slurred speech
- * Hyperactive reflex
- * Slow deep respiration

MX

- Vital signs are measured & recorded every 4 hours
- Serum ammonia level is monitored daily.
- Avoid constipation
- Enema to reduce ammonia absorption
- Sterilization of intestine (Neomycin sulphate)
- Lactulose is administered to reduce blood ammonia
- If sign of impending hepatic encephalopathy occur, reduce protein intake.

Portal Hypertension

- is the increased pressure throughout the portal venous system that results from obstruction of blood flow into and through the damaged liver.
- Commonly associated with hepatic cirrhosis, it can also occur with noncirrhotic liver disease.
- Although splenomegaly (enlarged spleen) with possible hypersplenism is a common manifestation of portal hypertension, the two major consequences of portal hypertension are ascites and varices.

Ascites

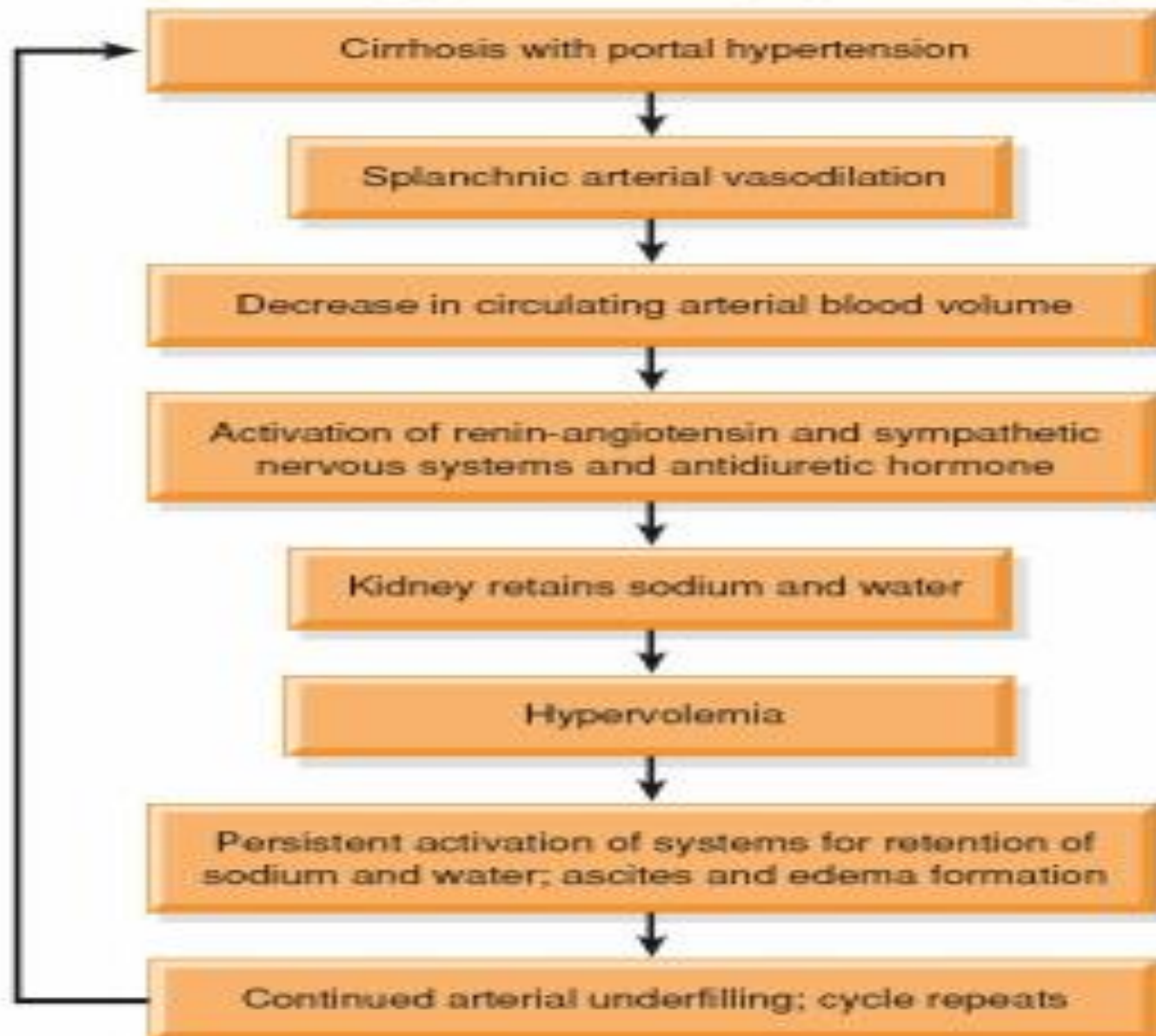
Pathophysiology

- not completely understood.
- **Portal hypertension** and the resulting **increase in capillary pressure and obstruction of venous blood flow** through the damaged liver are contributing factors.
- The **vasodilation** that occurs in the splanchnic circulation (the arterial supply and venous drainage of the GI system from the distal esophagus to the midrectum, including the liver and spleen) is also a suspected causative factor.
- The **failure of the liver to metabolize aldosterone** increases sodium and water retention by the kidney.

...

- Sodium and water retention, increased intravascular fluid volume, increased lymphatic flow, and decreased synthesis of albumin by the damaged liver all contribute to the movement of fluid from the vascular system into the peritoneal space.
- The process becomes self-perpetuating; **loss of fluid into the peritoneal space causes further sodium and water retention** by the kidney in an effort to maintain the vascular fluid volume. As a result of liver damage, large amounts of albumin-rich fluid, **20 L or more**, may accumulate in the peritoneal cavity as ascites
- With the movement of albumin from the serum to the peritoneal cavity, the osmotic pressure of the serum decreases. This, combined with increased portal pressure, results in movement of fluid into the peritoneal cavity

Physiology/Pathophysiology



Clinical Manifestations

- Increased abdominal girth
- Rapid weight gain,
- Short of breath,
- Uncomfortable from the enlarged abdomen,
- Striae and distended veins,
- Umbilical hernias,
- Fluid and electrolyte imbalances
- Abdominal hernias due to increased intra-abdominal pressure
- Tympany

Assessment and Diagnostic Test

- The presence of fluid can be confirmed either by percussing for shifting dullness,
- by detecting a fluid wave
- Ultrasonography of the liver and abdomen
- Diagnostic paracentesis

Medical and Nursing Management

Dietary Modification

- **Dietary restriction of salt** intake is the mainstay of treatment of ascites to achieve a negative sodium balance to reduce fluid retention.
- The recommended daily sodium intake of patients with ascites should be 2,000 mg or less

• • •

- Nutritional education and guidance on foods that contain high sodium
- avoid the use of added salt in meals and food preparation

Diuretics

- The addition of diuretics with sodium restriction:
Spironolactone, and furosemide
- Possible complications of diuretic therapy are dehydration, volume depletion, and electrolyte abnormalities such as hypokalemia or hyperkalemia, hyponatremia, renal impairment, and hepatic encephalopathy.
- Gynecomastia is a common side effect caused by spiro lactone. If patients with ascites develop dilutional hyponatremia (serum sodium 134 mmol/L), diuretics should be discontinued and fluid restriction initiated.
- Paracentesis

...

Transjugular intrahepatic portosystemic shunt (TIPS)

- is a method of treating ascites in which a cannula is threaded into the portal vein by the transjugular route to reduce portal hypertension, an expandable stent is inserted to serve as an intrahepatic shunt between the portal circulation and the hepatic vein.
- This diverts blood flow from a high-pressure vascular bed (portal) to a lower-pressure vascular bed (hepatic veins inferior vena cava), allowing for the return of blood to the heart and decompressing portal hypertension

Complications

- An important complication of ascites is the development of **spontaneous bacterial peritonitis** (SBP), in which infection develops in the ascitic fluid, usually from ***Escherichia coli*, a gram-negative bacteria.**

Esophageal varices

- Gastroesophageal varices is present in 50% of patients with cirrhosis and **is the most life-threatening complication in patients with chronic liver disease.**
- Esophageal varices are **dilated, tortuous veins** that are usually found in the submucosa of the lower esophagus but may develop higher in the esophagus or extend into the stomach.
- **Portal hypertension is the primary mechanism by which collateral circulation develops.**
- Because of increased obstruction of the portal vein, **venous blood from the intestinal tract and spleen seeks** an outlet through collateral circulation (new pathways for return of blood to the right atrium).

...

- The effect is increased pressure, particularly in the vessels in the submucosal layer of the lower esophagus and upper part of the stomach.
- These collateral vessels are not very elastic; rather, they are tortuous and fragile, and they bleed easily
- Factors that may precipitate hemorrhage are alcohol intake, physical exercise, and any activity that would increase intra abdominal pressure (e.g., lifting, straining at stool, vomiting, coughing)
- Bleeding esophageal varices are life-threatening and can result in hemorrhagic shock that produces decreased cerebral, hepatic, and renal perfusion.
- In turn, there is an increased nitrogen load from bleeding into the GI tract and an increased serum ammonia level, increasing the risk for encephalopathy.

...

C/M

- **hematemesis, melena**, or general deterioration in mental or physical status and often has a history of alcohol abuse.
- Signs and symptoms of shock

Mgt

- Endotracheal intubation may be necessary to prevent aspiration of blood
- **Oxygen**
- IV fluids with electrolytes and volume expanders are provided to restore fluid volume and replace electrolytes.
- **Vasopressin** (Pitressin) may be the initial mode of therapy, because it produces constriction of the splanchnic arterial bed and decreases portal pressure.

...

- Ciprofloxacin, or ceftriaxone in patients with advanced cirrhosis) to prevent infection
- Antacids, histamine-2, proton pump inhibitors
- Initial pharmacological therapy is vasopressin and nitroglycerin for the first 24 hours, after which somastatin
- Esophageal Banding Therapy (Variceal Band Ligation)
- **Balloon tamponade** may be used as a temporary measure to control bleeding in an active hemorrhage. In this procedure, pressure is exerted on the cardia (upper orifice of the stomach) and against the bleeding varices by a double-balloon tamponade

Cancer of the liver

Classification

- 1) Primary
- 2) Metastatic tumor

Primary tumor - It can be benign or malignant

• Origin	Benign	Malignant
• Hepatocytes	- Adenoma	- HCC
• Connective tissue	-Fibroma	-Sarcoma
• B/d vessels	-Hemangoma	- Hemangioendothelioma
• Bile duct	-Cholangioma	-Carcinoma

- Primary liver tumors usually occur in patients with chronic liver disease.
- Hepatocellular carcinoma (HCC) is by far the most common type of primary liver cancer.
- HCC grows rapidly & metastasizes elsewhere.
- Other types of primary liver cancer include cholangiocellular carcinoma (CCC) and combined HCC & CCC.

Metastatic carcinoma

- It arises from GIT (particularly colon), breast & the lung.
- is more common than primary tumors.
- Malignant tumour reaches the liver by way of portal system or lymphatic channels or direct extension from abdominal tumour
- Liver is a frequent site of metastasis

Causes

- Cirrhosis
- Hepatitis B & C
- Exposure to certain chemicals toxins
- Cigarette smoking & alcohol.

C/M

- Early manifestations are s/s of impaired nutrition: Such as
- Recent loss of weight
- Loss of strength
- Anorexia
- Anaemia
- Abdominal pain
- Enlargement of liver
- Jaundice & Ascites some times

MX

- Surgery (hepatic lobectomy), following surgery the patient should be observed closely for haemorrhage.
- Chemotherapy , Radiation

Nursing Care

- Rest & Positioning
- Psychological support
- Mouth care
- Patient education
- Pre & post operation care

Liver abscess

- Formation of pus in the liver.
- Two Categories of liver abscess have been identified
 - **Amoebic liver abscesses: most common.**
 - Cause: Entamoeba histolytica
 - **Pyrogenic liver abscesses: Less common**
 - Cause: Staphylococcus, streptococcus, E. Coli
 - Mechanism - GIT infection → Liver → Abscess cavity formation

C/M

- fever
- Chillness
- Dull abdominal pain
- Tenderness
- Hepatomegally
- Jaundice
- Anaemia

RX

- Intravenous antibiotic therapy (Depending on the organism identified)
- Continuous supportive care
 - Blood transfusion
 - Nutrition & fluid
 - Rest & positioning
 - Psychological support

Trauma of the Liver

- It is injury to the liver

Cause

- Stab
- Bullet injury
- Accidental puncture

MX

- Blood transfusion
- Surgical repair
- Control bleeding
- The area may be sutured or packed
- A drainage tube into the abdominal cavity is attached.

Complication - Peritonitis

-Impaired liver function

Disease of the Gall - bladder

Cholecystitis

- It is an inflammation of the gall bladder wall.
- It may be acute or chronic

Acute Cholecystitis: It is acute inflammation of gall-bladder wall.

1) Acute calculus Cholecystitis (infection with stone) is most common.

2) Acute acalculus Cholecystitis (Infection without stone)

Cause

- ✓ Gall stones (Obstruction of a cyst duct by a stone is the usual cause of acute Cholecystitis).
- ✓ Colon bacilli
 - E. coli
 - Streptococcus faecalis.
 - Staphylococcus
 - Salmonella or complication of typhoid fever

S/S

- Sudden onset
- Severe nausea & Vomiting
- Pain & tenderness in the **RUQ of abdomen or mid epigastrium**
- Fever , Jaundice

DX

- Murphy's sign on physical examination.
- Ultrasonography – is the best investigation
- x-ray
- WBC count

MX

- Complete bed rest
- IV fluid
- Analgesics
- Antibiotics (Ampicillin , gentamycin, cephalosporin)

- If the condition persists or worsen it may indicate suppuration (Empyema) of the gallbladder.
 - Surgery for Empyema - Cholecystostomy
 - Cholecystectomy

Chronic Cholecystitis

- It is almost always **associated with gallstones**

S/S

- Long history of vague digestive complaints
- Abdominal discomfort & flatulence after a large meal or one high intake.
- A dull aching pain
- Nausea & vomiting

- It affects middle-aged & older obese women. Female to male ratio is 3:1
- Chronic inflammation results in scarring & thickening of the wall of the gall bladder & Cholestasis.
- Moderate fever, dyspepsia
- If calculi are present, they progressively increase in size or number.

DX

- Ultrasonography

MX

- Surgery: Cholecystostomy
- Low fat diet, weight reduction
- sedatives

Complication

- abscess
- Pancreatitis
- Cholangitis (inflammation of bile duct)
- Rapture of the gall bladder
- Biliary cirrhosis

Cholelithiasis

- It is a formation of stone with in the gall bladder
- Consistency of the stone
 - Cholesterol
 - Bile pigment

Aetiology: Not well known

- **The possible reason for stone formation can be as follows:**
 - Change in **bile composition (high concentration cholesterol)**
 - **Bile stasis** (total parenteral nutrition)
 - **Infection**

Incidence

- May occur in both sexes
- More often occurs in middle aged female, higher in individual with regional enteritis
- Diabetes
- Cirrhosis of liver
- Obesity & genetic factors.
- Oestrogen administration.

S/S

- Gallstone causes no manifestation unless complicated.
- Pain which starts in the **upper midline area. It may radiate to the back & right shoulder.**
- The client is restless, changing position frequently to relieve intensity of the pain.
- Digestive disturbance following fat ingestion.
- Inflammation of gall bladder, nausea, vomiting, dyspepsia.
- Cholestasis (stasis of the bile within the liver leading to impaired functioning of the organ)
- Obstructive jaundice

DX

- Endoscopy & U/S

RX & Nursing Intervention

a. Medical management

- Administration of parenteral analgesics.

b. Endoscopy

- For stone removal from common bile duct

c. Cholesterol dissolving agents

- Ursodeoxycholic acid

d. Dietary management

- Avoid eating a fatty meal, consumption of a large meal after fasting
- In acute attack the patient should be NPO.
- IV fluid for maintaining fluid & electrolyte balance.
- NG tube insertion & aspiration if vomiting & abdominal distension occurs.
- Observe for sign of jaundice

e. Surgical Mgt

- Cholecystolithotomy

Complication of Cholelithiasis

- Cholangitis
- Biliary Cirrhosis
- Carcinoma & Peritonitis
- Choledocholithiasis (**Stone in the common bile duct**)

Disorders of the pancreas

Pancreatitis

- It is an inflammation of the pancreas
- It may be acute or chronic

Acute Pancreatitis :

- It is an inflammation of the pancreas that may **result in auto-digestion of the pancreas by its own enzymes.**
- Potentially serious disease
- If it is mild the pancreas becomes swollen & oedematous with treatment the patient recovers within a few weeks
- If it is severe & persists – necrosis & haemorrhage.
- Blood & enzymes may seep into the surrounding tissue & peritoneal cavity
- Peritonitis, Paralytic ileus, or ascites may develop

Aetiology & risk factors

- In 90% of the cases it is due to **biliary disease & excessive alcohol consumption**
- **blockage of ampulla of vater by stone** causing activation of pancreatic enzyme
- Alcoholism
- Infection & injury
- Drugs, pancreatic ischemia during episodes of hypotensive shock
- Idiopathic

DX

- Determination of serum
- Amylase increased
- Lipase increased
- CT scan

C/M

- * Sudden onset
- * Penetrating pain to the back (burning or boring pain) on the upper abdomen is the first presentation later on the pain is generalized.
- Nausea/Vomiting
- Constipation
- Abdominal distension & rigidity as a result of peritonitis
- Peristalsis diminished
- Shock due to the pain & loss of blood
- The skin became pale & cold
- Vital sign = B/P decrease, P/R increase
- Confusion, coma, transient hyperglycaemia
- Jaundice

RX & Nursing Intervention

1. Anti-pain -usually treated with narcotic analgesics
2. Keep the patient NPO
3. IV - Plasma or whole blood
 - Electrolyte & glucose solution
4. Gastric aspiration
5. Mouth care
6. Medication
 - Antibiotics
 - Analgesics

Surgical Intervention

- Surgery is not usually undertaken during on acute attack of Pancreatitis unless there is increasing obstruction jaundice due to impacted stone.
- **Choledocholithotomy**

Chronic Pancreatitis

- It is an inflammatory disorder characterized by **progressive anatomic & functional destruction of the pancreas.**

Patho-physiology

- Repeated attack of acute pancreatitis causes **scarring & calcification pancreatic tissue & irreversible damage** to the endocrine & exocrine pancreatic function.

Cause

- Alcohol consumption (excessive & prolonged consumption) (75%)
- Protein malnutrition
- Pancreatic trauma

C/M

- Recurring attacks of severe **upper abdominal & back pain accompanied by vomiting**
- Anorexia, nausea, flatulence, constipation, abdominal distension.
- Weight loss,
- steatorrhea (stool with high fat content)
- **Hyperglycaemia**

RX & Nursing Intervention

- Analgesics
- Low fat diet
- Avoid alcohol consumption
- Surgery: Surgical intervention may be used to **drain pseudo cysts or to relief intractable pain** partial pancreatectomy or anastomosis of the pancreatic duct with the jejunum, may be done.

Thank
Very
Much

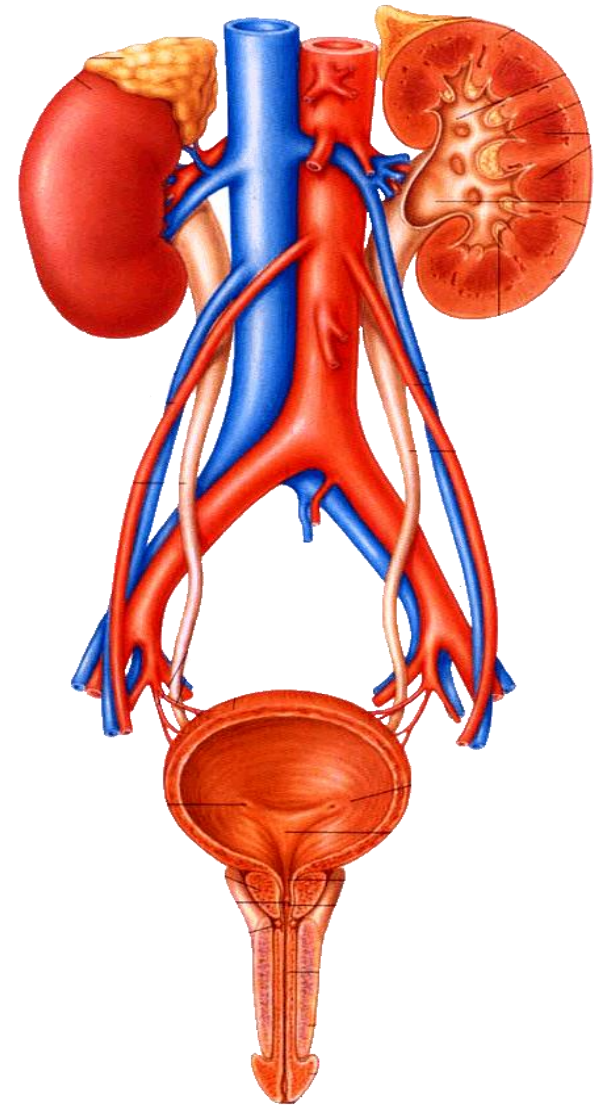
Genito - Urinary System disorders

By Afework E.

Anatomy & physiology overview

Composition

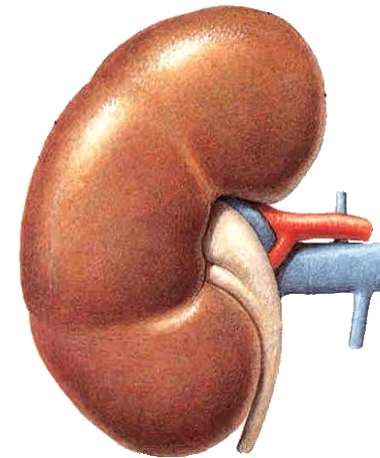
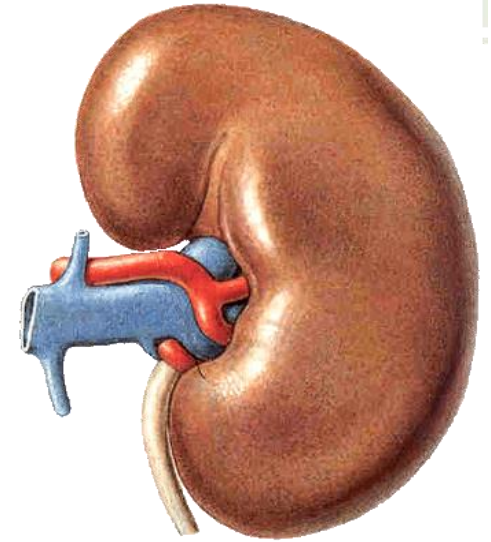
- **Two kidney**: form urine
- **Two ureter**: conduct urine from kidneys to bladder
- **Bladder**: receives and stores urine
- **Urethra**: conducts urine from bladder to exterior of body (discharged)



Kidney

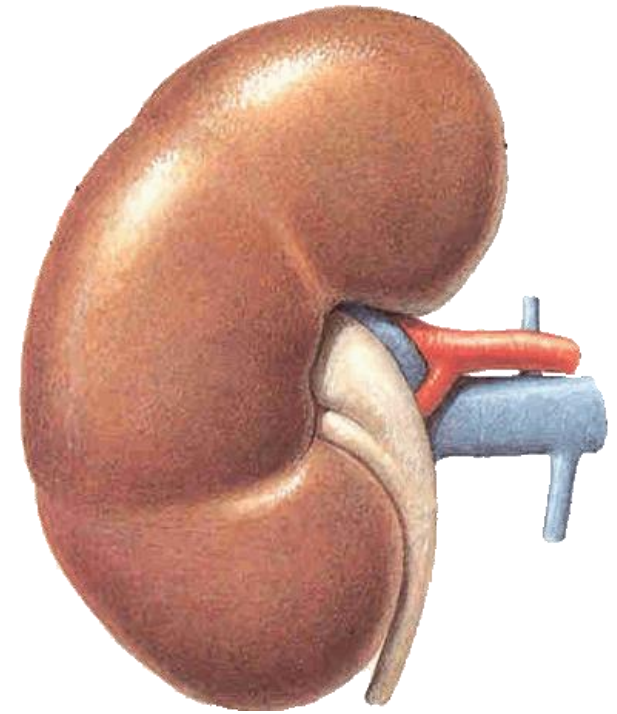
General features

- **Bean shaped, reddish-brown organs**
- **Superior extremity:** broad and thin
- **Inferior extremity:** narrow and thick
- **Anterior surface:** convex
- **Posterior surface:** flat
- **Lateral border:** convex
- **Medial border:** concave



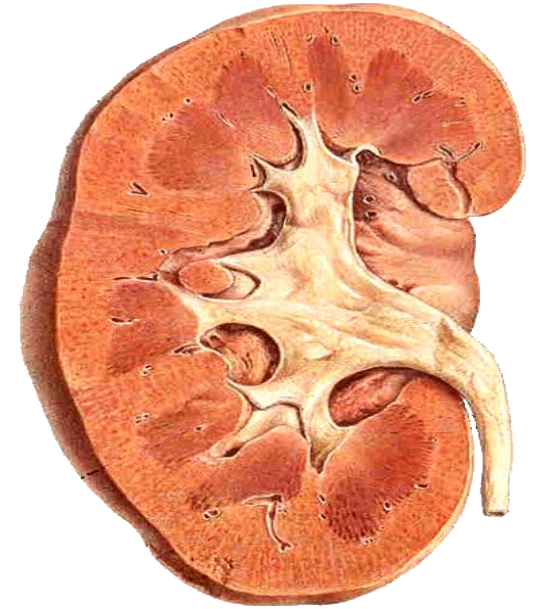
Renal hilum

- **a vertical slit** on the medial border of kidney, the **structures** which enter and leave the renal hilum is called **renal pedicle** including the renal vein, renal artery, renal pelvis, lymphatic vessels and nerves



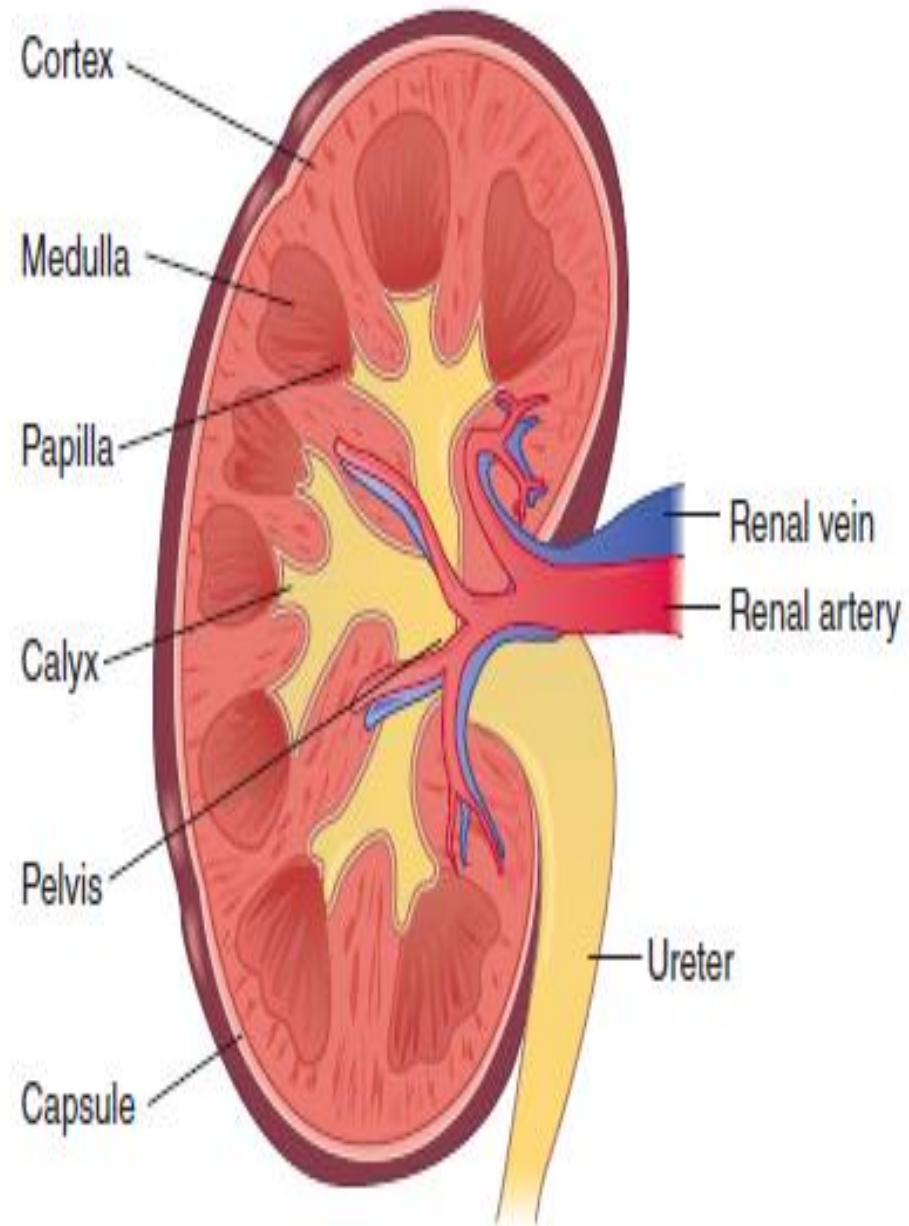
Renal sinus

- which is the renal hilum leads into a space within the kidney, the renal sinus is **occupied** by the renal vessels, minor renal calices, major renal calices, renal pelvis and some adipose tissue



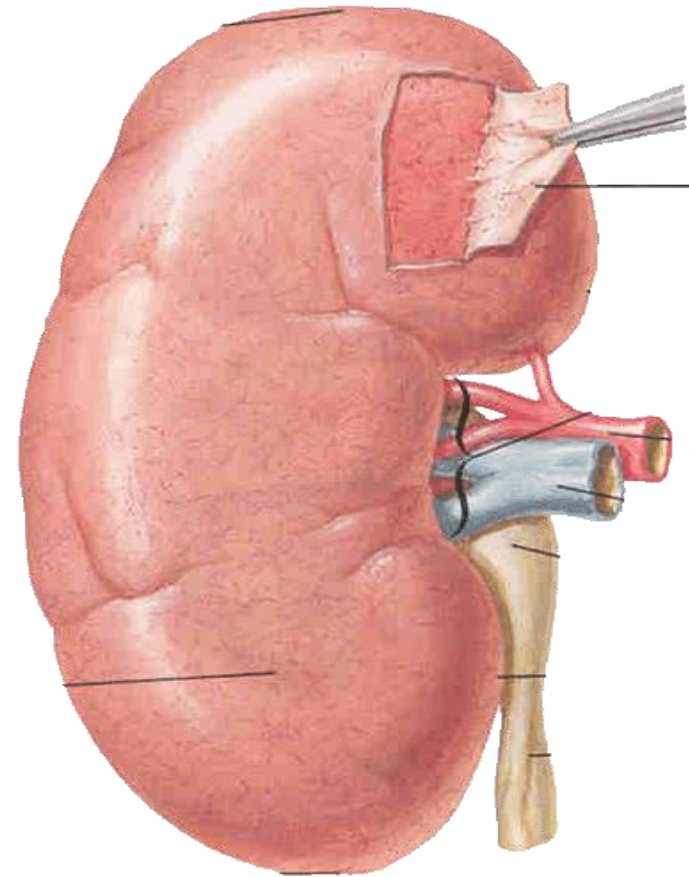
Structure of the kidney

- **Renal cortex**
 - renal columns
- **Renal medulla**
 - Renal pyramids
 - Renal papilla
 - Papillary foramina
- Minor renal calices (7—8)
- Major renal calices (2~3)
- Renal pelvis



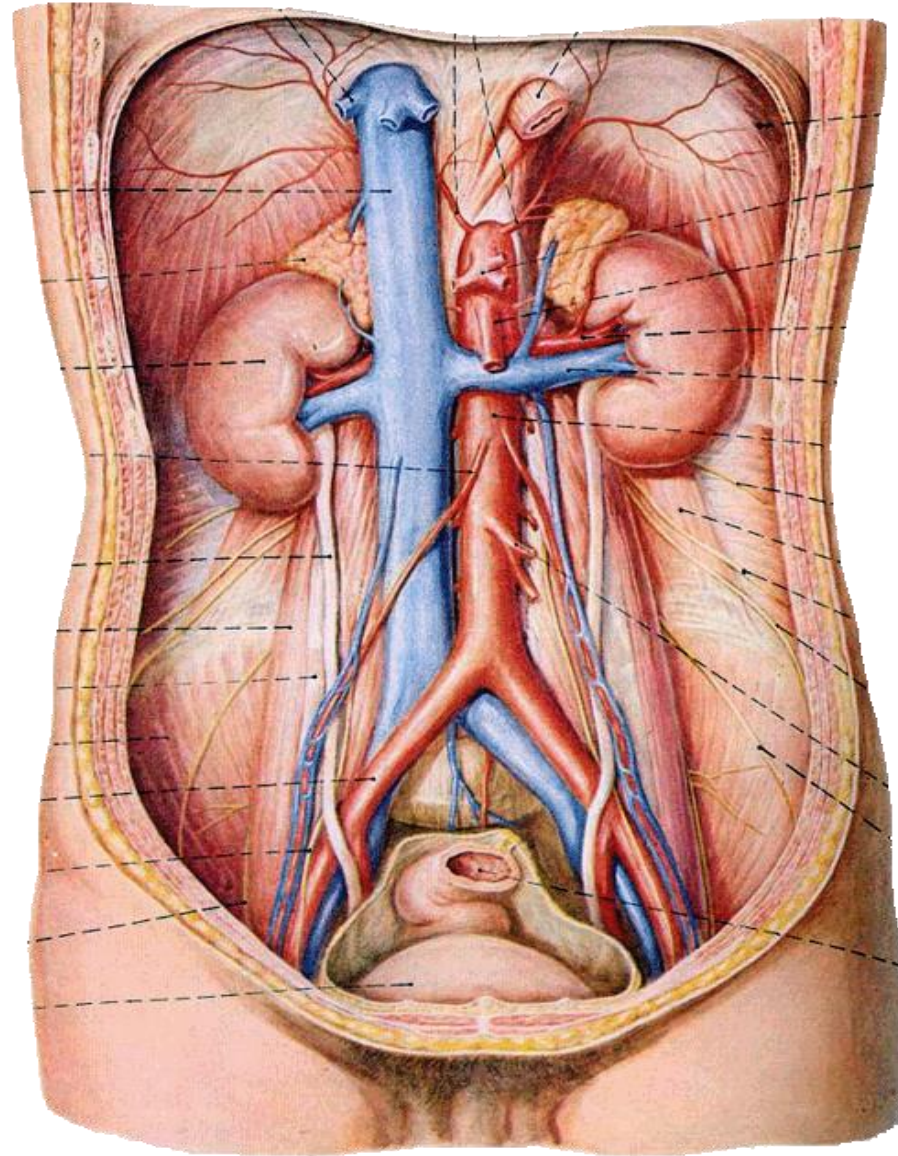
Coverings

- **Fibrous capsule:** a strong fibrous capsule which strips easily from a normal kidney surface but adheres firmly to an organ that has been inflamed
- **Fatty renal capsule:** a **thick** adipose connective tissue capsule, **surrounds the fibrous capsule**. It acts as a shock absorber to **protect the kidney from jolting and jarring**
- **Renal fascia:** on outside, surrounds both the kidney and suprarenal gland, **holding these organs in place**

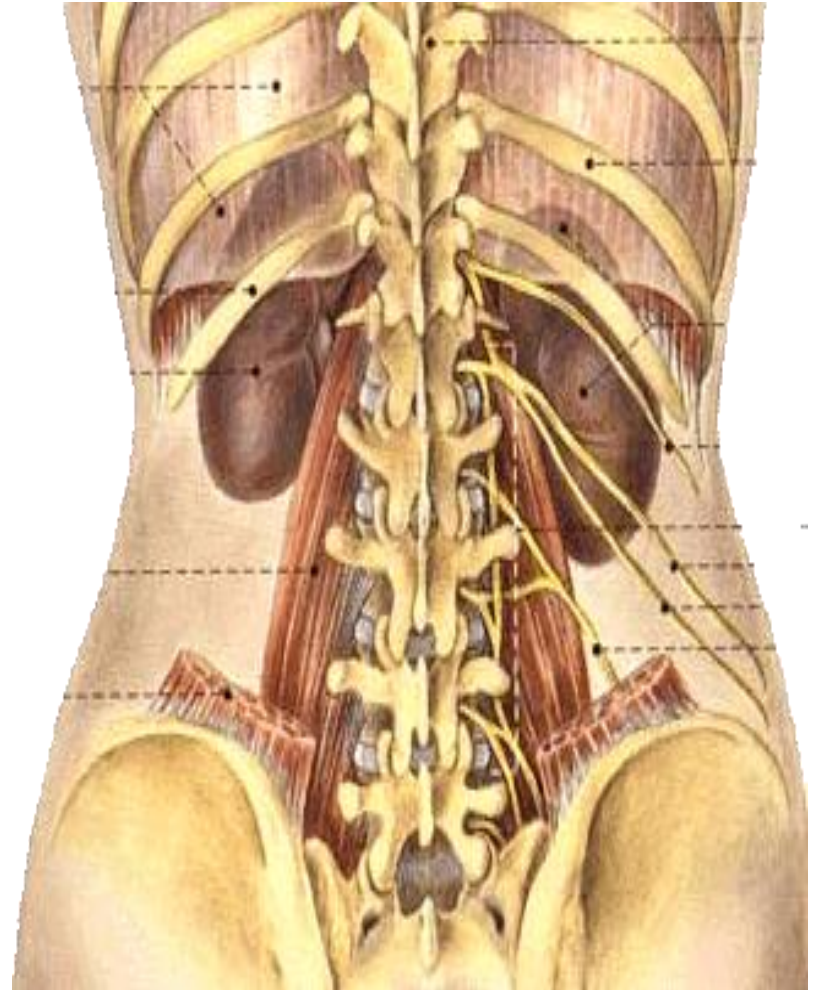


Position

- Lie **behind peritoneum** one on each side of the vertebral column high up on the post abdominal wall.

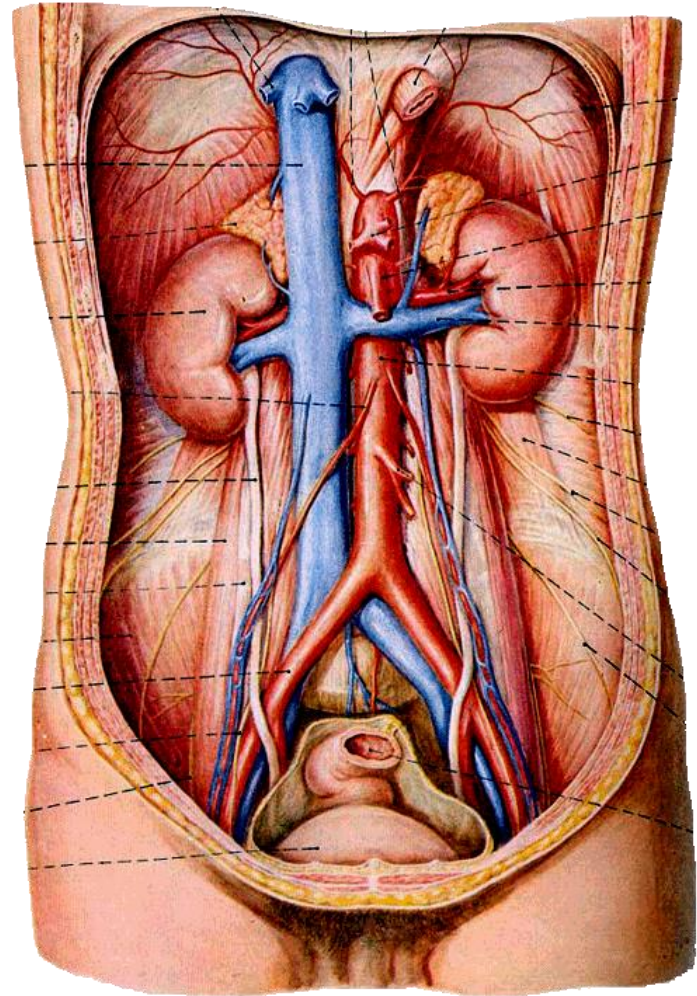


- **Left kidney:** lies at the level from the lower border of **T11 to L2**; the **12th rib** is behind its **middle part** of the post surface
- **Right kidney:** lies slightly **lower than the left kidney**, at the level from the lower border of **T12 to L3**; the 12th rib is **behind its upper** part of the post surface
- Renal hilum- at the level of **L1**, is about 5cm from the posterior median line



Ureters

- Muscular tube, about 25cm long
- Three parts
 - **Abdominal part:** descend on the psoas major behind the peritoneum
 - **Pelvic part:** in females, passes 2cm lateral to the neck of uterus and lies below the uterine artery
 - **Intramural part** — passes obliquely through the bladder wall for 2cm long



Bladder

General features

- Pyramidal in shape when empty, having
 - Apex,
 - Fundus
 - Body of bladder
 - Neck

Location

- In the adult, it lies in the lesser pelvis, **behind the pubic symphysis, in front of seminal vesicle, ampulla ductus deferentis and rectum in the male**, and **in front of uterus and vagina in the female**.
- In the young child the empty bladder projects above the pelvic inlet

Urethra

Female urethra

- Relatively short (about 5cm long), **wide and straight**
- Opens into vaginal vestibule

Male urethra : 15 to 25 cm

Assessment of Urinary System

1. History: Signs and Symptoms of genitourinary diseases

❖ Renal pain

- **Ureteric pain:** it is **severe colicky pain** which originates in the costo-vertebral angle. It may **radiate** into the lower quadrant of the abdomen and possibly to the upper thigh and testicle or labium

❖ Hematuria

❖ Anuria: is the complete **absence** of urine output.

- **Retention** of urine should be excluded before a patient is considered to have anuria.

❖ Polyuria: on the basis of 24 hours urine output of **more than 3L per day**

❖ Urinary frequency

❖ Nocturia: Implies the need to rise during hours of sleep to empty the bladder

❖ Dysuria

❖ Urgency: Is the loss of the normal **ability to postpone micturation** beyond the time when the desire to pass urine is initially perceived.

...

- ❖ **Incontinence:** Refers to an **involuntary loss of urine** that has become a social or hygienic problem
- ❖ **Hesitancy:** Is **difficulty initiating the process of micturation**
- ❖ **Terminal dribbling:** is **difficulty of completing micturation** in a clean stop fashion
- ❖ **Urethral discharge**
- ❖ **Vaginal discharge**
- ❖ **Menstrual History**

...

- ❖ **Postmenopausal bleeding:** is defined as bleeding that occurs after 6 months without periods and warrants further investigation.
- ❖ **Amenorrhea-** refers to the absence of periods
- ❖ **Dysmenorrhea:** is pain with menstruation, and is usually felt as a bearing down, aching, or cramping sensation in the lower abdomen and pelvis
- ❖ **Dyspareunia:** is pain on **sexual intercourse**. It can be superficial or deep.
- ❖ **Vaginismus:** refers to an **involuntary spasm of the muscles** surrounding the vaginal orifice that makes penetration during intercourse painful or impossible.
- More commonly, this sexual problem is related to situational or **psychosocial factors**

2. Physical examination

➤ Kidneys

- Inspect the **flanks for bruising or swelling**
- Assess **each kidney for tenderness.**

➤ Bladder

- Inspect the lower abdomen (supra pubic area).
- palpate for enlarged bladder arising from the pelvis
- Look for enlargement or distention.
- Palpate for tenderness or rigidity

❖ Genital ulcer

Scrotal Edema



Tumor of the Testis



Hydrocele



Scrotal Hernia



3. Diagnostic Studies of Urinary System

- Diagnostic studies are important in locating and understanding problems of the urinary system.

A. Urinalysis

- **General examination of urine** to establish baseline information or to provide data in establishing a tentative diagnosis & determine further studies to be ordered.
- Findings are: Color, Smell, protein, glucose, ketones, specific gravity, osmolality, PH, WBC, RBC, casts, culture for organisms, etc...

- **Creatinine clearance:** Creatinine is waste product of protein breakdown. It is an **excellent** measure of renal function.

$CC = \frac{\text{urine volume/minute} \times \text{urine creatinine(mg/dl)}}{\text{serum creatinine}};$

serum creatinine (NR 0.5 - 1.5 mg/dl)

- **Urine Culture:** **Confirm** suspected urinary tract infection and identity causative organisms
- normally bladder is sterile, but urethra contains bacteria and a few WBC .

...

B. Blood Chemistry

- BUN (Blood urea nitrogen) **most commonly used to diagnose renal problems.** Concentration of urea in blood is repeated by the rate at which kidney excrete urea (NR 10-30 mg/dl)
- RFT (Renal Function Test) is used it evaluate the **severity** of kidney disease and to **follow** the patient's clinical progress
- This test also give information concerning the kidneys effectiveness in caring out their excretory function.

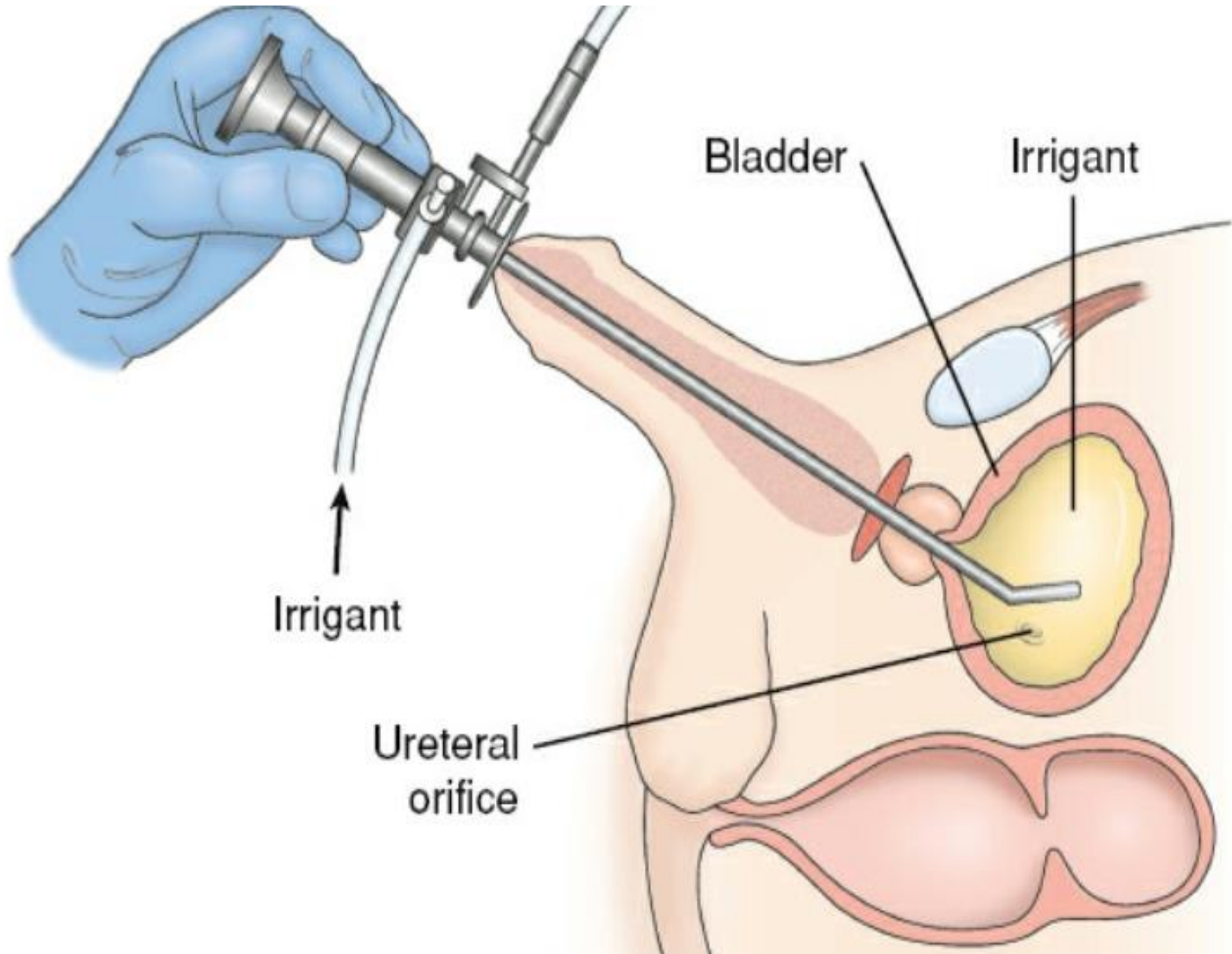
C. Radiological studies

- **KUB (KIDNEY, URETER AND BLADDER)**
 - Flat plate of abdomen and pelvis delineates (shows) size, shape and position of the organs, especially the kidney.
 - ✓ **IVP (intravenous pyelography)** or excretory urogram - **x-ray visualizes urinary tract after I.V injection of radio opaque dye.**
 - ✓ **Nephro-tomogram** - x-ray taken with rotating tubes, **delineates segments** of the kidney at different levels. I.V injection of radio opaque dye will be given

...

- ✓ **Retrograde Pyelography** - x-ray of **urinary tract** following injection of radio opaque dye in to the kidneys.
- cystoscope and urethral catheter are insert through it in to renal pelvis and dye is injected through catheter .
- ✓ **Renal arteriogram** (angiogram) - Injecting radiopaque dye in to artery **to visualize the renal blood vessel.**
- ✓ **Cystoscopy**
 - Involves use of tubular tighten scope to **inspect bladder.**
 - It used lithotomic position
- ✓ **Renal biopsy**
 - Insertion of biopsy needle in to the lower lab of the kidney.
 - Purpose: To obtain renal tissue from examination to determine type or renal disorder to follow progress of renal pathology.

Cystoscopy



✓ **ULTRASOUND** ...

- Small external ultrasound probe and conductive attached to the patient.
- Computer interprets tissue density based on sound wave & displays it in picture form.
- Simple explanation is required

Regulation of body fluids

- **Two physiologic mechanisms** assist in the regulation of body fluids.
- **Thirst level:** primarily regulates intake.
- Occurs when an **increase in the extracellular osmolarity** causes osmoreceptors (nerve cells in hypothalamus) **to shrink**.
- Cause of thirst
 - increased the osmolality of the ECF
 - hypovolemia
- **Renal concentrating mechanism:** the kidney **controls the concentration** of most of the constituents in body fluids including water & electrolytes.

Fluid imbalances

Fluid Deficits(FVD)

❖ Causes

- **Inadequate fluid intake**
 - Unconsciousness/coma or inability to express thirst
 - Oral trauma or inability to swallow
 - Impaired thirst mechanism
 - Withholding of fluid for therapeutic reason

➤ Excessive fluid losses

- **GI losses**

- vomiting
- diarrhea
- GI suctioning
- fistula drainage

- **Urine losses**

- diuretics therapy
- osmotic diuresis (hyperglycemia)
- salt wasting renal diseases

- **Skin losses**

- Exposure to hot environment
- Fever
- Burns and wounds that remove skin

➤ Other risk factors

- hemorrhage

C/M

- Acute weight loss (% body weight)
 - Mild FVD: 2% loss
 - Moderate: 2-5% loss
 - Sever: 6% or more loss
- Thirst, anorexia, nausea
- Decrease urine output (oliguria)
- Increase urine osmolality
- Increase specific gravity
- Increase serum osmolality

- Increased hematocrit
- Increased BUN
- Decreased vascular volume
- Tachycardia, weak and thready pulse
- Postural hypotension
- Hypotension & shock
- Volume in extracellular space decreased
- Depressed fontanel

- Sunken eyes & soft eyeballs
 - b/c of loss of ICF
- Dry skin (decreased skin turgor) & mucous membrane
- Cracked & fissured tongue
- Decrease salivation & lacrimation
- Neuromascular weakness & cramps
- Fatigue b/c of increased temprature
- Cool calmy skin related to peripheral vasoconstriction

Diagnosis

- History
- Physical exam
- Increased BUN
- Increased BUN to creatinine ratio ($>20:1$)
- Increased Hct
- Electrolyte changes may occur
- Urine osmolality increased as kidney attempt to conserve water

Medical mgt

- **Isotonic fluid replacement** e.g. 0.9%Nacl solution, ringers lactate
- After the patient becomes normotensive, a **hypotonic** solution (E.g. 0.45% Nacl solution often used to provide both electrolytes & water for **renal excretion of metabolic wastes**

Nursing mgt

- Monitoring **intake & output** at least every 8 hr & sometimes every hr.
- Monitoring daily body weight at the same time of day
- Monitoring vital signs
- Monitoring skin & tongue turgor
- **Do not allow the patient to sit or stand up quickly** as long as circulation is compromised to avoid orthostatic hypotension or possible syncope.

Fluid volume excess (FVE)

- FVE refers to an **isotonic expansion of the ECF** caused by the abnormal retention of water & sodium in approximately the **same proportion** in which they exist in the total body fluid.

Cause

- **Excessive sodium & water intake**
 - dietary intake
 - ingestion of medications or home remedies containing sodium
- **Inadequate renal losses**
 - renal disease(renal failure)
- **CHF**
- **Cirrhosis of liver**

C/M

- Acute wt gain (in excess of 5%)
- Pitting edema of extremities
- Puffy eyelid
- Pulmonary edema
 - SOB
 - wheezing
 - dyspnea
 - cough
- **Tachycardia full & bounding pulse**
- Increased BP & pulse pressure
- Distended neck vein
- Increased urinary out put

Diagnosis

- Hx
- Physical exam
- Decreased BUN
- **Hct may be decreased**
- Increased urine specific gravity (b/c of urine sodium level)
- Decreased serum osmolarity
- Chest x-ray reveals pulmonary congestion

Medical mgt

- Mgt is directed towards the cause:
 - ✓ if related to excessive administration, **discontinuing the infusion**
 - ✓ diuretics(thiazides)
 - ✓ restricting fluid and sodium intake
 - ✓ hemodialysis/peritoneal **dialysis** if pharmacologic & dietary mgt can not act effectively

Nursing mgt

➤ Monitoring

- daily input & output
- daily body wt
- degree of edema in most dependent body parts

Electrolytes

- Minerals in your blood or other body fluid that **carry an electric charge.**
- It can be acid, base, salts & gases.
- ❖ $\text{Na}^+ = 135\text{-}145\text{meq/L}$
 - = **main cat ion of ECF**
 - = essential for depolarization's.
- ❖ $\text{K}^+ = 3.5 - 5.0 \text{ meq/L}$
 - = **main cation of Icf**
 - = essential for Repolarization.
- ❖ $\text{Cl}^- = 95\text{-}108 \text{ meq/L}$
 - = **main anion of ECF**
 - = Influence acid base balance

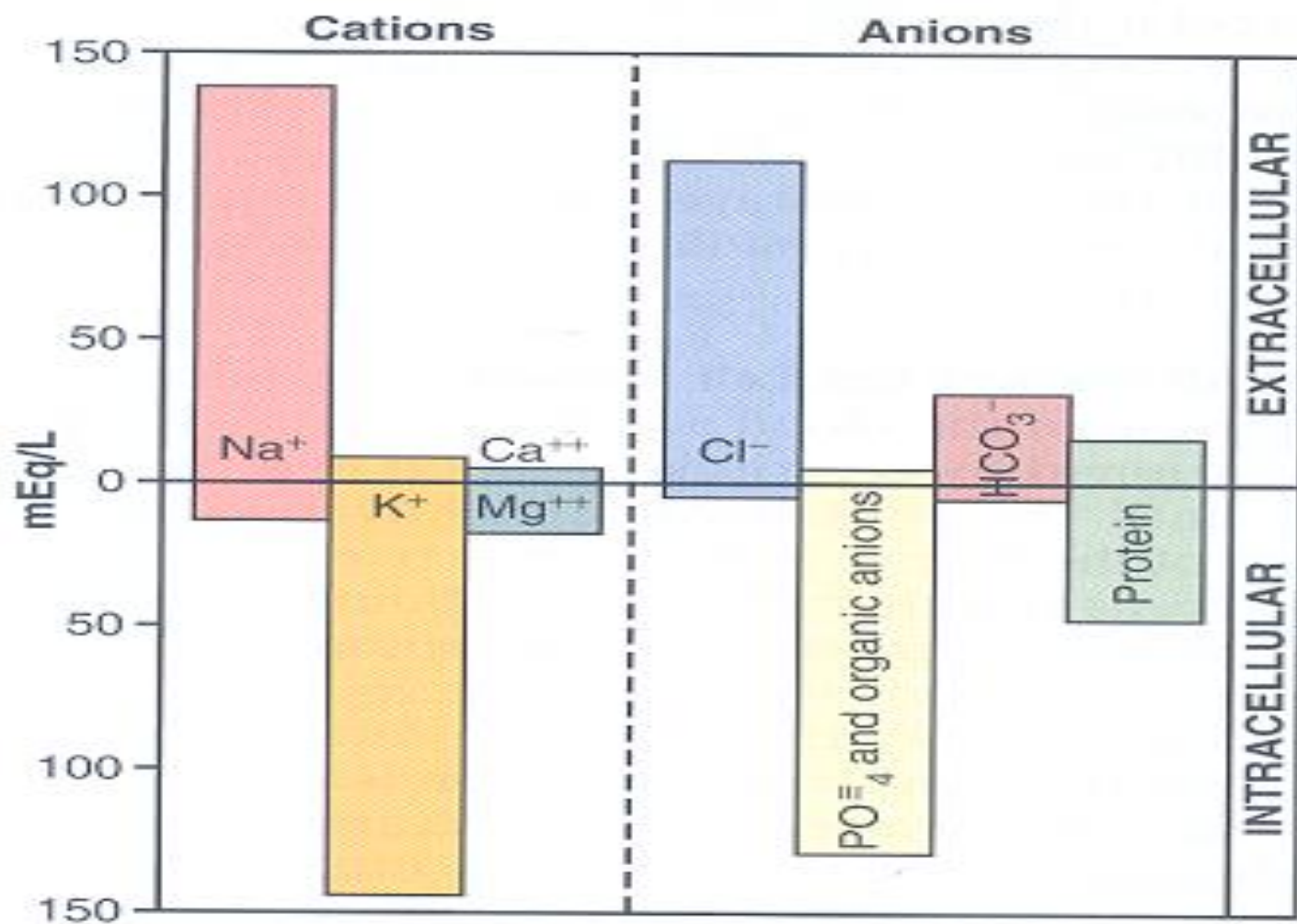


Fig..Major cations & anions of ECF &ICF

❖ HCO_3^- = 22-26 meq/L

= used for acid base balance

= major form is CO_2 is carried to lungs.

❖ Ca^{++} = 8.5 – 10.5 meq/L; 4.5- 5.1 mmol/l(ionized)

= essential for **muscle contraction**.

= “ “ **nerve action potential**.

= “ “ **clotting of blood**

❖ Mg^{++} = 1.8-3.0 mg/dl

= used for cofactor of many critical enzyme.

= metabolism, neuromuscular, enzyme activator

❖ (HPO_4^{-2}) = 2.5- 4.5 meq/dl

= Bone mineralization's

= Acid base balance

= Energy transfers with in cells like ATP

Electrolyte disorders

- Electrolytes are ionized molecules found through out blood, tissue and cells of body.
- These molecules **conduct an electric current** & help to **balance pH & acid base level in the body**.
- It is an **imbalance of certain ionized salts**.

Causes

- ✓ Medications
- ✓ Chronic disease
- ✓ Trauma

Sodium

✚ Hypernatremia: Elevated serum Na level.

Cause

- . Fluid loss
- . Sodium retention
- . Certain drugs
- . Inadequate H₂O intake(**common**)

S/S

- | | |
|---------------------------|----------------|
| . Thirst | . irritability |
| . hypotension | . fatigue |
| . dark concentrated urine | . lethargy |
| . dry mucous membrane | . muscle twit |
| . loss of elasticity skin | |

Management

- **Hypotonic saline solution**
- **Non saline eg. D5W**

TYPE OF HYPERNATREMIA

FLUID VOLUME AND SODIUM (NA) STATUS

CAUSES

Hypovolemic (most
common)

TBW decreases in a greater
proportion than Na is lost

Nonrenal: Fever, vomiting, diarrhea, exercise,
heat exposure, severe burns, insensible loss
from mechanical ventilation, profuse diaphoresis

Renal: Diuresis, severe hyperglycemia, increased
production of urea (high-protein diet), intrave-
nous (IV) administration of mannitol

Hypervolemic (least
common)

TBW is normal with in-
creased Na

Overadministration of saline solutions, particularly
in patients with diabetes ketoacidosis and os-
motic diuresis; overadministration of hyper-
tonic salt solutions; overingestion of salt

Euvolemic

TBW is decreased relative to
a normal total body Na

Acute diabetes insipidus; hypodipsia in infants,
elderly persons, and debilitated adults

Hypонатермия: Low Na serum level .

- most common electrolyte disorders.

Cause

- Water intoxication
- Drugs
- Impairment of adrenal gland & kidney
- Low dietary intake of Na

S/S

- paralysis
- Gastro intestinal symptoms
 - seizure
- Headache
 - coma
- Edema
 - Muscle weakness
- Slowed breathing

Mgt

- ✓ Sodium replacement
- ✓ Water restriction
- ✓ Hypertonic saline

Potassium

 **Hyperkalemia:** High potassium level.

Cause

- ❖ Ketoacidosis
- ❖ MI
- ❖ Sever burn
- **Kidney failure to excrete**
- Diuretics & drugs
- **Hyponatremia**

S/S

- Weakness
- Nausea & diarrhea
- Arrhythmia, **absence/flat p wave, peaked T wave**
- Muscle pain

Mgt :

- Restriction
- Ca gluconate
- NaHCO₃,
- **Insulin**

Hypokalemia - low k level

Cause

- Cushing syndrome
- kidney disease
- Aldosteronism
- Long term diuretic user
- DHN

S/ S

- Weakness
- paralysis
- Arrhythmia, **flat T wave, prominent U wave**
- Increased urinations
- anorexia, vomiting, nausea

Mgt

- Potassium supplement

Calcium

✚ Hyper calcemia - high Ca level.

Cause

- Parathyroid disorder
- Paget's disease
- Multiple myeloma
- Medications like antacids

S/S

- ❖ Fatigue
 - ❖ Constipation
 - ❖ Depression
 - ❖ Confusion
- Muscle pain,
 - N/V
 - DHN
 - Irregular heart beat

Mgt

- Restriction
- Calcitonin

Hypo Calcemia: low serum ca level.

Cause

S/S

- | | |
|------------------------|-----------------------------|
| ➤ parathyroid disorder | - Dry skin |
| ➤ Kidney failure | - Brittle nails |
| ➤ Sever burns | - Facial twitching |
| ➤ Medication | - Tetany/ convulsion |
| ➤ Thyroid ectomy | - Mood change |
| ➤ IBD | - muscle cramp |

Mgt :

- Calcium supplement
- Vitamin D

Magnesium

✚ Hypermagnesemia - excess mg

Cause

- Chronic kidney disease
- Addison's disease
- Over dose of Mg salt

S/S

- ❖ Lethargy
- ❖ Diminished reflex
- ❖ Hypotension
- ❖ Muscle weakness

Mgt

- restriction
- Ca gluconate
- hemodialysis

Hypomagnesemia - low mg level

Cause

- ✓ Low dietary intake
- ✓ Malabsorbtion
- ✓ Burns
- ✓ Diuretics

Mgt

- MgSO₄
- diet

S/S

- . Leg & foot cramp
- . Weight loss, vomiting
- . muscle spasm
- . Seizure
- . Muscle weakness
- . Arrhytemia

Chloride

 **Hyperchloremia**- excess in Cl.

Cause S/S

- ❖ Kidney failure - headache
- ❖ Hemodialysis - nausea
- ❖ Aldosteronism - Cardiac arrest
- ❖ Drugs - weakness
- ❖ Metabolic acidosis

Mgt: restriction, NaHCO_3 - hasten excretion Cl

Hypochloremia- low cl level.

Cause	S/S
➤ Hyponatremia	. Mental confusion
➤ Metabolic alkalosis	. Slowed breathing
➤ Hypokalemia	. Paralysis
➤ Prolonged vomiting	. Muscle tension & spasm
➤ GI suctioning.	
➤ Mgt: supplement(e.g. tomato), KCl, ammonium chloride(to decrease metabolic alkalosis)	

Phosphate

 **Hyper phosphatemia-** excess in HPO_4

Cause

S/S

- Kidney failure
 - Hypoparathyroidism
 - Ketoacidosis
 - Systemic infections
 - Blood transfusion
 - Over intake
- . Tingling sensation
 - . **Muscle spasm(tetany)**
 - . convulsion
 - . Cardiac arrest

Mgt: restriction, hemodialysis

Hypophosphatemia: low phosphate ion.

Cause

S/S

- | | |
|--------------------|-------------------|
| ➤ Sever burn | . Muscle weakness |
| ➤ Kidney disease | . Weight loss |
| ➤ Malnutrition | . irritability |
| ➤ Vit D deficiency | . Bone deformity |

Mgt

- Oral: 0.5–1 g elemental phosphorus PO bid – tid
- IV: Phosphate infusion IV over 6 hr

Acid Base Imbalance

- Normal Acid Base Balance

Normal PH 7.35-7.45

Bicarbonate: carbonic acid = 20:1

Compatible with life (if PH= 6.8-8.0)

- **Acid** = substance that contain H^+ ions.
- As CO_2 increases, carbonic acid increase, H^+ ion increase and PH drops (≤ 7.35).
- **Base** = substance that can accept H^+ ions
- As HCO_3^- increase, H^+ decreases and PH rises (≥ 7.45).

Acid- Base Imbalance

- Ratio of 20 to 1 is out of balance.
- **Acidosis** =
 - PH falls below 7.35
 - Increase in blood H_2CO_3 or
 - Decrease in
- **Alkalosis** - PH greater than 7.45
 - $\uparrow \text{HCO}_3^-$ or
 - $\downarrow \text{H}_2\text{CO}_3$

Causes

- **Metabolic**: change brought by systemic alterations
- **Respiratory**: change brought by respiratory alteration.

Regulations

1. **Respiratory regulations:** it is **a rapid mechanism** of control includes
 - Hyperventilation – blowing off CO_2
 - Hypoventilation - retain of CO_2
 - Measured by $\text{PaCO}_2 = 35\text{-}45\text{mmHg}$ (normal).
2. **Renal regulation:** it is slow (hours to days to take change PH). Mechanisms include like;
 - Excretion and retention of H^+ & HCO_3^-
 - Normal serum HCO_3^- level is = $22\text{-}26\text{mEq/L}$

Types of imbalance

1. Respiratory Acidosis

- ✓ ratio is changing
- ✓ excess CO_2 that results excess H_2CO_3
- ✓ exhaling of CO_2 inhibited
- ✓ Cause - hypoventilations
- ✓ Compensation: **Kidney retains HCO_3^-**

Causes

- **Depression of respiratory center in the medulla:** head injury, drug ingestion (anesthetics, opiates, barbiturates, ethanol)
- **Decreased amount of functioning lung tissue:** bronchial asthma, chronic bronchitis, emphysema, pneumonia, hemothorax, pneumothorax, pulmonary edema
- **Airway obstruction:** foreign body aspiration, sleep apnea, bronchospasm, laryngospasm
- **Disorders of the chest wall:** flail chest, impaired diaphragm movement (pain, splinting, chest burns, tight chest or abdominal dressings)
- **Abdominal distention:** obesity, ascites, bowel obstruction
- **Disorders of respiratory muscles:** severe hypokalemia, Guillain-Barre' syndrome, poliomyelitis, myasthenia gravis

C/M

- ↓RR
 - Shallow breathing
 - Dyspnea
 - Tachycardia
 - Headache
 - Decreased responsive
 - Diminished reflexes
 - Nausea & vomiting
- Disorientation
 - restless
 - agitation
 - fatigue
 - weakness
 - seizure

Mgt

- Oxygen therapy is required to treat the hypoxemia
- Sodium bicarbonate
- Bronchodilators
- Antibiotics are prescribed for respiratory infections

2. Respiratory alkalosis

- ↑ exhaling of CO_2
- ↓ H_2CO_3
- Cause: hyperventilation
- Compensation: Problem is excess blowing off CO_2 and result decrease in H_2CO_3 & increase HCO_3^- . **Response is kidney excretes excess HCO_3^-**

Causes

- **Hyperventilation** due to hypoxemia (a decrease in the oxygen content of blood):
- **Mechanical overventilation.**
- **Anemia;**
- **Hypotension;**
- **High altitudes;** and
- **Pulmonary disease**, such as pneumonia, interstitial lung disease, pulmonary vascular disease, and acute asthma
- **Direct stimulation of the central respiratory center:** anxiety, pain, fever, sepsis, salicylate ingestion, head trauma, central nervous system (CNS) disease (inflammation, lesions)

C/M

- Hyper ventilation
 - Breathlessness
 - Cardiac dysrhythmias
 - Hypotension
 - Syncope
 - Vertigo
 - Anxiety
 - Nervousness
 - Decreased level of consciousness
- ↓ psychomotor
 - **Hyperreflexia**
 - Muscle cramp
 - twitching
 - tetany
 - seizure
 - Confusion

Mgt

- Antianxiety medications
- If the cause of the hyperventilation is hypoxemia, oxygen therapy is needed

3. Metabolic acidosis

➤ HCO_3^- deficient

➤ Low PH level

➤ Cause :

- Relative gain in H^+ (lactic acidosis, keto acidosis)
- Actual loss of HCO_3^- (renal failure, diarrhea).

➤ Compensations :

- Problem: low HCO_3^- or high H^+ ion
- Response: **Lungs hyperventilate** result get rid of CO_2 leads to (decrease PaCO_2 and therefore raise level of HCO_3^-).

Causes

- **Decreased acid excretion**: chronic renal disease results in decreased acid excretion and is the most common cause of chronic metabolic acidosis
- **Excessive acid production**: oxygen tissue deprivation with **shock** and cardiopulmonary arrest, vigorous exercise (transient), prolonged periods of fever, ketoacidosis in insulin-dependent diabetics, alcoholic ketoacidosis,
- **Underproduction of bicarbonate**: **pancreatitis**
- **Excessive loss of bicarbonate**: severe diarrhea; intestinal obstruction; small bowel, pancreatic, ileostomy, or biliary fistula drainage
- **Hyperchloremic acidosis**, an increase in the extracellular concentration of chloride, **also promotes bicarbonate loss**

C/M

- ❖ Tachypnea
 - ❖ Kussmal breathing
 - ❖ Dysrhythmias
 - ❖ Hypotension
 - ❖ Poor perfusion
 - ❖ Unresponsiveness
 - ❖ Decrease pulse
 - ❖ Decreased urine out put
 - ❖ CHF
- Fatigue
 - Drowsiness
 - confusion
 - Apathy
 - seizure
 - nausea and vomiting
 - Distension & pain

Mgt

- Sodium bicarbonate

4. Metabolic Alkalosis

- **HCO_3^- excess**
- High PH
- Loss of H^+ ion or gain of HCO_3^-
- Compensations: Problem: too much base. Response: **lungs compensate by hypoventilates.** (Retain CO_2 & $\uparrow \text{paCO}_2$ = $\uparrow$ blood acid level)

Causes

- Most common: **vomiting and nasogastric (NG) suctioning**
- Other:
 - **ingestion of bicarbonates**, carbonates, acetates, citrates, and lactates found in total parenteral nutrition solutions,
 - **Ringer's lactate, and sodium bicarbonate administration;**
 - **Excessive intake of antacids**, which are composed of sodium bicarbonate or calcium carbonate;
 - **Loss of acids (gastric fluid loss**, diuretic therapy, excessive mineralocorticoid release);
 - **hypercalcemia;**
 - **Hypokalemia**

C/M

- Hypoventilation
 - Dysrhythmia
 - Decreased perfusion
 - Hypotension
 - Confusion
 - Lethargy
 - Unresponsiveness
 - Muscle cramp
 - twitching
- Hyper reflexia
 - Tetany
 - seizure
 - N&V
 - diarrhea

Mgt

- IV saline solutions to replace lost volume and chloride ions
- Potassium supplements,
- Carbonic anhydrase inhibitors: Acetazolamide promotes the renal excretion of bicarbonate.
- Potassium chloride is used to treat hypokalemia in a patient with metabolic alkalosis
- Reduce the production of hydrochloric acid in the stomach

Disorder of Urinary Tract

Infection of urinary tract

- It is classified as:
 - ✓ **Upper UTI**: Pyelonephritis
 - ✓ **Lower UTI**: cystitis, prostatitis & urethritis
- **Risk factor**
 - Failure to empty bladder
 - Obstructed urine flow
 - Low host defense
 - Instrumentation
 - Gender
 - Sexual activity

Lower UTIs

- Most UTIs result from fecal organisms.

Causes

- Gram negative bacteria
- Staphylococcus
- Klebsiella
- Enterobacter

Route of infection

- ✓ Up the urethra
- ✓ Hematogenous
- ✓ Direct extension

1. Cystitis

- An inflammation of the urinary bladder

Causes:

- Bacteria
- Toilet hygiene
- Catheterization
- BPH
- Pregnancy
- Honey moon cystitis **(increased frequency of sexual activity)**
- Veneral disease
- Parasities

C /m

- Dysurea
- Frequency
- **Cloudy or foul smelling urine**
- **Supra pubic pain**
- Urgency, nocturia , incontinency
- Hematuria & back pain may also present

○ For children <5 Yrs

- loss of appetite
- Irritability
- Weakness
- vomiting

DX:

- Hx & p/E
- U/A - colony count
- urine culture

2. Urethritis

- inflammation of urethra is usually caused by infections.

C/M

- Burning sensation or pain on urination
- Frequency - N/V
- Hematuria - lower back pain
- Fever & chills

Medical management for lower UTIs

If uncomplicated:

- Ciprofloxacin, 500mg P.O., BID, for 3 days *OR* Norfloxacin, 400mg P.O., BID, for 3 days.
- Cotrimoxazole 160/800mg P.O, BID for 3 days

 **If complicated:** similar to uncomplicated UTIs but needs prolonged duration and response should be closely followed as gram positives could be the cause

Nursing management

- Analgesics & applying heat to the perineum
- Encourage the patient to take enough amount of H₂O.
- Avoiding urinary tract irritants.
- Frequent voiding
- During urination , empty the bladder completely

Upper urinary tract infection

Pyelonephritis

- Bacterial infections of
 - * renal pelvis
 - * Tubule
 - * Interstitial tissue

Two types

1. Acute pyelonephritis:

- sudden inflammation which is caused by bacteria, primarily renal pelvis & interstitial tissue.
- Most common renal disease
- Known as acute infective tubulointerstitial nephritis.

Cause:

- bacteria like
 - E. coli
 - Proteus
 - Pseudomoniasis
 - Staphylococcus aureus
 - Streptococcus
- **More common in women's B/c of**
 - Shorter urethra
 - Proximity of the urinary meatus to the vagina .

Risk factors

- Sexually active women
- Pregnant women
- Diabetics
- People with other renal disease.
- Inability to empty the bladder
- Urinary obstructions

Sign and symptoms

- ✓ Urgency, frequency
 - ✓ Dysuria , nocturia
 - ✓ **Cloudy urine**
 - ✓ Pyrexia
 - ✓ Shaking chills
 - ✓ Flank pain
- anorexia
 - **CVA tenderness**
 - Hematuria

Dx

- Requires urinalysis and culture
 - Polyuria
 - low specific gravity
 - Bacteriuria
 - alkaline pH
- X-rays
- Ultrasound or CT scan

Management

- For out patient a 2 weeks course of antibiotics
 - Amoxicillin, cephalosporine, levofloxacin and ciprofloxacin.

2. Chronic pyelonephritis

- It is **persistent kidney inflammation** that can scar the kidneys and may lead to chronic renal failure.

Cause

- ❖ Persistent acute pyelonephritis (result from)
 - * Urinary tract anomaly
 - * Urinary obstruction
 - * Vesicoureteral reflux

C/M

- Fatigue
- Headache
- Poor appetite
- Polyuria
- Excessive thirst
- Weight loss

DX

- HX & P/E
- Creatinine Clearance
- BUN
- Bacteriuria

Complication

- **Hypertension**
- Kidney disease(chronic)

Medical Mgt:

- Antimicrobial agent based on identified pathogen
- Increased fluid intake

Nursing management

1. Monitor intake & out put
2. Assess temperature every 4 hours and administer antipyretics & antibiotics as prescribed.
3. Patient education on prevention of UTI through
 - adequate fluid intake.
 - Empty the bladder regularly
 - Perineal hygiene
 - Wiping from front to back
 - Cleaning after sexual intercourse

Uretric disorder

- Primary disorder of the ureters occurs **less frequently** than disease of the other part of the urinary system.
- A **congenital anomaly** or rarely **a neoplasm** may occur in ureters.
- The **ureters vesicle defect permits a urinary reflux** a back flow of urine in to the ureter from the bladder this predispose pyelonephritis.
- **Surgery** is carried out to correct the defect primary neoplasm of the ureters is quite rare and on tails extensive surgery.
- **Ureteritis occurs with pyelonephritis.**
- **Calculus** may become lodged a ureters causing an obstruction of the flow of urine as well as severe pain.
- **Surgical Rx** may be necessary to remove obstruction.

Immunologic renal disorders

- They are primary glomerular disease.
 - Acute glomerulonephritis
 - Chronic glomerulonephritis and
 - Nephrotic syndrome.
- The glomerular capillaries are primarily involved.
- Injury to the glomeruli is because of the two immune mechanisms:
 - Injury resulting from circulating **antigen antibody complex**.
 - Injury resulting from **antiglomerular antibodies** reacting with fixed glomerular antigen.

1. Acute Glomerulonephritis(AGN)

- **sudden onset of hematuria, proteinuria, and red blood cell casts** in the urine, and it may be accompanied by hypertension, edema (sodium and water retention), and decreased glomerular filtration.
- A specific set of renal disease in which an **immunologic** mechanism **triggers inflammation and proliferation of glomerular tissue.**

Cause

- Post infectious streptococcus species
- Bacteria other than group A streptococci
- Other specific agent like viruses and parasites.

C/M

- **Hematuria b/c of RBCS**
- **Protien plugs & erythrocytes.**
- **Proteinurea**
- **BUN & serum creatinine may rise**
- **Edema in dependent area.**
- **Hypertension**
- Headache, malaise, flank pain in sever case.

DX

- HX & P/E
- Lab studies
- ↓ed GFR to 50ml/min
- ↑ed BUN
- **↓ed serum albumin**
- renal biopsy

Complication

- ✓ Hypertensive encephalopathy
- ✓ CHF
- ✓ Pulmonary edema
- ✓ Acute & chronic renal failure.

Medical mgt

- Loop diuretics
- Na & water restriction
- dietary protein restriction

Nursing management

- Carbohydrate are given to provide energy & to **reduce catabolism of protein**. Input and out put measurement.
- Personal hygiene
- Basic infection control principles.
- Follow up evaluation of BP, U/A for proteinuria, serum BUN & creatinine level.

2. Chronic glomerulo nephritis

- It is advanced stage of a group of kidney disorders, resulting in inflammation and **gradual, progressive destruction of the glomeruli.**

Cause

- Repeated attack of AGN

C/M

- ❖ Elevated BUN
- ❖ Sudden, severe nose bleeding
- ❖ Feet slightly swollen at night
- ❖ Chronic renal failure may be developed.

DX

- Hx & P/E
- U/A
- Protein urea (usually $< 2\text{g /in a 24 h}$).
- RBC casts
- Serum creatinine level increase
- increased BUN
- X-ray: may show cardiac enlargement and pulmonary edema.

Mgt

- Depend on symptom guide
 - If the patient has hypertension
 - sodium & H₂O restriction
 - antihypertensive agent.
 - Monitor daily weight
 - Diuretics for fluid over load
 - Dialysis or transplantation to prevent death.
 - Emotional supports.
 - Observing the patient for changes in fluid and electrolytes status of determination of renal function.

Nephrotic syndrome

- Is **not specific glomerular disease** but a constellation of clinical finding result from **increased permeability to the plasma protein.**
- Generally considered a *disorder of childhood* but may be occur in adults & elderly.

It is characterized by:

- Massive protein urea ($>3.5\text{g/dl}$).
- Hypoalbuminemia ($<3\text{g/dl}$).
- Edema
- Hyperlipidemia(300mg/dl)

Causes

- ❖ Chronic glomerulo nephritis
- ❖ DM with intracapillary glomerulosclerosis
- ❖ Amyloids of kidney
- ❖ Multiple myeloma
- ❖ Renal vein thrombosis

Pathophysiology

- Damaged glomerular capillary membrane → ↑ permeability of glomerular capillaries to protein → Protein urea (>3.5 g/24hr) → Hypoproteinemia (albumin <3 g/100ml) → ↓ plasma oncotic pressure → compensatory synthesis of protein including lipoprotein by liver → **hyperlipidemia**
- ↓ plasma pressure → ↓ plasma volume → ↓ GFR → activation of rennin angiotensin system → ↑ aldosterone secretion → Na & H₂O retention → Edema
- ↓ plasma pressure → Fluids escape into tissue retention

C/M

- **Hypoalbuminemia**
- **Edema**
- **Hyperlipidemia**

DX

- HX & P/E
- U/A
- Needle biopsy of the kidney.

Complication

- Infection
- Thromboembolism
- Accelerated atherosclerosis
- Acute renal failure

Mgt

- ❖ Loop diuretics and low sodium Administration (2-3g/day).
- ❖ Corticosteroid drugs
- A major nursing intervention for patient's with nephritic syndrome is edema management

Renal failure

- Renal failure is severe impairment or total lack of kidney functions (inability to remove waste and metabolic end products).
- It can be Acute & chronic

A. Acute renal failure

- It is sudden & almost loss of kidney function ($\downarrow$ GFR), with progressive azotemia.

Causes

1. **Prerenal** (hypoperfusion of kidney): 50-70 %
- Factors outside the kidneys that **impair renal blood flow**:

- Hypovolemia - Hemorrhage. - Burns
- dehydration - GI loss
- Vasodilatation - sepsis - vasodilators
- Anaphylaxis
- ↓cardiac output
 - MI - dysrhythmias, hypertension
 - HF - Cardiac shock

II. **Intrarenal cause:** 20-30%

- It leads to **actual damage to the renal tissues** (parenchymia) resulting in malfunctioning of nephrons.

It may include nephrotoxic injury caused by:

- Drugs (like aminoglycoside, **NSAID** & **ACEI**)
- Heavy metals
- Hemolytic blood transfusion reaction
- Primary renal disorders (APN, AGN)
- Toxemia of pregnancy /eclampsia
- Malignant hypertension

III. **Post renal causes** (1-10%) involve **mechanical obstruction of urinary out flow** some where distal to the kidney.

- Prostate cancer
- BPH
- Calculi
- Trauma
- Tumors
- Possible pathologic process involved in ARF include:
 - Renal vasoconstriction
 - Cellular edema
 - Decreased glomerular capillary permeability.
- Clinical course of ARF
- It may progress through the phases of **initiation, oliguria, diuresis and recovery.**

1. **Initiation phase**: begins with the initial insult and ends when oligurea develops.
2. **Oliguric phase**:
 - Urinary volume less than 400 to 600 ml/day
 - Is the **most cardinal symptom**
 - There may be ARF without oliguria (High output failure)
 - Azotemia (abnormal collection of nitrogenous wastes)- rise in BUN and creatinine
 - Circulatory congestion
 - Hyperkalemia, hyperphosphatemia, hyperuricemia
 - **Metabolic acidosis**
 - **Anemia** due to depressed erythropoiesis, uremic GI lesion and ↓ RBC lifespan.
 - Lasts approximately 10 days

3. Diuretic phase

- The pt experiences gradually **increasing urinary output**, which signals that glomerular filtration has started to recover.
- Although urinary output may reach normal, renal function may be markedly abnormal.
- Urinary output doubles from oliguric phase
- **Hypokalemia is danger in this phase.**

4. Recovery phase

- Signals improvement of renal function
- Marked by fall in BUN and creatinine
- May take 3 to 12 months.

C/M

- Lethargic
- Nausea, vomiting and diarrhea
- Dry skin and mucous membrane
- breath odor of urine
- CNS manifestation like drowsiness, headache, convulsion.
- Acidosis
- Hyperkalemia, dysrhythmia & cardiac arrest.
- Anemia

Complications

- Cardiac - HPN, arrhythmia, pericarditis & CHF
- Neurologic - coma
- GI – hemorrhage

Management of ARF

Depends on the causes

- Maintain Sodium and water balance
- Measure body wt.
- Increase carbohydrates to spare catabolism of protein (oliguric phase).
- Restrict food & fluids containing potassium and phosphorous during **oliguric phase**

- After diuretic phase, the pt is placed on a high protein, high calori diet.
- ***IV glucose & insulin*** as an emergency and temporary measure to treat hyperkalmia.
- NaHCO_3 to treat acidosis
- Dialysis if there is excess serum urea, creatinine, hyper kalemia, pericarditis, neuropathy, encephalopathy.
- Measure daily urine output & input
- No electrolyte is permissible in oliguric phase but during diuresis phase
- Treat HPN, if any

Nursing Intervention

- Assess fluid and electrolyte status
 - Daily weight changes
 - precise intake and output balance
 - skin turgor & presence of edema
 - Distension of neck veins
 - BP, PR, RR
 - Identify potential sources of fluid

- Explain to patient and family rationale for restriction of certain foods and fluids.
- Assess nutritional status
- Weight change
- Anthropometric measures
- Encourage ***high calori, low protein*** (the protein given should be of having high biological value i.e animal origin), ***low sodium and low potassium***.
- Provide oral hygiene before meals

B. Chronic Kidney Disease(CKD)

- It is a **progressive, irreversible** deterioration in renal function in which the body's ability to maintain metabolic and fluid and electrolyte balance fails, **resulting in uremia.**

Cause

- **Hypertension(Leading)**
- Glomerulonephritis
- diabetis
- Pyelonephritis
- obstruction of UTI
- vascular disorder
- Drugs
- cystic kidney disease

Stages of Chronic Kidney Disease

Stage	Description	GFR* mL/min/1.73m ²
1	Slight kidney damage with normal or increased filtration	More than 90
2	Mild decrease in kidney function	60-89
3	Moderate decrease in kidney function	30-59
4	Severe decrease in kidney function	15-29
5	Kidney failure	<15 (or dialysis)

C/M

- 35 to 50% reduction in renal function is tolerated and pts are symptom free
- When GFR falls to 20-35% of normal, azotemia appears
- GFR of 10-20% of normal, azotemia is pronounced and there is overt renal failure.
- Electrolyte disturbance
 - Hyponatremia
 - Hyperkalemia
 - Acidosis
 - Hypocalcemia
- Endocrine disturbances
 - Renal osteodystrophy → due to change in Ca, P & parathormon imbalance.
 - Osteomalacia → due to ↓ 1, 25 dihydroxy - cholecalciferol
- Metabolic disturbances
 - Hyperuricemia
- Impotence, amenorrhea

- Gastro intestinal
 - Anorexia, nausea and vomiting
 - Peptic ulcer
 - GI bleeding
- Haematologic disturbances
 - Anemia
 - Increased susceptibility to infection
- Neuromuscular
 - Peripheral neuropathy
 - Fatigue, head ache & insomnia
- CVS disturbance
 - HPN, pericarditis
- Dermatological disturbances
 - Pruritus

Management

- Reduced protein intake
- Salt restriction
- Restrict potassium containing foods
- Calcium supplement along with synthetic vit.D to minimize symptoms of renal osteodystrophy.
- Antihypertensive
- H2 - blockers → to minimize peptic ulceration & bleeding
- Dialysis
- Renal transplantation

Renal cancer

- It is very **rare** tumor
- Wilm's tumor - it occurs in children
- Renal cell carcinoma
- It may metastasize to lungs, bone, liver, brain and lateral kidney
- 1/4 to 1/2 of pts will have metastatic disease at the time of diagnosis

Clinical Manifestation

- Classic triads, occurring late in the course of the disease is:
- Hematuria (**painless hematuria**)
- Pain → dull pain in the back
- Mass in the flank
- colicky pain if clot of tumors pass down the ureter
- Other - unexplained wt loss, Increased weakness, anemia

Rx

- Entirely depends on surgery. **It is not sensitive** to radiotherapy and chemotherapy.
- **Radical nephrectomy is the preferred treatment if the tumor can be removed- this include removal of.**
 - i. The kidney (and tumor)
 - ii. Adrenal gland
 - iii. Surrounding perinephrotic fat
 - iv. Lymph nodes
 - v. Radiation therapy, hormonal therapy, or chemotherapy may be **used along with surgery**
 - Vi. Immunotherapy may also be helpful

Nursing intervention

- The patient with a renal tumor may undergo

A. Extensive diagnostic and therapeutic procedures including

- Surgery, radiation and chemotherapy

B. Following surgery, the patient usually has catheters and drains in place to maintain patient urinary tract, to remove drainage, and to permit very accurate measurement of urine out put.

Based on the above, the patient requires

- A. Frequent **analgesia** during the postoperative period .
- B. Assistance with turning (**frequent positioning**)
- C. Encourage the patient in turning, **coughing, and taking deep breaths .**
- D. Assist and support the patient and family to cope with the diagnosis and uncertainties about the prognosis
- E. Follow up care is essential to detect signs of metastases
- F. Reassure the patient and family about the patient continued well-being

Cancer of the bladder

- It is **more common at age of 50 to 70%**
- M:F, 3:1 & frequent in people exposed to chemical carcinogens.
- It can be transitional cell carcinoma (account 80 - 90%) or squamous cell carcinoma.
- Cigarette smoking is **predominant cause**

Form of bladder cancer

1. Superficial bladder cancer

- Account 70% tends to recurrence

2. Invasive

- solid tumor
- metastasis to lung, bone and liver

Clinical manifestation

- urinary retention form blood clot
- pain, deep in pelvis, indicating spread out of bladder territory
- **painless hematuria**
- recurrent UTI

Dx

- urine for WBC & RBC
- Blood creatinine, BUN
- Cystoscopy

Management

- Surgery
- Chemicals

Obstruction of the urinary system

Ureteral constriction

- Narrowing of the ureters

Cause

- Infection, foreign body
- congenital anomaly and tumors.

Clinical Manifestation

- Patient shows all the sign and symptom of infection Eg.
Pain

Mx

- Surgical intervention - Uretroplasty & anastomosis
- Analgesics / Antispasmodics

Urethral Stricture

- Urethral Stricture is a narrowing of the lumen of the urethra due to **scar tissue and contraction**

Causes

- Injury: insertion of surgical instruments during transurethral surgery, indwelling Catheter or cystoscopic procedure.
- Straddle Injuries
- Automobile Accident
- Untreated Gonorrhea
- Congenital Abnormality

Clinical Manifestations

- The force & size of **urinary stream is diminished**
- symptoms of urinary infection and **retention** occur.
- Stricture cause urine to back up resulting in **cystitis, prostatitis** and **pyelonephritis**

Mx

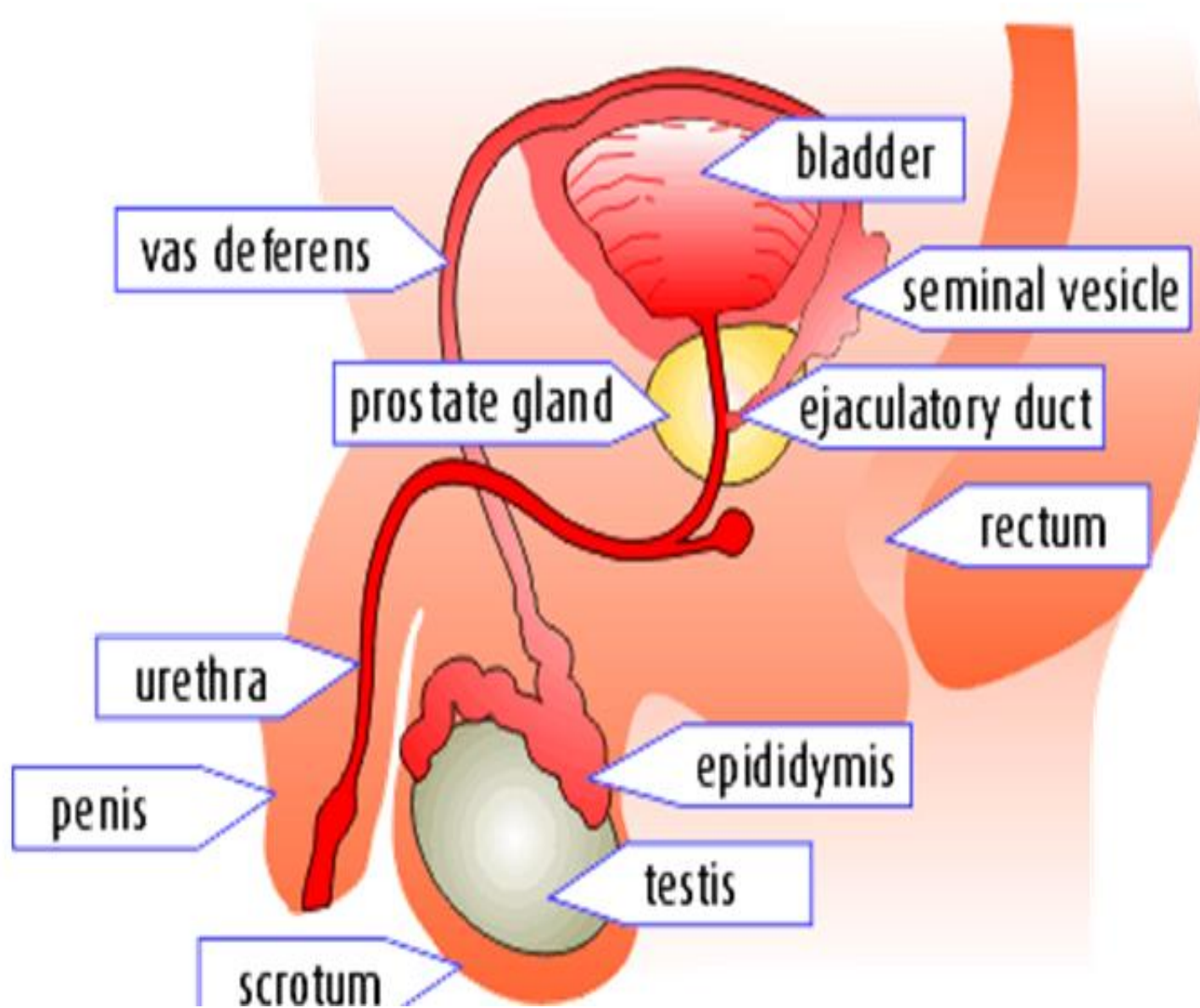
- Dilatation of the narrowed area with metal sounds or buggies.
- Operation under direct vision internal urethrotomy
- Fixation of boogies passes beyond the stricture in to the bladder - urine will drain from the bladder.
- Hot sitze bath and non - narcotic analgesics are given to control pain.
- Antimicrobial drugs are given for several days after dilatation to minimize infection.
- Surgical excision or urethroplasty may be necessary for sever cases.
- Rarely a temporary cystostomy is necessary, b/c of sever retention.

Disorder of Prostate Gland

- **The Prostate Gland:** The prostate is a gland of the male reproductive system.
- The prostate produces some of the **fluid for semen, which transports sperm during the male orgasm.**
- it is nearly the same size and shape as a **chestnut.**
- It is located in **front of the rectum,** just below the bladder, and wraps around the urethra, the tube that carries urine from the bladder out through the tip of the penis.

...

- The prostate is made up of approximately 30% muscular tissue, and the rest is glandular tissue.
- 45% of men in their 60s age are affected by erectile dysfunction, nearly 30% of all men will require intervention for benign prostatic disease during their lifetime.
- **Prostate cancer** is the most commonly diagnosed non-skin cancer in men.



1. Prostatitis (male reproductive system)

- Prostatitis is **inflammation of prostate gland** caused by infectious agent (bacteria, Fungi and mycoplasma) or by a variety of other problems.

Cause

- Infectious agents - bacterial, fungi & mycoplasma
- Urethral stricture & hyperplasia of prostate
- Microorganisms usually are carried to the prostate from the urethra.

...

Clinical Manifestation

- Perineal pain and discomfort
- Urethrtitis, Urgency, frequency and dysuria
- **Prostatodynia (pain in the prostate) on voiding.**
- Acute bacterial Prostatitis may produce **a sudden on set or fever & chills** and perineal, rectal, **low back pain** and dysuria may be evident.

...

Mgt

- The goal of treatment is to avoid the complication of abscess formation and septicemia.
- A broad spectrum antimicrobial drugs for 10 - 14 days, I.V administration of the drug may be necessary to achieve high serum and tissue level.
- **Bed rest, antispasmodics, laxatives** to soften stool and sitz bath
- Patient education, therapy, fluid intake but not force fluid, diet and drinks which have diuretic action e.g. Coffee, Tea, alcohol, coca etc--- **increase prostatic secretion should be avoided, avoidance of sexual intercourse, prolonged sitting also be avoided** and medical follow up for at least 6 months to 1 year.

2. Benign Prostatic Hypertrophy (hypertrophy) BPH

- The **most common problem** of the adult male reproductive system is benign prostates hyperplasia (BPH).
- This problem occurs in about 50% of men over 50 years of age and 75% of men over 70 years.
- The prostate gland **enlarges extending up ward** in to the bladder and **obstructing the outflow of urine** by encroaching on the vesical orifice.

...

Cause (Etiology)

- Uncertain but evident suggests **a hormonal cause** as initiating hyperplasia of the supporting stromal tissue and a glandular element in the prostate.

...

Clinical Manifestation

- Increasing potency of urination, nocturia, hesitancy in starting urination
- a sensation of incomplete emptying of the bladder,
- urine dribbles out after urination,
- an acute urinary retention (infection)
- fatigue secondary to nocturia, anorexia, nausea and vomiting due to impaired renal function
- **epigastria** discomfort due to distended bladder
- hematuria, uremia at the **later stage**.

Assessments and Diagnosis

- Digital rectal examination → reveals a prostate gland that is **large, rubbery, and non tender**
- Signs and symptoms
- Cystoscopy
- RFT to assess renal impairment

Management

- The plan of treatment depends on the cause, the severity of obstruction and the condition of the patient (Mx also depends on age)

- **Catheterization** to treat an acute urinary retention.
- Some times a **suprapubic cystostomy** to give adequate drainage.
- Water and electrolyte **replacement** if necessary.
- Antimicrobial drugs may be necessary to treat UTI

Medication

- ❖ **α blockers** - relax the smooth muscle of prostate and provide a larger urethral opening (Hytrin, Doxaben, Harnalidge)
- **5 α reductase inhibitor:** Shrink the prostate gland (Proscar, Avodart)
- .

- **Surgery (prostatectomy):** to remove the hyper plastic prostate tissue to provide permanent relief of the obstruction
- Because **hemorrhage** is a major complication of prostate surgery, all clotting defects must be corrected Surgical approaches Includes
- ✓ TURP (Trans urethral resection of prostate) is a removal of prostatic tissue by instrument introduced through urethra.
- ✓ Invasive “open” procedures

3. CANCER OF THE PROSTATE GLAND

- Cancer of the prostate is the **second** most common cause of cancer.
- The most prevalent cancer overall in black men with increasing number of men in the old age group greater attention will be focused on this condition .

Clinical manifestation

- Early Ca of the prostate does not usually produce symptoms
- if the neoplasm is large enough to encroach on the bladder neck and **cause obstruction of urine**, there are signs and symptoms of obstruction namely.
- Difficulty & frequent urination
- Urinary retention and
- Decreased size and force of urinary stream

Method of Diagnosis

- Finger rectal examination; palpable of it is **advanced stony hard**.
- Histological examination by surgically transurethral resection, open prostatectomy or needle biopsy perennial or trans rectal.
- **Serum acid phosphate level is frequently increased.**
- Urogram to demonstrate changes from urethral obstruction
- RFT

Mx

- Radical Prostatectomy radiation as a palliative therapy
- hormonal therapy may be selected to suppress all androgenic stimulation to the prostate due to orchiectomy or administration of estrogen
- blood transfusion, analgesics, strict observation.
- V/S, sign of anemia, shock, fluid balance etc...

Hydrocele

- A hydrocele is a collection of fluid generally in the **tunica vaginalis of testis**.
- The tunica vaginalis become widely distended with fluid.

Cause

- Occurs in association with acute infectious disease of the epididymic infectious disease such as mumps .
- The cause of chronic hydrocele is unknown

Mx

- Usually therapy is not required.
- Treatment is necessary only if the hydrocle become tense and comprise testicular circulation or if the scrotal mass **becomes large, uncomfortable or embarrassing, withdrawing the fluid through a large needle or removing the sac of the fluid.**
- **Surgical incision** through the wall of the scrotal down to the distended tunica vaginalis
- Some time sclerosis substance is injected in to the sac after aspirating fluid to cause the wall of hydrocle to become inflamed and disappear.
- Eventually post operative scrotal support is done.

Reading Assignment

- **Orchitis, epididymitis**
- **Priapism**
- **Paraphimosis**
- **Phimosis**
- **Hypospadias and**
- **Epispadias**

Thank
Very
Much